

BOX 17-8 HTLV-I-ASSOCIATED MYELOPATHY/TROPICAL SPASTIC PARALYSIS (HAM/TSP) AND SUBACUTE MYELO-OPTICO-NEUROPATHY (SMON)

HAM/TSP and SMON, characterized predominantly by spastic paraparesis, are noteworthy because there has been a dramatic drop in the number of new cases after the discovery of the etiology in both diseases. HAM/TSP used to be especially prevalent in Martinique and Japan. It was shown to be associated with chronic infection with the human T-lymphotropic virus type I (HTLV-I), the virus known to cause adult T-cell leukemia, and its pathomechanism was clarified by Osame (2002) and others (Aghajanian et al, 2020; Nozuma et al, 2020). It manifests with symmetric spastic paraparesis associated with only mild hyperreflexia in the arms, clinical and electrophysiologic evidence of moderate posterior column involvement at the thoracic level, and urinary frequency and urgency associated with the detrusor-urethral sphincter dyssynergia, suggesting a diffuse white matter lesion predominantly involving the thoracic cord (Shibasaki et al, 1988). SMON has been endemic in Japan since 1955 and was proven to be due to the chronic use of clioquinol, the drug used for diarrhea (Igata, 2010). It manifests with severe painful dysesthesia in the distal portion of the legs and perineum, and spastic paraparesis with or without visual loss. The central distal axonopathy of the dorsal root ganglion was shown to cause severe sensory symptoms pathologically and electrophysiologically (Shibasaki et al, 1982).

BOX 17-9 MALIGNANT SYNDROME AND MALIGNANT HYPERTHERMIA

Rigidity of acute or subacute onset can be seen in **malignant syndrome (syndrome malin)**. This condition commonly occurs with chronic use of neuroleptics or after the patient suddenly stops taking anti-parkinsonian medications (**neuroleptic malignant syndrome**). This is especially common for the antipsychotic drugs blocking the dopamine D2 receptor but may also be seen with other neuroleptics (Nisijima et al, 2007). Clinically, in addition to marked rigidity, it is characterized by hyperthermia, consciousness disturbance, various autonomic symptoms, and elevated serum creatine kinase levels and leukocytosis in the peripheral blood (Tao et al, 2021). It is treated by water, physical cooling, and administration of dantrolene sodium. In terms of hyperthermia, it is similar to serotonin syndrome (see Chapter 4-4, p. 27) and malignant hyperthermia. **Malignant hyperthermia** is induced by inhalation anesthesia and presents with muscle rigidity and respiratory as well as metabolic acidosis in addition to hyperthermia. It is commonly encountered in patients with various kinds of myopathy, among others central core disease. In terms of its pathogenesis, it is considered that the inhalation anesthetics act on muscles to cause excessive muscle contraction and hyperthermia. The mechanism of the malignant syndrome has not been fully elucidated, but a decrease in the central dopaminergic function, block of the dopamine receptors, and abnormal transmission of other catecholamines are also considered possibilities (Nisijima et al, 2007).

discharges following demyelination (Li et al, 2020; Shibasaki & Kuroiwa, 1974). In this case, pain is induced by a somatic stimulus applied to the skin and spreads rostrally as well as caudally along with painful spasm.

A. HEMIFACIAL SPASM

Hemifacial spasm is manifested as repetitive twitches or continuous spasm of the orbicularis oculi and oris muscles or either one of the two muscles. This is due to abnormal electrical discharge of the facial nucleus or facial nerve. The underlying cause is, like trigeminal neuralgia, considered to be demyelination of the facial nerve secondary to its compression by tortuous arteries in the posterior cranial fossa, but it has also been shown that excitability of the facial motor nucleus is increased (see Chapter 11-1A, p. 83). Decompression surgery can be done and can be curative,

but nowadays the local injection of botulinum toxin has been effectively used and may be more common.

B. TETANUS

The α motor neuron is particularly vulnerable to **tetanus toxin** produced by an anaerobic bacterium, *Clostridium tetani*. **Tetanus** is caused by infection of injured skin with the bacteria, and the toxin produced there is transported retrogradely through the α motor fibers to reach the anterior horn or brainstem motor nuclei. At the synaptic terminal, the inhibitory neurotransmitters glycine and GABA are blocked by the toxin, causing massive vigorous spasm. As the toxin has special affinity for the motor nucleus of the trigeminal nerve, the most common initial symptom is inability to open the mouth due to marked spasm

BOX 17-10 WHAT CAN OCCUR IN PARKINSON PATIENTS DURING L-DOPA TREATMENT?

A variety of untoward symptoms can occur in patients with Parkinson disease during long-term treatment with L-dopa or dopa agonists (Aquino & Fox, 2015; Aradi & Hauser, 2020). The most common phenomenon is the **wearing-off phenomenon**, in which the symptoms show diurnal fluctuation as a result of shortened duration of the drug effect (Antonini et al, 2011). By contrast, bradykinesia and rigidity might suddenly worsen and suddenly improve with no relation to the time of drug administration (**on-off phenomenon**). **Dyskinesia** (see Chapter 19-7, p. 177) is seen in about half of patients at 5 years after the initiation of L-dopa treatment (Gottwald & Aminoff, 2011). Dyskinesia and wearing-off occur earlier after the treatment onset especially in patients taking a higher L-dopa dose and those with a younger age at onset (Olanow et al, 2013). In addition, a variety of behavioral abnormalities may appear. In particular, patients may be unable to control their impulses for gambling, shopping, sexual behavior, and appetite (**impulse control disorders**) (Béreau et al, 2019; Corvol et al, 2018; Voon et al, 2011; Weintraub et al, 2015). These disorders are mediated by the nigrostriatal dopaminergic system and another dopaminergic pathway from the ventral tegmental area to the ventral pallidum and nucleus accumbens, which are both related to reward and addiction (Wise, 2009). **Pathologic gambling** is related to functional abnormality of the ventral pallidum, nucleus accumbens, hippocampus, amygdala, insula, and orbitofrontal cortex, and it is more common in patients receiving treatment with dopa agonists than with L-dopa (Djamshidian et al, 2011). As for the tendency to repeat the same behavior (**punding**) and **compulsive behavior**, it has not been determined whether those are related to L-dopa treatment or the pathology of Parkinson disease itself (Spencer et al, 2011). For dopa-induced dyskinesia and punding, a favorable effect of amantadine has been reported (Kashihara & Imamura, 2008).

of the masseter (**trismus**) (see Chapter 10-2A, p. 78, and Chapter 33-4B, p. 289).

C. STIFF-PERSON SYNDROME AND PROGRESSIVE ENCEPHALOMYELITIS WITH RIGIDITY AND MYOCLONUS

Stiff-person syndrome, previously called stiff-man syndrome, is characterized by continuous painful spasms involving the trunk and limb muscles that is dramatically

aggravated by movement or various kinds of stimuli. This condition is considered to be caused by excessive excitability of the anterior horn cells through an autoimmune mechanism, showing antibody diversity (Bernardo et al, 2020). In many cases, levels of antibodies against glutamic acid decarboxylase (GAD) are elevated in the serum and cerebrospinal fluid, which is expected to cause deficiency of GABA, thus impairing the inhibitory transmissions. In some cases of this condition, antibodies against GABAA-receptor-associated protein (GABARAP) are elevated instead of anti-GAD antibody (Raju et al, 2006). Furthermore, stiff-person syndrome can occur as a paraneoplastic syndrome in association with breast cancer and small cell carcinoma of lung, and in these cases, levels of antibody against amphiphysin might be elevated (Murinson & Guarnaccia, 2008). Detection of amphiphysin-IgG autoimmune neuropathy is important because it is often associated with breast cancer (Dubey et al, 2019).

Progressive encephalomyelitis with rigidity and myoclonus (PERM) is similar to stiff-person syndrome but more severe, with additional brainstem or other neurological defects (Balint et al, 2014). A relationship of this syndrome with glycine receptor antibodies or anti-GAD antibody has drawn special attention (Carvajal-Gonzalez et al, 2014; Jazebi et al, 2019).

Isaacs syndrome is also characterized by muscle spasm, but in contrast with stiff-person syndrome, the spasm in this condition is more pronounced in the distal limb muscles. Isaacs syndrome is caused by an autoimmune abnormality of the voltage-gated potassium channel (VGKC) of α motor fibers (see Chapter 28-2C, p. 265).

It is important to note that spasm or cramp is not infrequently seen in patients with motor neuron diseases, especially in their early clinical stage, probably due to increased excitability of the lower motor neurons.

D. PAINFUL MUSCLE SPASM (CRAMP)

The most common form of cramp is **idiopathic cramp**, of which no cause is identified. **Tetany** is caused by hypocalcemia in association with hypoparathyroidism, metabolic alkalosis, and hyperventilation syndrome. In tetany, the needle EMG shows repetitive discharges of doublets or triplets. In this case, the excitability of motor nerves is increased so that tapping of the facial nerve will cause spasm in the facial muscles (**Chvostek sign**), and tapping of the median nerve will cause spasm in the hand muscles (**Trousseau sign**). In practice, the Trousseau sign can be noted while a

tourniquet is tied around the arm for the purpose of blood pressure measurement.

In **glycogen storage disease**, among others **McArdle disease** (muscle phosphorylase deficiency), painful spasms are induced by vigorous physical exercise, and in fact this is the most common chief complaint in this condition. **McArdle disease** is an autosomal recessive hereditary disease, but it is more common in males. Vigorous physical exercise mobilizes anaerobic glycolysis, which produces a large amount of lactic acid. In this disease, as the aerobic glycolysis is impaired due to phosphorylase deficiency, lactic acid cannot be metabolized and accumulates in the muscle, which then causes painful spasm (Ellingwood & Cheng, 2018). **Pompe disease** (acid maltase deficiency) may present with limb-girdle muscular weakness associated with hyperCKemia (Lukacs et al, 2016).

6. MYOTONIA

Myotonia is manifested as stiffness of muscles and difficulty in muscle relaxation due to repetitive depolarization of sarcolemma. The main feature of myotonia is difficulty in relaxing the contracted muscle, and in the resting state of muscles, stiffness may or may not be present (Figure 17-10a). Furthermore, myotonia is induced by tapping the thenar muscle or tongue (**percussion myotonia**) (Figure 17-10b). The needle EMG of myotonia shows rhythmic discharges at 50 to 100 Hz, and its frequency fluctuates so that it sounds like a motorbike (also called the dive bomber sound).

In ordinary myotonia, muscle relaxation becomes increasingly easier as the patient repeats voluntary muscle contraction (**warm-up phenomenon**), which is seen in patients with **myotonic dystrophy** and **myotonia congenita**. In special conditions like **paramyotonia congenita**, muscle relaxation becomes increasingly difficult as the patient repeats muscle contraction. This phenomenon is called **paramyotonia**, which becomes characteristically worse in a cold environment.

In myotonic dystrophy, the patient has muscle atrophy and weakness in addition to myotonia, but in some kinds of myotonic disorders, myotonia is the sole manifestation. Furthermore, myotonia is closely related to periodic paralysis. Especially in hyperkalemic periodic paralysis, the patient shows myotonia during the intervals of paralytic attacks. Mutation of ion channel genes has been identified in many of these conditions (see Chapter 28-1C, p. 262).

In myotonic dystrophy, myocardial conduction abnormality and arrhythmia are commonly present, and they may be a cause of sudden death (Groh et al, 2008). Whether myotonia involves the extraocular muscles is an open question. In some patients, the return from lateral saccade looks slow, but in other cases, saccades look relatively normal even when there is marked myotonia in the hands.

7. COORDINATION

For testing the coordination of the upper extremities, the **nose-finger-nose test** (also called the **finger-nose-finger test**) is used to detect **dysmetria** and **dyssynergia** (see p. 127). These tests can be done in either a sitting or a supine position. If a patient has visual disturbance, it may be replaced by the **finger-to-nose test** without any visual target. Then **rapid alternating movement** is tested. For testing the distal upper limbs, finger tapping is useful. Patients are asked to tap the index finger rhythmically on the metacarpal-phalangeal joint of the thumb (Fisher, 1960). For the lower limbs, patients are asked to tap the shin with the opposite heel (Fisher, 1961). When these tasks are poorly performed, it is called **dysdiadochokinesis** or **adiadochokinesis**. For testing the coordination of the lower extremities, the **heel-knee test** is used to detect dysmetria and dyssynergia. For this test, the patient is placed in a supine position and asked to put a heel on the knee of the other leg and slide it down over the anterior aspect of the lower leg smoothly. The patient may be also asked to repeat tapping the middle of the shin with the heel to see rhythmic performance. These tests of lower extremities may be also done in a sitting position.

Cerebellar ataxia due to a unilateral cerebellar lesion is seen in the ipsilateral extremities, because the cerebellar hemisphere is functionally connected to the contralateral motor cortex (see p. 127) (see Figure 16-7, p. 116). There is some somatotopic organization in the cerebellar hemisphere, because the degree of ataxia is not infrequently discrepant between the upper and lower extremities.

While testing the strength of the forearm flexor muscles by manual muscle testing, if the examiner suddenly releases the resistance, some patients cannot stop the muscle contraction and may hit their own chest, which a healthy subject can avoid. This phenomenon is called **rebound phenomenon**. It is generally considered to be specific for cerebellar lesions, but if this is the only abnormality, it is not necessarily pathognomonic of a cerebellar lesion.

(a)



(b)



Figure 17-10 Myotonia seen in the thumb during an attempt to relax the grasped hand in a patient with myotonic dystrophy (a) and percussion myotonia elicited by tapping the tongue in another patient with myotonic dystrophy (b).
(b) by courtesy of Dr. A. Ikeda, Kyoto University.

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TENDON REFLEXES AND PATHOLOGIC REFLEXES

1. PHYSIOLOGIC MECHANISM OF THE TENDON REFLEX

The tendon reflex (**deep tendon reflex, muscle stretch reflex, myotatic reflex**) is a monosynaptic proprioceptive reflex. The afferent arc of the reflex starts from the **stretch receptor** in the muscle called the **muscle spindle**. The receptor is activated by sudden stretch, and an electric impulse generated there is conducted through **Ia fibers** to the anterior horn cells of the same segment, where the α motor neurons are monosynaptically activated. The efferent impulse is conducted through the α motor fibers to contract the muscle fibers from which the afferent impulse originated (see Figure 17-9, p. 147). An electric impulse generated in the tension receptor in the muscle tendon called the **tendon organ** as a result of the reflex muscle contraction is conducted through the **Ib fibers** to the anterior horn of the same segment and via the interneurons inhibits the α motor neurons disynaptically (post-synaptically). Furthermore, the anterior horn cell is also inhibited by its own recurrent branch via an interneuron called the **Renshaw cell** (Figure 18-1).

As the conduction velocity of both the afferent and efferent fibers of the tendon reflex is as fast as 50 m/s, the latency of the tendon reflex of the hand is about 25 ms. When recorded with electromyography (EMG), the tendon reflex is called the **T reflex** ("T" for tendon). A similar reflex can be produced by electrical stimulation of the afferent nerve trunk; this is called the **H reflex** ("H" for Hoffmann).

The tendon reflex is decreased or lost by a lesion of either the afferent arc or the efferent arc of the reflex. In addition, as the tone of the muscle spindle is modulated by the γ motor fibers, the tendon reflex can also be decreased by impairment of the γ motor fibers. Therefore, patients who are considered to have impairment of the peripheral nervous system with loss of tendon reflexes and yet no abnormality in muscle strength or somatosensory function

may have involvement of the γ motor fibers. There is some evidence for this situation with acute cerebellar lesions.

As the lower motor neuron is inhibited by a component of the supraspinal drive presynaptically via spinal interneurons, impairment of the supraspinal drive is expected to cause an increase in the excitability of the anterior horn cells and consequently an enhancement of the tendon reflex (see Chapter 17-4A, p. 147).

2. EXAMINATION OF TENDON REFLEXES

There is no general rule in terms of the order of examining tendon reflexes, but it is practical to start with the jaw jerk, and then to examine the biceps brachii, triceps brachii, radial and finger flexor muscles for the upper limbs, and the quadriceps femoris, gastrocnemius-soleus, and toe flexor muscles for the lower limbs, in this order, each comparing the left and right. Hamstring reflex and abdominal muscle reflex may also be examined as necessary.

To examine a tendon reflex, the muscle to be tested should be placed in a relaxed state with an intermediate length (Figure 18-2). To test the **biceps brachii reflex**, for example, the elbow should be placed in a mildly flexed position (Figure 18-2b). When testing the **quadriceps femoris reflex (patellar tendon reflex, knee jerk)** in a supine position, the knees should be mildly flexed and gently supported from below by the examiner's hand (Figure 18-2d). Then the tendon should be gently tapped by a tendon hammer toward the direction so that the muscle can be stretched. The tendon reflexes of the lower extremity can be also tested in the sitting position. To elicit the **masseter reflex (jaw jerk)**, for example, the jaw should be tapped downward (Figure 18-2a). Practically speaking, however, for most muscles of the extremities, straight tapping of the tendon at a right angle has a sufficient effect for stretching the muscle spindle. Generally speaking, for the flexor muscles like the biceps brachii, it is easier to put

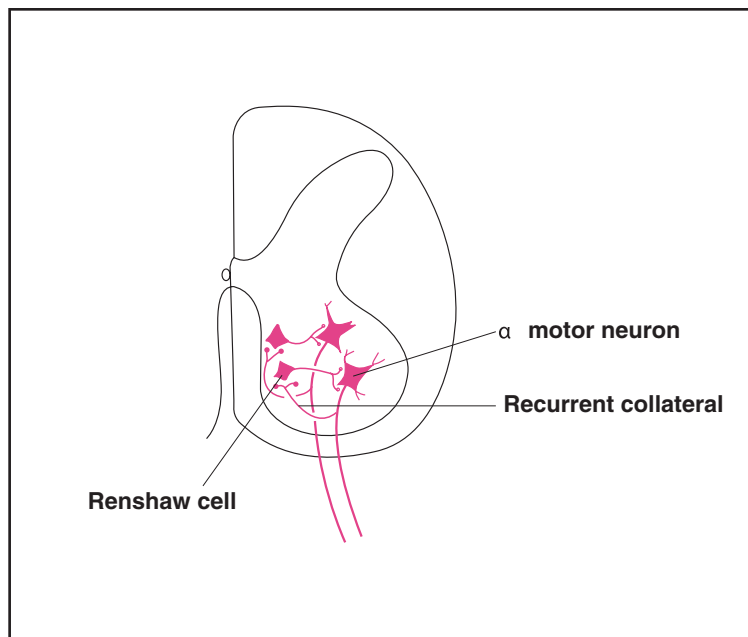


Figure 18-1 Schematic diagram showing the recurrent inhibitory pathway of the spinal cord. Although it is not shown in this figure, the Renshaw cell also inhibits other motor neurons that supply muscles with similar action. Cited from Brodal, 2010, with permission.

BOX 18-1 INVERTED TENDON REFLEX

This is a tendon reflex elicited in the antagonist muscle with respect to the stimulated muscle, which therefore appears paradoxical. In a localized lesion of the cervical cord, either within the cord substance (intramedullary) or compressing the spinal cord (extramedullary), the tendon reflex of the affected segment is decreased or lost while the tendon reflexes of the immediately caudal segments may be increased. For example, if the biceps brachii reflex is lost and the triceps brachii reflex is overactive due to a lesion at the C6 segment, a tendon tap to the biceps muscle might elicit elbow extension instead of flexion because of the hyperactive triceps reflex. This phenomenon is called the **inverted biceps reflex**. In contrast, if a segmental lesion at the C7 segment is associated with disseminated lesions, as seen in the neuromyelitis optica spectrum disorders (see Box 17-4, p. 148), a tendon tap to the triceps muscle might elicit elbow flexion instead of elbow extension because of the hyperactive biceps reflex due to the presence of a more rostral lesion (**inverted triceps reflex**).

the examiner's thumb on the tendon and to tap the thumb instead of directly tapping the tendon (Figure 18-2b). By contrast, for the extensor muscles like the triceps brachii

and the quadriceps femoris, it is easier to tap the skin over the tendon directly (Figure 18-2c, d).

Regarding tendon hammers, a slightly flexible hammer is ideal, but a metal hammer is satisfactory. What is most important is that the head is made of a relatively soft material. A hard material is painful to the patient and is not effective for eliciting the tendon reflex efficiently. As for the way of tapping, leaving the hammer on the tendon for a moment after tapping is more effective than tapping in a bouncing way. To examine the tendon reflexes in newborns and infants, it is more practical to tap the skin with the examiner's finger instead of using a hammer.

A. TENDON REFLEX OF THE DISTAL EXTREMITY MUSCLES

To examine the **finger flexor reflex** (Box 18-2), with the examiner's fingers placed on the palmar aspect of the gently flexed fingers, tapping should be applied on the examiner's fingers. As this reflex may not be clearly visible even in healthy subjects, it is difficult to prove a loss of this reflex clinically. By contrast, if this reflex is abnormally enhanced and accompanied by flexion of the thumb, it is called the **Wartenberg reflex** and provides an important clue to the presence of pyramidal signs. A similar reflex can be elicited by grasping the patient's middle finger with the examiner's thumb and middle

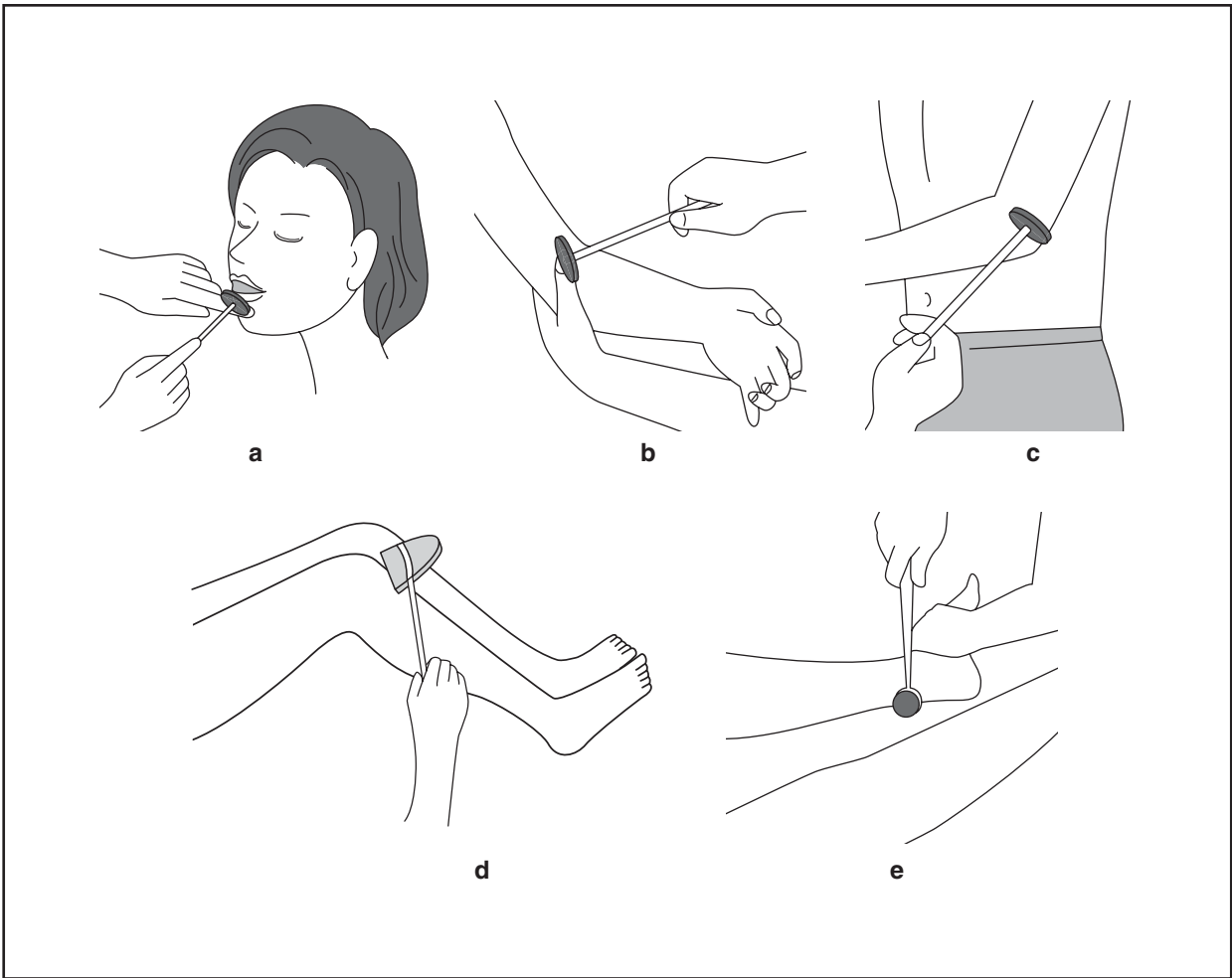


Figure 18-2 Methods for testing the representative tendon reflexes: (a) masseter reflex (jaw jerk), (b) biceps brachii reflex, (c) triceps brachii reflex, (d) knee jerk (quadriceps femoris reflex), (e) ankle jerk (Achilles tendon reflex). The ankle jerk can be also tested with the patient in a sitting position with the ankle free. In this figure, different kinds of tendon hammer are shown just for the sake of convenience.

fingers and then flipping a quick extension of the distal phalanx. An enhanced state of this reflex is called the **Hoffmann reflex**.

The **ankle jerk (Achilles tendon reflex)** is often decreased in healthy aged subjects and may not be detectable by visual inspection. To elicit the **toe flexor reflex**, the

BOX 18-2 SIGNIFICANCE OF FINGER FLEXOR REFLEX

The most common pathology of cervical spondylosis is the narrow spinal canal and intervertebral foramen at the C5-C6 level. In this case, the biceps brachii reflex is lost because the C6 spinal root is affected, and the triceps brachii reflex (mainly innervated by the C7 segment) and all the tendon reflexes caudal to it are enhanced (see Chapter 16-1B, p. 112). Although the brachioradialis muscle is located distal to the biceps brachii muscle, those two muscles are innervated by the same segment (C6), and thus the brachioradialis reflex is also lost. If the main lesion is at the C6-C7 level, the triceps brachii reflex is lost, and the finger flexor reflex (innervated by the C8-Th1 segment) and all the tendon reflexes caudal to it are enhanced. In this case, therefore, the Wartenberg reflex and the Hoffmann reflex are quite important diagnostically.

The Wartenberg reflex and the Hoffmann reflex, like the Mendel–Bechterew reflex and Rossolimo reflex, are enhanced tendon reflexes, and not pathologic reflexes in the narrow sense. Pathologic reflexes are the reflexes seen in healthy newborns but not in healthy adults, like the Babinski sign and Chaddock sign (see Chapter 18-3).

metatarsal bones can be tapped on their dorsal or plantar aspect. If this reflex is abnormally enhanced, it is called the **Mendel–Bechterew reflex** for the dorsal tap and the **Rossolimo reflex** for the plantar tap. Like the finger flexor reflex, it is often difficult to prove a loss of the toe flexor reflex clinically.

B. HOW TO CONFIRM LOSS OF ANKLE JERK

To confirm the loss of ankle jerk in patients with a possible diagnosis of peripheral neuropathy, it is important to apply a **reinforcement** technique. For example, while the patient is asked to clasp hands in front of the chest either in a sitting or supine position, the examiner gives a verbal order to pull them apart hard, and the Achilles tendon is tapped immediately after the verbal order is given. In fact, even in healthy older subjects, the ankle jerk or knee jerk may be elicited only by this maneuver (**Jendrassik maneuver**). In case the ankle jerk is not elicited even by this reinforcement in a sitting or supine position, the tendon tap can be tried while the patient is kneeling down on a chair with the feet protruded backward from the edge of the chair. If the ankle jerk is not elicited even after this change of position, then the reinforcement technique can be applied in this new position. If the ankle jerk is not elicited after all those trials, then it is judged to be lost. In this regard, if the ankle jerk is unilaterally absent, like a case of lumbar disc herniation, then its diagnostic significance is much higher.

BOX 18-3 ACHILLES TENDON REFLEX AND PERIPHERAL NERVE LESIONS

Some physicians might rule out polyneuropathy based on the presence of ankle jerk (Achilles tendon reflex), but theoretically this is not correct. That is because the receptor of the afferent reflex arc (muscle spindle) and the effector of the efferent arc (neuromuscular junction) are both situated in the middle of the calf muscle (halfway between the knee and ankle). Therefore, although the tendon tap is applied to the ankle, we are just looking at function of a reflex circuit proximal to the middle of the lower leg. In axonal polyneuropathy and dying-back neuropathy, the pathologic lesion is more severe in the more distal part of the nerve. In this case, therefore, in order to prove loss of tendon reflexes, the more distal tendon reflex, namely the toe flexor reflex, has to be examined. Practically speaking, however, as the toe flexor reflex may be difficult to elicit even in healthy subjects, electrophysiologic studies may become necessary.

C. SIGNIFICANCE OF HYPERACTIVE TENDON REFLEXES

If a symmetric increase of tendon reflexes is the only pyramidal sign, its diagnostic significance may not be specific. If it is accompanied by spasticity like clasp-knife phenomenon, clonus, and/or pathologic reflexes like the Babinski sign, then hyperreflexia is certainly regarded as a sign suggestive of a lesion of the upper motor neurons or pyramidal tracts. Of course, if the hyperreflexia is seen asymmetrically, its diagnostic significance is high.

As was described in relation to reinforcement, excitability of the motor neurons is increased throughout the body by intended movement of any part of the body. Therefore, if a patient is nervous or emotionally excited, the excitability of motor neurons might be increased even in the resting condition, thus lowering the threshold of motor neurons. In that case, therefore, it is worth examining the tendon reflexes after the patient is relaxed. However, no matter how nervous the patient is, hyperreflexia is never asymmetric in healthy conditions.

3. PATHOLOGIC REFLEXES AND THEIR GENERATING MECHANISMS

Some reflexes seen in healthy newborns are pathologic when seen in adults. There are two groups of pathologic reflexes: (1) the **pathologic reflex** in a narrow sense, the presence of which indicates an impairment of the upper motor neurons or the pyramidal tract, and (2) the primitive reflex (see Chapter 10-2E, p. 81). Representative forms of the pathologic reflex are the Babinski sign, Chaddock sign, and Strumpell sign of the lower extremities. Overactive tendon reflexes called by the investigator's name like Wartenberg, Hoffmann, Trömner, Mendel–Bechterew, and Rossolimo do not belong to the pathologic reflexes. The pathologic reflex is a polysynaptic reflex formed of complex somatosensory afferents and multisegmental motor efferents. The **plantar reflex** is elicited by stroking the sole of a foot with the blunt end of the handle of a tendon hammer or a key; it is manifested as toe flexion in healthy adults. The **extensor plantar reflex** consists of extension of the big toe associated with fanning and/or extension of other toes (Figure 18-3). It is elicited by stroking the lateral aspect of the sole from the heel toward the toes (**Babinski sign**) or by stroking the lateral dorsal aspect of a foot also from the heel toward the toes (**Chaddock sign**). Some patients are extremely sensitive to stimulation of the sole, in which case the Chaddock sign is better tolerated.

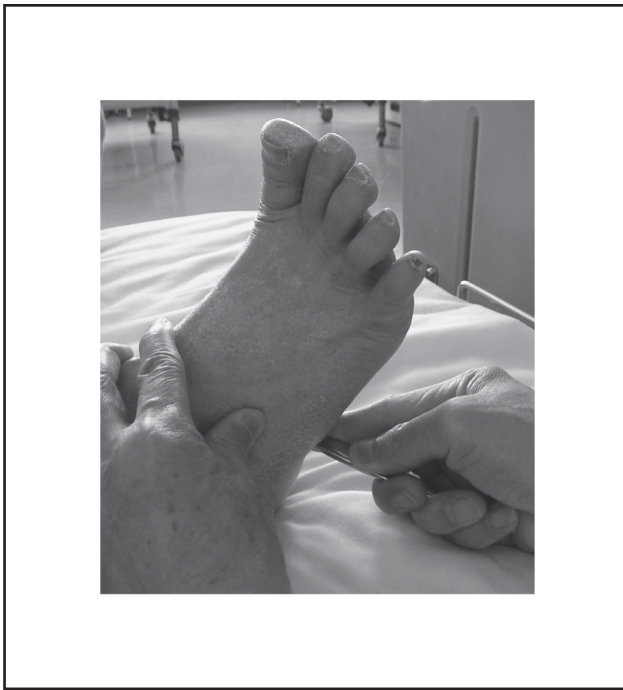


Figure 18-3 Typical Babinski sign elicited in a patient with multiple sclerosis. Mainly the big toe is dorsiflexed while the other toes show fanning. When it is difficult to elicit the sign, it is useful to grasp the dorsum of foot with the examiner's other hand while stroking the sole, as shown in this photo.

To elicit the Babinski sign, it is useful to grasp the dorsum of the patient's foot with the examiner's hand while stroking the sole. When the Babinski sign is exaggerated, gentle pressure of the sole by the handle of the hammer is just enough to elicit the sign. Furthermore, when the pyramidal sign is exaggerated, a **flexor withdrawal reflex** may be seen in the proximal leg on the side of the stimulated sole (Box 18-4). In patients whose upper and lower motor neurons are both involved, if the extensor hallucis longus muscle is completely paralyzed as a result of the lower motor neuron lesion, the big toe is not expected to extend in response to the plantar stimulus. In that case, the proximal leg muscles might contract, eliciting the flexor withdrawal reflex, which can be interpreted to suggest the presence of upper motor neuron involvement.

The Babinski sign is the most specific of all the neurological signs. Its presence always indicates a lesion of the pyramidal tract or the upper motor neurons. However, there are many false negatives and false positives (Isaza Jaramillo et al, 2014). Another problem with the Babinski sign is considerable interobserver variability. Although this is an important sign in neurology, the mechanism underlying its generation has not been clearly elucidated. It is seen in healthy infants under the age of 6 months, and it disappears as myelination

BOX 18-4 COMPLETE MOTOR PARALYSIS OF TOES AND BABINSKI SIGN

In patients with complete motor paralysis of the toes, special caution is necessary for judging the presence or absence of the Babinski sign. If the toe extensor muscles are completely paralyzed due to impairment of the anterior horn cells or the α motor fibers innervating those muscles, the Babinski sign is not expected to occur even in the presence of the upper motor neuron impairment, because the reflex circuit is interrupted. In that case, however, if the muscle strength of the pelvic girdle and proximal legs is preserved to a certain extent, those muscles will contract in response to the plantar stimulus and produce the flexed posture of the legs. This is called **flexor withdrawal reflex** and may be interpreted as a pyramidal sign. By contrast, if the toe extensor muscles are paralyzed by a lesion of the corticospinal tracts or the upper motor neurons, the Babinski sign is expected to be positive even if the degree of motor paralysis is complete. An exception to this is the so-called **spinal shock** seen in the acute stage of a severe vascular or inflammatory insult to the central nervous system. In spinal shock, the tendon reflexes and superficial reflexes are all lost.

of the pyramidal tract is completed. Although other reflexes such as snout, sucking, and grasping are also seen in healthy infants, the occurrence of these reflexes in adults has

BOX 18-5 ALTERNATIVES TO THE BABINSKI SIGN; THE BING SIGN

There are many ways of producing an extensor response of the great toe other than the Babinski sign. The Chaddock sign is the most frequently suggested alternative. Others include the Bing sign, Oppenheim sign, and Gordon sign. The Bing sign elicits extension of the toe by pricking the dorsum of the toe. This is clearly a pathologic response as the toe moves toward the noxious stimulus. Curiously, Bing apparently never described the procedure exactly this way but instead referred to pricking the dorsum of the foot or "over the 4th and 5th metatarsal" (Davis et al, 2018). One of the authors (M.H.) once used this sign to advantage in an unconscious post-traumatic patient with his foot in a cast revealing only the great toe; the Bing sign clearly documented corticospinal tract damage.

nothing to do with a pyramidal lesion, and they belong to the primitive reflexes (see Chapter 10-2E, p. 81).

4. LOSS OF SUPERFICIAL REFLEX AS A PYRAMIDAL SIGN

Among the superficial reflexes, the abdominal reflex and the cremasteric reflex can be lost by a lesion of the pyramidal tract.

A. ABDOMINAL REFLEX (ABDOMINAL WALL REFLEX)

The **abdominal reflex** is a reflex shift of the umbilicus elicited by a superficial stimulus to the abdominal wall. To elicit this reflex, the abdominal wall is stroked or lightly scratched by the handle of a hammer from lateral toward the umbilicus. As a normal response, the umbilicus shifts toward the side of stimulus presentation. However, if the abdominal wall is stroked away from the umbilicus, it may confuse the judgment because the umbilicus might be mechanically pulled laterally. It is useful to test the upper, middle, and lower segments of the abdomen, each comparing the left and right.

BOX 18-6 ABDOMINAL REFLEX AND ABDOMINAL MUSCLE REFLEX

The **abdominal reflex (abdominal wall reflex)** is a superficial reflex elicited by stroking the abdominal wall from lateral to medial and judged by observing a lateral shift of the umbilicus. By contrast, the **abdominal muscle reflex** is elicited by tapping the rectus abdominis muscle on either side and judged by observing the contraction of that muscle. In a lesion of the pyramidal tract above the lower thoracic segment, the abdominal reflex is lost whereas the abdominal muscle reflex is exaggerated. Therefore, it is conceivable that the abdominal muscle reflex is a muscle stretch reflex whereas the abdominal reflex is a polysynaptic superficial reflex involving the central nervous system, at least above the lower thoracic cord. If the thoracic cord or the thoracic nerves of the segments Th8-Th12 are involved by a lesion, both of these reflexes are lost, because the final common pathway is interrupted. Both reflexes may be difficult to elicit in aged subjects, obese subjects, multiparas, and those having a past history of abdominal surgery. Loss of the abdominal reflex or enhancement of the abdominal muscle reflex is especially meaningful when it is asymmetric.

Tapping the rectus abdominis muscle causes a reflex contraction of that muscle (**abdominal muscle reflex**), and this reflex is enhanced in a lesion of the pyramidal tract (Box 18-6). However, this reflex might not be easily elicited even in healthy subjects depending on the condition of the abdominal wall, and its diagnostic significance is not as high as the tendon reflexes of extremities, unless there is a clear asymmetry.

B. CREMASTERIC REFLEX

This is a superficial reflex that can be tested only in males. The stimulus is given by stroking the skin of the proximal medial aspect of the thigh with the handle of a hammer downward and the reflex is observed as an ascent of the testis. Like the abdominal reflex, its loss indicates a lesion of either the spinal cord or spinal nerves of the corresponding segments or a lesion of the pyramidal tract. The stimulus can be applied upward as well, but it is easier to judge the reflex if it is applied toward the opposite direction from the

BOX 18-7 HOW IS THE RESTING TREMOR IN PARKINSON DISEASE SUPPRESSED BY VOLUNTARY MOVEMENT?

Involuntary movements are generally induced or enhanced by intended movement of the corresponding part of the body, and the resting tremor seen in Parkinson patients is an exception in this regard (see p. 165). In fact, resting tremor can be suppressed even by an act of will (Blakemore et al, 2019). The following physiological concept might be related to this phenomenon. Namely, the rhythmic oscillation of the motor cortex tends to be increased during the relatively inactive (idling) state whereas the oscillation tends to be lost when that part of cortex is activated might be related. Assuming that the rhythmic activity of the motor cortex is driving the resting tremor, intended movement of the corresponding part of the body is expected to suppress the rhythmic activity and to suppress the tremor rhythm. Then, the question arises as to why the intended movement of other parts of the body or mental excitation enhances the resting tremor. For this phenomenon, distraction to other activities might put the part of the tremor-related motor cortex into a kind of idling state and thus increase the rhythmic oscillations, including the tremor pacemaker (Shibasaki, 2012). This hypothesis, however, does not explain why essential tremor and action tremor (see p. 168) are enhanced by intended movement of the corresponding part of the body even though both are mediated by the sensori-motor cortex.

response. The diagnostic significance is higher if this reflex is lost just on one side.

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INVOLUNTARY MOVEMENTS

Involuntary movements, or hyperkinetic movement disorders, are defined as abnormal unintended movements. The movement should be defined as “abnormal” because not all unintended movements are abnormal. For example, movements like blinks, spontaneous eye movements, chewing, swallowing, yawning, breathing, walking, and habits can be executed voluntarily, but they may occur spontaneously without involving “will to move.” Involuntary movements are grouped into several categories depending on their characteristic clinical features. They include tremor, chorea, ballism, athetosis, dystonia, myoclonus, dyskinesia, tics, and asterixis, and many of them were described more than 100 years ago based on clinical observation. Usually it is possible to classify involuntary movements by visual inspection, but electrophysiologic tests are useful in confirming the diagnosis of some of them (Shibasaki, 2012). It should be kept in mind that many of the involuntary movements can occur as the side effect of pharmaceutical drugs.

1. EXAMINATION OF INVOLUNTARY MOVEMENTS

Involuntary movements can be examined systematically as shown in the flow chart (Shibasaki et al, 2020) in Figure 19-1. The physical examination of involuntary movements may be started by paying attention to their rhythmicity: Do the movements repeat at a steady rhythm (**rhythmic**) or is there a steady interval between them (**periodic or irregular**)? The distinction between rhythmic and periodic may not be clear in some cases, particularly when the movements are repeated at relatively high frequency. A typical example of rhythmic involuntary movements is **tremor**, and a typical example of periodic involuntary movements is periodic myoclonus as seen in Creutzfeldt–Jakob disease. **Myoclonus** is usually irregular, but it may appear rhythmic when it repeats with short intervals. In this case, however, each individual movement may look shock-like and can be confirmed by palpating the contracting muscle,

whereas each movement of tremor is associated with relatively slow muscle contraction. Recording of the surface electromyogram (EMG) is helpful for distinguishing the two conditions.

Most other involuntary movements occur irregularly. **Ballism** or **ballismus** is a violent movement and is usually seen unilaterally (hemiballism or hemiballismus). Among the involuntary movements associated with relatively slow muscle contraction, **athetosis** and **dystonia** are characterized by their twisting or writhing nature. **Dyskinesia** is composed of complex movements, and **chorea** is made up of relatively simple movements.

It is not always necessary to try to classify an involuntary movement into a known category. Some patients may have a mixed feature of more than one involuntary movement (for example, coexistence or mixture of tremor, myoclonus, and dystonia) and some patients may have a hitherto undescribed involuntary movement. What is most important is to precisely describe what was observed. If the examiner can describe the clinical manifestation appropriately, that will often lead to a correct diagnosis.

2. TREMOR

Tremor is characterized by repetition of the same movements at a certain pace (**rhythmicity**). Tremor was first described in the 17th century, but its concept was established when Parkinson described **paralysis agitans** in 1817. In most cases of tremor, the surface EMG correlate of muscle contraction is usually a brief burst of action potentials. It commonly shows alternating contraction of agonist and antagonist muscles, but in some cases the agonist and antagonist contract at almost the same time, although not exactly synchronously. Furthermore, these EMG findings may vary depending on the clinical stage and the position of extremities. Even in a single case, reciprocal contraction may be seen in one hand and simultaneous contraction in the other hand at the same time

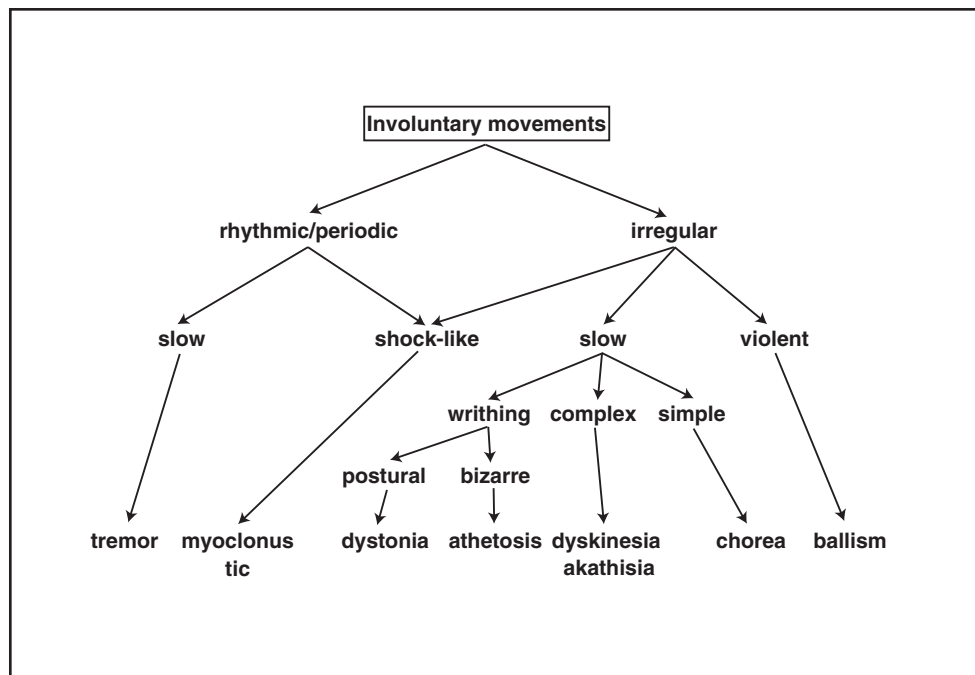


Figure 19-1 Classification of involuntary movements based on the clinical features, and a sample flow chart of the steps for visual inspection.

(Figure 19-2). Tremor is largely classified into **physiologic tremor** and **pathologic tremor**, and pathologic tremor is classified into resting tremor, postural tremor, and action tremor (Deuschl et al, 2001). Recently a new classification of tremor was proposed based on the syndrome (Axis 1) and the etiology (Axis 2) (Bhatia et al, 2018).

A. RESTING TREMOR

Tremor seen in the resting condition is called resting tremor. Resting tremor is commonly seen at about 5 Hz, and the typical resting tremor is seen in **Parkinson disease**. Similar resting tremor may be seen in parkinsonism other than idiopathic Parkinson disease, and its presence

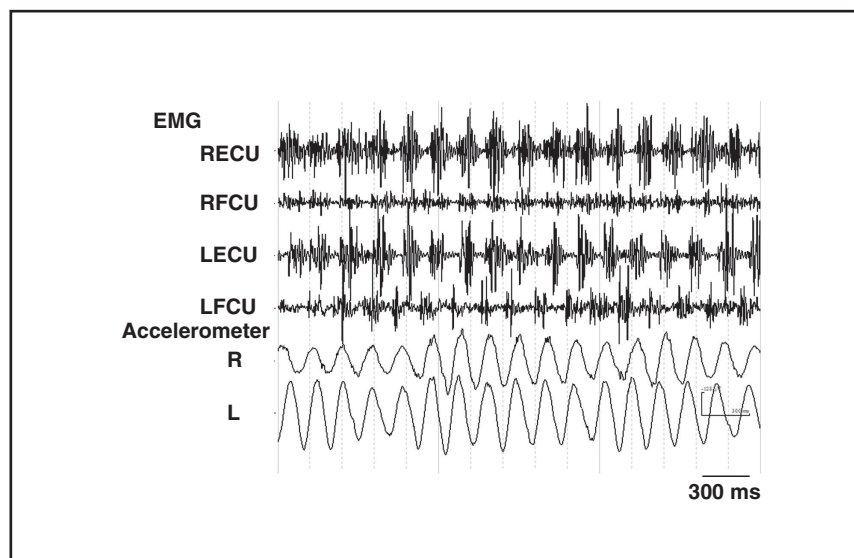


Figure 19-2 Records of the surface EMG and accelerometer in a patient with essential tremor. ECU: extensor carpi ulnaris muscle, FCU: flexor carpi ulnaris muscle. In the right hand (R), the wrist flexor and extensor muscles are contracting nearly at the same time while in the left hand (L), they are contracting alternately.
By courtesy of Dr. Zoltan Mari.

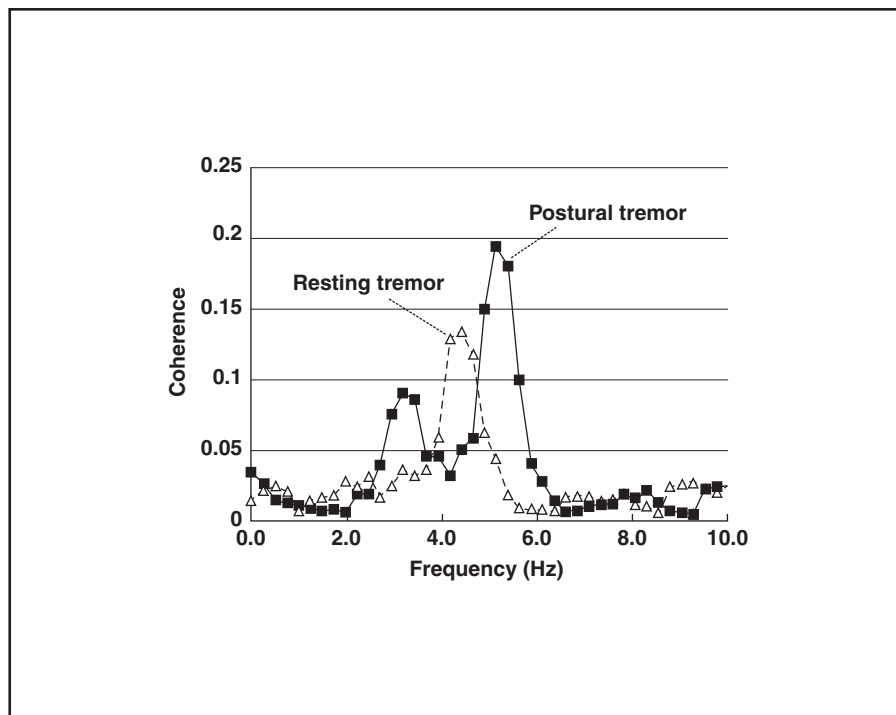


Figure 19-3 Cross spectrum between EEG and EMG in a 78-year-old patient with essential tremor who developed Parkinson disease 50 years after the clinical onset of essential tremor. EMG discharges associated with resting tremor of a hand show a peak of correlation with EEG recorded from the contralateral central region at 4.3 Hz (open triangles), and those associated with postural tremor show a peak of correlation at 5.2 Hz (filled squares). Cited from Inoue et al., 2007, with permission.

indicates a lesion or dysfunction of the basal ganglia. “Resting” does not imply psychological or emotional resting at all, because the resting tremor is actually enhanced by mental activity, emotional excitation, and physical exercises involving other parts of the body. For example, a patient with Parkinson disease may have no tremor when at complete rest, but once he or she starts talking or walking, the tremor often appears in the hands. For another example, in a Parkinson patient having resting tremor in the right hand, the tremor may stop once he or she starts moving the right hand while the resting tremor may appear in the left hand. Physiologically, therefore, resting tremor is defined as a tremor seen in the muscles that are not actively contracting. The resting tremor may be seen in the lips, which may be called **rabbit syndrome** because of its resemblance to the lips of a rabbit.

Although the mechanism that generates tremor in Parkinson disease has not been completely clarified, it is considered to be mediated by the sensori-motor cortex. There is a significant correlation of frequency between the cortical rhythmic activities of the hand area of the motor cortex recorded by electroencephalogram (EEG) or magnetoencephalogram (MEG) and the EMG discharges of tremor of the contralateral hand (Figure 19-3). This

phenomenon is called **cortico-muscular coherence**. Here a question arises as to whether this correlation might be due to proprioceptive feedback from the peripheral muscle to the sensori-motor cortex as a result of the tremor. However, as a result of mathematical analysis, this cortical rhythm is believed to be a cause of tremor rather than its result. Namely, it is postulated that the tremor is driven by the cortical rhythm. Resting tremor may be responsive to levodopa in some cases of Parkinson disease but not in others. The dopamine-resistant resting tremor may be explained by increased cerebellar and reduced somatosensory influences on the cerebellar thalamus (Dirkx et al, 2019).

Although there is cortico-muscular coherence in Parkinson tremor, the pacemaker of the tremor is thought to be subcortical with involvement of the basal ganglia but not in the sensori-motor cortex. In patients with Parkinson disease who have resting tremor, positive correlation has been found at the tremor frequency between the tremor of a hand and the neuronal discharge recorded from the contralateral subthalamic nucleus during deep brain stimulation (DBS) (Figure 19-4) (Wang et al, 2006). In a functional magnetic resonance imaging (MRI) study, functional correlation between the subthalamic nucleus and the ipsilateral sensori-motor cortex has been found to be greater

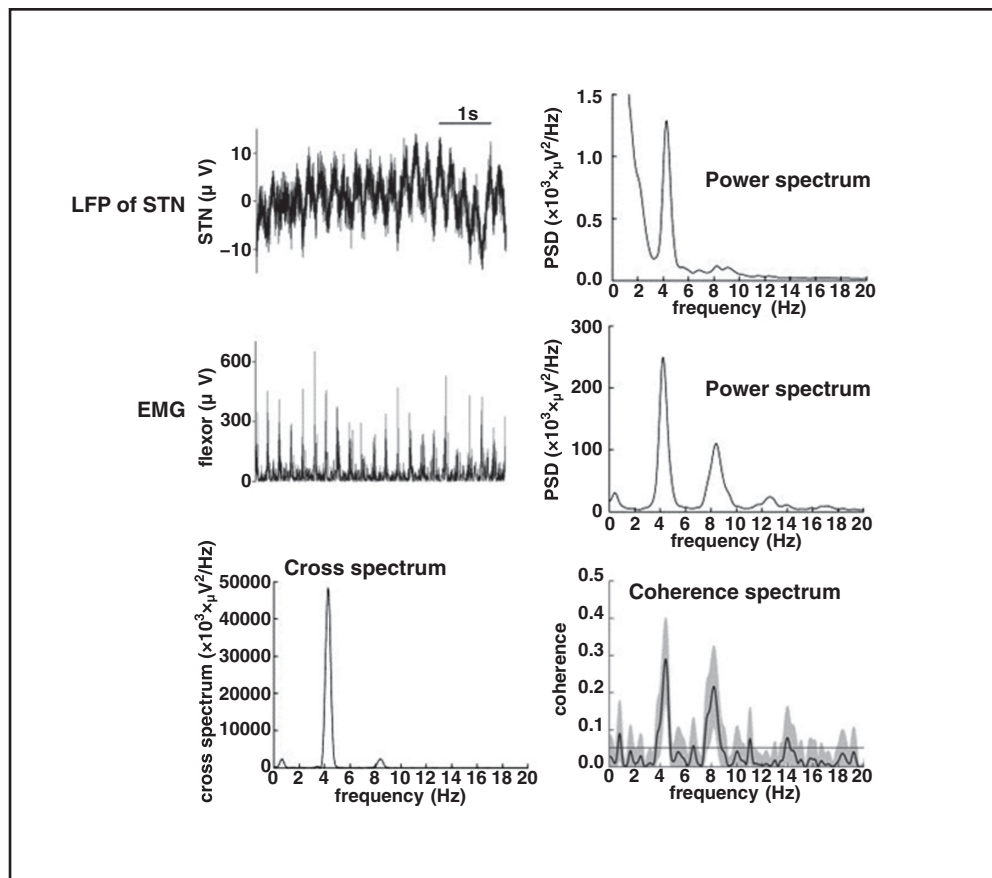


Figure 19-4 Local field potential (LFP) of the subthalamic nucleus (STN) and the tremor EMG of the contralateral hand (EMG) in a patient with Parkinson disease. Both LFP and EMG have the maximum power at the peak frequency of 4.5 Hz, and there is a correlation between the two at that peak frequency.
Cited from Wang et al., 2006, with permission.

in the Parkinson patients with tremor than those without tremor (Baudrexel et al, 2011).

A role of the cerebellum in generating tremor has drawn increasing attention. Clinically, DBS of the ventralis intermedius (Vim) nucleus of the thalamus (cerebellar relay nucleus) is more effective for Parkinson tremor than that of the ventralis oralis posterior (Vop) nucleus (striatal relay nucleus). In fact, DBS of Vim was shown to be an effective long-term treatment for Parkinson tremor and essential tremor (Cury et al, 2017). A functional MRI study of Parkinson patients with tremor by Helmich et al. (2011) showed that activity in the contralateral sensorimotor cortex, Vim, and ipsilateral cerebellum is increased in proportion to the magnitude of resting tremor. A study with transcranial magnetic stimulation (TMS) showed that the inhibitory effect of the cerebellum on the contralateral motor cortex is decreased in Parkinson patients (Ni et al, 2010). Helmich et al. (2021) explained parkinsonian tremor as resulting from the combined actions of two circuits: the basal ganglia that trigger tremor episodes

and the cerebello-thalamo-cortical circuit that produces the tremor.

B. POSTURAL TREMOR

Postural tremor is defined as a tremor seen while the corresponding extremity is assuming a certain posture—in other words, during the isometric contraction of the corresponding muscles. Typical examples are seen in the hands while the upper extremities are kept extended forward or in the arms while they are kept in front of the chest with the elbows flexed, and in the legs while the knees are kept flexed in the supine position. It usually starts a few seconds after taking a certain posture and increases in amplitude. In contrast, in patients with Parkinson disease, the resting tremor disappears once after taking a certain posture and some seconds later it might reappear. This phenomenon is called **re-emergent tremor** (Jankovic et al, 1999). There are two types of postural tremor in Parkinson disease, one dopamine-responsive and the other dopamine-resistant (Dirkx et al, 2018).

Tremor seen in the hands while sitting in a chair with the hands placed on the knees is usually the resting tremor, but the essential tremor (see the next paragraph) involving contraction of the proximal upper extremities may mechanically cause shaking of the hands, which may resemble resting tremor of the hands. In this case, it is useful to find out which muscles are actually contracting.

i. Essential tremor

Essential tremor is the most common cause of postural tremor (Bhatia et al, 2018). It is commonly seen in both hands at the frequency of 5 Hz on average. In the typical case of essential tremor, although the tremor may continue throughout motor execution like the nose–finger–nose test, it is especially pronounced when the movement stops at the nose or at the examiner's finger.

Essential tremor often appears to be an autosomal dominant hereditary disease with high penetrance, but finding genes is proving very difficult. The age of onset varies, from young adults to aged people. There are no significant movement abnormalities other than the tremor, although there might be a little ataxia or dystonia in some cases. Its

BOX 19-1 ASYMMETRY OF TREMOR

It might be thought that hereditary diseases like essential tremor would present with symmetric symptomatology, but that is not correct. In fact, essential tremor and Parkinson tremor start unilaterally in most cases. Amplitude of tremor can be markedly asymmetric in both essential tremor and Parkinson disease. By contrast, the peak frequency of tremor is relatively symmetric in essential tremor while, in Parkinson disease, it may be also asymmetric and the variability of tremor frequency is different between the two sides (Farkas et al, 2006). In other words, Parkinson tremor is asymmetric in terms of both the amplitude and the peak frequency, but essential tremor may be asymmetric in amplitude but not much so in frequency. It is important to point out that while the frequency is similar in the two arms in essential tremor, it is never exactly the same. In contrast, the tremor seen in hyperthyroidism and the asterix seen in hepatic encephalopathy are usually symmetric, and so is physiologic tremor. It is noteworthy that, for all kinds of tremor, the pattern and the distribution may not be constant, and they tend to vary even within an individual patient depending on the clinical stage and the position of the extremities.

BOX 19-2 INVOLUNTARY MOVEMENTS SUPPRESSED BY ETHANOL INTAKE

It is well known that essential tremor is decreased by ethanol intake. Other conditions for which ethanol may be effective include hereditary myoclonus–dystonia syndrome (see Box 19-7, p. 176), post-hypoxic myoclonus (Lance–Adams syndrome), and paramyotonia congenita (see Chapter 28-1C, p. 262). Instead of giving ethanol for treatment, however, there is a drug that has similar effects on the involuntary movements but does not cause drunkenness: sodium oxybate, which is approved in the United States for narcolepsy (Frucht et al, 2005). Another agent shown experimentally to improve essential tremor is octanoic acid (Voller et al, 2016).

progression is very slow, but it can cause significant disability in daily life. Characteristically, essential tremor is ameliorated by ethanol intake. Furthermore, about 20% of the patients with essential tremor have also migraine.

Pathophysiology of essential tremor has not been completely elucidated, but it is postulated, like the resting tremor in Parkinson disease, that it is also mediated by the sensori-motor cortex based on the demonstration of cortico-muscular coherence (see Figure 19-3).

The pacemaker of essential tremor is postulated to exist in the cerebellum (Pan et al, 2020). In patients with essential tremor, the neuronal discharge of the Vim nucleus of thalamus (cerebellar relay nucleus) is increased with the same peak frequency as the tremor EMG of the contralateral hand, and both activities show a significant correlation at the same frequency (Figure 19-5). Furthermore, the tremor is suppressed by DBS of that thalamic region. A patient with essential tremor was reported in whom tremor disappeared and was replaced by ataxia and slow action tremor after hemorrhage in the ipsilateral cerebellar hemisphere (Rajput et al, 2008). Kinesiologic analysis of patients with essential tremor disclosed intention tremor and hypermetria suggesting cerebellar dysfunction. Furthermore, a functional neuroimaging study with positron emission tomography (PET) in patients with essential tremor revealed increased blood flow in the cerebellum and red nucleus at rest, and increased glucose metabolism in the cerebellum. MR spectroscopy revealed a decrease in the ratio of *N*-acetyl aspartate to creatine in the cerebellar cortex. These findings are interpreted to suggest impairment of the cerebellar cortex and disinhibition of the cerebellar efferent pathway. Furthermore, tremor induced in animals by harmaline is known to be suppressed by ethanol and β

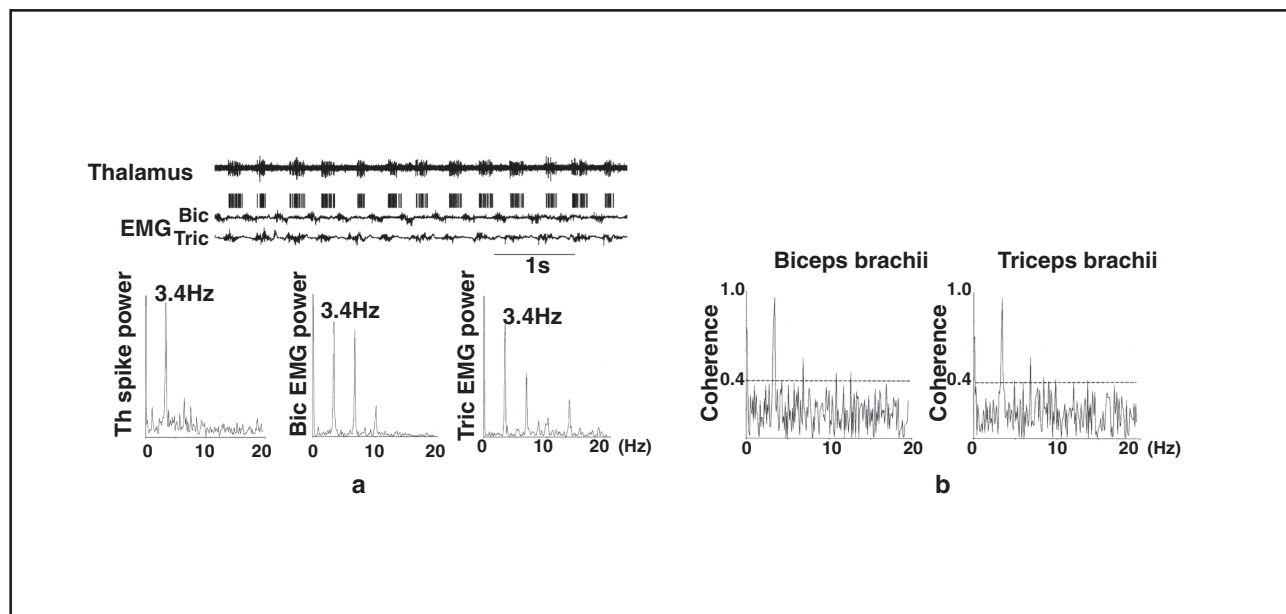


Figure 19-5 Simultaneous recording of neuronal discharge of the cerebellar relay nucleus of thalamus (Th) and the surface EMG associated with tremor of the contralateral hand and their frequency analysis (a) and correlation between the thalamic activity and the tremor EMG (b) in a patient with Parkinson disease. In (a), the thalamic data are shown both as a raw recording and then with another record below showing the firing of the identified neuron within the burst. Both the thalamic activity and the tremor EMG show peak frequency at 3.4 Hz, and the two activities show a significant correlation at the same frequency.

By courtesy of Prof. Y. Katayama and Dr. K. Kobayashi of Department of Neurosurgery, Nihon University School of Medicine.

blockers, and in this condition increased neuronal activity was shown in the olivo-cerebellar system.

Now sufficient evidence has been derived from clinical, neuroimaging, and postmortem studies suggesting that cerebellar dysfunction is a pathophysiology of essential tremor (Louis & Faust, 2020).

There is increasing evidence suggesting an overlap between essential tremor and Parkinson disease (Jiménez-Jiménez et al, 2012), but the nature and possible pathogenic mechanisms of the relationship are not well understood (Tarakad & Jankovic, 2019).

ii. Other conditions presenting with postural tremor

Tremor seen in **Wilson disease** is most commonly a typical postural tremor (Wilson, 1912). In this condition, the tremor mainly involves the distal upper extremities in the early clinical stage, but in the advanced stage it shows gross tremor in the arms, which is called **wing beating tremor**. The wing beating tremor may not be detected unless the arms are held in flexion in front of the chest, the legs are kept in flexion in the supine position, or the legs are held apart in the sitting position.

In contrast, irregular tremor seen with hepatic encephalopathy is formed of repetitive drops of extended hands

or extended fingers, and the term **asterixis** was coined by Adams and Foley in 1953 (Young & Shahani, 1986). This is caused by repetitive interruptions of EMG discharges of extensor muscles and is also called negative myoclonus. It is especially common in the state associated with hyperammonemia. The asterixis is different from the wing beating tremor of Wilson disease. The term “flapping” may be used to mean both wing beating tremor and asterixis, but this term may not be very appropriate.

Hyperthyroidism is also known to cause postural tremor. Thus, when a patient has tremor of undetermined cause, it is worth checking liver and thyroid functions. Whenever the wing beating tremor is encountered in young subjects, it is particularly important to test the blood level of ceruloplasmin and urinary copper. In Wilson disease, ceruloplasmin levels are decreased in the serum, and urinary excretion of copper is elevated (see also the section on dystonia, p. 172). The total serum copper might be low, but the free serum copper is generally high.

Complex tremor comprising resting, postural, and action tremor at a frequency of less than 4.5 Hz is called **Holmes tremor**. This tremor is seen with a lesion involving the medial part of the midbrain (Raina et al, 2016). Tremor with the characteristics of essential tremor and additional neurological signs of uncertain significance such

as impaired tandem gait, questionable dystonic posturing, memory impairment, or other mild neurological signs might be called **essential tremor plus** (Bhatia et al, 2018).

Bobble-head doll syndrome is head tremor at 2 to 3 Hz seen in patients with suprasellar cysts, third ventricular cysts, or aqueductal obstruction (Saracoglu et al, 2019).

iii. Titubation

Oscillation of the head in the anteroposterior or vertical direction in the standing position is called **titubation**. This phenomenon is seen in association with cerebellar lesion and is considered to be a kind of postural tremor because it involves the neck and axial muscles during isometric contraction.

iv. Physiologic tremor

In healthy subjects, fine tremor of the fingers might be seen at a frequency of about 10 Hz when they hold the upper limbs straight forward, especially when the subjects are excited or nervous. Normally, physiologic tremor is difficult to see and is a pure **mechanical tremor** due to a resonance of the peripheral structures without involving muscle contraction; thus, it can be recorded with an accelerometer but not with EMG. In **enhanced physiologic tremor**, not only the mechanical component but also a reflex component with the spinal cord is involved. This can be recorded with both accelerometry and EMG, but it behaves like a mechanical tremor in that the frequency is influenced by weighting. By contrast, Parkinson tremor and essential tremor are **central tremors**, implying a central oscillator and no influence of weighting on the tremor frequency.

C. ACTION TREMOR

Tremor involving a muscle that is changing its length by isotonic contraction is called **action tremor**. Action tremors occur during kinetic or intentional movement (**kinetic tremor** or **intention tremor**). Intention tremor may be related to the fact that the excitability of the motor cortex starts increasing before the actual movement starts. A typical example of action tremor is cerebellar tremor due to a lesion of the cerebellar efferent pathway or the dentatothalamic tract. It is usually seen in the upper extremities as a tremor of large amplitude that increases as the target is approached. Its frequency varies but is most commonly about 3 Hz. The frequency tends to be low for the action tremor involving the proximal muscles and higher

BOX 19-3 TREMOR CAUSED BY MIDBRAIN LESION

Infarction involving the medial part of the midbrain causes complex tremor composed of resting, postural, and kinetic tremor at the frequency of less than 4.5 Hz in the contralateral upper extremity. Resting tremor is caused by a lesion of the nigrostriatal pathway in the midbrain tegmentum, and postural and kinetic tremor is due to a lesion of the dentothalamic tract or red nucleus. This tremor complex corresponds to Holmes tremor (Gajos et al, 2010). It may also be called peduncular rubral tremor (Remy et al, 1995), but this terminology might be a little confusing. The resting tremor in this condition responds to L-dopa in some cases. If unilateral oculomotor nerve palsy is associated with the involuntary movement in the contralateral extremity, it is called Benedikt syndrome (see Chapter 9-1A, p. 58, and Table 8-1, p. 53).

BOX 19-4 FRAGILE X-ASSOCIATED TREMOR/ATAXIA SYNDROME

This syndrome occurs in males of age over 50 years with the initial symptom of postural and action tremor. This is due to a CGG triplet repeat of the fragile X mental retardation gene 1 (FMR1). When the repeat expansion is 55 to 200, only the postural and intention tremor and cerebellar ataxia are manifest, while, with the expansion of more than 200, a complete form of fragile X syndrome occurs with additional symptoms of parkinsonism, cognitive decline, polyneuropathy, and autonomic nervous impairment (Berry-Kravis et al, 2007). Repeat-associated non-AUG (RAN) translation at the CCG repeat was also shown to play a pathogenetic role in this condition, where AUG (adenine-uracil-guanine) is a DNA codon (Krans et al, 2016).

for that of distal muscles. In most cases, cerebellar ataxia is also seen, but dysmetria is difficult to confirm because of the presence of tremor. Common causes of a lesion of the superior cerebellar peduncle are multiple sclerosis and vascular lesions. A combination of scanning speech, intention tremor, and nystagmus is called the Charcot triad of multiple sclerosis.

Tremors seen during execution of a specific task include voice tremor, primary writing tremor, and orthostatic tremor. **Voice tremor** is characterized by oscillations in its amplitude while speaking. A recent study of 93 patients with voice tremor spectrum by multimodal functional

MRI in the United States revealed various overlap of pathophysiology between essential tremor of voice and dystonic tremor of voice (de Lima Xavier & Simonyan, 2020).

Primary writing tremor is seen in the hand only when writing. These conditions may be variants of focal dystonia (see p. 172). Shaking of the legs while standing is seen in patients with Parkinson disease, parkinsonism, essential tremor, and cerebellar ataxia and is called **orthostatic tremor**. For this condition, there seems to be a clinical and etiologic heterogeneity (Benito-León & Domingo-Santos, 2016), and orthostatic tremor with the underlying condition of undetermined etiology is called **primary orthostatic tremor**. The condition is primarily seen while standing still, and little if any during walking or other activities. Recording with the surface EMG discloses symmetric oscillation at 13 to 18 Hz not only in the legs but also in the upper limbs. Treatment with benzodiazepines, β blockers, and anticonvulsants, including gabapentin, has been reported to be effective (Benito-León & Domingo-Santos, 2016). Occurrence among siblings has been reported (Virmani et al, 2012).

3. CHOREA

Chorea is formed of irregular, relatively quick movements involving especially the hands, mouth, and tongue and is characterized in the beginning by a quasi-purposive appearance. In other words, the movement may appear purposeful and can be easily imitated. Sydenham in 1686 described small movements of the fingers, toes, and lips in children with rheumatic fever (**Sydenham chorea**). This form of chorea is different from the gross dancing movements described by Huntington in 1872.

There are a number of hereditary diseases presenting with chorea (Martinez-Ramirez et al, 2020). **Huntington disease** is a representative form of autosomal dominant hereditary disease and is characterized by chorea and progressive cognitive decline that often starts in the 30s (Box 19-5). It is due to a triplet repeat of huntingtin gene, and its main pathologic abnormality includes neuronal degeneration of the caudate nucleus and atrophy of the frontal lobe. In the patients with young onset, rigidity may predominate over other symptoms.

Neuro-acanthocytosis (chorea-acanthocytosis, Levine-Critchley syndrome) can be clinically similar to Huntington disease, but its mode of inheritance is variable. Its X-linked form is called McLeod syndrome. It is characterized by acanthocytosis in the peripheral blood, and the orolingual chorea is so

BOX 19-5 GENOTYPE AND PHENOTYPE OF HUNTINGTON DISEASE

Among the hereditary neurodegenerative diseases, Huntington disease is especially known for its **phenotypic heterogeneity**. Even in patients with genetically confirmed Huntington disease, the involuntary movement may not be typical chorea, and instead it may appear as partial dyskinesia or even cerebellar ataxia. Therefore, in a patient presenting with any kind of movement disorder and progressive cognitive impairment, this diagnosis should be considered. On the other hand, patients showing the typical clinical picture of Huntington disease may have a different genetic abnormality (**genetic heterogeneity**) or no genetic abnormality may be identified (Kim & Fung, 2014; Termsarasab, 2019; Wild et al, 2008).

severe that it often causes tongue bite (see Figure 3-15, p. 20). **Dentate-rubro-pallido-luysian atrophy (DRPLA)** is an autosomal dominant hereditary disease and shows myoclonus, generalized convulsions, cognitive decline, chorea, and cerebellar ataxia. It is caused by abnormal expansion of a CAG repeat, and its gene product is atrophin-1. The main clinical picture is different depending on the age of onset. Patients with childhood onset show myoclonus, generalized convulsion, and mental deterioration, whereas those with adult onset tend to show chorea and cerebellar ataxia in addition to myoclonus.

Chorea may be seen in a variety of systemic conditions such as vascular disease of the basal ganglia, rheumatic fever, pregnancy (chorea gravidarum), and thyrotoxic chorea. It is also important to note that chorea can occur as a side effect of neuroleptics. Chorea seen in association with rheumatic fever (**Sydenham chorea**) is considered to be caused by an autoimmune mechanism against striatal antigen in relation to streptococcal infection. A series of conditions seen in children in association with streptococcal infection is called **pediatric autoimmune neuropsychiatric disorders associated with streptococcal infection (PANDAS)** and **pediatric acute-onset neuropsychiatric syndrome (PANS)** (Gamucci et al, 2019), but many experts do not believe that this is really a definable separate syndrome.

Whether chorea is, like essential tremor and Parkinson tremor, mediated by the sensori-motor cortex or not remains an open question. As D2 blockers are effective on chorea in most cases, a relative hyperexcitable state of the dopaminergic system is postulated for

the underlying mechanism. The vesicular monoamine transporter type 2 inhibitors such as tetrabenazine can be effective for chorea in Huntington disease (Gibson & Claassen, 2021).

4. BALLISM

Gross violent movements of the extremities that look like throwing out or kicking away were first described as ballism(us) by Greiff in 1883. Ballism is commonly caused by a vascular lesion of the subthalamic nucleus or striatum and thus involves contralateral upper and lower limbs (**hemiballism**). Vigorous muscle contractions irregularly involve the shoulder and pelvic girdles and the proximal limbs. It may be so severe that the patient may fall on the floor across over the bedframe, and if hemiballism continues, it may even be life-threatening. However, it usually responds to medication with diazepam, a D2 blocker like haloperidol, or a VMAT2 inhibitor. In the chronic stage, each movement becomes less violent and resembles chorea, although it still involves primarily the proximal muscles.

Ballism-like involuntary movements can be seen as a manifestation of transient ischemic attack (TIA), and it is called **limb-shaking TIA**. It is commonly seen in patients with marked narrowing or occlusion of the internal carotid artery, and gross violent movements involve the contralateral upper or

lower extremity, lasting less than 5 minutes. It is often induced by physical activities like standing up and may be accompanied by mild hemiparesis. Its underlying mechanism is undetermined, but it is postulated to be related to a hemodynamic disturbance (Muraga et al, 2016; Persoon et al, 2010).

Hemichorea–hemiballism is seen in patients with uncontrolled diabetes mellitus, especially the **nonketotic hyperglycemic state** (Li & Chen, 2015). They may show high signal intensity lesions in the contralateral striatum on T1-weighted MRI (Figure 19-6).

5. ATHETOSIS

Athetosis was described by Hammond in 1871 as irregular, slow, writhing movements seen in the hands and feet. The movement is often strange and cannot be imitated. It is caused by an organic lesion of the striatum, including cerebral palsy, a residual state of encephalitis, or anoxic encephalopathy. In these conditions, athetosis is commonly seen in combination with dystonia. Indeed, there may not really be any important physiologic difference between athetosis and dystonia.

6. DYSTONIA

Dystonia is defined as a movement disorder characterized by sustained or intermittent muscle contractions causing abnormal, often repetitive, movements, postures, or both. Dystonic movements are typically patterned and twisting and may be tremulous. Dystonia is often initiated or worsened by voluntary action and associated with overflow muscle activation (Albanese et al, 2013). In many cases of dystonia, the affected part of the body shows abnormal posture of a writhing nature in the resting condition, and slow, writhing involuntary movements are superimposed on the abnormal posture upon voluntary movement. In many other cases, there is no dystonia at rest but only action dystonia. For childhood-onset patients, dystonia is generalized and mainly involves the trunk and extremities (**generalized dystonia**), but there is also **focal dystonia**, which is usually of adult onset. Focal dystonia includes **blepharospasm**; **oromandibular dystonia**; **cervical dystonia (spasmodic torticollis)**; **spasmodic dysphonia (laryngeal dystonia)**; **focal hand dystonia**, including **writer's cramp** and **musician's cramp** (which are not actually cramps, but dystonia); and so forth.

At the early clinical stage of blepharospasm and cervical dystonia, repetitive phasic movements like blinks and neck rotation, respectively, may predominate, resembling tremor. In the advanced stage, the eyelids cannot be opened and the

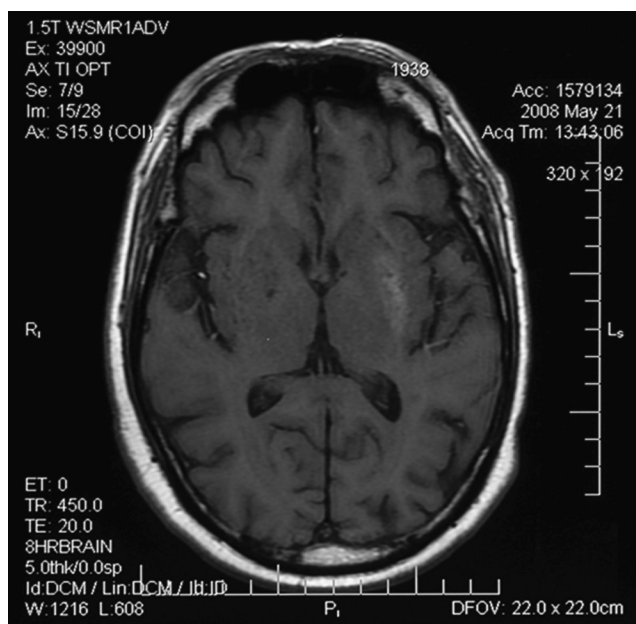


Figure 19-6 T1-weighted MRI in a patient with hyperglycemia presenting with right side chorea. Note an increased signal in the left striatum. By courtesy of Dr. Stephen G. Reich of University of Maryland.

neck might remain rotated. In writer's cramp, tremor and myoclonus may be superimposed onto the dystonia.

Writer's cramp and musician's cramp occur only when writing or only when playing a specific musical instrument, respectively, at least in the early clinical stage. This **task specificity** is characteristic of focal dystonia. For example, a patient with **embouchure dystonia** (lips) suffers from oral dystonia only when playing a certain kind of wind instrument but not when playing other kinds of wind instruments (Haslinger et al, 2010). Hand tremor occurring only when writing is called primary writing tremor, and this phenomenon might be caused by the same mechanism as focal dystonia (see p. 171). Abrupt interruption of hitting movement of a golf club (**yips**) may be another example of task-specific dystonia.

The condition in which blepharospasm and oromandibular dystonia are combined is called **cranial dystonia** (formerly called Meige syndrome). In this condition, the bulbar muscles may show abnormal contraction while speaking or swallowing, causing **spasmodic dysphonia** and **spasmodic dysphagia**, respectively (see Chapter 14-7, p. 104).

Local injection of botulinum toxin is now widely used for effective treatment of a variety of neurological conditions, not just the hyperkinetic movement disorders but also non-neurological conditions (Anandan & Jankovic, 2021; Chiu et al, 2021; Hallett et al, 2013).

A. FUNCTIONAL ABNORMALITY OF THE SENSORI-MOTOR CORTEX IN FOCAL DYSTONIA

Functional abnormality of the somatosensory cortex and the motor cortex or abnormality of functional coupling between the two cortical areas has been proposed as the pathophysiology of focal dystonia. For example, in patients with musician's cramp, fusion of finger areas in the primary somatosensory cortex has been demonstrated by an MEG study (Figure 19-7). A TMS study showed a decrease in the GABA-A-mediated inhibitory functions in the primary motor cortex. Whether these abnormalities of the sensorimotor cortex are a cause of focal dystonia or a plastic change secondary to dystonia remains an open question.

B. FIXED DYSTONIA

The condition in which an extremity is fixed in a dystonic posture is called **fixed posture dystonia** or **fixed dystonia** (Figure 19-8). Sometimes it is fixed so hard that it is impossible to passively return it to the normal position. It is common in young females, and it often follows trauma or complex regional pain syndrome (Schrag et al, 2004). It is usually localized, but it may spread widely. Its most common cause is considered to be functional (Hawley & Weiner, 2011), but it

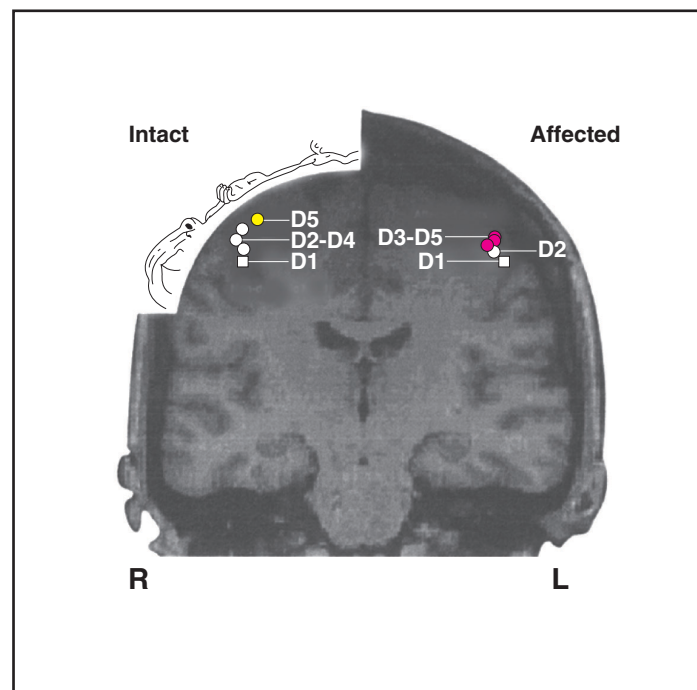


Figure 19-7 Abnormal somatotopic organization in the primary somatosensory cortex in a patient with musician's cramp, demonstrated by recording of the somatosensory evoked magnetic field following stimulation of each individual finger. Areas corresponding to digits 1 through 5 (D1-D5) are separate in the intact cortex but fused in the affected cortex. Cited from Elbert et al., 1998, with permission.

BOX 19-6 GENERATING MECHANISM OF DYSTONIA

Generalized dystonia is often considered to be caused by abnormal function of the sensorimotor cortex secondary to a lesion of the basal ganglia. DBS of the internal segment of the globus pallidus (GPi) is known to improve dystonia (Sako et al, 2008). In focal dystonia, by contrast, functional abnormality of the sensorimotor cortex as a result of plastic change is considered to play a major role (Abbruzzese & Berardelli, 2011; Hallett, 2011; Quartarone & Hallett, 2013). For example, writer's cramp is common among people who are engaged in writing, and musician's cramp commonly affects musicians who play a special kind of musical instrument over a long period of time. Furthermore, in patients with musician's cramp, fusion of finger areas in the primary somatosensory cortex has been demonstrated by an MEG study (see Figure 19-7), and a TMS study showed a decrease in the GABA-A-mediated inhibitory functions in the primary motor cortex (Hallett, 2011). In relation to these clinical observations, Merzenich and his group trained monkeys to make continuous hand movements, and they observed motor deterioration in the hand and physiologic abnormality in the sensorimotor cortex. These findings suggest development of plastic change in the sensorimotor cortex as a result of continuous movement. Although these changes may be secondary to the long-standing dystonia, it is more plausible to consider these plastic changes as a cause of dystonia.

Conte et al. (2019) proposed that focal dystonia might be a network disorder with the basal ganglia and cerebellum as the central contributors in the network.

Intramuscular injection of botulinum toxin is effective in many cases of focal dystonia (Hallett et al, 2013). This effect is likely produced both by weakening the muscles directly related to dystonia and by changing the excitability of the sensorimotor cortex (Dresel et al, 2011).

may be an organic disorder modified by a functional factor. In some cases of typical fixed dystonia, an excitability study of the motor cortex by TMS disclosed abnormal inhibitory function, which may suggest that those patients may have a predisposition to dystonia (Avanzino et al, 2008).

C. SENSORY TRICK

A phenomenon called **sensory trick** (*geste antagoniste*) is another characteristic feature of focal dystonia (Ramos et al, 2014). It is commonly seen in cervical dystonia. For example, as soon as the back of the patient's head is touched



Figure 19-8 Fixed dystonia of the right leg following post-traumatic complex regional pain syndrome. By courtesy of Dr. Takahiro Mezaki.

by the examiner's hand or by the patient's own hand, or as soon as the patient stands against a wall to which the head is attached, the head returns to normal position. Sometimes, the head might return to normal position even before the actual touching if the patient can predict the touching. In patients with blepharospasm, rubbing the forehead may help opening the eyelids. The mechanism underlying the sensory trick has not been fully elucidated. In some cases, intended movement of adjacent muscles might be effective (**motor trick**). For example, a patient with blepharospasm might learn to move his or her own mouth to help open the eyelids.

Furthermore, some paradoxical phenomena might be seen in patients with generalized dystonia. Patients who have great difficulty walking forward because of generalized dystonia might be able to run forward or to walk backward or sideways more easily. This might be another manifestation of task specificity.

D. CAUSES OF GENERALIZED DYSTONIA

Generalized dystonia is seen in organic lesions of the striatum such as cerebral palsy, Wilson disease, residual state of



Figure 19-9 Abnormal posture of a patient with generalized dystonia due to Wilson disease.
By courtesy of Department of Neurology, Kyushu University.

encephalitis, Creutzfeldt–Jakob disease, and hereditary or sporadic neurodegenerative diseases primarily affecting the basal ganglia. It is especially common in patients who have been taking antipsychotic drugs for a long period of time (**tar-dive dystonia**). It should be kept in mind that patients with focal dystonia might develop generalized dystonia during the clinical course, especially subjects having DYT1 gene mutation.

As **Wilson disease** is a treatable condition if the diagnosis is made early enough, its possibility should be considered whenever young subjects present with dystonia of undetermined etiology or postural tremor (Figure 19-9). Diagnosis can be made by finding a Kayser–Fleischer ring in the cornea due to copper deposit and a decrease in the blood ceruloplasmin level. Although extremely rare, patients with Wilson disease may have neither a Kayser–Fleischer ring nor a decreased ceruloplasmin level, in which case measurement of urinary copper or liver biopsy may be warranted. MRI can also be useful in the diagnosis.

E. HEREDITARY DYSTONIA

A number of hereditary disorders presenting with dystonia have been reported, and genetic abnormalities have been disclosed in many of them (Charlesworth et al, 2013; Larsh et al, 2020). **Torsion dystonia** was first reported by Oppenheim in 1911 as dystonia musculorum deformans. This is inherited in an autosomal dominant trait and is called **DYT1** with torsin gene mutation, but sporadic cases are not rare. The age of onset varies from childhood to young adults, and clinical forms are also variable. In some cases, the clinical symptoms may begin with focal dystonia and later spread to develop generalized dystonia. Muscles of the cranial nerve territory are rarely involved at first. DYT1 dystonia may present with a Holmes tremor–like phenotype (Rodriguez-Quiroga et al, 2018). Like the cases of hereditary myoclonus–dystonia syndrome (Box 19-7), DYT1 is often associated with depression. DYT6 due to mutation of THAP1 is also common.

There are three types of hereditary dopamine-responsive dystonia: **hereditary progressive dystonia with marked diurnal fluctuation (Segawa disease)**, **autosomal recessive dopamine-responsive dystonia with Segawa syndrome**, and **sepiapterin reductase deficiency**. Segawa disease is autosomal dominant due to mutation of GTP cyclohydrolase I gene, and it occurs in childhood (Figure 19-10). It is characterized by diurnal fluctuation (better in the morning and getting worse as the day goes on). There is also marked improvement of the symptoms immediately

BOX 19-7 HEREDITARY MYOCLONUS–DYSTONIA SYNDROME

This is an autosomal dominant hereditary disease characterized by myoclonus and dystonia occurring in childhood or puberty. It is not always progressive, and some cases might have a relatively benign clinical course.

Dystonia is generalized while myoclonus mainly involves the trunk and proximal extremities. In most cases, myoclonus and dystonia occur independently, but in some cases myoclonus and dystonia may appear in combination (myoclonic dystonia or dystonic myoclonus). **Mutation of the epsilon-sarcoglycan gene** has been found in many families with this condition (Menozzi et al, 2019; Nardocci et al, 2008). The symptoms may improve after ethanol intake. Patients with this disease may also have depression. Electrophysiologic studies of many patients with this syndrome disclosed no evidence of cortical origin for myoclonus (Roze et al, 2008). This condition might have been included among the cases previously diagnosed as familial essential myoclonus. **Tyrosine hydroxylase deficiency** is another cause of familial myoclonus–dystonia syndrome, but this is transmitted in an autosomal recessive fashion (Stamelou et al, 2012a).

after awakening from a nap. Sepiapterin reductase deficiency occurs in infancy or childhood with motor and language delays, axial hypotonia, dystonia, weakness, oculogyric crises, and diurnal fluctuation of symptoms with sleep benefit (Friedman et al, 2012).

X-linked recessive dystonia–parkinsonism (DYT3) seen in the Philippines is called Lubag (meaning “twisted”) syndrome. Clinically, the patients present with dystonia, but later in the clinical course parkinsonism predominates. Pathologically, Goto et al. (2013) found a decrease in the number of neuropeptide Y-positive cells in the striatum. **Pantothenate kinase–associated neurodegeneration (PKAN)** is known as a neurodegeneration with brain iron accumulation (NBIA) and includes a condition previously named Hallervorden–Spatz syndrome (see Box 16-5, p. 125). **Beta-propeller protein associated neurodegeneration (BPAN)**, another form of NBIA, is characterized by learning disability and progressive gait abnormalities since childhood followed by progressive dystonia in adulthood (Paudel et al, 2015). **Machado–Joseph disease** is a type of spinocerebellar degeneration (SCA3) with ataxin as the gene product, but in some cases dystonia may be a predominant clinical sign (see Chapter 16-2C, p. 130). **MEGDHEL syndrome** (3-methylglutaconic aciduria, dystonia–deafness,



Figure 19-10 Dystonic posture of the left leg in a patient with Segawa disease.
By courtesy of Department of Neurology, Kyushu University.

hepatopathy, encephalopathy, Leigh-like syndrome) is a progressive deafness–dystonia syndrome with neonatal liver involvement (Maas et al, 2017).

7. DYSKINESIA

The term “dyskinesia” is often used to encompass complex involuntary movements that do not fit into another category or is used for a particular situation of an involuntary movement that does fit into another category. Dyskinesia is characterized by relatively quick, irregular involuntary movements involving the lips, tongue, extremities, and trunk, and is often caused by medications. Focal dyskinesia is commonly seen in the lips and tongue (**orolingual dyskinesia** or **oral dyskinesia**). Orolingual dyskinesia is also commonly induced by drugs (**tardive dyskinesia**), but it

may also be due to multiple small infarctions in the striatum. This type of dyskinesia is also common among aged subjects, and its etiology is often undetermined.

The condition in which a patient cannot keep sitting still because of an urge to move is called **akathisia**. Akathisia is commonly induced by long-term administration of antipsychotic drugs or L-dopa. It is important to prevent this condition because, once it develops, it may be resistant to drug control.

A common side effect seen in patients with Parkinson disease during L-dopa treatment is dyskinesia (see Box 17-10, p. 151). Since this phenomenon is more common in relatively young patients with Parkinson disease, a dopamine agonist may be preferred for young patients instead of L-dopa. It is believed that L-dopa acts on both the D2 and D1 receptors whereas the commonly used dopamine agonists act predominantly on the D2 receptor.

Focal uncontrolled writhing abdominal movements are called **belly dancer syndrome** (Meyer et al, 2017; Shamin & Hallett, 2007). Recently an adenyl cyclase 5 (ADCY5) gene mutation has drawn increasing attention as a genetic cause of any hyperkinetic movement disorder, especially dyskinesia (Vijiaratnam et al, 2019).

8. MYOCLONUS

Myoclonus is defined as shock-like involuntary movements of the face and extremities. It was first described in 1881 by Friedreich, who reported a case of “Paramyoklonus multiplex” (Friedreich, 1881). Myoclonus is usually caused by abrupt, instantaneous contraction of muscles (**positive myoclonus**), but it can also be caused by transient interruption of the ongoing muscle contraction (**negative myoclonus**). In fact, the two kinds of myoclonus often occur in combination, namely an abrupt muscle contraction immediately followed by a silent period of muscle discharge, or vice versa. Negative myoclonus is also called asterixis. Asterixis was originally described for hand tremor seen in patients with hepatic encephalopathy, but nowadays this term is also used for non-rhythmic negative myoclonus.

Involuntary movements fulfilling the criteria of myoclonus can also occur in physiologic situations such as **hiccup** and **nocturnal myoclonus**. Nocturnal myoclonus is

a shock-like movement seen unilaterally during the early stage of sleep. It is distinguished from periodic limb movement in sleep (PLMS) (see Chapter 27-5, p. 257) in that nocturnal myoclonus is more shock-like and not periodic.

Myoclonus can be classified into three types depending on its estimated sites of origin (cortical, brainstem, and spinal), but some myoclonia remain unclassified (; Shibasaki & Hallett, 2005; Shibasaki, 2007a,b; Shibasaki & Thompson, 2011; Merchant et al, 2020; Shibasaki et al, 2020). There might be myoclonus originating from the thalamus, basal ganglia, and cerebellum, but if any, those are estimated to be mediated by the sensori-motor cortex, and thus those might be classified into a category of cortical myoclonus.

A. MYOCLONUS OF CORTICAL ORIGIN

Cortical myoclonus in most cases is considered to be based on the abnormal increased excitability of the primary motor cortex (M1). Physiologically, myoclonus of the hand is preceded by a cortical spike generated in the hand area of the contralateral M1 and is caused by abrupt contraction of the hand muscle via the fast-conducting corticospinal tract (Figure 19-11) (Shibasaki, 2012; Shibasaki & Hallett, 2005). Thus, cortical myoclonus is commonly seen in the fingers, feet, and face, which are innervated by a large area of the M1. If multiple sites of the M1 are activated, myoclonus involves the corresponding multiple sites, and as the

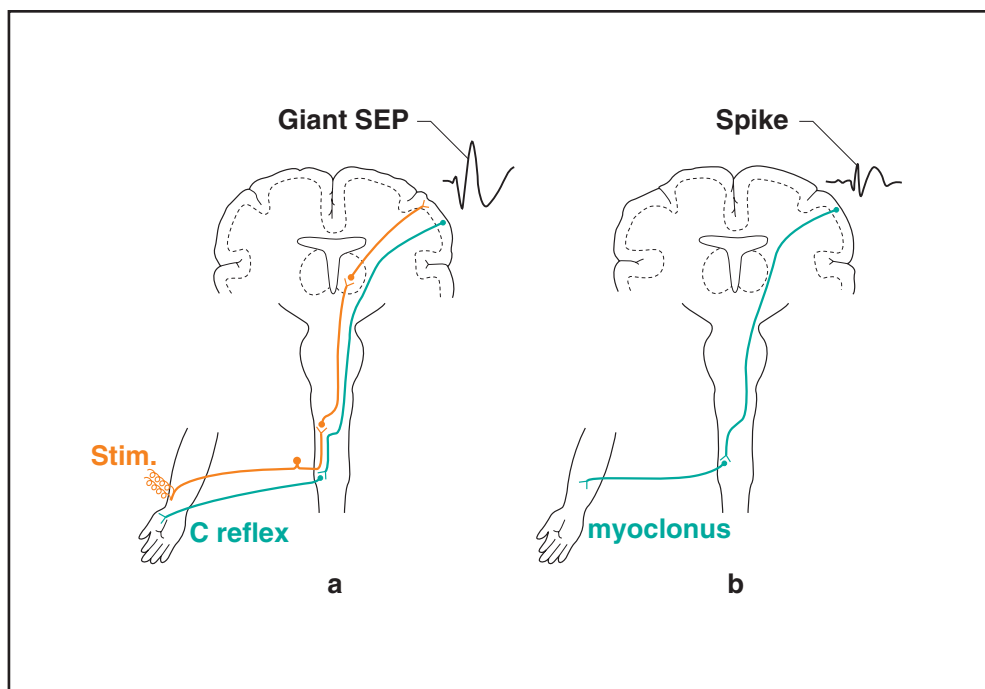


Figure 19-11 Schematic illustration showing generating mechanism of cortical reflex myoclonus (a) and spontaneous cortical myoclonus (b). Giant SEP: giant somatosensory evoked potential, Stim.: stimulus, C reflex: cortical reflex mediated by the transcortical reflex loop.

cortical discharge spreads within the M1, myoclonus may also spread to adjacent muscles rapidly.

Excitability of the M1 can be studied by TMS. By using a paired stimulus with an interstimulus interval of several milliseconds, impairment of GABA-A-ergic intracortical inhibition has been demonstrated in cortical myoclonus. A similar phenomenon is seen also in focal dystonia, but Hanajima et al. (2008) found a qualitatively different abnormality in the inhibitory system between the two conditions.

i. Cortical reflex myoclonus

In most cases of cortical myoclonus, the excitability of the primary somatosensory cortex (S1) is also abnormally enhanced, and the incoming stimulus from the periphery produces a **giant evoked potential** and a reflex myoclonus through a **transcortical reflex** mechanism (Shibasaki & Hallett, 2005). This is called **cortical reflex myoclonus** as against **spontaneous cortical myoclonus**, and both conditions are considered to be caused by involvement of the M1 either spontaneously or by the reflex mechanism (see Figure 19-11). Most myoclonus epilepsies are considered to be caused by involvement of the whole strip of M1 by this epileptogenic discharge. Thus, cortical myoclonus may be called **epileptic myoclonus**. Reflex myoclonus may be also elicited by photic stimulus (**photic cortical reflex myoclonus**) (Shibasaki & Neshige, 1987).

ii. Causes of cortical myoclonus

Cortical myoclonus is seen in a variety of conditions (Box 19-8), and there is no disease specificity. Its representative condition is a group of diseases categorized into **progressive myoclonus epilepsy**, which includes a variety of hereditary neurodegenerative diseases and metabolic abnormalities (Box 19-9). Recently, gene abnormalities have been identified in many of these diseases (Table 19-1) (Franceschetti

BOX 19-8 CAUSES OF CORTICAL MYOCLONUS

1. Progressive myoclonus epilepsy
2. Juvenile myoclonic epilepsy
3. Post-anoxic myoclonus
4. Alzheimer disease
5. Creutzfeldt–Jakob disease (in advanced stage)
6. Metabolic encephalopathy (especially uremia)
7. Corticobasal degeneration
8. Olivo-ponto-cerebellar atrophy
9. Rett syndrome

BOX 19-9 CAUSES OF PROGRESSIVE

MYOCLONUS EPILEPSY

1. Unverricht–Lundborg disease
2. Lafora disease
3. Neuronal ceroid lipofuscinosis
4. Mitochondrial encephalomyopathy*
5. Sialidosis
6. Dentato-rubral-pallido-luysian atrophy (DRPLA)
7. Benign adult familial myoclonus epilepsy (BAFME)
8. Angelman syndrome

*Myoclonus epilepsy associated with ragged-red fibers (MERRF, Fukuhara disease)

et al, 2014; Shahwan et al, 2005; van den Ameele et al, 2018; van der Veen et al, 2019).

iii. Negative myoclonus

Cortical myoclonus and cortical reflex myoclonus are usually positive, but negative myoclonus is also seen in some cases, either alone or in combination with positive myoclonus.

TABLE 19-1 GENE ABNORMALITIES OF PROGRESSIVE MYOCLONUS EPILEPSY

Disease	Inheritance	Chromosome	Gene	Protein
Unverricht–Lundborg	AR	21q22.3	CSTB	Cystatin B
Lafora	AR	6q24 6q22	EPH2A NHLRC1	Laforin
Myoclonus epilepsy assoc. with ragged-red fibers	Maternal	mtDNA	MTTK	tRNA ^{Lys}
Sialidosis				
Type 1	AR	6p21.3	NEU1	Sialidase 1
Type 2	AR	20	NEU1	Sialidase 1
Dentato-rubral-pallido-luysian atrophy (DRPLA)	AD	12p13.31	DRPLA	Atrophin 1
MEAK	AD		KCNC1	
NCL (5 types)	AR			

MEAK: myoclonus epilepsy and ataxia due to KCNC1 mutation (Oliver et al, 2017), NCL: neuronal ceroid lipofuscinosis (Williams & Mole, 2012, for classification of NCL; Berkovic et al, 2016, for adult-onset NCL), AR: autosomal recessive, AD: autosomal dominant.

(Modified from Shahwan et al, 2005)

Negative myoclonus of cortical origin is often generated by a transcortical reflex mechanism (**cortical reflex negative myoclonus**) (Shibasaki et al, 1994). It has to be kept in mind that negative myoclonus is often caused as a side effect of medications, especially anticonvulsants. Unilateral negative myoclonus is seen in patients with infarction in the contralateral thalamus (**thalamic asterix**) (Tatu et al, 2000) (Box 19-14).

Post-anoxic myoclonus (Lance–Adams syndrome) may also present with a mixture of positive and negative myoclonus and is mostly considered to be of cortical origin (Lance & Adams, 1963; Shibasaki, 2010). This condition is commonly encountered and is difficult to treat, but anti-convulsants, including levetiracetam, can be effective. Post-anoxic myoclonus should be distinguished from myoclonus seen in the acute phase of anoxic encephalopathy (Elmer et al, 2016).

BOX 19-10 HEREDITARY PAROXYSMAL DYSKINESIA

This is an autosomal dominant hereditary disorder characterized by paroxysmal attacks of dystonic, choreic, and ballistic movements and was previously called paroxysmal kinesigenic choreoathetosis. The age at onset is commonly in puberty, and it is more common in males than in females. It is often the case that the patient's father might not have been aware of his own trouble until he sees the same problem in his son. Three forms are recognized in this category. One is a brief episode induced by sudden movement or intention to move (**paroxysmal kinesigenic dyskinesia, PKD**). The second form is **paroxysmal non-kinesigenic dyskinesia (PNKD)**, in which the attacks are triggered by coffee, alcohol, mental stress, and fatigue. The third form is **paroxysmal exercise-induced dyskinesia (PED)**, which is induced by prolonged exercise. At least four genes have been identified, including proline-rich transmembrane protein 2 (PRRT2), which typically causes PKD, but there is no complete phenotype–genotype correlation (Erro et al, 2014; Friedman et al, 2017; Huang et al, 2015; Méneret et al, 2012; Wang et al, 2011). Glucose transporter-1 deficiency syndrome (GLUT-1DS) exhibits paroxysmal eye movements and epileptic seizures in early infancy and later may cause PED (Ito et al, 2015). A functional MRI study disclosed increased activity in the putamen during the intervals, suggesting a possibility of epileptic dysfunction (Zhou et al, 2010). A small dose of carbamazepine is effective except for PED. These conditions are now classified as synaptopathies (Erro et al, 2017).

BOX 19-11 UNVERRICHT–LUNDBORG DISEASE

This is an autosomal recessive hereditary disease previously also called ataxia cerebellaris myoclonica or Ramsay Hunt syndrome. It is characterized by progressive postural and action myoclonus, cerebellar ataxia, generalized convulsions, and mental deterioration starting between 6 and 14 years of age. Mutation of the cystatin-B (CSTB) gene (previously also called EPM1) on chromosome 21q22 was found (Hyppönen et al, 2015). Some patients also show ocular motor apraxia, dystonia, and marked dementia (Chew et al, 2008).

BOX 19-12 BENIGN ADULT FAMILIAL MYOCLONUS EPILEPSY

This is an autosomal dominant hereditary disease characterized by progressive myoclonus epilepsy of late onset and has a relatively benign clinical course. The initial family was reported by Inazuki et al. in Japan in 1990 as “familial essential myoclonus and epilepsy (FEME),” and in 1991 Yasuda et al. reported a large family with “benign adult familial myoclonic epilepsy (BAFME).” In this condition, myoclonus of the hands is quite rhythmic and resembles tremor, so that it is often misdiagnosed as familial essential tremor. Physiologically, it fulfills the diagnostic criteria of cortical myoclonus, but generalized convulsion is rare, and mild if any. Ikeda et al. called this condition “cortical tremor: a variant of cortical reflex myoclonus” in 1990, and Terada et al. called it “familial cortical myoclonic tremor as a unique form of cortical reflex myoclonus” in 1997. The same disorder is called “familial cortical myoclonic tremor with epilepsy (FCMTE)” in Europe and China (Cen et al, 2016).

Two autopsy cases of this hereditary disease reported from the Netherlands showed loss of dendritic trees and abnormal arborization of Purkinje cells (van Rootselaar et al, 2007). Gene abnormality was found on chromosome 8q23.3-q24.11 in Japanese families, on 5p15.31-p15.1 in European families (Depienne et al, 2010), and on 3q26.32-q28 in a Thai family (Yeetong et al, 2013). A large family with a similar clinical picture was reported from Canada, in which mutation of the NOL3 gene on chromosome 16q21-22.1 was identified (Russell et al, 2012). It is possible that these families may belong to hereditary ion channel disorders (see Chapter 28-1A, p. 259). In Japanese families with BAFME, abnormal expansions of TTTCA and TTTTA repeats in intron 4 of *SAMD12* were found (Ishiura et al, 2018).

An autosomal recessive family with similar clinical picture has been reported, although the age of onset was much younger compared with the autosomal dominant form (Stogmann et al, 2013).

BOX 19-13 MITOCHONDRIAL ENCEPHALOMYOPATHY AND ABNORMALITY OF MITOCHONDRIAL FUNCTIONS

Mitochondrial encephalomyopathy is a group of disorders with functional abnormality of the mitochondrial respiratory chain due to mutation of the mitochondrial gene (mtDNA) or nuclear DNA (nDNA). It occurs in infants, children, or young adults and clinically manifests with epileptic attacks, vascular lesions, progressive cerebellar ataxia, neuropathy, and so on. In this category are included **Kearns–Sayre syndrome** or **progressive ophthalmoplegia** (see Chapter 9-5B, p. 68), **myoclonus epilepsy associated with ragged-red fibers (MERRF, Fukuhara disease)** (Table 18-2), **mitochondrial encephalomyopathy with lactic acidosis and stroke (MELAS)** (see Box 2-1, p. 6), “neuropathy, ataxia, retinitis pigmentosa,” Leigh syndrome, **Leber hereditary optic atrophy**, and mitochondrial neurogastrointestinal encephalomyopathy (MNGIE). **Leigh syndrome**, also known as infantile subacute necrotizing encephalomyelopathy, is a genetically heterogeneous disorder (Lake et al, 2016) and is characterized by neurodevelopmental regression and progressive dystonia with brainstem involvement. **Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE)** is an autosomal recessive disease and presents with gastrointestinal symptoms, followed by blepharoptosis, progressive external ophthalmoplegia, emaciation, polyneuropathy, and leukoencephalopathy (Garone et al, 2011). Many of the mitochondrial diseases cause movement disorders of various kinds (Schreglmann et al, 2018).

Recently, abnormal mitochondrial functions other than the respiratory chain have drawn increasing attention. Those functions include the transport of mitochondrial protein, the dynamics of mitochondrial shift, its division, fusion and distribution, and programmed cell death. Namely, the question of how mitochondria are related to pathogenesis of neurodegenerative diseases in the aged population such as Alzheimer disease, Parkinson disease, and amyotrophic lateral sclerosis is a hot topic (DiMauro, 2013; Yang et al, 2021).

It is important to note that cortical myoclonus is not necessarily due to organic brain lesion. In uremic encephalopathy, for example, typical positive myoclonus and negative myoclonus of cortical origin are seen, and they may improve completely after the treatment with dialysis. There might be cases of uremia whose myoclonus is of brainstem origin.

BOX 19-14 UNILATERAL ASTERIXIS DUE TO THALAMIC LESION

Patients with vascular lesion in the lateral thalamus may show asterixis in the contralateral hand. Many of those patients also show ataxia and loss of superficial as well as proprioceptive sense in the contralateral extremities, suggesting the presence of a lesion in the motor thalamus (Vim nucleus) and the sensory relay nucleus (see Chapter 30-1,2, p. 271). Two patients with thalamic asterixis were reported in whom the β EEG rhythm was increased over the contralateral central region in association with each asterixis, which might suggest an increased inhibitory mechanism of the corresponding motor cortex (Inoue et al, 2012b). These findings suggest that thalamic asterixis is mediated by the sensori-motor cortex, probably as a result of impairment of the input from the cerebellar dentate nucleus to the motor cortex.

B. MYOCLONUS OF BRAINSTEM ORIGIN

Palatal tremor (previously called palatal myoclonus) (Box 19-16), reticular reflex myoclonus, startle disease, opsoclonus, and intractable hiccup belong to this category.

Reticular reflex myoclonus was first described by Hallett et al. (1977) in post-hypoxic patients. In response to auditory or somatosensory stimulus, muscle contraction starts in the sternocleidomastoid and trapezius muscles and spreads rostrally to involve the facial muscles and caudally to involve the extremity muscles (Figure 19-12).

BOX 19-15 TRANSIENT MYOCLONIC STATE WITH ASTERIXIS IN ELDERLY PATIENTS

Hashimoto et al. (1992) reported seven aged patients who showed transient episodes of asterixis with the name “transient myoclonic state with asterixis in elderly patients.” In this condition, both positive and negative myoclonus are seen mainly in the upper extremities and spontaneously disappear in a week or two but may recur. It responds well to clonazepam. No metabolic abnormalities or history of drug use was identified. A short burst of spike-and-wave complexes is seen on the EEG over the central region in association with the asterixis, suggesting a cortical origin. This condition is common in Japan, but so far there has been no report of similar conditions from other countries. Although there is no familial occurrence, it may be a kind of ion channel disorder just like benign adult familial myoclonus epilepsy (Box 19-12).

BOX 19-16 PALATAL TREMOR AND GUILLAIN-MOLLARET TRIANGLE

Palatal tremor involves rhythmic vertical oscillation of the soft palate at a frequency of about 3 Hz. It used to be called palatal myoclonus, but in a symposium of the first international congress of the Movement Disorder Society organized by one of the authors (M.H.) in Washington, DC, in 1990, another author (H.S.) presented a case of palatal myoclonus due to brainstem lesion on video and suggested that the movement might fit the characteristics of tremor rather than myoclonus. At that time, Professor C.D. Marsden, as the chairman of the symposium, motioned for voting, and as the result the new term was agreed on by a majority of the audience. Palatal tremor is divided into symptomatic and essential groups. **Symptomatic palatal tremor** is caused by a lesion involving any part of the Guillain-Mollaret triangle, but most commonly by a lesion of the central tegmental tract in the pons, and then by a cerebellar lesion (Deuschl et al, 1994; Jang & Borruat, 2014). In this case, enlargement of the inferior olive is commonly observed on MRI, which is considered to be due to trans-synaptic degeneration. Familial causes of progressive ataxia and palatal tremor include Alexander disease, which is a leukodystrophy predominantly affecting the brainstem (Martinez-Poles et al, 2020); polymerase gamma mutations; SCA20; and GM2 gangliosidosis, and two sporadic cases were found to have tauopathy (Gao et al, 2017).

Palatal tremor is sometimes associated with synchronous vertical oscillation of the eyes (ocular myoclonus) (see Chapter 9-6C, p. 71), and in some cases it is also associated with rhythmic involuntary movements at the same frequency in the neck and/or upper extremities. Repetitive, rhythmic, slow (1 to 4 Hz) movement mainly affecting the cranial and limb muscles is sometimes called **myorhythmia** (Une et al, 2020), and this condition is due to various etiologies, including Whipple disease (Baizabal-Carvallo et al, 2015).

The **Guillain-Mollaret triangle** is a circuit connecting the cerebellar dentate nucleus with the contralateral inferior olive via the superior cerebellar peduncle and the central tegmental tract and back to the dentate nucleus via the inferior cerebellar peduncle. It is known that the parvocellular neurons of the red nucleus receive main afferents from the contralateral dentate nucleus and send main efferents to the ipsilateral inferior olive through the central tegmental tract (see Brodal, 2010). However, it is also believed that there is a direct pathway from the dentate nucleus to the contralateral inferior olive, which passes by the red nucleus without forming synapses (Lapresle & Ben Hamida, 1970). In view of the occurrence of

pseudohypertrophy of the inferior olive following not only a lesion of the central tegmental tract but also a lesion of the cerebellum, involvement of the direct dentato-olivary tract is more likely to play a primary role in generating symptomatic palatal tremor.

Patients with **essential palatal tremor** often complain of hearing ear clicking due to periodic contraction of the tensor veli palatini muscle (Deuschl et al, 1994). It has now been reported that many cases of essential palatal tremor are likely functional (Pirio Richardson et al, 2006; Stamelou et al, 2012b). Local injection of botulinum toxin can be used for its treatment.

Startle disease is an excessive startle reaction elicited by an unexpected, sudden stimulus and is considered to be an abnormal enhancement of an ordinary startle reflex. In this case, the response begins in the orbicularis oculi muscles and spreads generally. Familial occurrence of this condition by autosomal dominant transmission is called **hyper-ekplexia**. In many of these families, mutation of the gene coding the $\alpha 1$ or β subunit of inhibitory glycine receptor has been demonstrated (see Chapter 28-1A and Table 28-1, p. 260).

Myoclonus of brainstem origin tends to be resistant to medical treatment. **Opsoclonus** (see Chapter 9-6D, p. 71) has been treated symptomatically with clonazepam and gabapentin with some effect. Opsoclonus caused by an autoimmune pathogenesis including a paraneoplastic syndrome may be treated with plasmapheresis or intravenous immunoglobulin (IVIg). For **essential palatal tremor**, the ear click can be symptomatic. No drug therapy has been consistently effective, but injection of botulinum toxin may be useful (see Box 19-16); this entity is usually functional. Symptomatic palatal tremor usually does not have an ear click, but it is typically associated with ataxia and appears to be caused by lesions in the brainstem affecting the dentato-rubral-olivary pathway. **Hiccup** is a physiologic myoclonus involving the diaphragm, but long-lasting, medically intractable hiccup is certainly abnormal and may be caused by a lesion of the medulla oblongata. Various drugs, including gabapentin, baclofen, olanzapine, and midazolam, have been tried for intractable hiccup with some effect.

C. MYOCLONUS OF SPINAL CORD ORIGIN

There are two kinds of spinal myoclonus: spinal segmental myoclonus and propriospinal myoclonus. **Spinal**

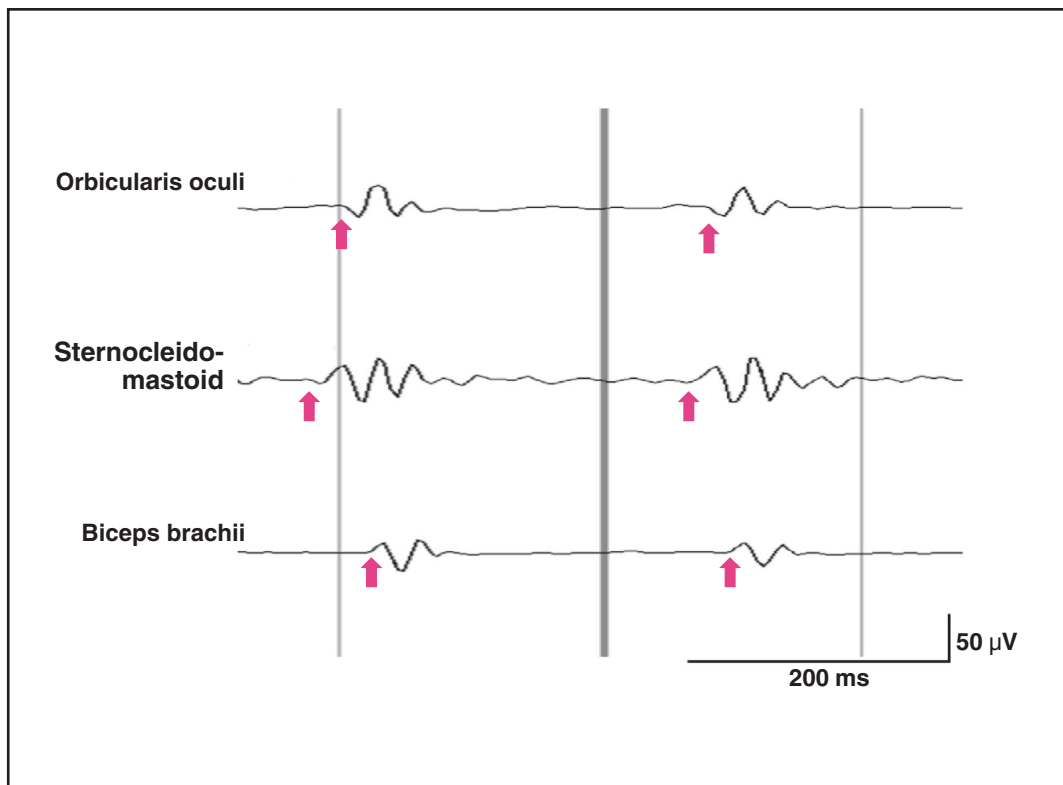


Figure 19-12 Surface EMG of reticular reflex myoclonus recorded from three muscles in a 60-year-old woman who showed spontaneous as well as reflex myoclonus in the face, neck, and upper extremities following an episode of anoxic encephalopathy. Arrows indicate the onset of each EMG discharge. Cited from Inoue et al., 2012a, with permission.

segmental myoclonus is irregular or periodic positive myoclonus localized to the muscles innervated by one or two spinal segments, and it is stimulus-sensitive in some cases. Although its etiology is not always identifiable, some patients are thought to have myelitis. Prognosis is favorable with good response to clonazepam.

Propriospinal myoclonus manifests irregular shock-like movements of the trunk, mainly causing abrupt flexion of the trunk. Physiologically, it is characterized by slow spread from a certain thoracic segment to rostral as well as caudal segments. It may be stimulus-sensitive and often requires differentiation from a functional disorder, which now seems the most frequent etiology (Esposito et al, 2014; Kang & Sohn, 2006; van der Salm et al, 2014; Williams et al, 2008).

D. MYOCLONUS OF UNDETERMINED ORIGIN

Source of myoclonus seen in the following conditions has not been fully determined. Those include periodic myoclonus seen in Creutzfeldt–Jakob disease and subacute sclerosing panencephalitis (SSPE), hereditary myoclonus–dystonia syndrome (see Box 19-6), and drug-induced myoclonus.

BOX 19-17 DIAPHRAGMATIC FLUTTER

Rhythmic abdominal movements may be a manifestation of diaphragmatic flutter. Diaphragmatic flutter is also called van Leeuwenhoek disease after Antonie van Leeuwenhoek, a Dutch businessman and scientist in the 17th century who is known as an inventor of the microscope. After noticing abnormal rhythmic movements in his own abdomen, he first consulted a doctor because he thought they might be due to cardiac disease, but nothing else suggestive of heart disease was found. Then he examined his own body and discovered something wrong in the diaphragm. Diaphragmatic flutter is now considered to be a kind of myorhythmia involving the respiratory muscles, predominantly the diaphragm (Bohlhalter et al, 2008).

i. Creutzfeldt–Jakob disease

Periodic myoclonus is seen in the initial stage of this disease (see Box 23-8). It repeats with an interval of about 1 second, although the interval varies to a considerable degree within the same patient. EMG discharge associated with

BOX 19-18 STARTLE REFLEX IN PROGRESSIVE SUPRANUCLEAR PALSY

The reflex center of startle reflex is considered to be a reticular nucleus in the caudal pons. Since this nucleus is involved by neurodegeneration in progressive supranuclear palsy (PSP), loss of startle reflex has been reported in cases of PSP. In the clinical setting, this reflex can be demonstrated but requires two persons. Someone stands behind the patient and makes a loud clap. A second person in front of the patient observes whether there is a blink or not. By giving unexpected electrical stimulus to the supraorbital nerve and by recording EMG from the cervical muscles (trigemino-cervical reflex), Bartolo et al. (2008) found loss of this reflex in many cases of PSP.

the myoclonus also varies in its length, and the sites of involvement may also vary. It is commonly accompanied by periodic synchronous discharge (PSD) on EEG, but either periodic myoclonus or PSD may appear independently. The relationship between the myoclonus and PSD also varies (Figure 19-13). In this disease, therefore, there is no causative relationship between the cortical activity and the peripheral motor phenomenon.

ii. Subacute sclerosing panencephalitis (SSPE)

This is chronic encephalitis that occurs years after infection with measles virus. Periodic involuntary movements seen in

this condition repeat at an interval of 4 to 13 seconds, and the frequency is quite constant at each stage of disease. Each movement is much longer in duration than ordinary myoclonus, and thus it might be called **periodic dystonic myoclonus**. It is commonly seen in the extremities, but it may also be seen in the eyes. On EEG, characteristic PSDs are seen in this condition. Unlike Creutzfeldt–Jakob disease, PSDs in this condition are quite constant in waveform and duration and are constantly associated with the involuntary movements, suggesting cortical participation in their generation (Figure 19-14) (Oga et al, 2000).

9. MOTOR STEREOTYPES

Repetitive occurrence of the same movements may be called motor stereotypies. Movements commonly encountered in this condition range from simple movements like shaking arms, nodding, and swinging the body to complex movements, and they may look purposive, but actually they have no purpose or function (Edwards et al, 2012; Martino & Hedderly, 2019). They tend to occur in clusters, each cluster lasting several seconds to several minutes. The patient's consciousness is usually preserved, but the movements do not appear to be intentional, and distraction of the patient's attention to other objects may decrease the movements. They are called **motor stereotypies** in order to distinguish them from stereotypic behavior encountered in psychiatric diseases.

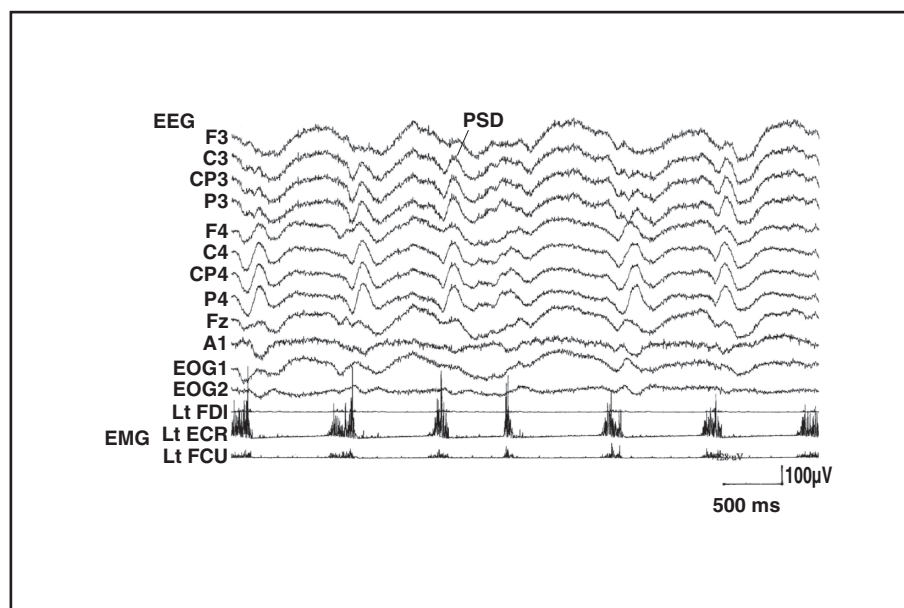


Figure 19-13 Polygraphic record of EEG and EMG in a patient with Creutzfeldt–Jakob disease. EOG: electrooculogram, Lt: left, FDI: first dorsal interosseous muscle, ECR: extensor carpi radialis muscle, FCU: flexor carpi ulnaris muscle.

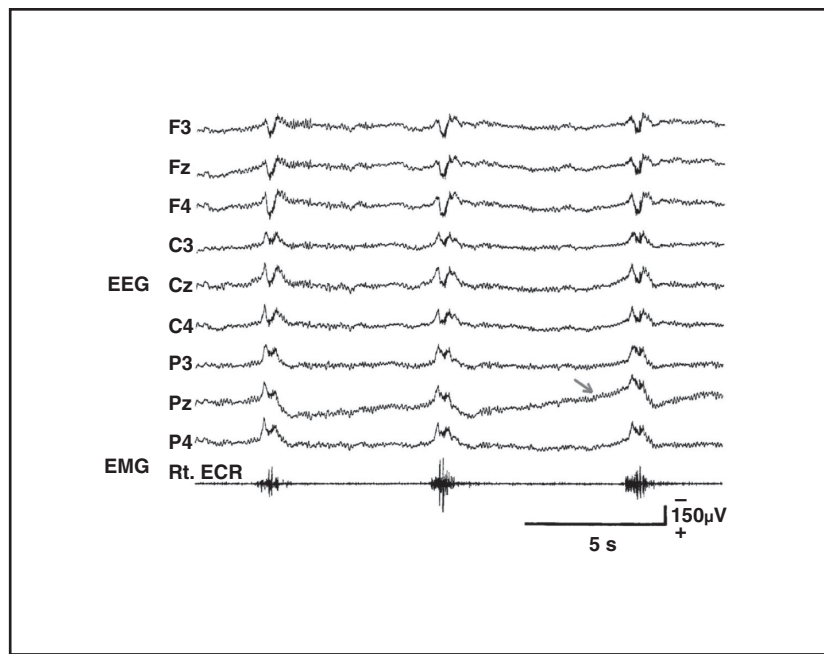


Figure 19-14 Polygraphic record of EEG and EMG in a patient with subacute sclerosing panencephalitis (SSPE). Rt. ECR: right extensor carpi radialis muscle. There is a fixed relationship between dystonic myoclonus and periodic synchronous discharge on EEG. Cited from Oga et al., 2000, with permission.

Motor stereotypies are commonly observed in children with Asperger syndrome and other automatisms, and mental retardation. In adults, this abnormal behavior is seen in severe infectious encephalitis, autoimmune encephalitis like limbic encephalitis, cerebrovascular diseases, and neurodegeneration, among others frontotemporal lobar degeneration (see Chapter 23-2B, p. 223) (Mateen & Josephs, 2009). It is postulated that motor stereotypies are seen when the circuit connecting the dorsolateral prefrontal cortex and the anterior striatum is interrupted.

Repetitive movements may be also seen in complex tics, obsessive-compulsive neurosis, and habits, but distinguishing motor stereotypies from these conditions is not difficult from the history and other accompanying symptoms and signs. In motor stereotypies, reciprocal innervation of agonist and antagonist muscles may be preserved because the movement appears quasi-purposive.

Rett syndrome is a developmental disorder due to mutation of the methyl-CpG binding protein 2 (MECP2) gene (Percy et al, 2010). It primarily affects female infants. Girls with this syndrome develop normally until 12 months of age and then manifest various neurological symptoms, including stereotypic movements of both hands such as washing, squeezing, and clapping. These stereotypic movements subside in several years, but the patients may show other involuntary movements such as tremor, chorea, myoclonus, and dystonia, and also rigidity and ataxia. Dystonia

may progress to cause scoliosis during the course of illness (Temudo et al, 2008).

10. INVOLUNTARY MOVEMENTS THAT CAN BE SUPPRESSED MOMENTARILY

Some involuntary movements can be suppressed for a short period of time if the patient wishes to do so. Those are tics, some functional involuntary movements, and restless legs syndrome.

A. TICS

Tics are irregular brisk movements ranging from shock-like simple movements resembling myoclonus (**simple tic**) to complex movements (**complex tic**). Patients with tics tend to repeat certain movements like blinking, grimacing, shrugging of the shoulders, and twisting of fingers, but in patients with **Gilles de la Tourette syndrome**, tics appear as various kinds of movements. Patients often pout, smack, and vocalize (**vocal tic**), and utter obscene words (**coprolalia**). They can stop the movements for several seconds if they try, but it is often followed by rebound. The patients often feel a **premonitory urge** to move before a bout of tics and feel release after the bout (Cavanna et al, 2017). In fact, there may be overlap between tics and functional

tic-like movements (Ganos et al, 2019). Many families suggest a possibility of autosomal dominant transmission, but the genes have been elusive (Yu et al, 2019). Tic disorders are usually associated with comorbid symptoms such as attention-deficit hyperactivity and obsessive-compulsive disorder/behavior, and multiple cerebral structures have been shown to be involved (Yael et al, 2015).

B. FUNCTIONAL (PSYCHOGENIC) INVOLUNTARY MOVEMENTS

Functional involuntary movements are not infrequently encountered (see Chapter 29, p. 268). They may look like tremor, myoclonus, or focal dystonia. Characteristics of functional involuntary movements include the following: (1) it is difficult to explain as an ordinary involuntary movement (**incongruous**), (2) it fluctuates in pattern, degree, and distribution during the clinical course (**inconsistent**), (3) it decreases or disappears when the patient's attention is drawn to others (**distraction**), and (4) there is a **psychogenic background** in the patient's character or environment (Hallett et al, 2012; Perez et al, 2021). In unilaterally predominant tremor, if the patient is requested to repeat voluntary movements with the intact or less affected hand at a certain pace, the tremor frequency might be replaced by the frequency of the voluntary movement (**entrainment**). When it is difficult to confirm the functional nature, it is useful and important to follow up the clinical course with repeated observations.

In cases showing repetitive shock-like movements with some time intervals in between, it may be worthwhile to record EEG and EMG simultaneously and to back average the EEG with respect to the EMG correlates of the involuntary movements. If a Bereitschaftspotential (readiness potential) is demonstrated before those movements, it suggests involvement of the mechanisms responsible for normal voluntary movement (Shibasaki, 2012; Shibasaki & Hallett, 2006). However, it should be noted that in conversion the movement is involuntary; hence, there is no intention.

C. RESTLESS LEGS SYNDROME

In this condition, patients complain of an unpleasant sensation in the lower extremities and trunk, and they feel a strong urge to move their legs. It tends to occur while resting at night, and the symptoms are transiently relieved by moving the legs. It is said to be seen in 3% to 10% of the population. About 60% of cases are said to be familial, and

increasing attention has been paid to this condition from a genetic point of view (Winkelmann et al, 2008; Ylikoski et al, 2015).

Many patients suffering from this phenomenon may also have periodic limb movement in sleep (Chapter 27-5, p. 257) (Högl & Stefani, 2017). Because this phenomenon is suppressed by L-dopa or dopamine agonists, its possible relationship with Parkinson disease has been discussed, but the current consensus is that these two conditions are not related in pathogenesis (Fulda & Wetter, 2008). A recent multicenter genetic study of restless legs syndrome and Parkinson disease found no evidence for a causal relationship or genetic correlation between the two (Estiar et al, 2021).

Restless legs syndrome was considered in relation to peripheral nerve lesions. A study by Hattari et al. (2009) involving a large number of patients with various neuropathies showed that this phenomenon is seen more commonly in patients with hereditary neuropathies than in the control group, but not in the group with acquired neuropathies. Restless legs syndrome also occurs in association with various conditions, including iron deficiency, pregnancy, uremia, and celiac disease.

11. INVOLUNTARY MOVEMENTS OF PERIPHERAL NERVE ORIGIN

Fasciculation and **myokymia** are abnormal muscle contractions due to an increase in the spontaneous excitability of anterior horn cells (see Chapter 17-2B, p. 138). Repetitive muscle contraction seen in **hemifacial spasm** is myoclonus of peripheral nerve origin (see Chapter 11-1A, p. 83). In **painful legs and moving toes syndrome**, irregular and slow involuntary movements are seen in the toes associated with pain in the legs. This condition is often associated with peripheral nerve lesions like sciatica. A similar condition seen in the upper extremities is called **painful hand and moving fingers syndrome**.

Some patients with inflammatory or hereditary polyneuropathies may show finger tremor, although its pathogenesis is not well known. Involuntary movements resembling athetosis might be seen in patients with proprioceptive loss if they keep their eyes closed, regardless of a lesion in the central or peripheral nervous system. This is called **pseudoathetosis** (see Chapter 20-2B, p. 196). However, tremor associated with polyneuropathy is also seen while the eyes are kept open. A retrospective large case-control study by Ahlskog et al. (2012) showed that tremor was more

frequently seen in patients with neuropathy associated with IgM-monoclonal gammopathy than in those with neuropathy of other causes. Disabling tremor has been also reported in patients with CIDP positive for neurofascin IgG4 antibodies and poor response to IVIg (Querol et al, 2014).

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SOMATOSENSORY FUNCTION

Sensation (sense) is defined as the primary reception of input from the external environment and that from the internal condition of one's own body, whereas **perception** is defined as recognition (awareness) of these inputs. With this distinction, sensation is divided into exteroceptive (somatic) sensation and interoceptive (visceral) sensation. **Somatic sensation** is further divided into general somatic sensation, which receives sensory input from the skin, skeletal muscles, and joints, and special somatic sensations such as visual, auditory, and equilibrium. Among these sensations, general somatic sensation is described in this chapter as the somatosensory system. There are three kinds of sensation in the somatosensory system to be examined: **superficial**, **deep/proprioceptive**, and **cortical**. Each sense is further classified into several modalities (Box 20-1).

BOX 20-1 MODALITIES OF SOMATOSENSORY SYSTEM

1. Superficial sensation
 - Pain
 - Warm sense
 - Cold sense
 - Touch, tactile sense
2. Deep/proprioceptive sensation
 - Vibratory sense
 - Joint position sense
 - Movement sense, kinesthesia
 - Deep pressure pain
3. Cortical sensation
 - Stereognosis
 - Graphesthesia
 - Two-point discrimination
 - Baroesthesia
 - Touch localization

1. STRUCTURE AND FUNCTIONS OF THE SOMATOSENSORY SYSTEM

A. SOMATOSENSORY RECEPTORS

Different kinds of receptors are known for the somatosensory system. Receptors for superficial sensation are free nerve endings for pain as well as temperature sense; hair follicle endings, Krause corpuscles, Ruffini corpuscles and free nerve endings for light touch; and Meissner corpuscles, Merkel disks and Pacinian corpuscles for touch. **Modality specificity** of these receptors is not mutually exclusive, and it is determined by the lowest threshold of each receptor. In other words, all receptors can be activated to a certain degree if stimulated with sufficiently high intensity. The kind of stimulus that activates a receptor with the lowest threshold is an adequate stimulus for that receptor. Receptors for deep/proprioceptive sensation are muscle spindles, Golgi tendon organs, and joint receptors (see Chapter 18-1, p. 157).

Sensory receptors generate a local potential called **receptor potential** or **generator potential** in response to the stimulus. When the receptor potential exceeds a certain threshold, an action potential is generated in the nerve terminal attached to the receptors. The discharge rate of action potentials is proportional to the stimulus intensity. Physiologically, the sensory receptors are classified into two types depending on the mode of excitation: rapidly adapting and slowly adapting. **Rapidly adapting receptors** are phasic receptors that rapidly diminish their receptor potentials in response to continuous stimulus presentation, thus rapidly diminishing the discharge rate of action potentials generated in the afferent nerve fibers. **Slowly adapting receptors** are tonic receptors that slowly diminish the receptor potentials and the nerve discharge.

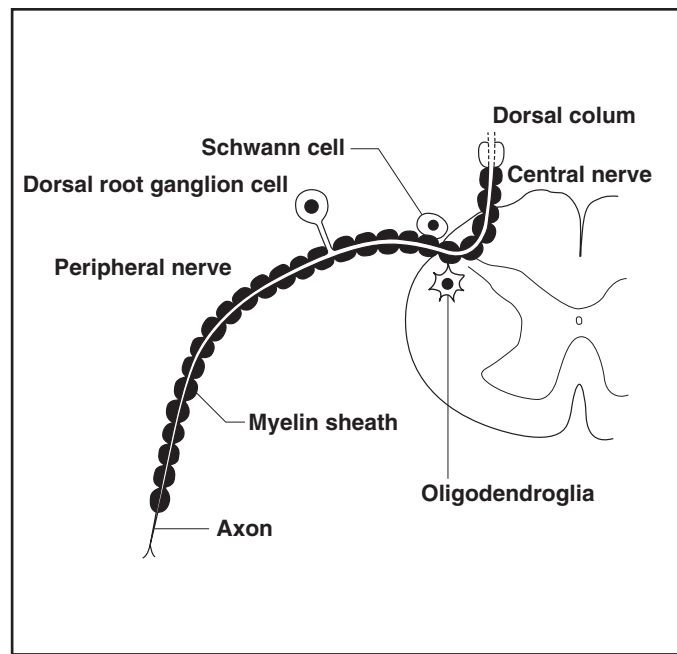


Figure 20-1 Schematic illustration of a peripheral somatosensory nerve fiber. The dorsal root ganglion cell is a bipolar cell, and its peripheral axon and the extramedullary part of the central axon have their myelin sheath made of Schwann cells, whereas the intramedullary part of the central axon has its myelin sheath made of oligodendroglia.

B. PERIPHERAL SOMATOSENSORY NERVES

Primary neurons of the somatosensory system for the limbs and trunk are in the **dorsal root ganglion**. For the cranial nerve territory, the primary neurons are in the semilunar ganglion of the trigeminal nerve (see Figure 10-1, p. 75), the geniculate ganglion of the facial nerve (see Figure 11-1, p. 83) and the superior ganglion and petrous ganglion of the glossopharyngeal nerve (see Figure 14-6, p. 102). All these neurons are bipolar cells, and impulses are conducted through the peripheral axon and the central axon to enter the spinal cord or brainstem. As described in each relevant

chapter, the extramedullary axons are surrounded by myelin sheath of Schwann cell origin, whereas the intramedullary axons are made by myelin sheath of oligodendroglia (Figure 20-1). Histologically, therefore, a peripheral somatosensory nerve belongs to the central nervous system once it enters the spinal cord or brainstem.

The peripheral nerves are classified into four groups based on their diameter, conduction velocity, and function (Table 20-1). The largest fibers are A α fibers, to which Group I and Group II fibers belong. Proprioceptive fibers (Ia and Ib) and α motor fibers belong to Group I. Group II

TABLE 20-1 CLASSIFICATION OF PERIPHERAL NERVE FIBERS BASED ON DIAMETER

Nerve fibers	Diameter (μ)	Conduction (m/s)	Functions
A α fiber			
Group I	12–20	72–120	Proprioception (Ia, Ib), α motor fibers
Group II	6–11	30–72	Light touch, touch, secondary terminal of muscle spindle, γ motor fibers
A δ fiber			
Group III	1–5	4–30	Fast pain
C fiber			
Group IV	0.3–1.5	0.4–2.0	Slow pain

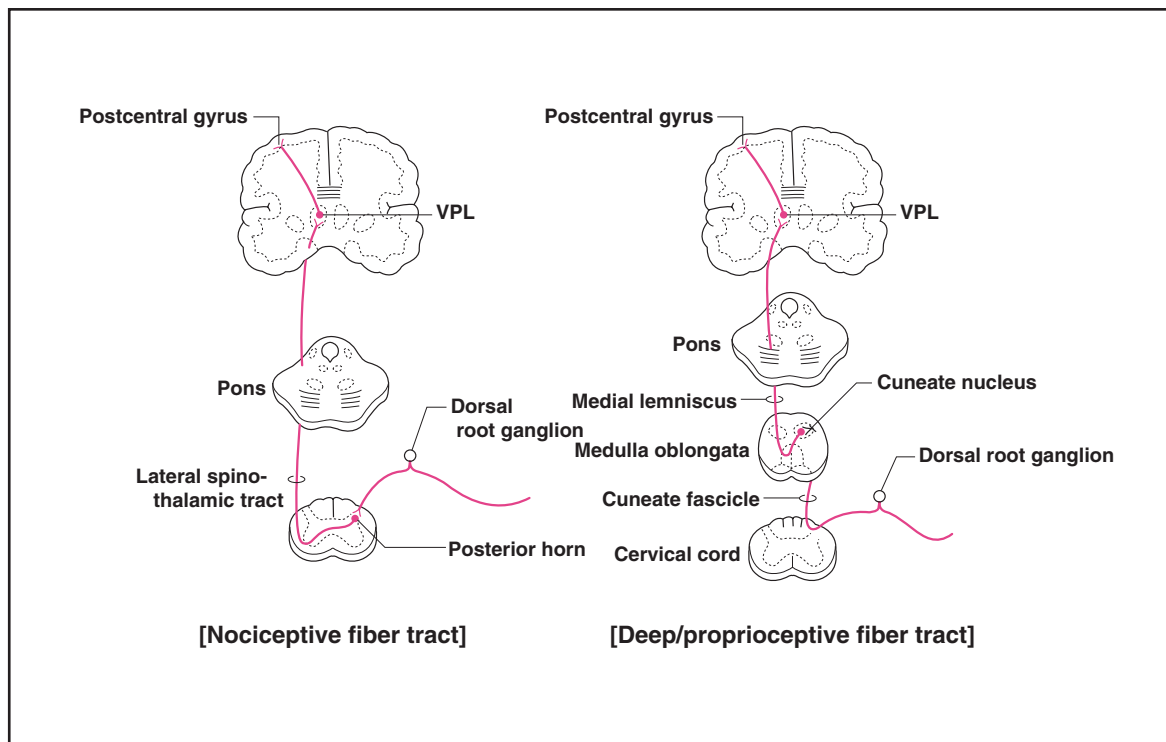


Figure 20-2 Schematic diagram of the somatosensory pathways comparing the nociceptive and deep/proprioceptive tracts. VPL: ventral posterolateral nucleus of thalamus.

includes the sensory fibers for light touch and touch, fibers from the secondary sensory ending of muscle spindle, and γ motor fibers. A δ (Group III) fibers and C (Group IV) fibers are related to the nociceptive sensation. A δ fibers are myelinated, and C fibers are unmyelinated. The larger the fiber diameter, the higher the conduction velocity.

C. SOMATOSENSORY PATHWAY IN THE SPINAL CORD

After entering the spinal cord through the dorsal root entry zone, the somatosensory fibers are separated into two groups (Figure 20-2). Nociceptive small fibers send their branches up and down for one to three segments through the Lissauer tract to enter the dorsal horn gray matter, where they synapse with the secondary neurons. Deep/proprioceptive large fibers enter the **dorsal column** from the dorsal root entry zone and ascend to the lower medullary tegmentum, where they synapse with the secondary neurons. Fibers related to the sensation of the lower limbs and lower trunk ascend through the medial part of the dorsal column (**gracile fascicle**, fasciculus gracilis, **Goll column**) and reach the **gracile nucleus** (nucleus gracilis) at the medulla. Fibers related to the sensation of the upper limbs and upper trunk ascend through the lateral part of the dorsal column

(**cuneate fascicle**, fasciculus cuneatus, **Burdach column**) and reach the **cuneate nucleus** (nucleus cuneatus) at the medulla (see Figure 20-2 and also Figure 5-3, p. 32). There is a laminar structure in the dorsal column, with the fibers innervating the caudal part of the body located medially and those innervating the rostral part of the body located laterally (see Figure 16-3, p. 111).

D. NOCICEPTIVE PATHWAY

The secondary neurons of the nociceptive pathway are in the substantia gelatinosa of dorsal horn (layer II and III of Rexed), where they receive synaptic transmission from the central axon of the primary neurons. Substance P is considered to be the neurotransmitter for this synapse. Axons of the secondary neurons either make synaptic contact with other neurons in the dorsal horn or cross through the anterior white commissure to the anterior part of the lateral column of the other side, where they ascend as the **lateral spinothalamic tract**. This tract ascends through the lateral part of the brainstem to reach the **ventral posterolateral nucleus (VPL)** of the thalamus (see Figure 19-2). In the transverse section of the lateral spinothalamic tract, fibers are arranged in laminar layers, with the caudally originating fibers located laterally

and the rostrally originating fibers located medially (see Figure 16-3, p. 111). Therefore, when the spinal cord is compressed from the lateral side, numbness and nociceptive loss appear first in the distal part of the contralateral lower limb and spread proximally as the compression increases. In an intramedullary lesion, by contrast, those sensory symptoms are limited to a certain segment of the body depending on which layers are damaged. If the most lateral layer of the spinothalamic tract is spared by an intramedullary lesion, the perianal skin innervated by the sacral segments may remain unaffected. This phenomenon, called **sacral sparing**, suggests the presence of an intramedullary cord lesion. Thus, even in a brainstem lesion, if only a certain layer of the spinothalamic tract is involved, it might cause nociceptive loss restricted to the corresponding parts of the trunk or limbs. Even in this case, distinction from the dissociated segmental sensory loss due to a lesion of the spinal cord gray matter is possible from other accompanying symptoms and signs.

E. PROPRIOCEPTIVE PATHWAY

The secondary neurons of the proprioceptive pathway are in the gracile nucleus and cuneate nucleus located in the medullary tegmentum. Axons arising from those nuclei cross at the same level and ascend the ventral medial part of the brainstem tegmentum as the **medial lemniscus** to reach the VPL (see Figure 19-2 and Figures 5-2 and 5-3, p. 32). As the medial lemniscus is located near the midline of the brainstem, it is not damaged by external compression of the brainstem. Thus, selective impairment of tactile or proprioceptive sense in the contralateral limbs in patients with a posterior fossa lesion suggests the presence of an intramedullary brainstem lesion.

F. THE THIRD SENSORY NEURON AND CORTICAL RECEPTIVE AREAS

The third sensory neurons are in the VPM and VPL of the thalamus (see Chapter 30-1, p. 271), and their axons pass through the posterior limb of internal capsule and reach the **primary somatosensory cortex (S1)** in the postcentral gyrus as the thalamocortical fibers. In the S1, like the primary motor cortex, there is **somatotopic organization** in the upside-down homunculus (see Figure 10-2, p. 77). In the clinical setting, it is rare to encounter a localized lesion in the postcentral gyrus. Based on experimental studies using evoked potentials and evoked magnetic fields, it is postulated that the S1 plays a role in the sensation itself and its localization and perception of intensity.

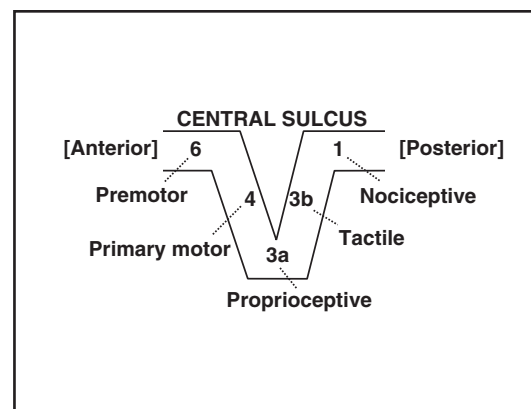


Figure 20-3 Functional localization around the central sulcus in relation to Brodmann area. This diagram shows the main function of each area because this functional localization is not highly specific.

As for the functional localization in the S1, the tactile input is mainly received in the posterior wall of the central sulcus (area 3b), the proprioceptive input in its trough (area 3a), and the nociceptive input in the crown of the postcentral gyrus (area 1) (Figure 20-3). For somatosensory function, however, functional specificity of these cortical areas may not be very high.

Nerve fibers originating from the S1 project bilaterally to the **secondary somatosensory cortex (S2)** located in the superior bank of the sylvian fissure, and also to the ipsilateral areas 5 and 7 located posterior to the S1. The S2 is known to play a role in selective attention, and areas 5 and 7 are thought to be involved in somatosensory perception and recognition.

Nociceptive cortical areas are unique because the nociception is closely related to the affect and emotional reaction. The S1 is primarily related to perception of the intensity and localization of pain, and the S2 is engaged in distinction of the nature of pain. In addition, the amygdala, insula, and cingulate gyrus are related to the emotion-related nociceptive perception and the emotional reaction to pain (Shibasaki, 2004) (Figure 20-4). In a lesion of these limbic areas, the patient becomes indifferent to pain even if the pain can be perceived (**indifference to pain**), or the patient becomes unable to express the pain in language or behavior (**asymbolia for pain**). This condition, however, has to be distinguished from **congenital insensitivity to pain**, which is due to an ion channel disorder of the peripheral nerve (see Chapter 28-1B, p. 262).

In relation to pain, itch is a complex sensory and emotional experience, and functional neuroimaging studies disclosed that multiple cortical areas similar to those related to pain perception are involved in its perception and emotional response (Mochizuki & Kakigi, 2015).

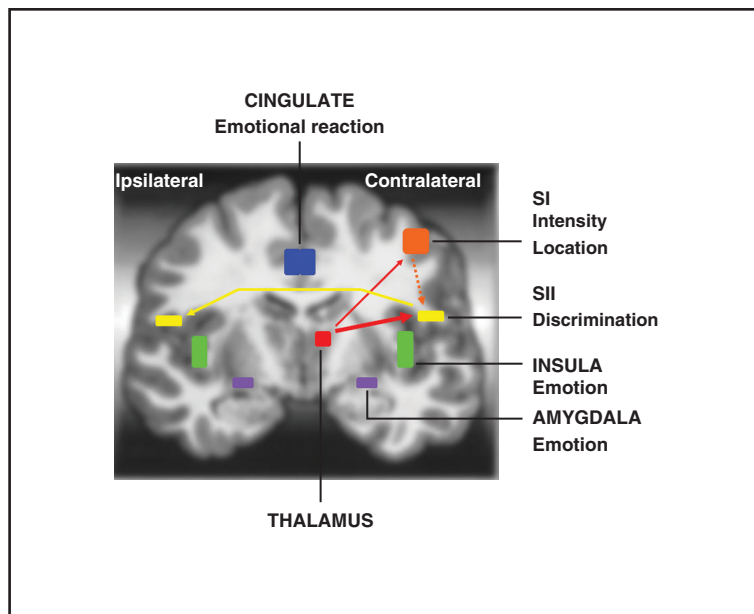


Figure 20-4 Nociceptive cortical areas and their functions. S1: primary somatosensory cortex, S2: secondary somatosensory cortex. Based on Shibasaki, 2004.

2. EXAMINATION OF SOMATOSENSORY FUNCTION AND ABNORMAL FINDINGS

A. EXAMINATION OF SOMATOSENSORY FUNCTION

Sensory examination of the face was described in the section on the trigeminal nerve (Chapter 10-2B, p. 79). For the sensory examination of extremities, pain and temperature sense, light touch, vibration sense, and joint position sense are usually examined. The detail in which the sensory examination should be performed differs depending on the results of history taking. If there is no sensory symptom in the history of the present illness, it may be sufficient to screen some or all of these modalities in the extremities.

Pain sense may be examined with sterile pins (one per patient) in order to avoid infection and injury. Breaking the thin wooden stick holding a cotton swab sometimes reveals a sharp wooden point that can also be used to test pain. **Temperature sense** is tested by using flasks filled with hot or cold water. To test cold sensation, often the side of a tuning fork is used; this is not actually cold but feels cold due to rapid absorption of heat. **Tactile sense** can be tested in two ways: light touch and a little stronger touch. Light touch can be tested with tissue paper, and a little stronger touch may be tested by rubbing the skin with a soft material. **Vibration sense** is usually tested by placing a vibrating tuning fork over the bone. Vibration sense differs depending on which part of the body is tested and on the age of the subject. **Joint position sense** is tested by flexing or

extending a finger or a toe while the subject keeps their eyes closed and is asked to tell the position of the finger or toe. It is important to grasp the finger or toe from both sides instead of front and back in order to avoid the effect of pressure sense. In relation to joint position sense, joint movement sense (**kinesthesia**) can be tested by asking a patient to report any movement of the finger or toe regardless of the direction of movement. Occasionally there is dissociation between the results of joint position sense and kinesthesia. In sensory testing, it is always important to compare the left and right homologous areas.

Examination of cortical sensation is important because its loss might be the only abnormality of the sensory functions in some cases of a localized lesion in the postcentral area or in patients with corticobasal degeneration (see Box 22-2, p. 213). In order to mask the visual information, all cortical senses must be tested while the subject's eyes are kept closed. **Stereognosis** is tested by asking a patient to touch an object with a hand and to tell its name or shape. If a patient is weak or clumsy and cannot manipulate the object themselves, then the examiner should move it across the hand. **Graphesthesia** is tested by asking a patient to read a letter or number scripted on the skin. **Two-point discrimination** is tested by asking a patient to judge one or two by presenting a single stimulus or two stimuli simultaneously to the skin with various distances between the two. The distinguishable distance between the two stimuli is longer in the affected extremity than in the homologous part of the contralateral extremity. **Baroesthesia** is tested

by asking a subject to judge the weight of an object placed in the supported hand. **Touch localization** is tested by asking a patient to report what part of the body was touched. Again, it is important to compare each sense between the two homologous sites.

B. SOMATOSENSORY SYMPTOMS AND SIGNS

If joint position sense or joint movement sense is lost, voluntary movement cannot be executed with appropriate sensory feedback, resulting in incoordination of movement. This condition, called **sensory ataxia**, can be corrected by visual input. In this condition, if the patient holds the arms straight ahead and closes the eyes, the fingers may show an athetosis-like movement (**pseudoathetosis**). In the same principle, a patient with loss of joint sense in the lower limbs tends to fall if the eyes are closed in the standing position. This is called a **positive Romberg sign** (Figure 20-5). Likewise, the patient may have to hold on to a basin to keep balance when washing the face (**basin phenomenon**).

When a patient complains of sensory symptoms like pins and needles, it is better to describe what the patient is actually feeling instead of just saying “numbness.” Spontaneous abnormal sensation is called **dysesthesia**, and unusual sensation caused by stimulus is called **paresthesia**. Hypersensitivity to a painful stimulus is called **hyperalgesia**. Uncomfortable painful sensation elicited by a light

stimulus such as exposure to a breeze is called **hyperpathia**. In this condition, the pain induced by a pin prick might spread much wider and last much longer than usually expected. Hyperpathia is characteristically seen in association with **central pain** such as thalamic pain, although it may also be seen in a peripheral nerve lesion. Decrease in the somesthetic sense is called **hypesthesia**, and loss of sensation is called **anesthesia**—likewise, **hyp(o)algesia** and **analgesia** for decrease and loss of pain sense, respectively.

Pain is one of the most common symptoms in clinical medicine. Pain is often caused by a lesion of the nervous system, but it is also induced by algogenic (algogenic) substances without a direct lesion of the nervous system.

Instantaneous, radiating, shock-like pain along the pathway of a peripheral nerve is called **neuralgia**, such as trigeminal neuralgia (see Chapter 10-2B, p. 80), glossopharyngeal neuralgia (see Chapter 14-2, p. 101), intercostal neuralgia, and postherpetic neuralgia (Box 20-2). Radiating pain seen in cervical spondylosis and lumbar disc herniation is also a typical neuralgia (**root pain**). In cervical spondylosis, radiating pain is induced or aggravated by flexing the neck posterolaterally because the mechanical compression of the nerve root by the narrowed intervertebral foramen is worsened by this maneuver (**Spurling sign**). When the fifth lumbar or the first sacral root is involved by lumbar disc herniation, passive straight leg raising causes a radiating pain along the sciatic nerve (**Lasègue sign**).

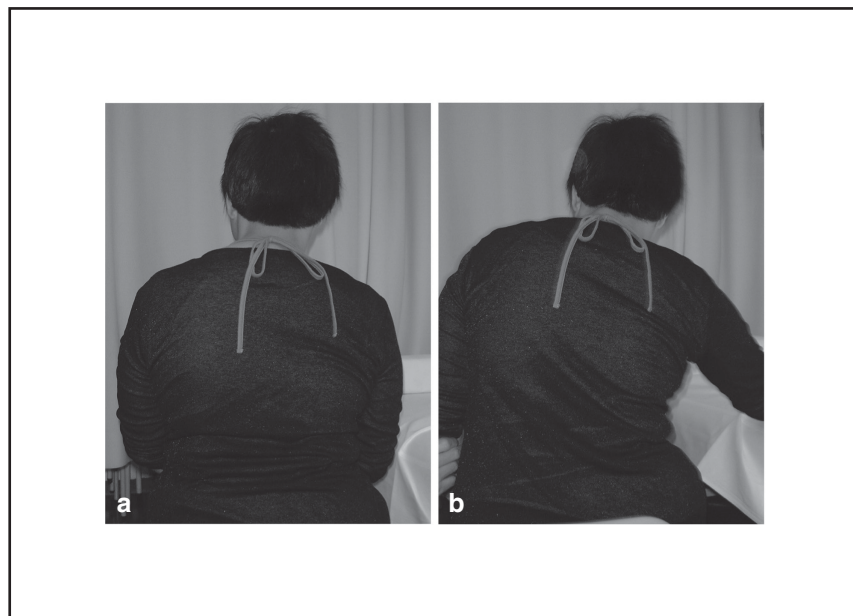


Figure 20-5 Romberg sign seen in a patient with congenital sensory neuropathy. (a) Eyes kept open, (b) eyes closed. As the proprioceptive sense is totally lost below the neck in this patient, the Romberg sign is positive even in the sitting position. Cited from Shibasaki et al., 2004, with permission of the patient.

BOX 20-2 PAIN ASSOCIATED WITH HERPES ZOSTER

Neuralgia may appear after the skin rash of herpes zoster disappears (**postherpetic neuralgia**). However, there are also various kinds of pain in the acute phase of the disease. In fact, the disease may start with lightning pain in the skin region corresponding to the distal axon of the affected nerve. Then painful dysesthesia might appear in the skin region innervated by the nerve. Furthermore, dull aching occurs in the region where the affected sensory ganglion is located, which is probably inflammatory in nature. Moreover, the skin rash itself is painful in the acute phase. Thus, there are at least four different kinds of pain in herpes zoster.

In the mechanical entrapment of a peripheral nerve like carpal tunnel syndrome (Box 20-3) and tarsal tunnel syndrome, tapping of the entrapped site elicits radiating pain in the distal part of the nerve (**Tinel sign**). Pain, burning sensation, and numbness or sensory loss over the lateral aspect of the thigh is called **meralgia paresthetica**. This is due to mechanical compression of the lateral femoral cutaneous nerve and is commonly seen in obese, aged, or diabetic subjects (Parisi et al, 2011). This kind of neuropathy caused by mechanical compression is generally called **entrapment neuropathy**.

Radiating pain starting from the back and descending down along the posterior aspect of the legs induced by neck flexion is called **Lhermitte sign** (see Chapter 15-1, p. 107). This phenomenon is commonly seen in patients with demyelinating diseases such as multiple sclerosis and is postulated to be due to nonsynaptic (ephaptic) transmission of impulses across the demyelinated nerve fibers within the dorsal column.

Painful dysesthesia symmetrically involving the distal limbs, especially the distal lower limbs, is a common symptom in polyneuropathy, often due to **diabetic polyneuropathy** or **alcohol vitamin deficiency polyneuropathy**. In these conditions, burning pain in the feet is commonly experienced at night. Sensory dominant polyneuropathy due to a deficiency of vitamins such as thiamin, cobalamin, and folate is not uncommon (Koike et al, 2015). Pain was thought to be relatively uncommon in Guillain-Barré syndrome (see Chapter 17-3F, p. 142), but in fact it may not be so rare, and pain may even be the initial symptom of this condition (Ruts et al, 2010). Pain in this condition includes root pain, painful dysesthesia, and myalgia.

BOX 20-3 CARPAL TUNNEL SYNDROME

One of the most common **entrapment neuropathies**, carpal tunnel syndrome is due to mechanical compression of the median nerve in the carpal tunnel at the wrist. Clinically it is characterized by atrophy of the thenar muscles and sensory impairment of the radial aspect of the palm. As the border line of sensory innervation by the median and ulnar nerves exists on the midline of the fourth finger, the sensory symptoms in this condition are typically seen on the palmar aspect of the first three fingers and only the radial half of the fourth finger (**finger splitting**). Tapping on the wrist causes a radiating pain toward the tips of the first to fourth fingers (**Tinel sign**). It is commonly seen in people who are engaged in occupations or sports in which the wrist is frequently used. It also tends to occur in pregnant women and in association with obesity. Systemic diseases that are known to cause carpal tunnel syndrome include diabetes mellitus, myxedema due to hypothyroidism, acromegaly due to pituitary adenoma, amyloidosis, rheumatoid arthritis, and deposit of β 2-microglobulin as a result of long-term hemodialysis.

Following a vascular lesion in the thalamus or brainstem, an extremely uncomfortable pain might occur in the contralateral limbs. This condition, called **thalamic pain**, is often associated with hyperpathia (see p. 299). This is a cardinal symptom of **Dejerine-Roussy syndrome** (Box 20-4).

When the distal part of a limb is lost due to trauma or a surgical procedure, the patient might feel as if the lost part still exists (**phantom limb**). If pain is felt in the lost part of the limb, it is called **phantom limb pain**. Its mechanism has not been clarified. Injection of botulinum toxin was reported to be effective for phantom limb pain (Kollewe et al, 2009), but the reason for its efficacy is obscure.

C. IMPAIRMENT OF THE SOMATOSENSORY NERVOUS SYSTEM AND DISTRIBUTION OF THE SENSORY SYMPTOMS

Distribution of the somatosensory symptoms gives an important clue to the anatomic diagnosis. Abnormality of facial sensation was described in Chapter 10-2B (p. 9). Symmetric sensory impairment of the distal lower limbs or distal four limbs (**glove-and-stocking distribution**) suggests diffuse involvement of the peripheral nerves (**polyneuropathy**) (Figure 20-6). Sensory symptoms usually start in the toes or in both the fingers and toes, and they tend to

BOX 20-4 THALAMIC PAIN AND DEJERINE-ROUSSY SYNDROME

According to Henry et al. (2008), the first case of thalamic syndrome was reported by Dejerine and Egger in 1903. A 76-year-old woman suddenly developed left hemiparesis and a pins-and-needles sensation over the left half of her body. The hemiparesis improved in 2 weeks, but severe burning pain developed on the left half of her body, especially in the temple, ear, nose, and tongue. Tactile sense was lost in the left face and the left distal limbs and decreased in the left proximal limbs. The left face was sensitive to cold. Vibration stimuli given to the left limbs caused an unpleasant burning sensation. Joint position sense was lost in the left upper limb and decreased in the left lower limb. The patient could not tell the texture of an object by touching it with her left hand. Tendon reflexes were hyperactive on the left. There was athetoid shaking in the left hand. Later Dejerine reported another case, and two similar cases were reported by Thomas and Chiray in 1904. The first patient reported by Dejerine died in 1905, and the autopsy disclosed a lesion in the posterior lateral part of the right thalamus.

Dejerine and Roussy in 1906 coined the term “le syndrome thalamique.” Thus, this syndrome is characterized by central pain, loss of deep/proprioceptive sense, decrease of or hypersensitivity to superficial sense, mild hemiparesis, ataxia, and choreoathetosis. The pathologic findings of these two cases were recently reinvestigated, and the responsible lesions involving the VPL nucleus of the contralateral thalamus were confirmed (Guédon et al, 2019). Thalamic pain is a main symptom of Dejerine-Roussy syndrome.

spread proximally. The sensory symptoms are usually more severe in the distal part of the limbs and they may be associated with muscle weakness of the distal limbs, and tendon reflexes are usually lost.

i. Polyneuropathy primarily presenting with sensory symptoms

Sensory predominant polyneuropathy is caused by a variety of etiologies. Among metabolic diseases, diabetes mellitus is most common and is characterized by severe burning dysesthesia. Toxins primarily involving the dorsal root ganglion include arsenic, organic mercury, cisplatin, and *n*-hexane (see Chapter 17-3I, p. 145). Malnutrition also causes sensory-dominant polyneuropathy. Examples are vitamin deficiency polyneuropathy due to gastrointestinal diseases,

BOX 20-5 FABRY DISEASE

Although this is an extremely rare disease, it is important to consider it whenever a patient presents with severe pain of unknown etiology. It is an X-linked recessive hereditary disease involving a deficiency of α -galactosidase A, which causes deposition of ceramide-trihexoside and ceramide-digalactoside in the vascular wall and in the central as well as peripheral nervous system (Lenders et al, 2015; Löhle et al, 2015). It occurs in puberty, manifesting lack of sweating, attacks of severe pain, renal failure, myocardial damage, and angiokeratoma of the skin, especially on the scrotum (see Chapter 3-1, p. 10). It is also known as a cause of cerebrovascular disease in young subjects (see Chapter 2-5, p. 6). Although it occurs in males, mild symptoms due to a peripheral nerve lesion might occur also in females. In the peripheral nervous system, small myelinated fibers (A δ) and unmyelinated fibers (C) are affected (see Box 19-9). Marked enlargement of the dorsal root ganglia was shown by MR neurography, suggesting glycolipid accumulation in the dorsal root ganglia (Godel et al, 2017). Pain attacks may be ameliorated by carbamazepine. Enzyme replacement therapy by liver transplantation was shown to have some effect on the peripheral and autonomic nervous symptoms (Toyooka, 2011). In the future, gene therapy and application of induced pluripotent stem (iPS) cells might be warranted.

chronic alcoholism, and niacin deficiency. Vitamin B₁ (thiamin) deficiency causes **beriberi**, which is characterized by axonal sensory-dominant polyneuropathy with severe burning dysesthesia of the feet and marked tenderness of the calf muscles, associated with congestive heart failure. Niacin deficiency causes **pellagra**, which is characterized by ruby erythema of the tongue, diarrhea, dermatitis (see Figure 3-14), and dementia in addition to sensory polyneuropathy. Vitamin B₁₂ deficiency may also cause polyneuropathy in addition to the subacute combined degeneration of the spinal cord (see Chapter 16-2C, p. 131). Folate deficiency also causes sensory-dominant axonal polyneuropathy (Koike et al, 2015).

In case the sensory symptoms are localized to a certain part of the limbs or trunk, it is important to figure out whether the affected region corresponds to the territory innervated by a single peripheral nerve, a dermatome innervated by a certain spinal segment, or a region innervated by the brachial plexus for the upper limb or the lumbosacral plexus for the lower limb. For this purpose, diagrams of the

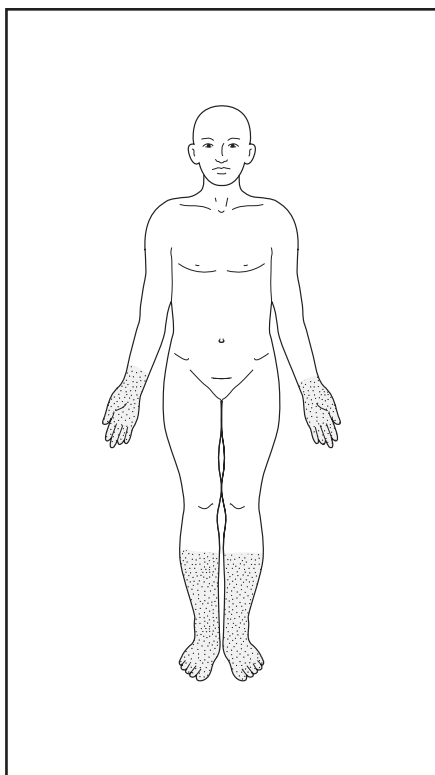


Figure 20-6 Glove-and-stocking distribution of the sensory symptoms due to polyneuropathy.

sensory nerve innervation (Figure 20-7) and the plexus (see Figures 17-6, and 17-7, p. 143) are useful.

ii. Segmental sensory impairment

Sensory impairment localized to a dermatome innervated by one or more segments of the spinal cord is caused by a lesion of the spinal cord or the corresponding nerve roots. If the pain and temperature senses are selectively impaired in that distribution, it is called **segmental dissociated sensory loss**. The lesion responsible for this condition is in the gray matter of the spinal cord, because a lesion of the dorsal root entry zone or roots is expected to impair the tactile and deep/proprioceptive senses as well. A typical example manifesting segmental dissociated sensory loss is **syringomyelia** (Figure 20-8).

When a cervical cord lesion is localized to the central region around the central canal as shown in Figure 19-8, dissociated sensory loss is seen in the dermatomes innervated by the corresponding segments bilaterally, but the tendon reflexes of the biceps and triceps brachii muscles are preserved. This is because the reflex circuit passes through the lateral part of the spinal gray matter and thus is not damaged by the lesion. If the lesion extends a little more laterally, the corresponding tendon reflexes are lost. However,

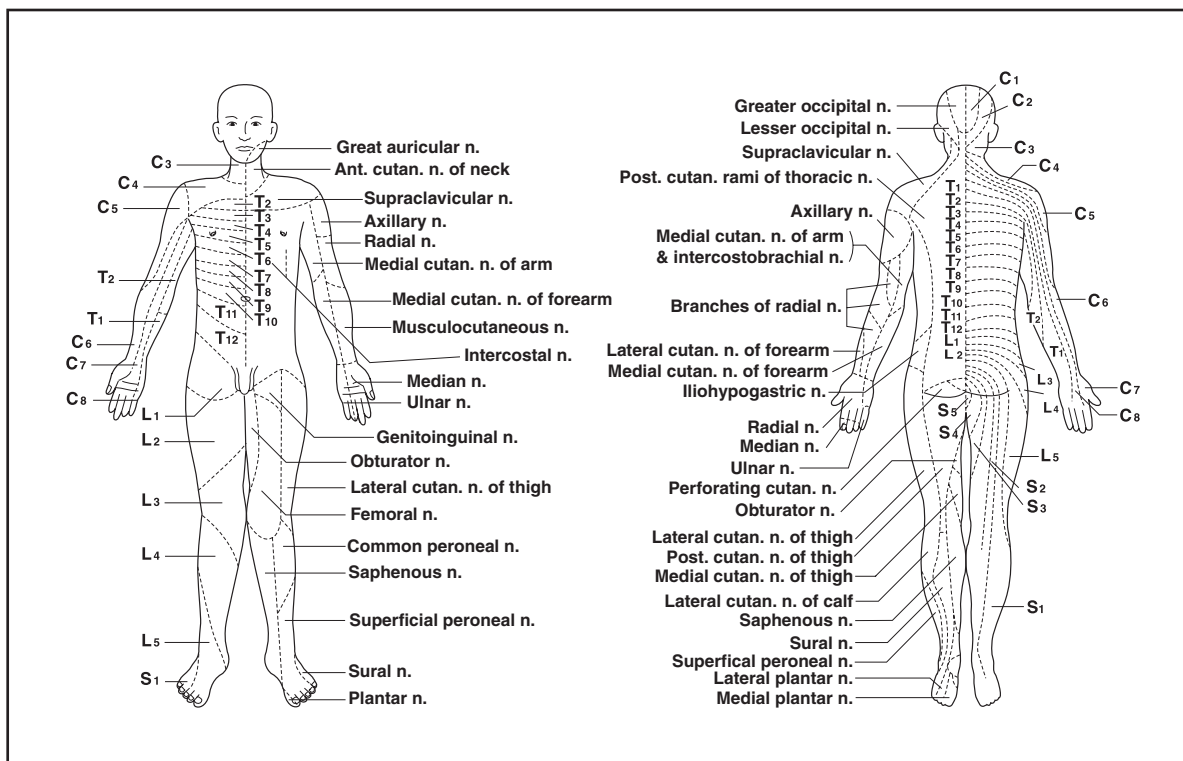


Figure 20-7 Somatosensory innervation by the peripheral nerves and spinal dermatomes. (a) Frontal view, (b) back view.

BOX 20-6 IMPAIRMENT OF DORSAL ROOT GANGLION CELLS

Dorsal root ganglion cells are vulnerable to certain drugs and toxins, especially anticancer drugs such as vincristine and cisplatin, and acrylamide (**toxic sensory ganglionopathy**). In this case, degeneration first occurs in the distal end of the axon and progresses proximally (**dying-back neuropathy**). Cisplatin selectively damages large sensory neurons but does not affect motor neurons or autonomic nervous system (Krarup-Hansen et al, 2007). The sensory symptoms usually start from the toes, because the longer axons are more vulnerable. In subacute myelo-optico-neuropathy (SMON) due to clioquinol intoxication, the dying-back neuropathy occurs not only in the peripheral axon but also in the central axon of the dorsal root ganglion cells (Igata, 2010). Therefore, degeneration occurs in the distal end of the gracile fascicle (Goll column) (see p. 193), because the gracile fascicle is longer than the cuneate fascicle (Burdach column) (Shibasaki et al, 1982). Dorsal root ganglion cells may be selectively affected by autoimmune inflammation (Kusunoki et al, 2008). Dorsal root ganglionopathy may be associated with Sjögren syndrome (Hongo et al, 2012). One of the authors (H.S.) reported siblings who were considered to have congenital aplasia of large neurons in the dorsal root ganglion (see Figure 20-5) (Shibasaki et al, 2004).

this principle does not apply to the thoracic cord, because there is no testable tendon reflex in the thoracic segments.

iii. Sensory impairment in a transverse spinal cord lesion

In a transverse lesion of the spinal cord, somatic senses of all modalities are diminished below the affected segment, because both the spinothalamic tracts and dorsal columns are transected at that level (Figure 20-9). In addition, as the gray matter is also damaged in the affected segment(s), pain and temperature senses are lost at the corresponding level in a sash-shaped distribution. In this condition, the patient may complain of a tight sensation around the trunk at the corresponding segments (**girdle sensation**). Based on the length of the analgesic area or of the girdle sensation, it is possible to estimate the length of the transverse lesion. As the corticospinal tracts are also damaged in this case, the trunk and limbs caudal to the affected segments are paralyzed. The autonomic pathways are also impaired, resulting in sphincter disturbance and loss of sweating below the affected level (see Chapter 21-3, p. 207).

iv. Brown-Séquard syndrome

If either the left or right half of the spinal cord is damaged at a certain segment, motor paralysis is seen below the affected

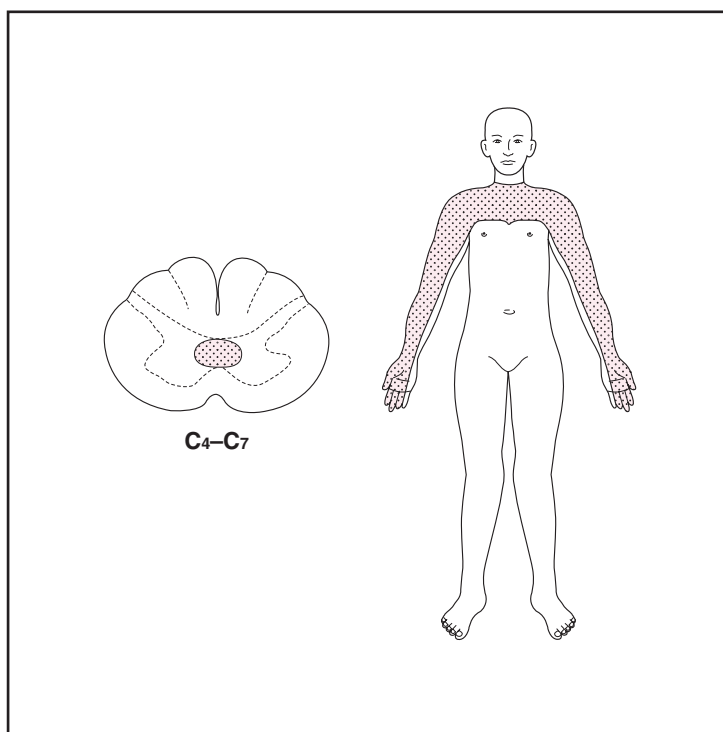


Figure 20-8 Segmental dissociated sensory loss due to a lesion of the central gray matter at the fourth to seventh cervical segment of the spinal cord.

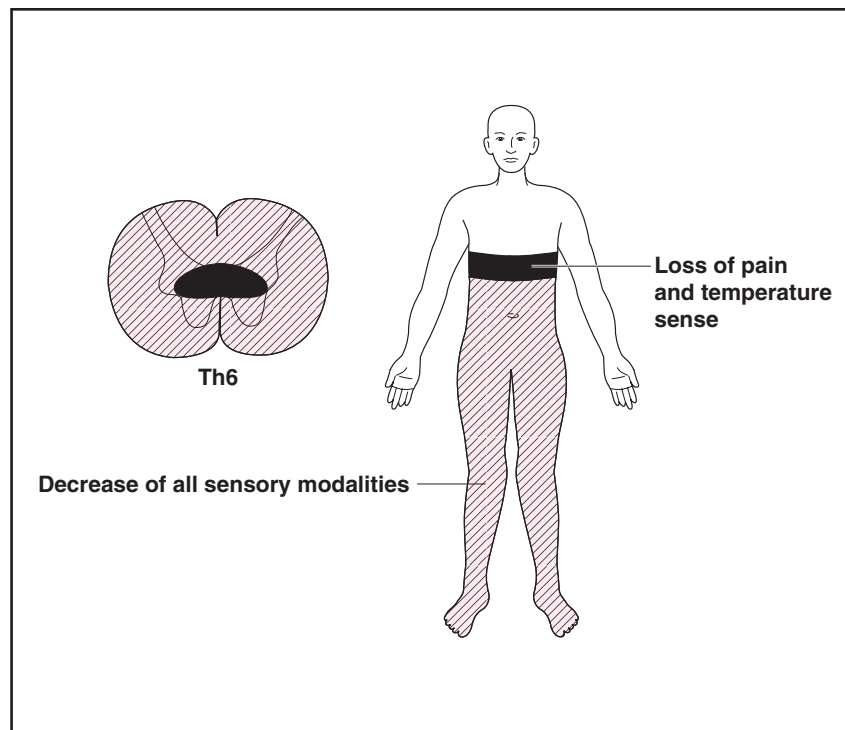


Figure 20-9 Distribution of sensory impairment in a transverse lesion of the sixth thoracic cord.

segment only on the affected side due to a lesion of the corticospinal tract, and vibratory sense and joint position sense are diminished below that level also on the affected side due to a lesion of the dorsal column. By contrast, pain and temperature senses are diminished below the affected segment on the contralateral side, because the fibers forming the spinothalamic tract already crossed from the other side below the affected segment. This clinical picture is called Brown–Séquard syndrome (Figure 20-10). In this case, since the gray matter is also damaged on the affected side, pain and temperature senses may be segmentally lost at the dermatomes on the affected side, again corresponding to the length of lesion.

In fact, it is quite rare to see a transection that is exactly limited to either half of the spinal cord; it is more common to see a bilateral lesion with considerable asymmetry. In the clinical setting, a modified form of Brown–Séquard syndrome is commonly encountered in an extramedullary compression of the spinal cord, because lateral compression of the spinal cord from one side has a contrecoup effect on the other side of the cord. Namely, motor paralysis is present bilaterally but is more severe on the compressed side, and likewise for the proprioceptive sensory impairment, but the nociceptive impairment is more severe on the contralateral side. In an extramedullary compression, motor paralysis starts from the distal lower limb,

because, in the corticospinal tract, the more caudally projecting fibers are located more laterally (see Figure 16-3, p. 111). Likewise, the nociceptive impairment on the contralateral side starts distally and ascends as the compression increases in its degree because of the laminar structure of the lateral spinothalamic tract (Figure 20-11, see also Figure 16-3).

In a Brown–Séquard syndrome due to an external cord compression, segmental sensory loss at the affected segment may not be just for the pain and temperature sense but also might include other modalities, because the dorsal root may be directly compressed.

When the spinal cord is compressed from anteriorly, the corticospinal tracts and the lateral spinothalamic tracts may be bilaterally affected (see Figure 20-11). This is considered to be due to the fact that the spinal cord is bilaterally fixed to the spinal canal with the denticulate ligament, so that anterior compression causes mechanical distortion of the cord (see Chapter 16-1B, p. 112).

The sympathetic tract of the autonomic nervous system descends through the medial part of the lateral column close to the gray matter (see Chapter 21-1, p. 204). Therefore, if sphincter disturbance is present in a case of external cord compression, it suggests that the compression is strong enough to affect the intramedullary structure. In fact, the presence or absence of sphincter

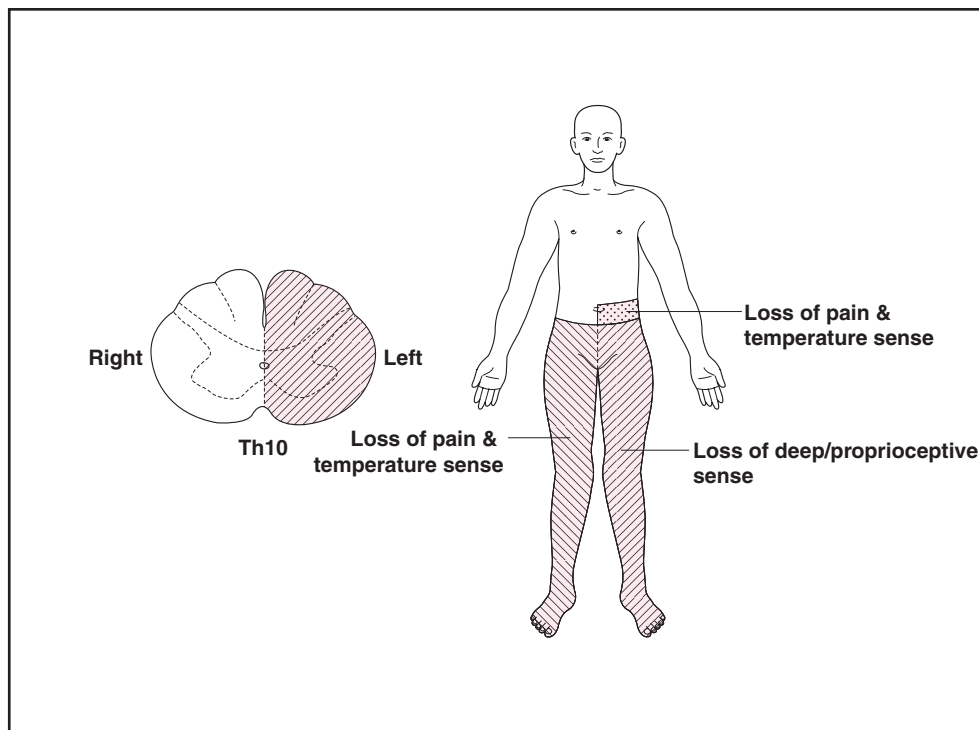


Figure 20-10 Brown-Séquard syndrome due to a lesion of the left half of the 10th thoracic cord.

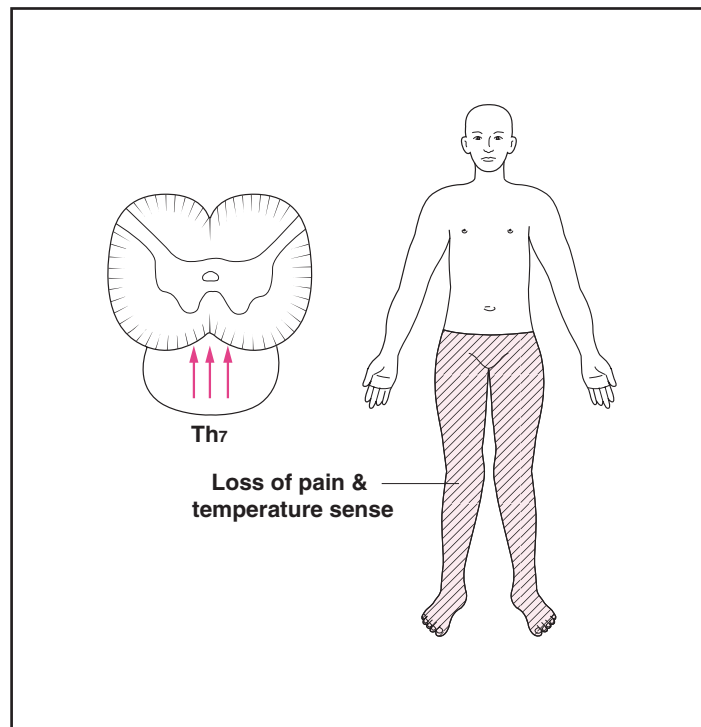


Figure 20-11 An example of distribution of nociceptive loss due to anterior compression of the spinal cord. Although the cord is compressed at the seventh thoracic segment in this case, pain and temperature sense is lost below the 11th thoracic segment due to the laminar structure of the spinothalamic tract (see Figure 16-3, p. 111).

disturbance provides an important clue, in addition to the degree of motor weakness, for judging the need for surgical decompression.

Hemianalgesia is commonly seen in functional disorders, which will be described in Chapter 29 (p. 269).

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AUTONOMIC NERVOUS SYSTEM

The functions of the autonomic nervous system are (1) control of vital signs, (2) visceral control, and (3) adjustment to the external environment. The autonomic nervous system consists of sympathetic and parasympathetic nervous systems, which are different from each other anatomically as well as pharmacologically. Both systems contain afferent and efferent systems.

1. STRUCTURE AND FUNCTION OF THE AUTONOMIC EFFERENT SYSTEM

The center of the autonomic nervous system is in the hypothalamus for both the sympathetic and parasympathetic systems (see Chapter 31-1, p. 276).

A. SYMPATHETIC NERVOUS SYSTEM

The **sympathetic tract** arising from the hypothalamus descends through the brainstem tegmentum and the medial part of the lateral column of the spinal cord and enters the **intermediolateral nucleus**, which extends from the first thoracic (Th1) segment to the first lumbar (L1) segment (see Figures 8-4, p. 51, 14-1, p. 98, and 14-5, p. 101). It is important to note that the whole body, including the head and face, is innervated by the sympathetic nervous system through the intermediolateral nucleus. For example, the intermediolateral nucleus at the Th1 segment, called the **cilio-spinal center of Budge** (see Figure 8-4), projects its preganglionic fibers to the superior cervical ganglion, which innervates the pupils. For another example, the urinary bladder and genital organs are innervated by the intermediolateral nucleus at the Th12-L1 segment.

Thus, if the sympathetic tract is unilaterally damaged in the brainstem or cervical cord, sweating is lost and the skin temperature is elevated on the whole body (from head to feet) on the affected side. If the sympathetic tract

is unilaterally damaged in the thoracic cord, the loss of sweating and the elevation of skin temperature are seen on the affected side below the affected segment. If the intermediolateral nucleus is damaged at a certain level between the Th1 and L1 segments, the loss of sweating and the elevation of skin temperature may be seen segmentally on the dermatomes corresponding to the affected segment.

Preganglionic fibers originating from the intermediolateral nucleus join the ventral root at each segment, exit the spinal canal through the intervertebral foramen, and make synaptic contact with the postganglionic neurons in the **sympathetic ganglia** (Figure 21-1, see also Figure 16-6, p. 115). **Postganglionic fibers** are unmyelinated fibers and innervate pupils, blood vessels, sweat glands, endocrine glands, and visceral organs.

In the sympathetic nervous system, the preganglionic fibers are much shorter than the postganglionic fibers, because the sympathetic ganglia are situated near the spinal canal (Figure 21-2). An exception to this principle is the preganglionic fibers for the pupils, which make synaptic contact with the postganglionic neurons in the superior cervical ganglion (see Figure 8-4, p. 51). By contrast, in the parasympathetic nervous system, the preganglionic fibers are much longer than the postganglionic fibers, because the parasympathetic ganglia are located near their target organs (see Figure 21-2).

The neurotransmitter of the preganglionic fibers is acetylcholine with nicotinic transmission for both the sympathetic and parasympathetic nervous systems. The neurotransmitter for the parasympathetic postganglionic fibers is acetylcholine with muscarinic transmission, and that for the sympathetic postganglionic fibers is norepinephrine. The only exception for the latter system is the postganglionic fibers for sweat glands, for which acetylcholine is the transmitter with muscarinic transmission (see Figure 21-2).

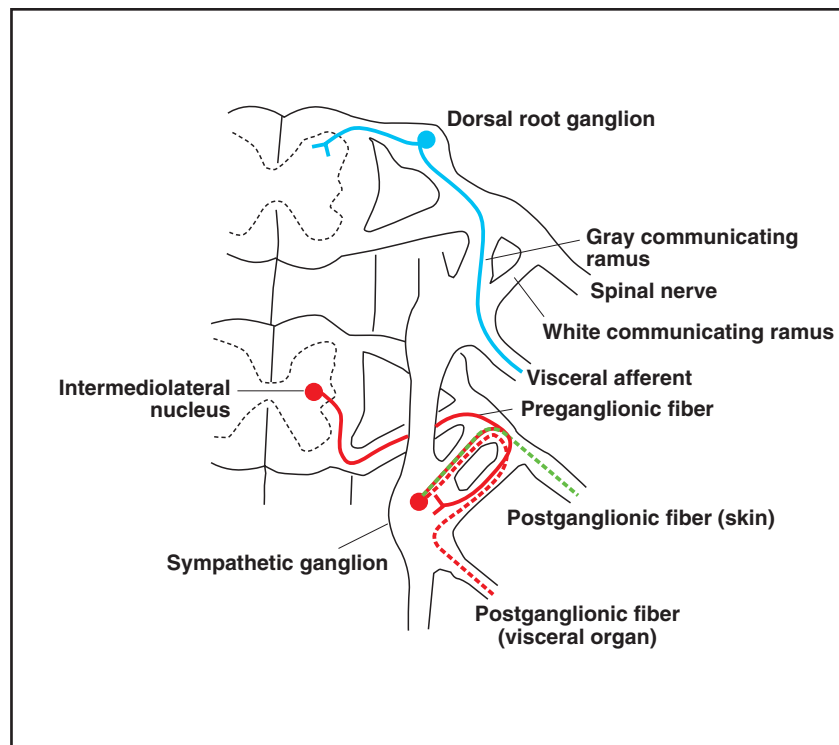


Figure 21-1 Schematic diagram showing the efferent pathway (in red) and the afferent pathway (in blue) of the spinal sympathetic nervous system. The preganglionic fibers (solid red line) pass through the white communicating ramus to enter the sympathetic ganglion. The postganglionic fibers for the skin (dotted green line) pass through the gray communicating ramus and join the spinal nerve, and those for the visceral organs (dotted red lines) form the visceral nerve. The afferent sympathetic fibers are the peripheral axons of small bipolar cells in the dorsal root ganglion, which transmit the impulse from the visceral organ through the gray communicating ramus to the dorsal root ganglion, and its central axons go to the dorsal horn of the spinal cord.

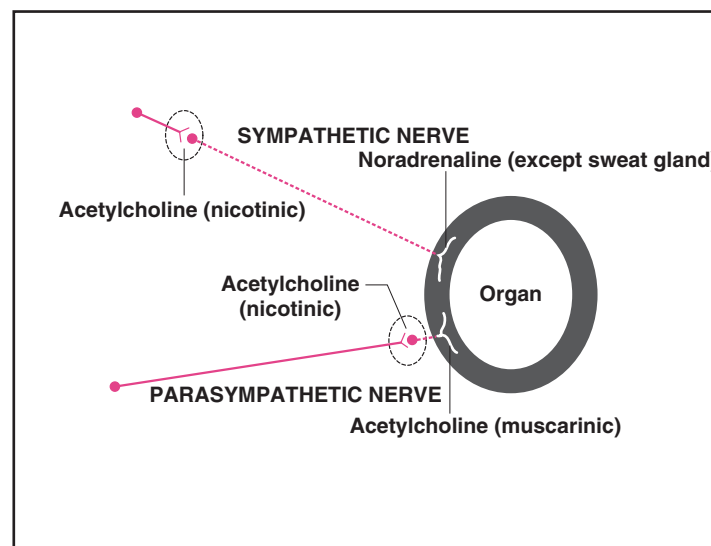


Figure 21-2 Schematic illustration of the sympathetic and parasympathetic ganglia with neurotransmitters. The preganglionic fiber (solid line) is myelinated, and the postganglionic fiber (dotted line) is unmyelinated. The preganglionic fibers are much longer than the postganglionic fibers for the parasympathetic nervous system, and vice versa for the sympathetic nervous system.

B. PARASYMPATHETIC NERVOUS SYSTEM

The preganglionic neurons of the parasympathetic nervous system are located in the brainstem and in the second to fourth sacral segments of the spinal cord. In the brainstem, the parasympathetic nuclei of four cranial nerves (III, VII, IX, and X) project their preganglionic fibers to their respective ganglia. The Edinger-Westphal nucleus of the oculomotor nerve controls pupils and accommodation (see Chapter 8, p. 49). The lacrimal nucleus and the superior salivatory nucleus of the facial nerve control lacrimation and salivation of the sublingual and submandibular glands, respectively (see Chapter 11-1D, p. 85). The inferior salivatory nucleus of the glossopharyngeal nerve controls salivation of the parotid gland (see Chapter 14-4, p. 101). The dorsal motor nucleus of the vagus controls visceral organs such as the heart, lung, and gastrointestinal organs (see Chapter 14-5, p. 101).

The parasympathetic nucleus of the sacral cord controls the urinary bladder and urinary tract, genital organs, and colon from the middle transverse colon to the rectum. As described in relation to the sympathetic nervous system (see earlier in the chapter), the neurotransmitter for the parasympathetic preganglionic fibers is acetylcholine with nicotinic transmission and that for the postganglionic fibers is acetylcholine with muscarinic transmission (see Figure 21-2).

When treating patients with myasthenia gravis with cholinesterase inhibitors, symptoms like miosis (small pupils), increased salivation and upper respiratory secretion, and abdominal pain may occur because the drug enhances not only the cholinergic transmission at the neuromuscular junction but also the muscarinic transmission of the parasympathetic postganglionic fibers. In this case, sweating is also increased even though the sweat glands are innervated by the sympathetic nervous system. This is because the sweat glands are exceptionally controlled by cholinergic transmission instead of noradrenergic transmission (see earlier in the chapter). Furthermore, when giving atropine sulfate to myasthenic patients to prevent the autonomic side effects of cholinesterase inhibitors, there is no need to worry about its untoward effect on the neuromuscular junction, because atropine inhibits only muscarinic transmission and does not inhibit nicotinic transmission of acetylcholine at the neuromuscular junction.

C. DISORDERS OF THE PERIPHERAL AUTONOMIC NERVES

Some peripheral neuropathies affect primarily the autonomic nervous system. **Familial amyloid polyneuropathy**

BOX 21-1 FAMILIAL AMYLOID POLYNEUROPATHY

This is an autosomal dominant hereditary disease of high penetrance seen in Europe and the United States, and large families with this disease are known in Portugal and Japan. The symptoms commonly start in young adults, but cases of late onset have also been reported (Tozza et al, 2021). In addition to sensory polyneuropathy starting from the feet and loss of tendon reflexes, severe autonomic symptoms such as alternating diarrhea and constipation and orthostatic hypotension dominate the clinical picture. As amyloid tends to deposit in small nerve fibers, pain and temperature senses are impaired first, and deep/proprioceptive senses may be lost later. Mutation of the transthyretin (TTR) gene was found to cause this familial disease (Ando et al, 2005). The clinical course may show rapid progression of polyneuropathy (Adams et al, 2015). Yamashita et al. (2012) reported that liver transplantation prolonged the survival rate of this condition. A TTR-stabilizing agent such as tafamidis was suggested as a treatment option (Sekijima et al, 2018). Now patisiran, an RNA interference therapeutic that inhibits TTR production, is the preferred treatment for hereditary TTR-mediated amyloidosis with polyneuropathy (Adams et al, 2021).

Similar symptoms may be seen in systemic amyloidosis as a complication of multiple myeloma. In this condition, as in the familial disease, nociceptive loss and autonomic failure also dominate the clinical picture, because small nerve fibers are primarily affected (Kaku & Berk, 2019).

(Box 21-1) and **systemic amyloidosis** associated with multiple myeloma are well known to cause severe impairment of autonomic functions.

Autonomic ganglia might be affected by an autoimmune pathology, and this condition acutely causes severe autonomic disturbance that is often intractable to medical treatment.

D. CORTICAL CENTER OF THE AUTONOMIC NERVOUS SYSTEM

Although no particular cortical areas have been identified as an autonomic center, the limbic areas including the amygdala, insula, and cingulate gyrus are connected to the hypothalamic autonomic center. For urination, the **paracentral lobule** at the posterior mesial aspect of the frontal lobe is believed to play an important part in its voluntary control. In relation to this, bilateral compression of the mesial

frontal lobes by a **falx meningioma** is known to cause bladder disturbance in addition to spastic paraparesis, thus resembling spinal cord compression. Urinary incontinence seen in patients with **idiopathic normal-pressure hydrocephalus** might be due to dysfunction of the descending fibers from the mesial frontal cortex passing by the dilated lateral ventricles (see Box 22-3, p. 213).

Patients with partial epilepsy may show cardiac arrhythmia or transient cardiac arrest during an ictal episode. The mesial temporal lobe or insula is believed to be responsible for this paroxysmal phenomenon, but the dominant hemisphere for this autonomic function has not been determined.

E. NEURAL CONTROL OF URINATION

Although urination (micturition) is a representative autonomic function, it can be controlled by intention as well. Thus, a complex neuronal circuit from the frontal cortex through the brainstem to the sacral cord is involved in its control (Benarroch, 2010). Among the cortical areas, the mesial prefrontal area, anterior cingulate gyrus, insular cortex, and motor cortices are considered to be involved in this function. In the spinal cord, the preganglionic neurons at the S2 to S4 segments, Onuf nucleus, and the intermediolateral nucleus at the L1-L2 segments play major parts. Furthermore, the periaqueductal gray substance and the pontine micturition center are considered to play a role for connecting the cortex with the caudal spinal cord. Unilateral infarction in the lateral medulla or the dorsal lateral pontine tegmentum can cause urinary retention (Cho et al, 2015).

2. STRUCTURE AND FUNCTION OF THE AUTONOMIC AFFERENT SYSTEM

There are two kinds of autonomic afferent systems: visceral pain and visceral sensation. The primary neuron for **visceral pain** is a bipolar cell located in the dorsal root ganglion. Its peripheral axon transports impulses from a visceral organ to a ganglion along the sympathetic postganglionic fibers through the gray communicating ramus (see Figure 21-1). The central axon enters the dorsal horn of the gray matter at the Th1 to L1 segments, from where the secondary neuron projects its axon centrally through the spinothalamic tract.

As the nociceptive impulses originating from the heart, for example, enter the dorsal horn at the Th2 segment, the pain is felt as if it came from the Th2 somatosensory

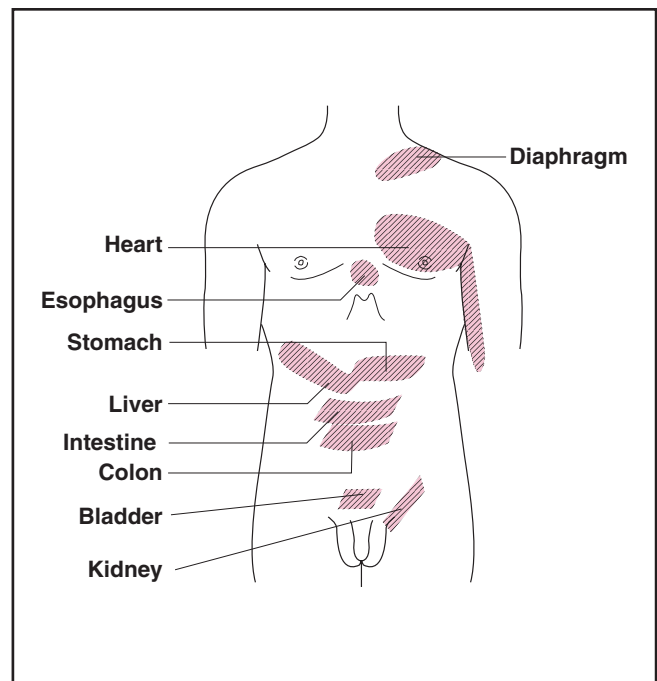


Figure 21-3 Dermatomes of referred pain originating from representative visceral organs.

dermatome as a **referred pain**, namely the left upper chest and the medial aspect of the left arm (Figure 21-3). For another example, the pain due to urinary stones is felt in the inguinal region and the anterior aspect of the proximal thigh, because the sympathetic afferent fibers from the urinary tract enter the spinal cord at the L1 dermatome.

Visceral sensation like fullness of the abdomen and bladder distention is centrally transmitted through the parasympathetic nerve fibers. For example, fullness of upper abdomen is transmitted through the vagus nerve to the solitary nucleus in the medulla, and the fibers from the solitary nucleus project to the thalamus and the brainstem reticular formation. Bladder distention, which will eventually lead to micturition desire, is transmitted through the pelvic nerve to the S2 to S4 segments of the sacral cord, and then transmitted centrally through the lateral and posterior columns of the spinal cord. Bladder afferent information is considered to be partially transmitted also through the sympathetic nerve.

3. EXAMINATION OF THE AUTONOMIC NERVOUS FUNCTION

Examination of pupils can test an important autonomic function in the cranial nerve territory, but because it

was described in Chapter 8-2 (p. 53), it will not be repeated here.

A. AUTONOMIC SYMPTOMS OF SKIN

It is important to examine the color, temperature, and moisture of the skin. The skin temperature and moisture can be tested by palpating the skin by hand. In a unilateral lesion of the brainstem tegmentum like **Horner syndrome**, decreased sweating and elevated temperature of the skin are often seen ipsilaterally (see Chapter 8-1, p. 51). Elevation of the skin temperature in this case is due to dilation of capillaries and arterioles due to a lesion of the sympathetic tract, causing increased blood flow. In testing the skin moisture for Horner syndrome, the examiner should move the hand from the possible dry side to the moist side. Movement in the other direction is less sensitive since fluid could be moved along with the hand to the dry side. It is also possible to compare the homologous areas of the two sides using both hands of the examiner.

In **complex regional pain syndrome (CRPS)**, previously called reflex sympathetic dystrophy or causalgia, the affected part of the skin is pale, with increased sweating and decreased temperature. CRPS is subdivided into two types. CRPS I is intended to encompass reflex sympathetic dystrophy and similar disorders without a nerve injury, and CRPS II occurs after damage to a peripheral nerve (Birklein et al, 2015; Borchers & Gershwin, 2014). In a transverse lesion of the spinal cord, sweating is decreased on the skin below the affected level. In a segmental lesion of the spinal cord, **dermographia** (raised red wheal similar to hives caused by scratch) and abnormal **pilomotor reflex** (reflex pilomotor erection like goose bumps) might be seen at the corresponding dermatome. For all these tests, it is important to compare the homologous areas of the two sides; in case of a spinal cord lesion, the skin regions above and below the affected segment should be compared.

B. MICTURITION AND SEXUAL FUNCTIONS

Micturition/defecation and sexual functions are important autonomic functions. Careful history taking is most useful for checking these functions. Increased frequency of urination (**urinary frequency**) and **urge incontinence** are caused by a lesion of the central nervous system proximal to the sacral segment of the spinal cord. By contrast, a lesion of the sacral cord or cauda equina causes difficulty of micturition or **urinary retention**. In a lesion of the sacral cord, the anal sphincter reflex is decreased or lost, and in male

patients, the bulbocavernosus reflex is also lost. The **anal sphincter reflex** is tested by stimulating the perianal skin with a sterile pin and observing or digitally palpating a reflex contraction of the anal sphincter muscle. Alternatively, anal contraction can be seen with cough or sniff. To test the **bulbocavernosus reflex**, the glans penis is pinched and a reflex contraction of the anal sphincter muscle is confirmed by digital examination.

As the patient may not actively complain of sexual dysfunction, it is important in men to confirm the presence or absence of disturbance of penile erection (**impotence**) or disturbance of ejaculation during the history taking as necessary. In a lesion of the spinal cord rostral to the sacral segment, excessive erection may spontaneously occur and continue (**priapism**). In relation to this phenomenon, touching the legs might cause reflex urinary incontinence or profuse sweating in the legs.

C. GASTROINTESTINAL SYMPTOMS

Diarrhea associated with continuous abdominal pain or continuous constipation of unknown etiology might be due to disorders of the autonomic nervous system. However, pure disturbance of defecation due to neurological disorders without any disturbance of urination is extremely rare. **Alternating occurrence of diarrhea and constipation** is often seen in disorders of the autonomic nervous system. A triad of dysphagia, chest pain, and regurgitation is seen in **achalasia**. A combination of achalasia, alacrima (loss of tears), and adrenal failure is called triple A syndrome.

D. ORTHOSTATIC HYPOTENSION

This is one of the most common symptoms suggestive of the presence of autonomic failure, and the history taking might give an important clue. Generally, the symptom is feeling faint or a sense of impending loss of consciousness. For testing this function at the bedside, blood pressure can be measured first in the supine position and then in the upright position. After the patient takes the upright position, it is important not only to measure blood pressure immediately after standing, but also to keep measuring intermittently for at least about 3 minutes. If the systolic pressure drops by more than 20 mmHg or the diastolic pressure by more than 10 mmHg, it can be judged to be orthostatic hypotension. It is important to measure the pulse rate simultaneously and to see whether or not the drop in blood pressure is compensated by an increase in the pulse rate. Since patients with movement disorders may require an excessive effort

BOX 21-2 DIAGNOSTIC CRITERIA OF MULTIPLE SYSTEM ATROPHY

The possibility of multiple system atrophy (MSA) is considered when a patient manifests autonomic disorders that are associated with parkinsonism or cerebellar ataxia, or both. According to the diagnostic criteria proposed by Gilman et al. (2008), diagnosis of definite MSA requires demonstration of the α -synuclein positive inclusion body in the glia in addition to neuronal degeneration in the nigrostriatal system, olive, and cerebellum. Sporadic cases of adult onset with progressive autonomic failure including urinary incontinence or orthostatic hypotension and dopa-resistant parkinsonism or cerebellar ataxia are diagnosed as probable MSA. Sporadic cases with progressive parkinsonism or cerebellar ataxia and at least one of the autonomic abnormalities are diagnosed as possible MSA. Cases of MSA with predominant parkinsonism are called MSA-P, and those with predominant cerebellar ataxia are called MSA-C (Meissner et al, 2019). Laryngeal stridor during sleep is especially common in patients with MSA (Cortelli et al, 2019). In fact, laryngeal movement disorders characterized by irregular arytenoid cartilage movements are highly prevalent in patients with MSA (Gandor et al, 2020). A recent neuropathologic study of 134 patients with a clinical diagnosis of MSA from the United States revealed that the clinical diagnosis was correct in 62% of the cases, and the most common misdiagnosis was dementia with Lewy bodies, followed by progressive supranuclear palsy and Parkinson disease (Koga et al, 2015). Oligodendroglial synucleinopathy is considered to play a primary role in the pathogenesis of MSA (Jellinger, 2014; Meissner et al, 2019; Wenning et al, 2008). Imaging and autonomic function tests were shown to contribute to increasing the accuracy of a diagnosis of MSA (Pellecchia et al, 2020). The clinical onset of MSA can be as young as the 30s (Batia et al, 2018). Some cases of MSA are familial, and recently a genetic susceptibility to MSA has been identified (Heckman et al, 2014; Multiple-System Atrophy Research Collaboration, 2013). Thus, MSA-C and hereditary cerebellar ataxia may cause some difficulty in the differential diagnosis (Giordano et al, 2017; Li et al, 2018).

for standing up from the bed, it may have some effect on the blood pressure. Therefore, to determine the postural change of blood pressure precisely, it is ideal to use a tilt table. Postural orthostatic tachycardia syndrome (POTS) was discussed in Box 4-3 (p. 27).

To test the physiologic change in pulse rate with breathing, the patient may be asked to take deep, slow breaths while the pulse rate is being measured. Quantitatively, the

R-R interval may be calculated by electrocardiography to measure its variability. Complete loss of the variability in the R-R interval suggests autonomic failure.

Orthostatic hypotension is caused by various etiologies, including neurodegenerative diseases such as **pure autonomic failure** and **multiple system atrophy (Shy-Drager syndrome)** (Box 21-2), and autonomic failure due to a lesion of the spinal cord or the peripheral nervous system. In fact, follow-up of patients presenting with idiopathic autonomic failure (pure autonomic failure) may often show other signs of α -synucleinopathies later in the clinical course (Coon et al, 2020; Giannini et al, 2018; Kaufmann et al, 2017). Orthostatic hypotension is also seen in Parkinson disease at various frequencies (up to 40%, depending on the report). In fact, a deficiency of transmitter uptake at the sympathetic nerve terminals of the myocardium has been demonstrated by single-photon-emission computed tomography (SPECT) using 123 metaiodobenzylguanidine (MIBG) as a tracer in patients with idiopathic Parkinson disease, although its relationship with orthostatic hypotension is not clear. It is the current consensus that **decreased uptake of MIBG in myocardium** is relatively characteristic of Lewy body-associated disorders, including Parkinson disease, Lewy body dementia, and pure autonomic failure, and it is rare in multiple system atrophy, progressive supranuclear palsy, Alzheimer disease, and frontotemporal dementia (King et al, 2011; Tiraboschi et al, 2016).

The Aschner eyeball pressure test and the carotid sinus pressure test are used to evaluate autonomic function by demonstrating reflex bradycardia in response to pressure on the eyeballs and carotid artery, respectively. However, these tests are not recommended especially for patients with autonomic failure, because these maneuvers may induce cardiac arrest in them.

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POSTURE AND GAIT

1. CENTRAL CONTROL MECHANISM OF GAIT

The spinal cord contains machinery for stepping, the locomotor generator. However, the spinal cord is under the control of supraspinal centers. In human walking, intention or attention takes part to various degrees, from fully attended walking to nearly automatic walking. Multiple structures of the central nervous system are known to be involved in the central control of walking (Figure 22-1) (Nutt et al, 2011; Shibasaki et al, 2004). In addition to the sensorimotor areas directly related to leg movement such as the pre-SMA and SMA proper (see Chapter 16, p. 109), anterior cingulate gyrus, lateral premotor areas, and the foot areas of the primary sensorimotor cortices, the cortical areas related to visual information processing,

such as the occipital cortex and the posterior parietal area, are also important. Furthermore, the brainstem gait center, cerebellum, and basal ganglia also play an important part. In animals, the locomotion center located in the dorsal part of the rostral brainstem plays an important role. The greater the intention and attention paid to the walking, the more widespread is the cortical activation.

2. EXAMINATION OF GAIT

Careful observation of walking provides important information in the clinical setting. Abnormal gait is classified into several typical patterns. **Spastic gait** is characterized by a stiff shuffling gait, like walking in waist-high water.

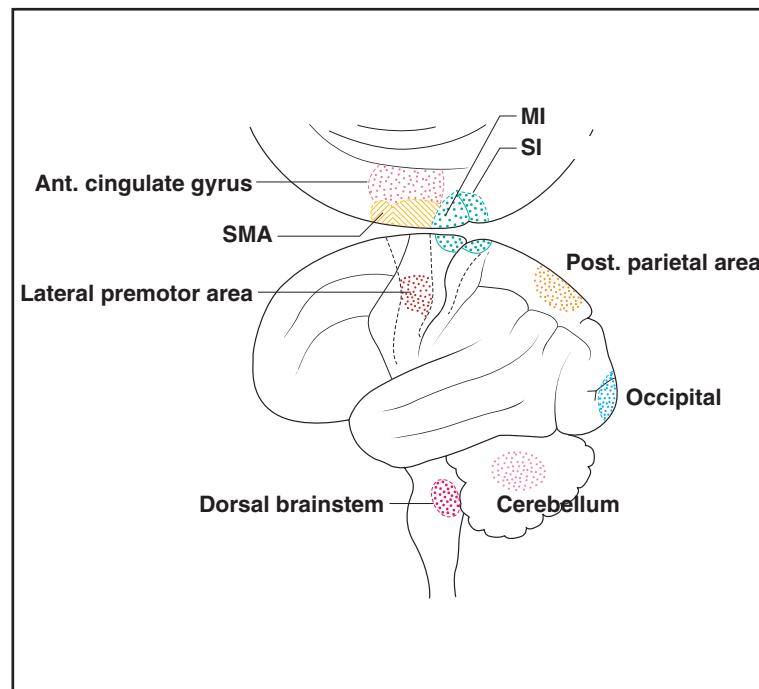


Figure 22-1 Schematic diagram showing the human brain areas activated during intended walking. Based on Shibasaki et al., 2004.

The knee does not bend, and the leg circumducts in swing. The patient walks slowly, dragging both legs, without lifting the feet completely off the ground. When severe, the legs tend to cross with each step because of increased tightness of hip adductors (**scissor gait**). The patient may complain of bouncing and trembling, especially when going down stairs. This is considered to be due to patellar and ankle clonus, which is evoked by stretching the leg muscles when stepping downstairs (see Chapter 17-4B, p. 147). Spastic gait is believed to be seen in patients with spasticity of the leg muscles due to a lesion of the pyramidal tract or upper motor neurons. Recently, however, lesser corticospinal influences and larger reticulospinal influences on spastic muscles were shown in patients with spasticity due to spinal cord injury (Sangari & Perez, 2019).

Parkinsonian gait is characterized by delayed initiation, slow walking with small steps, loss of arm swing, and tendency to accelerate involuntarily (**festinating gait, propulsion**). It is not rare to festinate backward and fall (**retropulsion**). Some patients might show markedly flexed (stooped) posture while standing or walking, but the posture improves when lying down. This condition is called **camptocormia** or **bent spine syndrome** (see Chapter 15-2, p. 107). Dual task conditions such as motor and cognitive tasks (for example, walking with serial 7 subtractions) or dual motor tasks (walking while carrying a tray) may be tested to bring out more abnormality, depending on the situation (Selge et al, 2018).

A. FREEZING OF GAIT

Patients with Parkinson disease may suddenly become unable to move, particularly upon gait initiation or turning (**freezing phenomenon**) (Fasano et al, 2017a, 2017b; Hallett, 2008; Snijders et al, 2016). The freezing phenomenon might improve with some types of visual input. For example, the patient can walk much better if transverse lines are drawn on the floor at short intervals so that the patient could walk across the lines. This phenomenon is called **kinésie paradoxale (paradoxical kinesia)** (Box 22-1). By taking advantage of this phenomenon, wearing technology of various kinds is under investigation for assessing, as well as managing, freezing of gait (Mancini et al, 2019; Weiss et al, 2020).

Paradoxical kinesia is also seen in patients with **pure akinesia**, of which progressive supranuclear palsy (PSP) is believed to be the most frequent cause (Park et al, 2009) (Box 22-2). As for the causes of frequent falls in patients with PSP, abnormalities of vestibulospinal function (which can be tested with vestibulo-ocular reflexes or

BOX 22-1 HOW DOES PARADOXICAL KINESIA OCCUR?

Patients who cannot walk on the flat floor because of frozen gait may be able to climb stairs. This phenomenon, called **paradoxical kinesia (kinésie paradoxale)**, may also be seen when the patient walks across the transverse lines drawn on the floor at short intervals, suggesting that the visual input of transverse lines in front of the body plays a primary role in improving the initiation of gait. A similar phenomenon is also seen in the upper limbs: The patient who has great difficulty in moving the arm in response to verbal instruction might be able to promptly catch a ball thrown toward him or her. There was a hypothesis that the cerebellum might take over the role of the basal ganglia immediately in response to visual input, but it has not been proved. Hanakawa et al. (1999a, 1999b) used a blood flow activation study with single photon emission computed tomography (SPECT) while patients with Parkinson disease walked on the treadmill, comparing the transverse lines and parallel lines, and demonstrated a greater activation in the right lateral premotor area in association with the transverse lines compared with the parallel lines. This result was interpreted by postulating that, in Parkinson patients, self-initiated gait is impaired because of a deficient excitatory input to the SMA, but the lateral premotor area, which is relatively preserved in Parkinson disease and which is known to receive abundant visual input, might be activated by the transverse lines (Shibasaki et al, 2004). However, a question as to why only the transverse lines, but not the parallel lines, are effective remains to be solved. In general, paradoxical kinesia occurs because external triggering of movement is more effective than internal triggering in Parkinson patients (Hallett, 2008).

vestibular-evoked myogenic potentials) have been reported (Liao et al, 2008).

It is noteworthy that Parkinson patients have greater difficulty in controlling the posture in the anteroposterior direction as seen in festinating gait, propulsion, and retropulsion, while patients with cerebellar ataxia have loss of equilibrium in all directions, especially in the lateral direction. Falls are not uncommon in patients with Parkinson disease (Fasano et al, 2017a), and this might be influenced by impairment of the pedunculopontine nucleus (Bohnen et al, 2009; Garcia-Rill et al, 2019) (see Box 16-3, p. 124).

To examine postural stability, the **pull test** is performed while the patient stands with the feet side by side and the examiner stands behind the patient with one foot slightly behind the other. The patient should be given an

BOX 22-2 PROGRESSIVE SUPRANUCLEAR PALSY AND CORTICOBASAL DEGENERATION

Among the neurodegenerative disorders presenting with progressive frontal dementia and parkinsonism, progressive supranuclear palsy (PSP; Steel–Richardson–Olszewski syndrome) and corticobasal degeneration (CBD) are the most commonly encountered. CBD is clinically referred to as corticobasal syndrome (CBS) due to ambiguity about the pathology. These two conditions commonly occur in adults over age 60. It is generally believed that PSP is characterized by supranuclear vertical gaze palsy (see Chapter 9-2, p. 62), retrocollis, postural reflex abnormality, and parkinsonian gait disturbance, whereas CBS is characterized by asymmetric rigidity and bradykinesia, tremulous myoclonus, ideomotor apraxia, primitive reflexes, and cortical sensory loss. In clinical practice, however, distinguishing these two conditions is not infrequently difficult, and in fact some patients show clinical features of both conditions. In fact, pathologic studies of clinically diagnosed cases of CBS revealed any of four conditions; CBD, Alzheimer disease, PSP, and frontotemporal lobar degeneration (FTLD) (Lee et al, 2011). Furthermore, remarkable phenotypic heterogeneity was suggested by a recent retrospective multicenter study of 100 autopsy-confirmed cases of PSP (Respondek et al, 2014) as well as by a literature survey (Respondek et al, 2017). According to the diagnostic criteria of PSP proposed by the Movement Disorder Society, specific combinations of four features (ocular motor dysfunction, postural instability, akinesia, and cognitive dysfunction) are important for its clinical diagnosis (Höglinger et al, 2017).

explanation in advance that the examiner is going to pull the patient abruptly backwards by the shoulders and that he or she is allowed to step backwards in order to maintain balance. However, since there is a risk of falls during this test, especially in patients with movement disorders, it should be done with great caution.

B. ATAXIC GAIT

Unsteady, staggering gait due to loss of equilibrium is called **ataxic gait**. Patients with ataxia usually stand or walk with a wide base (**wide-based gait**). In patients with mild ataxia, ordinary walking might appear normal, but **tandem gait** will disclose truncal ataxia. It is important to instruct the subject to put the heel of one foot just in front of the toes of the other foot. When testing the tandem gait, it may be

BOX 22-3 IDIOPATHIC NORMAL-PRESSURE HYDROCEPHALUS

Progressive gait disturbance associated with urinary incontinence and cognitive impairment in the elderly patients may be due to idiopathic normal-pressure hydrocephalus (INPH), which is a treatable condition. This condition was first reported by Adams et al. in 1965. As a ventricular shunt is effective in some cases of this condition, how to determine its indication has been a practical problem (Halperin et al, 2015; Torsnes et al, 2014). Having all three symptoms makes the diagnosis more likely. McGirt et al. (2008) investigated the effect of shunting in 132 patients who had at least two of these three symptoms, dilated ventricles on imaging study, and β waves (slow rhythmic oscillation) of the cerebrospinal fluid. Other causes of the secondary hydrocephalus such as subarachnoid hemorrhage, head trauma, and meningitis were excluded in those cases. After the mean follow-up period of a year and half, 75% of all patients showed improvement. It was found to be especially effective in patients in the relatively short period after the clinical onset and in those predominantly presenting with gait disturbance. Among the triad, improvement was found to be most prominent in gait disturbance. It is important to keep this condition in mind whenever elderly subjects suffer from gait disturbance and falls of unknown etiology (Jaraj et al, 2014). It remains controversial whether INPH really does exist or whether it is secondary to underlying neurodegenerative diseases (Nakajima et al, 2021; Oliveira et al, 2019; Saper, 2017).

anticipated that walking slowly might be easier than walking fast, but the opposite is actually true. In ataxic gait due to an organic lesion, tandem gait can be executed more easily at a faster speed than a slower speed, just like healthy subjects walking on a balance beam. Some patients with functional (psychogenic) gait disturbance of ataxic nature tend to execute the tandem gait poorly at a fast speed and much better with a slower speed.

Ataxic gait is due to a lesion of the cerebellum, vestibular system, or proprioceptive system. When a patient complains of vertigo, ataxic gait is often vestibular in nature. Presence of tinnitus, hearing loss, or nystagmus suggests vestibular ataxia (see Chapter 13-2, p. 94). Patients with ataxic gait of cerebellar origin also tend to have abnormalities in ocular movements, dysarthria, and limb ataxia. Occasionally, a localized lesion confined to the cerebellum may cause vertigo due to involvement of the vestibular input to the cerebellum (see Chapter 13-1, p. 93). **Sensory ataxia** is caused by interruption of proprioceptive input by

BOX 22-4 TANDEM GAIT AND ROMBERG TEST

Tandem gait and the Romberg test are important examinations for patients presenting with unsteady gait, but the principle of each test should be correctly understood when applying these tests.

In patients with mild ataxia, ordinary walking might appear nearly normal, but **tandem gait** will for the first time disclose truncal ataxia. In contrast, in patients with clear truncal ataxia, regardless of whether it is cerebellar, vestibular, or sensory in origin, tandem gait is expected to be severely abnormal, and it may even cause falling. In this case, routine use of tandem gait is unnecessary and should be avoided.

The use of the **Romberg test** should be limited to patients with possible somatosensory ataxia, because blocking visual information is expected to worsen the equilibrium irrespective of its cause. When the patient has clear proprioceptive loss on the physical examination, the Romberg test is no longer needed. Blocking visual input in this case might also increase the risk of falling.

a lesion of the dorsal column or the peripheral nervous system (see Chapter 20-2B, p. 196). As the effect of proprioceptive deficit may be compensated visually, interruption of visual input by closing the eyes provokes or worsens the truncal ataxia (**Romberg sign** and **basin phenomenon**) (see Figure 20-5, p. 196).

C. WADDLING GAIT AND STEPPAGE GAIT

Muscle weakness in the pelvic girdle causes excessive pelvic swing on walking (**waddling gait**). This phenomenon is typically seen in myopathy, especially in limb-girdle muscular dystrophy, but it may be also seen in neurogenic muscle atrophy of the proximal muscles like Kugelberg-Welander disease (see Chapter 17-2A, p. 136).

Weakness of the foot extensor muscles, especially the anterior tibial muscle, causes difficulty in lifting toes (**foot-drop**), and the patients lift legs high in order to avoid tripping during walking (**steppage gait**). A typical example of steppage gait is seen in Charcot-Marie-Tooth disease (see Chapter 17-3K, p. 145).

Falls/falling are a big social problem in our aging society. When taking the medical history of patients with gait disturbance, it is important to confirm any episode of falling within the past year and the presence or absence of any disorders that might cause falling (Thurman et al, 2008).

BOX 22-5 INTERMITTENT CLAUDICATION

In this condition, the patient develops pain and numbness in the lower extremities after walking some distance, which disappears after resting. It is not uncommon in the elderly subjects. There are two causes for this condition: (1) obstructive disorders of the peripheral arteries like **Buerger disease** and (2) neurological disorders related to the spinal cord or cauda equina. In the former condition, the skin of the affected limb appears pale with a cold surface temperature, and pulsation of the dorsal artery of foot is deficient. **Neurogenic or spinal intermittent claudication** is commonly due to spinal canal stenosis or an arteriovenous malformation. As the symptoms may appear only when walking, it is important to examine the patient when the symptoms appear after walking. Segmental loss of superficial sensation, muscle weakness, or loss of tendon reflexes might be observed during the episode. Neurological deficit during this episode might be due to direct mechanical compression of the lumbosacral cord or cauda equina, but it might also be due to the regional ischemia due to stealing of arterial perfusion from the narrowed artery to the skeletal muscles during walking (**steal phenomenon**). Whatever the cause, treatment of the underlying condition is important.

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MENTAL AND COGNITIVE FUNCTIONS

If a patient has any mental disorders or cognitive disturbances, those symptoms are often noticed while taking the history. Therefore, for evaluating the mental and cognitive functions in detail, it may be efficient to do so after completing the physical examination, instead of spending a long time for the interview at the beginning of the examination. On the other hand, a number of clinicians prefer to do the cognitive testing first. In fact, abnormalities of higher cortical functions, if any, are often detected while carrying out the physical examination, because verbal communication is necessary for physical examination.

There are various ways of quantitatively evaluating the mental state (Daffner et al, 2015). The **Mini-Mental State Examination (MMSE)** proposed by Folstein et al. (1975) is shown in Table 23-1 as an example. Another popular test is the **Montreal Cognitive Assessment (MoCA)** (Box 23-1) (Chiti & Pantoni, 2014). The MoCA includes executive functions testing. What is most important for the mental state examination is to obtain concrete information about the patient's daily life from the patient's family or to directly observe the patient's behavior in the clinic. The direct observation of actual behaviors often

provides far more important information than examining the patient in the outpatient clinic.

The mental state and cognitive functions can only be examined in conscious patients. Furthermore, it is important to confirm the patient's hearing ability, language comprehension, and spontaneous speech before starting the mental evaluation, because all the related tests are performed by using spoken or written language. A patient who has difficulty comprehending spoken language may still be able to communicate by using written language. If the patient has difficulty in both spoken and written language, then the testing has to be done by using gestures or pictures.

1. EXAMINATION OF MENTAL AND COGNITIVE FUNCTIONS

Cognitive testing is performed for the following items. The order of examination of these items can be freely chosen depending on the patient's condition and circumstances:

TABLE 23-1 MINI-MENTAL STATE EXAMINATION (MMSE)

Category	Possible points	Description
Orientation to time	5	Year, season, day of the week, month, and day
Orientation to place	5	State, county or prefecture, city, hospital, floor, and region
Registration	3	Memory of 3 unrelated objects
Attention and calculation	5	Serial 7s
Recall	3	Registration recall of the 3 objects presented above
Language	2	Naming a pencil and a watch
Repetition	1	Repeating a phrase forward and backward
Complex commands	6	6 complex commands, including drawing a presented figure

Total possible score: 30

(Modified from Folstein et al, 1975)

BOX 23-1 MONTREAL COGNITIVE ASSESSMENT (MOCA)

1. Alternating trail making
2. Visuoconstructional skills (cube)
3. Visuoconstructional skills (clock)
4. Naming
5. Memory
6. Attention: Forward digit span, Backward digit span, Vigilance, Administration
7. Sentence repetition
8. Verbal fluency
9. Abstraction
10. Delayed recall
11. Orientation

Orientation to time, place, and person

Memory: retention span, recent memory, old memory, and presence or absence of confabulation

Calculation

Common knowledge and judgment

Emotion and character

Presence or absence of illusion, hallucination, and delusion

State of daily living

A. ORIENTATION

Orientation involves testing how much a patient is aware of the situation in which he or she is placed in terms of time, place, and person. If the patient is suddenly asked what day it is, he or she may be unable to answer promptly because he or she is upset by the unexpected question. In that case, it is useful to make the patient relaxed before repeating the question. The presence of **disorientation** is often noticed while taking the history. In most patients with mental impairment, orientation is more impaired to time than to space.

Sometimes it may be difficult to distinguish whether the patient is disoriented to time or has disturbance of recent memory (see the next paragraph). If the patient answers a wrong day, it is useful to give the correct day and ask the patient to remember it. If the patient can recall the correct day a few minutes later, recent memory may not be a primary problem for that patient.

B. MEMORY

Memory is divided into **episodic memory** and **procedural memory**, and episodic memory is mainly composed of **semantic memory**. Acquisition (registration), retention, and retrieval of information are all necessary for each category of memory. According to the length of the information retention, memory can be divided either into short-term memory and long-term memory, or into immediate memory, recent memory, and remote memory. **Immediate memory** is a memory lasting seconds and almost equal to repetition. For example, it is tested by giving five digits and asking the patient to repeat them soon afterward. **Recent memory** is a memory lasting a few minutes. In **transient global amnesia**, recent memory is mainly impaired so that the patient repeats the same questions (Box 23-2). **Remote memory** is a memory lasting years, and it is tested by asking about past events related to the patient or from the news. In general, patients with memory disturbance can recall better the events that occurred in childhood than more recent events.

BOX 23-2 TRANSIENT GLOBAL AMNESIA

This episode suddenly occurs in healthy adults above middle age and is characterized by confusion—for example, repeating the same questions or comments during the episode. It is accompanied by retrograde amnesia of various lengths, but memory recovers completely, starting from the oldest memories. During such an episode, therefore, the patient is not only unable to acquire, retain, and retrieve the new information but also unable to retrieve the old memory. In contrast with a psychomotor seizure, there is no behavioral abnormality during the attack of retrograde amnesia. Excessive emotional excitation may provoke the attack in some cases, but a precipitating factor may not be found in others. A release of glutamate from the amygdala to the hippocampus by emotional excitation was thought to cause spreading depression in the hippocampus. Hypotheses for its pathogenesis also include partial epilepsy and transient ischemic attack of the hippocampus. A study of patients within 48 hours after the onset of the attack using diffusion MRI showed abnormality in the CA-1 region of the hippocampus, which disappeared within 10 days (Bartsch et al, 2007). Diffusion-weighted MRI showed hippocampal abnormalities in 70% of the patients, especially in 12 to 24 hours after the clinical onset (Szabo et al, 2020). These data suggest ischemia as the most likely cause (Bartsch et al, 2010). Relapse of this condition is relatively uncommon.

Working memory is the ability to hold and manipulate information in the mind for seconds to minutes while it is being used to achieve a particular goal. A typical example is a question-and-answer period when a speaker receives several questions: The speaker has to keep the remaining questions in mind while responding to the first question.

i. Examination of memory

The presence of memory disturbance is usually suspected when taking the history. In the clinical practice, three unrelated words can be given verbally or in written form, and the patient is asked to recall them after a short time. Patients can also be asked about current events. For quantitative measurement, 10 pairs of unrelated words are presented and the patient is asked to memorize them; after a short time, the patient is asked to provide the paired word for each given word. If this test was poorly performed, then 10 pairs of related words may be used. As described earlier in the chapter, if the patient is found to be disoriented to time, the correct information should be provided for memorizing.

In memory loss due to acute cerebral lesions such as strokes, acute encephalitis, and head trauma, patients often lose memory of events that occurred various lengths of time before the acute insult. In this case, the information preserved in the past cannot be retrieved (**retrograde amnesia**). How far back the memory is lost differs depending on the severity of retrograde amnesia. In severe retrograde amnesia, all the previously acquired information is lost back to childhood, and during recovery, the memory gradually returns, starting with the older events.

If consciousness is lost due to an acute brain insult due to trauma, for example, the patient is not expected to remember what happened when unconscious. Even in that case, however, if the memory center is not damaged by the insult, the patient is able to recall later what happened up to the exact moment of injury. Therefore, if retrograde amnesia is present after the recovery from unconsciousness, it suggests that the memory center has also been damaged.

Memory function is usually tested by using verbal or written language. However, some patients may retain memory of visual or auditory information but not in language. In this case, it is useful to present a set of objects or a set of sounds, and after a short time, to present one of the objects or sounds to ask whether it was one of the presented objects or sounds or not. Theoretically, it is also possible to do a similar test for gustatory and olfactory information.

ii. Korsakov (Korsakoff) syndrome

When a patient has disorientation in time and loss of recent memory, it is important to check if there is any **confabulation**. If a patient has confabulation, the patient not only gives a clearly wrong answer to the examiner's question but also believes that the given answer is correct. In an extreme situation, the patient might even agree with wrong information deliberately given by the examiner. For an example, depending on the situation, the examiner mimes holding a string and asks the patient if he or she sees it.

A combination of disorientation in time, loss of recent memory, and confabulation is called **Korsakov syndrome** (it may also be spelled Korsakoff), which is usually associated with a lesion of the Papez circuit, including the mammillary bodies and hypothalamus (Maillard et al, 2021). The most representative conditions manifesting this syndrome are **Wernicke encephalopathy** (Figure 23-1; Box 23-3) and limbic encephalitis (see Box 28-1, p. 265). In these conditions, the patients commonly show emotional abnormalities, character change, and a frivolous, unreliable personality. If this condition is further accompanied by abnormal behaviors like oral tendency and hypersexuality, it is called **Klüver–Bucy syndrome**. The latter syndrome is seen in association with bilateral lesions of the mesial temporal lobes like herpes simplex encephalitis, limbic encephalitis, and post-anoxic encephalopathy.

The neural circuit related to memory function has been known as the Papez circuit (Jang & Kwon, 2014) (Figure 23-2). This circuit starts from the hippocampus and uncus; then goes through the fornix, septal nucleus and mammillary body, **mammillothalamic tract of Vicq d'Azyr**, anterior nucleus of thalamus, and cingulum in this order; and finally returns to the hippocampus. In this circuit, a part of fibers from the septal nucleus is considered to enter the cingulate gyrus directly without going through the anterior nucleus of the thalamus. This pathway, together with the **dorsal medial nucleus of the thalamus** (see Chapter 30-4, p. 273), is known to play an important role in memory (Danet et al, 2015), and it is well known that bilateral lesions of the medial thalamus cause marked amnesia (Figure 23-3). Among the variants of the thalamic arteries, the artery of Percheron is known to supply the paramedian thalami and the rostral midbrain (Kichloo et al, 2019).

C. CALCULATION

For testing mental calculation, serial subtraction of 7 from 100 is commonly used. If the patient is suddenly given this order, he or she may be too upset to respond correctly.

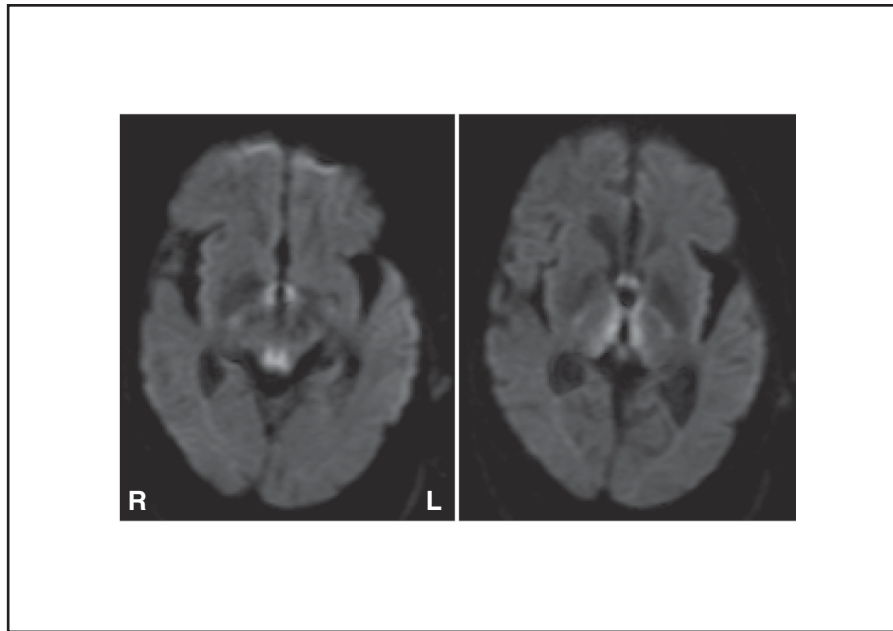


Figure 23-1 Diffusion MRI of a patient who acutely presented with diplopia, unsteady gait, delusion, and marked fluctuation of body temperature following massive alcohol intake. The MRI shows abnormal signal around the third ventricle and in the mammillary bodies, suggestive of Wernicke encephalopathy.

Therefore, it is important to take into account the patient's character and circumstances. Because this task requires working memory and attention, if the patient has difficulty responding verbally, it may be useful to give the same test in written form. When the patient has any difficulty in calculation, it is important to judge whether it is a part of generalized cognitive impairment or **acalculia**, which is a focal cortical sign due to a lesion of the left supramarginal gyrus and an important component of Gerstmann syndrome (see Box 24-4, p. 236) in right-handed subjects. It is usually possible to make this distinction by taking other mental functions into consideration, but it is believed that patients with acalculia tend to make more errors in the position of digits on vertically written calculations. By contrast, the patients with generalized cognitive impairment find it easier to make written calculations than mental calculations.

D. COMMON KNOWLEDGE AND JUDGMENT

Judgment is divided into two categories: cognitive judgment and social judgment. Cognitive judgment is directly related to the perception and recognition of external stimuli and internal milieu, whereas social judgment involves interpersonal relationship. For cognitive judgment of general sensory information, the primary and secondary sensory centers for each modality play a role. For special sensory modalities, special receptive areas in the parietal

and temporal cortex are important (see Chapter 23-1F, p. 194). Regarding social judgment, the orbitofrontal cortex and the dorsolateral prefrontal area play an important role for the judgment based on visual information, and the amygdala is related to the judgment of facial emotion (Mah et al, 2004). For moral judgment, the mesial prefrontal area is considered to be important (Avram et al, 2014).

Common knowledge and judgment can be evaluated during the history taking by talking to the patient about topics of common interest, depending on the patient's educational level and occupation. The topics may include recent news about society, sports, arts, and hobbies. It is also possible to evaluate the process of thinking and judgment by asking the patient how he or she feels about that particular topic.

E. EMOTION AND CHARACTER

These features can also be estimated by observing the facial expression and behavior of the patient during the history taking. It may also be judged by observing the emotional reaction of the patient by telling some jokes. Information obtained from the patient's family as to whether or not there has been any recent change in the patient's character is very useful (Figure 23-4).

BOX 23-3 WERNICKE ENCEPHALOPATHY

This is an acute encephalopathy affecting heavy drinkers and is characterized by symptoms and signs such as fluctuating disturbance of consciousness, delusion, Korsakov syndrome (see p. 218), ophthalmoplegia, nystagmus, ocular motor disturbance, and cerebellar ataxia (some or all of these may appear). In addition, the ankle jerk may be lost due to peripheral neuropathy, and Babinski sign may be positive. Pathologically, a necrotic lesion associated with small hemorrhages is seen in the mammillary bodies, periventricular region of the thalamus/hypothalamus, periaqueductal gray substance, base of the fourth ventricle, and cerebellar vermis (see Figure 23-1). Furthermore, an autonomic storm with excessive fluctuation of vital signs, especially body temperature (poikilothermia), may occur, and it may be fatal. Early diagnosis is extremely important for this condition, because early treatment with a large dose of active vitamin B₁ is effective. This treatable condition may be misdiagnosed as strokes of the rostral brainstem because the two conditions present a quite similar clinical picture.

Central pontine myelinolysis may present with similar clinical manifestations in addition to quadriparesis. This condition occurs when marked hyponatremia is acutely corrected by sodium supplementation therapy. Pathologically, it is characterized by a demyelinating lesion in the central part of the pons. This pontine lesion may be associated with demyelinating lesions in the cerebral white matter (**extrapontine myelinolysis**).

A rare condition seen in heavy drinkers is Marchiafava-Bignami disease, which is characterized by a white matter lesion especially involving the corpus callosum. Cranial MRI is helpful for diagnosis of this condition.

F. ILLUSION, HALLUCINATION, AND DELUSION

These are a group of positive psychiatric symptoms (see Chapter 1-1, p. 2). In an **illusion**, a given stimulus is sensed differently from what it is expected to be sensed. In contrast, an **hallucination** is a phenomenon of seeing or hearing something that is not there to be seen or heard, respectively. A typical example of a **visual hallucination** is seeing insects that are not actually present, and the patient often reacts to the image either verbally or behaviorally. Visual hallucinations are commonly seen in **delirious states (delirium)** such as those due to alcohol intoxication, other toxic encephalopathies, and metabolic encephalopathies. Visual hallucinations also occur in patients with Parkinson disease during L-dopa treatment or in patients with diffuse Lewy body disease (see p. 223), especially at night. Patients with

BOX 23-4 SUBACUTE AND FLUCTUATING BEHAVIORAL ABNORMALITY

There are several conditions that cause behavioral abnormalities of subacute onset and fluctuating course, many of which are treatable. Among others, **chronic subdural hematoma** causes cognitive dysfunction and gait disturbance a few months after a relatively mild head injury. Although it is easily detected by neuroimaging techniques, it is still an important differential diagnosis of dementia of unknown etiology. **Metabolic encephalopathy** due to liver cirrhosis or kidney failure also causes cognitive disturbance of subacute onset and fluctuating course. In the presence of a shunt between the portal vein and the systemic circulation, the patient may develop symptoms of encephalopathy after ingesting a protein-rich diet (**portal-systemic encephalopathy**). In this condition, ammonia levels are elevated in the blood and triphasic waves may be seen on the electroencephalogram (EEG) (see Figure 33-2, p. 290). Toxic encephalopathy is an important differential diagnosis for rapidly progressive dementia (toxic encephalopathy). Chronic use of drugs such as methyl bromide, bismuth subsalicylate (used for collagenous colitis), and disulfiram (used for alcohol dependency) may cause rapidly progressive dementia, often associated with seizures, myoclonus, and ataxia (de Souza et al, 2013, Ali et al, 2017, and Peddawad et al, 2018, respectively). Disorders of amino acid metabolism with an abnormal urea cycle such as **citrullinemia** are a common cause of metabolic encephalopathy. Citrullinemia results from deficiency of argininosuccinate synthetase. In this case, it is often discovered that the patient has been fond of beans since childhood, probably because beans are rich in arginine.

auditory hallucinations may hear voices of people who are not there and may even respond to them by speaking or acting.

A **delusion** is defined as an abnormal belief of false ideas, such as a delusion of persecution and a grandiose delusion. Auditory hallucinations and delusions are often seen in patients with psychosis and schizophrenia, among others, but they may also be seen in patients with intractable temporal lobe epilepsy during the interictal phase and other organic brain diseases.

G. STATE OF DAILY LIVING

Information about the patient's daily life obtained from the patient's family and housemates is most useful for evaluating

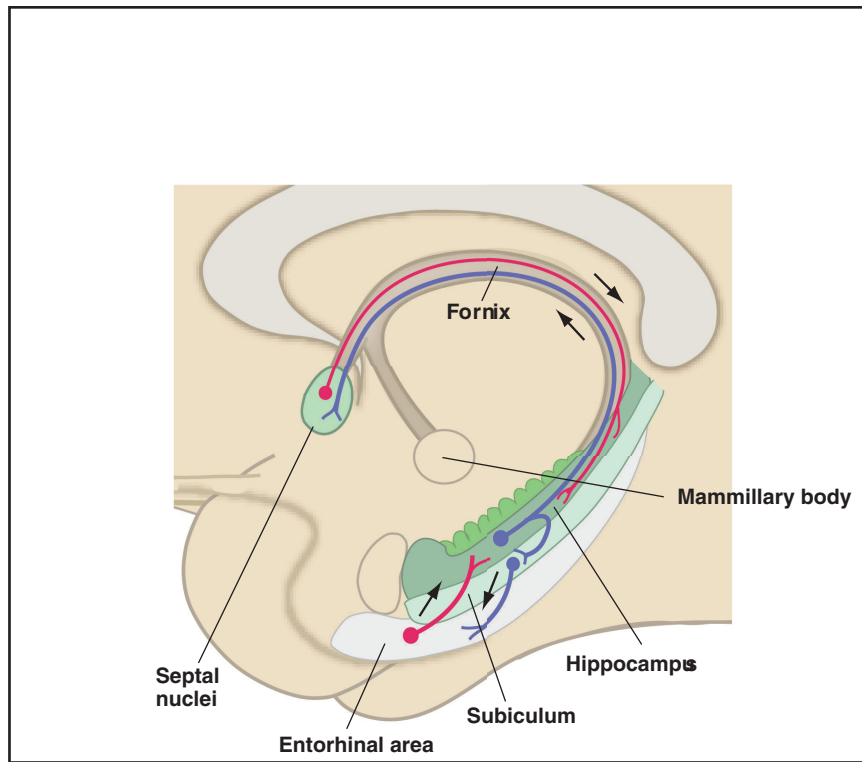


Figure 23-2 Schematic diagram showing the neuronal circuit related to memory.

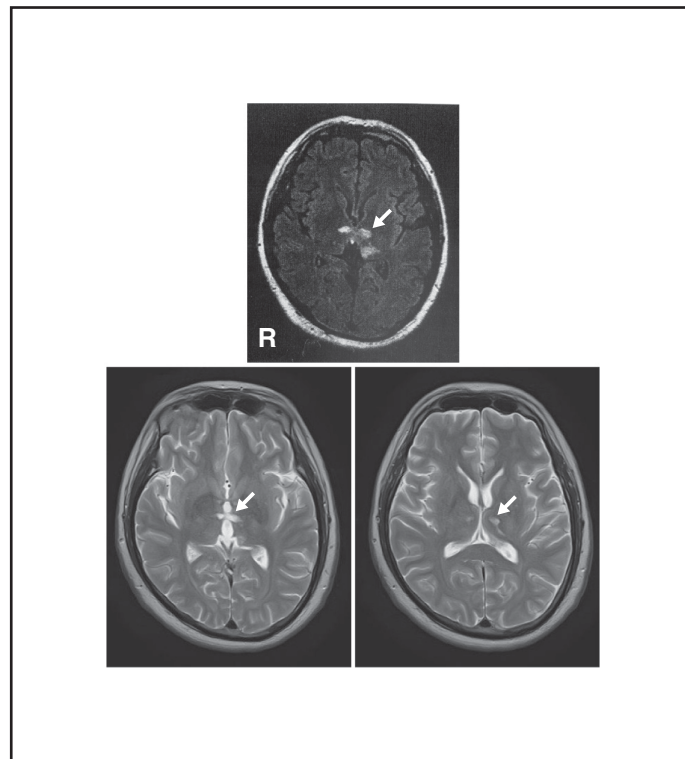


Figure 23-3 Diffusion MRI (top) and T2-weighted MRI (bottom) of a patient who presented with marked amnesia following bilateral medial thalamic infarction (arrows). There was consciousness disturbance and abnormal sensation on the right half of the body at the clinical onset.

BOX 23-5 CHANGE OF EMOTION AND CHARACTER BY A LESION OF THE RIGHT TEMPORAL LOBE

A 35-year-old man developed severe headache, and 12 hours later he lost consciousness. After recovery from the acute episode, he often got angry at home and showed a marked character change. On examination the patient was found to be in a state of abulia with little facial expression and slow motor initiation and execution. There was also left homonymous inferior quadrantanopsia. A large hemorrhage was found in the right temporo-parietal lobe (Figure 23-4a). In relation to this case, the French poet Apollinaire, after having sustained a war injury to the right temporal head (Figure 23-4b), became unable to control his emotions and stopped writing to his fiancée, to whom he used to write almost every day before the injury (Bogouslavsky, 2003).

mental functions. In case of someone who works in the home, for example, information about whether they make any mistakes in shopping, cooking, cleaning, and washing may disclose abnormalities, if any. Especially in Alzheimer disease, questions as to whether or not the patient has any difficulty in finding the way to his or her own house may provide useful information in relation to topographic abilities, which are commonly impaired in this disease.

2. DEMENTIA

Dementia is a condition in which all or some of these mental/cognitive functions are impaired to various degrees; in its advanced stage, it causes severe disability in social life. It is often accompanied by impairment of focal cortical functions (see Chapter 23, p. 216), and in the very early stage, focal cortical dysfunction may predominate the clinical picture. It is known that the prevalence of dementia increases with aging between ages 65 and 85. In a study of 911 subjects over age 90 in California, 45% of women were diagnosed with dementia versus 28% of men (Corrada et al, 2008). It was further shown that, in women over 90, the prevalence of dementia doubled every 5 years.

When examining patients with dementia, it is of utmost importance to consider the treatable medical conditions before making a diagnosis of neurodegenerative diseases and vascular dementia. Treatable conditions include nutritional disorders such as vitamin B₁₂ deficiency and niacin deficiency (pellagra), chronic hepatic diseases, kidney failure, hypothyroidism, poor cerebral perfusion due to cardiac failure, hypoxia due to chronic lung diseases, treatable structural lesions, and long-term drug use.

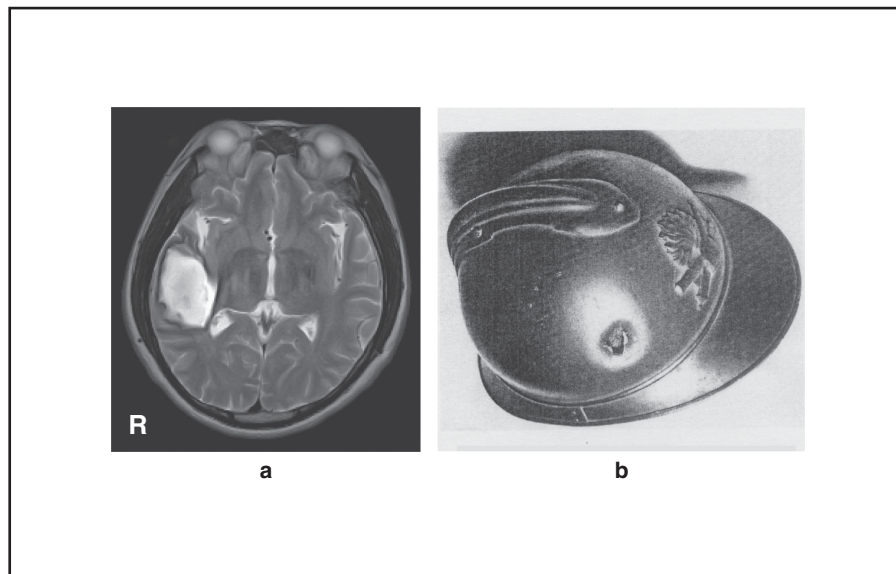


Figure 23-4 Character change due to a lesion of the right temporal lobe. (a) MRI of a patient who showed marked character change and left inferior quadrantanopsia following acute attack of severe headache and consciousness disturbance. A large hemorrhage is seen in the right temporal and parietal lobes. (b) Helmet used by the French poet Apollinaire during the war. His character changed dramatically after the head injury. Cited from Bogouslavsky, 2005, with permission.

A. ALZHEIMER DISEASE AND RELATED DISORDERS

i. Alzheimer disease

Alzheimer disease (AD) is the most common cause of dementia involving the elderly population. The disease affects both sexes but is more common in women than in men. The most common initial symptom is memory disturbance, and disorientation to time is also commonly seen. At the advanced stage, it causes severe disability in social life. Compared with dementia due to multiple cerebral infarctions, patients with AD tend to lose insight into the disease. Seizures have been reported in a small proportion of patients with AD; in fact, subclinical epileptiform electroencephalographic (EEG) activity was detected in about 40% of AD patients (Vossel et al, 2016). Based on a multicenter study in France, seizures were especially common in patients with autosomal dominant early-onset AD (Zarea et al, 2016).

Cognitive impairment associated with minimal impairment of instrumental activities of daily living is called **mild cognitive impairment (MCI)**. According to the guidelines of the American Academy of Neurology, the cumulative incidence for the development of dementia in individuals with MCI older than age 65 followed for 2 years is about 15% (Petersen et al, 2018). In a multi-institutional study of a large series of autopsied cases of MCI, MCI diagnosis was usually associated with comorbid neuropathologies, and less than one-quarter of MCI cases showed pure AD at autopsy (Abner et al, 2017).

In AD, neurodegeneration involves the hippocampus, amygdala, and parietal and temporal cortices. Histologically, it is characterized by neuronal degeneration and loss associated with neurofibrillary tangles and senile plaques. The neurofibrillary tangles are formed by deposits of tau protein, and the senile plaques are formed by deposits of amyloid.

ii. Lewy body dementia and Parkinson disease

Dementia characterized by accumulation of Lewy bodies is called **dementia with Lewy bodies (DLB)** or **diffuse Lewy body disease**. It clinically manifests with progressive cognitive disorders and parkinsonism. Pathologically, Lewy bodies are seen diffusely in the cerebral cortex. Visual hallucinations are commonly seen at night, and functional abnormalities of the occipital lobe have been suggested by EEG and functional neuroimaging studies (McKeith et al, 2017).

As cognitive impairment is seen in more than half of patients with idiopathic Parkinson disease at the advanced stage (**Parkinson disease with dementia, PDD**), this might be distinguished from DLB. Two possibilities have been proposed for the cause of dementia in Parkinson disease: DLB and coexistence of AD (Compta et al, 2011). In a prospective, neuropathologic study from Norway of 22 cases of Parkinson disease, the presence of Lewy bodies was the only finding that was positively correlated with the degree of dementia, and it was proposed that dementia in Parkinson disease was mostly due to DLB (Aarsland et al, 2005). Another study that used chemical imaging with positron emission tomography (PET) looking for L-dopa uptake, acetylcholine metabolism, and glucose metabolism found common abnormalities between the patients with Parkinson disease with dementia and those with DLB (Klein et al, 2010). A recent neuropathologic study of many cases of a spectrum of parkinsonian disorders disclosed concomitant pathologies, especially between DLB and AD (Coughlin et al, 2020; Dugger et al, 2014; Walker et al, 2015).

In relation to visual hallucinations in DLB, it should be kept in mind that hallucinations are also seen as a side effect of L-dopa treatment. Cholinesterase inhibitors may improve cognitive function and behavior in patients with DLB (Mori et al, 2012).

According to a comparative study of cognitive impairment among patients with clinically diagnosed DLB, PDD, and AD, it was shown that both DLB and PDD patients were more impaired and declined more rapidly than those with AD in the visuospatial domain as well as in the executive domain (Smirnov et al, 2020). Identification of DLB is important because zonisamide (adjunctive to levodopa) was shown to improve parkinsonism in DLB without worsening cognitive functions or psychiatric symptoms (Murata et al, 2018).

iii. Basal ganglia and cognitive functions

Degenerative disorders of the basal ganglia are often accompanied by dementia. Sawamoto et al. (2002, 2007) studied patients with Parkinson disease by presenting a day of the week and asking the patient to advance 1 to 3 days as quickly as possible; they showed poor performance (bradyphrenia) and lower blood flow activation in the basal ganglia in association with the task compared with age-matched healthy control subjects. As this study did not require a motor response, it was claimed that the results were not affected by bradykinesia.

B. FRONTOTEMPORAL LOBAR DEGENERATION

A group of neurodegenerative disorders characterized clinically by progressive behavioral abnormality and language impairment and pathologically by degeneration of the prefrontal cortex and the anterior temporal neocortex is referred to as **frontotemporal lobar degeneration (FTLD)**. FTLD affects adults between 45 and 65 years of age, and there is no sex predilection. It often seems sporadic, but familial cases have been reported, and now genetic abnormalities are often etiologic. Some cases of FTLD are also accompanied by motor neuron diseases and parkinsonism.

Various clinical forms are known in FTLD, including **frontotemporal dementia (FTD)**, **semantic dementia**, **progressive nonfluent aphasia**, and **FTD/Pick complex** (Kertesz et al, 2005; Lashley et al, 2015; Neary et al, 2005; Whitwell et al, 2020). There is also the **behavioral variant of FTD**, which is characterized by progressive impairment of personality, social behavior, and cognitive functions (O'Connor et al, 2017; Rascovsky et al, 2011). Each of these categories shows a characteristic clinical picture at the initial stage, but at the advanced stage, they may show other abnormalities and become indistinguishable from each other.

Pathologically, FTLD is largely classified into two groups: a group with tau inclusions (tauopathies) and another group without tau but with ubiquitin-positive inclusions (FTLD-U). Neuropathologic studies of patients who presented clinically with FTLD showed tauopathies in 46%, FTLD-U in 29%, AD in 17%, and others, including DLB (Forman et al, 2006). In fact, there is some overlap between cases of FTLD and familial cases of AD in terms of phenotype, inclusion protein, and genotypes (van der Zee et al, 2008). In the second group (FTLD-U), there is a mutation of the progranulin gene (Beck et al, 2008).

Recently a number of genes have been confirmed or suggested to cause or influence risk for FTD or FTD-spectrum disorders, and among others C9orf72, GRN, and MAPT are the most commonly implicated in FTD (Sirkis et al, 2019). It is also known that the autosomal dominant form of FTD and amyotrophic lateral sclerosis (ALS) occur in the same family, and these two conditions share common genetic risk loci on chromosome 9p21.2 and 19p13.11, suggesting a possible common pathogenesis in the two conditions (DeJesus-Hernandez et al, 2011; Diekstra et al, 2014; Hodges, 2012; Renton et al, 2011). In fact, C9orf72 (chromosome 9 open reading frame 72) hexanucleotide repeat expansions are the most common cause of familial FTD and familial ALS, and patients with ALS with these

BOX 23-6 DEMENTIA PUGILISTICA (CHRONIC TRAUMATIC ENCEPHALOPATHY)

It has been reported that boxers who have repeatedly received punches to the head tend to develop memory disturbance, behavioral abnormality, character change, and motor disturbance later (dementia pugilistica, punch-drunk encephalopathy). It is also known that a similar condition occurs in players of other sports, such as American football, hockey, soccer, and professional wrestling (**chronic traumatic encephalopathy**, CTE). Various pathologic abnormalities such as brain atrophy, abnormality of the cavum septi pellucidi, atrophy of the mammillary body, diffuse deposit of tau inclusion and amyloid, and inflammatory change with microglia have been reported (McKee et al, 2015). Diagnosis of traumatic encephalopathy syndrome (TES) requires (1) substantial exposure to repetitive head impacts from contact sports, military service, or other causes; (2) core clinical features of cognitive impairment (in episodic memory and/or executive functioning) and/or neurobehavioral dysregulation; and (3) a progressive course, and the clinical features cannot be fully accounted for by any other neurological, psychiatric, or medical conditions (Katz et al, 2021).

BOX 23-7 HIV-RELATED NEUROLOGICAL DISORDERS

In **acquired immunodeficiency syndrome (AIDS)** due to infection with the human immunodeficiency virus (HIV), a number of neurological complications are known to occur, involving the central and peripheral nervous system and muscles. Brain pathology is mediated by two different mechanisms: activation of microorganisms that were latent as a result of opportunistic infection on one hand and direct infection of the brain with HIV on the other. The microorganisms of the first group include toxoplasma, cryptococcus, cytomegalovirus, JC virus (Box 23-11), and syphilis. In the second group, a diffuse inflammatory change in the brain due to HIV infection causes dementia, depression, and parkinsonism (Arendt & Nolting, 2008; Gannon et al, 2011). HIV-associated neurocognitive disorders (HAND) may be due to synaptodendritic damage and disruption of neuronal networks (Irollo et al, 2021). With the introduction of highly active antiretroviral therapy (HAART), the prevalence and the severity of these neurological complications have decreased.

BOX 23-8 PRION DISEASE

This is a group of diseases causing fatal neurodegeneration due to deposition of prion protein. It was previously called subacute spongiform encephalopathy based on its pathological feature. Its representative form is **Creutzfeldt–Jakob disease (CJD)**. Most cases of CJD are sporadic, but 10% to 15% of cases are transmitted by autosomal dominant inheritance with mutation of the coding region of the human prion protein gene (Baldwin & Correll, 2019; Ironside et al, 2017; Roeber et al, 2008). Sporadic cases of CJD manifest with cognitive decline, periodic myoclonus, and periodic synchronous discharges on EEG, and the clinical course is rapidly progressive. Another form of familial prion disease is **Gerstmann–Sträussler–Scheinker syndrome**, which initially presents with mild gait disturbance, abnormal sensation, loss of tendon reflexes in the lower extremities, and truncal ataxia (Kang et al, 2019). **Fatal familial insomnia** is a familial prion disease characterized by insomnia and MRI abnormality in the pulvinar (see Chapter 27-7, p. 258). A variant form of spongiform encephalopathy (variant CJD) is known as mad cow disease because it first arose in cows in the United Kingdom. Clinically it starts with psychiatric symptoms, and myoclonus is rare, if any, in this condition.

Cases of rapidly progressive dementia tend to be diagnosed as CJD, but other conditions, including Alzheimer disease, vascular dementia, autoimmune encephalitis, brain tumor like lymphoma, infectious disease, and metabolic encephalopathy, could present with similar symptoms. Therefore, the diagnosis of CJD has to be made with caution (Chitravas et al, 2011).

expansions more frequently show behavioral and cognitive symptoms than those with classical ALS (Cammack et al, 2019; Rohrer et al, 2015).

Recently TDP-43 (transactive response DNA binding protein of 43kDa) proteinopathy has been demonstrated in some subtypes of FTLT (Mackenzie & Neumann, 2020). However, TDP-43 pathology has also been shown to be commonly observed in subjects over 80 years of age (Nag et al, 2017), and limbic-predominant age-related TDP-43 encephalopathy (LATE) has been proposed as one of the causes of cognitive impairment in the old subjects (Nelson et al, 2019).

The familial occurrence of **ALS/parkinsonism–dementia complex** has been reported in Chamorro of Guam, the Kii peninsula of Japan, and in western New Guinea. Clinical, pathologic, and genetic studies have been extensively carried out in those cases, but so far heterogeneous results have been reported (Kokubo et al, 2012;

Steele et al, 2015). By the pathologic as well as PET imaging study in the patients of the Kii peninsula, tau deposition was demonstrated in the cerebral cortex and white matter, and pontine basis, corresponding to the clinical phenotypes (Shinotoh et al, 2019).

C. LEUKOENCEPHALOPATHY

Diffuse lesions of the cerebral white matter are divided into congenital or hereditary leukodystrophy and acquired leukoencephalopathy. The most common form of the latter group is a diffuse ischemic lesion of the cerebral white matter causing **vascular dementia**. The cerebral lesions causing vascular dementia are classified into three groups: a large cerebral infarction, multiple small infarctions, and diffuse white matter lesions. White matter hyperintensities in the brain are the consequence of cerebral small vessel disease and can easily be detected on magnetic resonance imaging (MRI; Cannistraro et al, 2019; Prins & Scheltens, 2015).

Binswanger disease, previously known as progressive subcortical encephalopathy, occurs in hypertensive patients and is clinically characterized by progressive dementia and gait disturbance. MRI of this condition reveals diffuse abnormalities in the periventricular white matter (**leukoaraiosis**), but it is often accompanied by multiple small infarctions in the basal ganglia. The autosomal dominant hereditary form of Binswanger disease is called cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (**CADASIL**). This disease occurs in adults mainly between 40 and 60 years of age with progressive dementia and gait disturbance, and also convulsion in some cases. Mutation of the notch 3 gene has been identified in this condition. Families of autosomal recessive inheritance are also known as **CARASIL**, in which a mutation of the HTRA1 gene was identified. This form is accompanied by alopecia and lumbago due to lumbar spondylosis and is common among Asians (Fukutake, 2011; Nozaki et al, 2015).

Hereditary leukodystrophy usually occurs in infants and children, but some forms may occur in adults. **Adrenomyeloneuropathy** is an adult form of adrenoleukodystrophy (see Box 17-7, p. 149). **Krabbe disease (globoid cell leukodystrophy)** is a recessively inherited lysosomal storage disease due to deficiency of β -galactocerebrosidase, usually affecting early infants. Adult-onset Krabbe disease is characterized by MRI white matter hyperintensities, especially involving the precentral gyrus, corona radiata, and posterior internal capsule (Cousyn et al, 2019). **Alexander disease** due to mutations in the glial fibrillary acidic protein

BOX 23-9 VANISHING WHITE MATTER DISEASE

This is an autosomal recessive hereditary leukoencephalopathy that occurs in children with cerebellar ataxia. This disease is characterized by acute worsening of the symptoms in association with infection and head trauma. Characteristically, the follow-up studies with proton or FLAIR MRI reveal the vanishing white matter. Mutation of the *EIF2B* genes 1 to 5 was found in this condition, and its phenotype is heterogeneous (Matsui et al, 2007; van der Lei et al, 2010).

(GFAP) gene is pathologically characterized by the presence of Rosenthal fibers. Its infantile and juvenile forms are characterized by mental deterioration, but its adult form also presents with spastic paraparesis, cerebellar ataxia, and palatal tremor (Graff-Radford et al, 2014; Martinez-Poles et al, 2020). **Hereditary diffuse leukoencephalopathy with spheroids** (HDLS) is characterized by progressive cognitive and behavioral dysfunction followed by motor impairment such as gait disturbance and bradykinesia in adults under age 60. This is an autosomal dominant disease, and recently a gene encoding the colony stimulating factor 1 receptor (*CSF-1R*) has been identified as a causative gene (Konno et al, 2014; Tada et al, 2016). HDLS and pigmentary orthochromic leukodystrophy (POLD) are now genetically considered to be a single clinicopathologic entity (Adams et al, 2018). In patients with **vanishing white matter disease** (Box 23-9) and **membranous lipodystrophy** (**Nasu–Hakola disease**) (Box 23-10), symptoms may first

BOX 23-10 NASU–HAKOLA DISEASE

This is an autosomal recessive hereditary disease characterized by multiple fractures since puberty, progressive memory disturbance, and psychiatric symptoms such as character change, euphoria, and indifference, finally resulting in akinetic mutism. This disease was first reported by Nasu et al. from Japan in 1970, and later Hakola et al. coined the term “osteodysplasia polycystica hereditaria combined with sclerosing leukoencephalopathy” (Kaneko et al, 2010). The main pathologic abnormality of the cerebral hemispheres is sudanophilic leukodystrophy, and systemically there are membranocystic lesions in the bones and fatty tissues. Diagnosis is made by finding cystic lesions in the epiphysis of long bones on X-ray examination. Mutation of the *TREM2* or *DAP12* gene was identified in families with this disease (Ghezzi et al, 2017).

BOX 23-11 PROGRESSIVE MULTIFOCAL LEUKOENCEPHALOPATHY (PML)

This is a progressive demyelinating disease of the cerebral white matter due to destruction of the oligodendroglia via reactivation of latent **JC virus**. Clinically, the disease starts with focal cortical symptoms of the unilateral parietal lobe and rapidly progresses to dementia and motor paralysis associated with MRI findings revealing fusion of multiple small white matter lesions to form diffuse leukoencephalopathy. It is usually associated with immunodeficiency states such as **malignant lymphoma** and **AIDS**, but more recently, its occurrence during **immunosuppressive treatment** of the relapsing and exacerbating form of multiple sclerosis (MS), especially with natalizumab, and in the immune-deficient state following **organ transplantation** has drawn particular attention. The incidence of PML during natalizumab treatment increases in proportion to the total number of infusions, but PML in this condition is less severe than the ones caused by other etiologies, if discovered at the early stage (Chalkias et al, 2014; Schwab et al, 2017; Vermersch et al, 2011). It was reported that the anti-JC virus antibody index was higher in the serum or plasma of MS patients who developed PML even before natalizumab treatment, suggesting the possibility of predicting the risk of developing PML (Plavina et al, 2014). PML after organ transplantation may occur irrespective of the kind of transplanted organ and is more severe compared with other causes (Mateen et al, 2011). The latent period from transplantation to clinical onset varies from 1 month to more than 20 years. Treatment of PML is aimed at the underlying systemic diseases and improvement of the immunodeficiency state. PML associated with AIDS is treated by highly active antiretroviral therapy (HAART), which prevents the incidence of progressive multifocal leukoencephalopathy (PML) and suppresses its progression.

appear in childhood, but the patients may visit the adult neurology clinic at a later age. **Pelizaeus–Merzbacher disease** is an X-linked hypomyelinating leukodystrophy caused by mutations or rearrangements in the proteolipid protein 1 (*PLP1*) (Elitt et al, 2020; Lossos et al, 2015). **Retinal vasculopathy with cerebral leukoencephalopathy and systemic manifestations** (RVCL-S) is a familial microvasculopathy leading to early visual loss and progressive neurocognitive deterioration in the fourth to sixth decades. This is an autosomal dominant disease caused by a mutation in *TREX1* (Ford et al, 2020).

Adult-onset **neuronal intranuclear inclusion disease** is a slowly progressive neurodegenerative disease characterized by eosinophilic hyaline intranuclear inclusion in the central nervous system, peripheral nervous system, and visceral organs. Clinically it manifests with dementia, muscle weakness, and autonomic disturbances (Sone et al, 2016). Differentiating it from similar neurodegenerative diseases is often difficult, but demonstration of the intranuclear inclusion in the fibroblasts by skin biopsy is useful (Sone et al, 2011).

Acquired diffuse white matter lesions are also caused by anticancer drugs such as fluorouracil and as a delayed effect of acute carbon monoxide poisoning (**interval form of acute carbon monoxide poisoning**) (Tábuas-Pereira et al, 2017). Furthermore, leukoencephalopathy might occur after the whole-brain irradiation (**chronic post-radiation leukoencephalopathy**), and some patients might show acute transient episodes of impaired consciousness and neurological deficits several years after the irradiation (Di Stefano et al, 2013). In relation to leukoencephalopathy, it should be kept in mind that the primary malignant lymphoma of the central nervous system may show progressive white matter lesions on MRI.

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APHASIA, APRAXIA, AND AGNOSIA

1. NEURAL CIRCUITS RELATED TO LANGUAGE, PRAXIS, AND RECOGNITION

Several hypotheses have been proposed for the neural circuits related to language, praxis, and recognition based on the correlation between clinical symptoms and signs and pathological findings of patients who died of head trauma or vascular brain lesions, and more recently based on the correlation with the neuroimaging data in clinical cases (Figure 24-1). Moreover, as a consequence of the recent advances in the various noninvasive techniques for studying brain functions in healthy subjects (Shibasaki, 2008), and as the result of presurgical evaluation of patients with medically intractable partial epilepsy by intracranial recording, further information on the **functional localization (specialization)** as well as **inter-areal functional coupling** has been accumulated.

Functions of the nervous system are generally symmetric, but the hemispheric language dominance is of utmost importance for evaluating these higher cortical functions. Although there seems to be some racial difference in the language dominance, it is believed that about 95% of the population is right-handed, and the left cerebral hemisphere is dominant in the majority of those right-handed subjects. The remaining 5% of the population may be left-handed, but about half of the left-handed subjects are also believed to have dominance in the left hemisphere. Thus, only a very small proportion of the population has dominance in the right hemisphere.

An exception to this rule is the occurrence of aphasia in right-handed subjects with a lesion of the right hemisphere, although it is extremely rare. This phenomenon is called **crossed aphasia**. In this chapter, higher cortical functions are discussed for right-handed subjects who have language dominance in the left hemisphere.

A. FOCAL NEUROLOGICAL DEFICIT AND DISCONNECTION SYNDROME

It is considered that higher cortical functions are impaired either by a focal lesion of the association cortex (**focal neurological deficit**) or by interruption of the functional connection among different cortical areas (**disconnection syndrome**). The concept of disconnection syndrome was brought into prominence in modern times by Geschwind (Geschwind, 1965a, 1965b; Hallett, 2015). As a consequence of recent advances in electrophysiologic and neuroimaging techniques, functional connectivity has been documented in healthy subjects, and the notion of disconnection syndrome is drawing increasing attention (Catani & ffytche, 2005; Matsumoto et al, 2004b, 2012). In the clinical setting, it is reasonable and practical to consider the two concepts in combination.

B. LANGUAGE

When evaluating language function, it is important to consider both **spoken language** and **written language**. Anatomically, in addition to the **anterior language area** in the left inferior frontal gyrus and the **posterior language area** in the left superior temporal gyrus, it is useful to consider a third language area in the left parietal lobe (Figure 24-2). These areas cover a large region along the left sylvian fissure and receive blood supply from the left middle cerebral artery (see Figure 7-5, p. 41, and Figure 7-8, p. 44). In addition, participation of the left basal temporal lobe in language function was proposed by Lüders et al. (1986), and the **basal temporal language area** has been demonstrated electrophysiologically (Matsumoto et al, 2004b; Usui et al, 2009). The technique of cortico-cortical evoked study was developed for studying functional and pathologic connectivity in patients with epilepsy (Matsumoto et al, 2017).

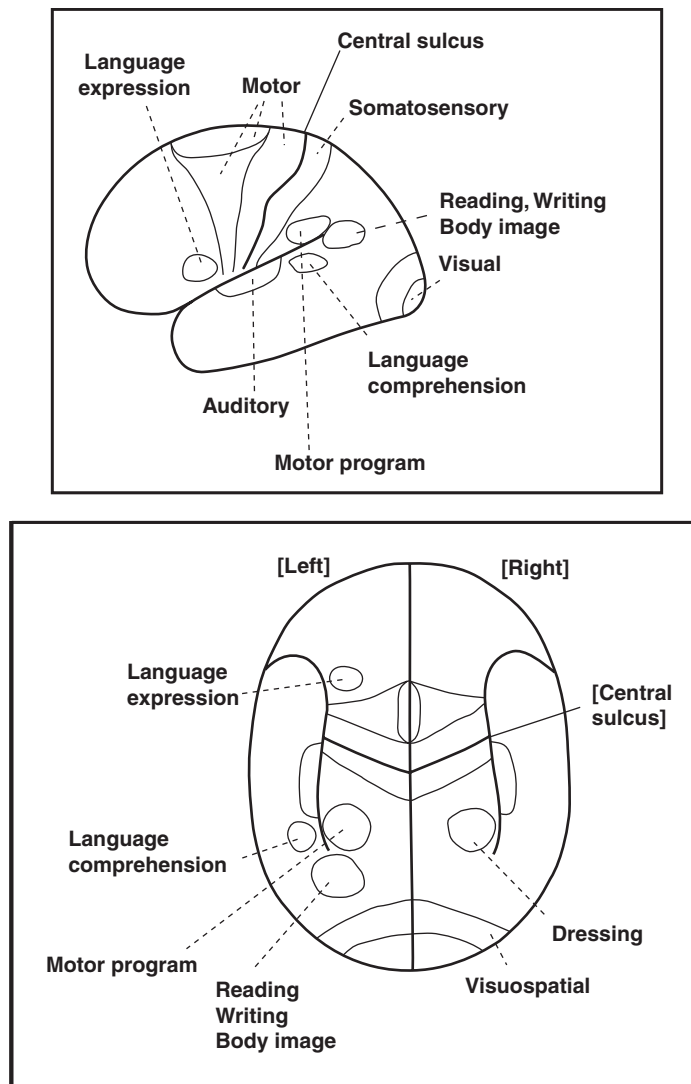


Figure 24-1 Schematic diagram showing functional localization in the cerebral cortex in right-handed subjects. (a) Lateral view of the left hemisphere. (b) Axial view.

Regarding spoken language, an auditory input is received by the auditory cortex located in the bilateral superior temporal gyri and is transmitted to **Wernicke's area** in the posterior part of the left superior temporal gyrus (Figure 24-3a). This information has been obtained primarily from the observation of patients with stroke (Hillis et al, 2017), but in neurodegenerative diseases presenting with primary progressive aphasia, it has been shown that word comprehension is actually located in the left anterior temporal lobe and sentence comprehension is distributed widely throughout the language network (Mesulam et al, 2015, 2019). In Wernicke's area, the words are understood with the aid of functional connectivity with other related cortical areas, and it plays a critical role in speech production (Binder, 2015). For repeating orally presented words (**repetition**), the information is conducted from Wernicke's area through the **arcuate**

fasciculus (Figure 24-3b) to **Broca's area** in the left inferior frontal gyrus (Figure 24-3c). From there, the words are pronounced through the bilateral motor cortices. For speaking spontaneously, Broca's area receives the relevant information from other related cortical areas, and from there the words are pronounced through the bilateral motor cortices.

By intraoperative stimulation studies in patients as well as functional magnetic resonance imaging (MRI) studies in healthy subjects, Exner's writing area was identified in a cortical area just anterior to the hand area of the left primary motor cortex (Roux et al, 2009).

C. CLASSIFICATION OF APHASIA

Impairment or loss of acquired language function is called **aphasia**. If Wernicke's area is damaged, the patient is unable

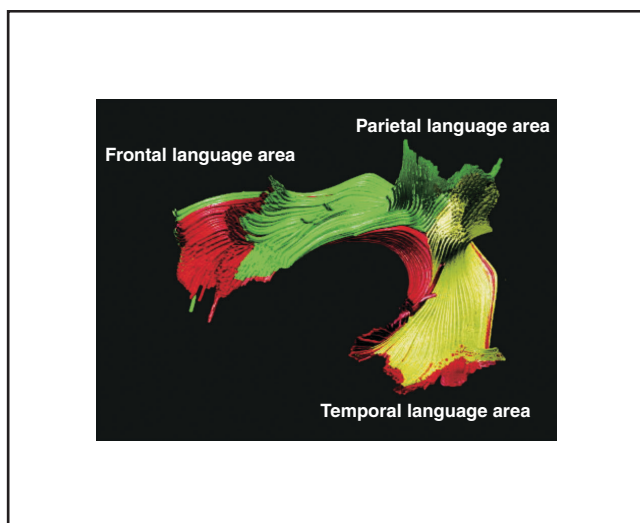


Figure 24-2 Functional connectivity related to language, reconstructed based on the diffusion tensor tractography (DTT). Cited from Catani & ffytche, 2005, with permission.

to understand spoken language and also unable to repeat spoken words. This condition is called **sensory aphasia** or **Wernicke aphasia**. In this case, patients are rather talkative and speak incomprehensible language (**jargon aphasia**) because they cannot understand their own spoken words. This condition is often called fluent aphasia, but the use of the term “fluent” is rather misleading in this situation.

In contrast, the patient with a lesion of Broca’s area and the neighboring region, including the underlying white matter (anterior language area), is not able to speak spontaneously or to repeat given words (**motor aphasia, Broca aphasia**) (Gajardo-Vidal et al, 2021). In this case, patients do not try to speak much or even become mute because they can understand their own mistakes. Speech in this condition is characterized by abnormal intonation (**dysprosody**) and abnormal sentence construction, which sounds like a telegraph (**telegraphic speech**). When repeating the given words, a syllable might be replaced by another different and yet similar syllable (**literal paraphasia**), and a word might be replaced by another different and yet similar word (**verbal paraphasia**).

A group of neurodegenerative syndromes characterized by progressive, selective decline in speech and language functions is called **primary progressive aphasia** (PPA). A neuropathologic study of 69 cases of PPA by Spinelli et al. (2017) disclosed frontotemporal lobar degeneration as the most common pathology, followed by Alzheimer disease.

D. SPOKEN LANGUAGE AND WRITTEN LANGUAGE

In most cases of sensory aphasia, comprehension is impaired not only for spoken language but also for written language.

It might be explained by postulating either that Wernicke’s area is involved in the comprehension of both spoken and written languages, or that the angular gyrus situated just posterior to Wernicke’s area is also affected by the same lesion (see Figure 24-3e). Likewise, in most cases of Broca’s aphasia, both speaking and writing are impaired. It might also be explained by postulating either that Broca’s area is engaged in speaking as well as writing, or that Exner’s area, situated just dorsal to Broca’s area, is also affected by the same lesion (see Figure 24-3g).

Some patients can understand written words but cannot understand spoken words (**pure word deafness**). In this condition, the patient usually can hear and understand the meaning of nonverbal sounds. Likewise, the condition in which the patient can write but cannot speak is called **pure word muteness** or **anarthria** (Schowalter et al, 2021). Some patients with Broca aphasia are unable to stick out their tongue in response to verbal command. This condition can be called verbal apraxia or apraxia of speech, but “lingual apraxia” might be the better terminology for this condition, because Broca aphasia itself might be considered as “apraxia of language.”

If the arcuate fasciculus is selectively affected while Broca’s area and Wernicke’s area are preserved, patients cannot repeat verbally given words even though they can understand the spoken language and speak relatively well.

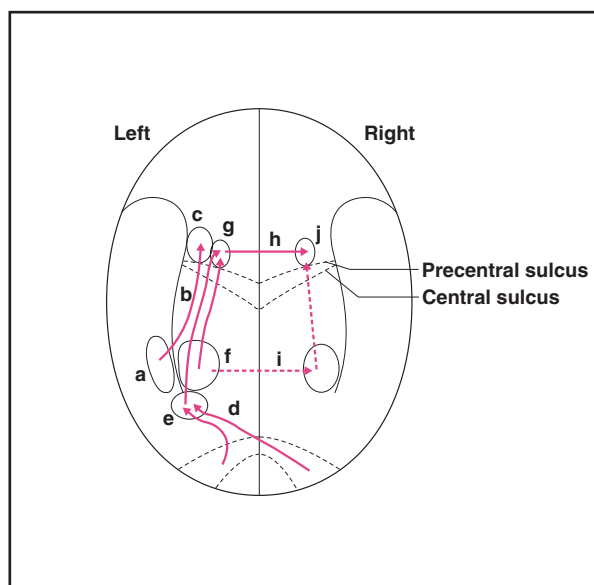


Figure 24-3 Schematic diagram showing the neural circuits related to language, praxis, and gnosis in right-handed subjects. View from the top of head. a: Wernicke’s area, b: arcuate fasciculus, c: Broca area, d: corpus callosum related to visual function, e: left angular gyrus, f: left supramarginal gyrus, g: left premotor area (Exner’s area), h and i: corpus callosum, j: right premotor area.

BOX 24-1 ALEXIA WITH AND WITHOUT AGRAPHIA

Most patients with aphasia have difficulty in both spoken language and written language. Some patients can speak and understand spoken language but can neither read nor write. For written language, the information is transmitted from the visual center in the bilateral occipital cortices to the reading center in the left angular gyrus (see Figure 24-3e), where the written words are understood with the aid of functional connectivity with other related cortical areas. For copying written words, the information from the left angular gyrus is transmitted through the arcuate fasciculus to the writing center (Exner's area) in the left middle frontal gyrus (see Figure 24-3g). Then, the words are written with the right hand through the corticospinal tract from the left motor cortex. For writing spontaneously, the above pathway from the left angular gyrus to the frontal writing center plays a part.

Therefore, patients with a lesion in the left angular gyrus can neither read nor write (**alexia with agraphia**). In contrast, patients with a lesion in the left occipital white matter and the splenium of the corpus callosum interrupting the input of visual information to the left angular gyrus can write but cannot read even the words written by themselves (**alexia without agraphia, pure alexia**) (see Figure 24-3d). The most common cause of this typical disconnection syndrome is an infarction in the left medial occipital white matter.

BOX 24-2 KANJI (MORPHOGRAM) AND KANA (SYLLABOGRAM) OF JAPANESE LANGUAGE

Japanese language is composed of two kinds of characters: *kanji* (morphogram) and *kana* (syllabogram). Clinical studies of Japanese patients with aphasia suggested different mechanisms in the processing of the two characters (Iwata, 1984). Studies of healthy Japanese subjects using functional neuroimaging techniques revealed that reading and writing of *kanji* are primarily processed in the posterior region of the left inferior temporal gyrus, whereas reading and writing of *kana* are primarily processed in the ventral part of the left parietal cortex (Figure 24-4) (Nakamura et al, 2000; Ueki et al, 2006).

The 71-year-old Japanese woman shown in Figure 24-5 had progressive memory disturbance for 2 years and could not write *kanji* but could write *kana* relatively well (Figure 24-5a). She could read and copy sentences including *kanji*. Blood flow was decreased in the left posterior inferior temporal region (Figure 24-5b).

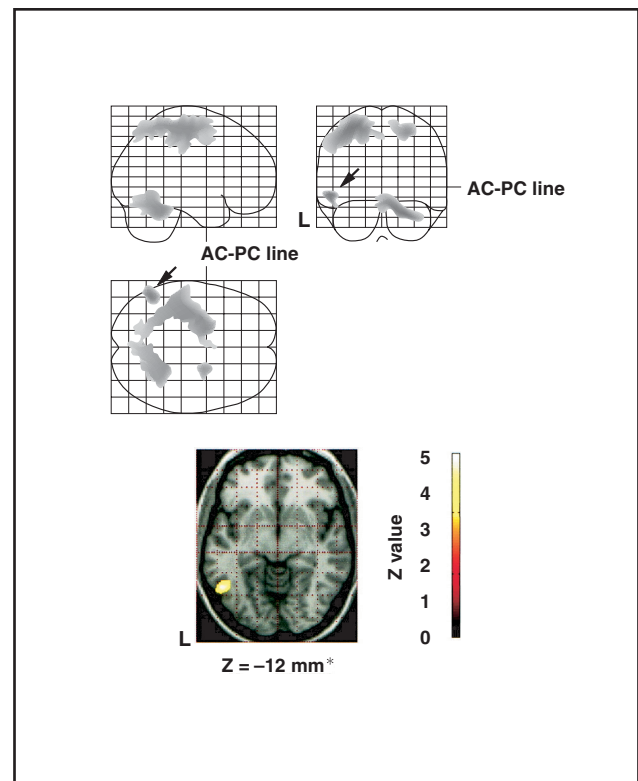


Figure 24-4 Activation of the left posterior inferior temporal cortex (arrow) associated with reading and writing of *kanji* (morphogram) of Japanese language, demonstrated by a functional MRI study. The transverse MRI on the bottom was obtained at 12 mm below the anterior commissure (AC)–posterior commissure (PC) line on the Talairach coordinate. Cited from Nakamura et al., 2000, with permission.

This phenomenon, called **conduction aphasia**, is another typical example of a disconnection syndrome (see p. 230). If the perisylvian area is widely affected, the patient shows features of all types of aphasia (**global aphasia**). Furthermore, in contrast with the ordinary Wernicke aphasia, in which repetition is also impaired, some patients with sensory aphasia may be able to repeat even though they cannot understand the spoken words. This is called **transcortical sensory aphasia**. Likewise, some patients with motor aphasia may be able to repeat given words although they cannot speak spontaneously. This condition is called **transcortical motor aphasia**. These phenomena may be explained by postulating that there are lesions in the language areas but they do not directly interrupt the neural circuits necessary for repetition (see Figure 24-3a, b, c).

E. PRAXIS

Based on clinical and experimental data, it is believed that the left parietal lobe, especially the supramarginal gyrus, plays a main role in programming daily activities like tool

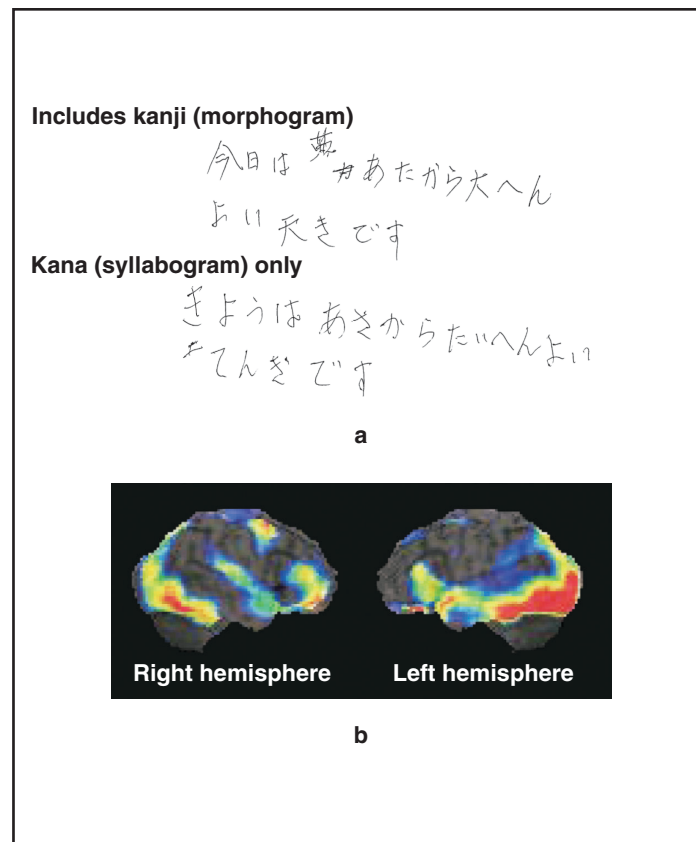


Figure 24-5 Disturbance of *kanji* (morphogram) writing seen with a dictation task in a Japanese patient with a clinical diagnosis of Alzheimer disease. (a) The patient can write *kana* (syllabogram) relatively well but can write only simple *kanji* (morphogram). (b) The blood flow study by three-dimensional stereotactic surface projections (3D-SSP) analysis of SPECT data in this patient revealed hypoperfusion in the left posterior inferior temporal region (shown in red).

use and gesture (see Figure 24-3f). The fibers from there project anteriorly through the arcuate fasciculus to the premotor area of the left middle frontal gyrus (Hattori et al, 2009; Matsumoto et al, 2012; Wheaton et al, 2005). The information from the left premotor area is conducted to the right premotor area through the corpus callosum, and the primary motor cortex is activated by the premotor area on each side, so that actual movements are executed.

F. CLASSIFICATION OF APRAXIA

If the left parietal lobe including the supramarginal gyrus is damaged, the patient is unable to program a motor task and finds it difficult to use given tools in an appropriate sequence. For example, if a set of cup, kettle, and coffee is presented, the patient may put coffee directly into the kettle instead of the cup. This phenomenon, called **ideational apraxia**, is often accompanied by difficulty in copying geometric forms of any complexity (**constructional apraxia**). In this condition, the patient may be able to draw individual

shapes but cannot synthesize them into a complex figure. This phenomenon can be also tested by asking the patient to imitate given forms by hands such as the posture of the examiner's fingers.

Reaching and grasping are important tasks for daily living, in which the left inferior parietal lobule takes part (Figure 24-6) (Hattori et al, 2009). Furthermore, a lesion of the fiber tracts projecting from the left parietal cortex to the left frontal cortex causes difficulty in using tools or making gestures (**ideomotor apraxia**). In contrast, a lesion of the premotor area makes the movement of the contralateral hand clumsy (**limb-kinetic apraxia**). In this case, however, it is a prerequisite that there is no spastic paresis, rigidity, or motor ataxia in the corresponding hand.

It is thought that the right premotor area is activated only through the left premotor area. Therefore, a lesion of the left premotor area makes not only the right hand movement clumsy but also the left hand movement. In this case, however, the right hand may be paralyzed at least to a certain degree due to involvement of the hand

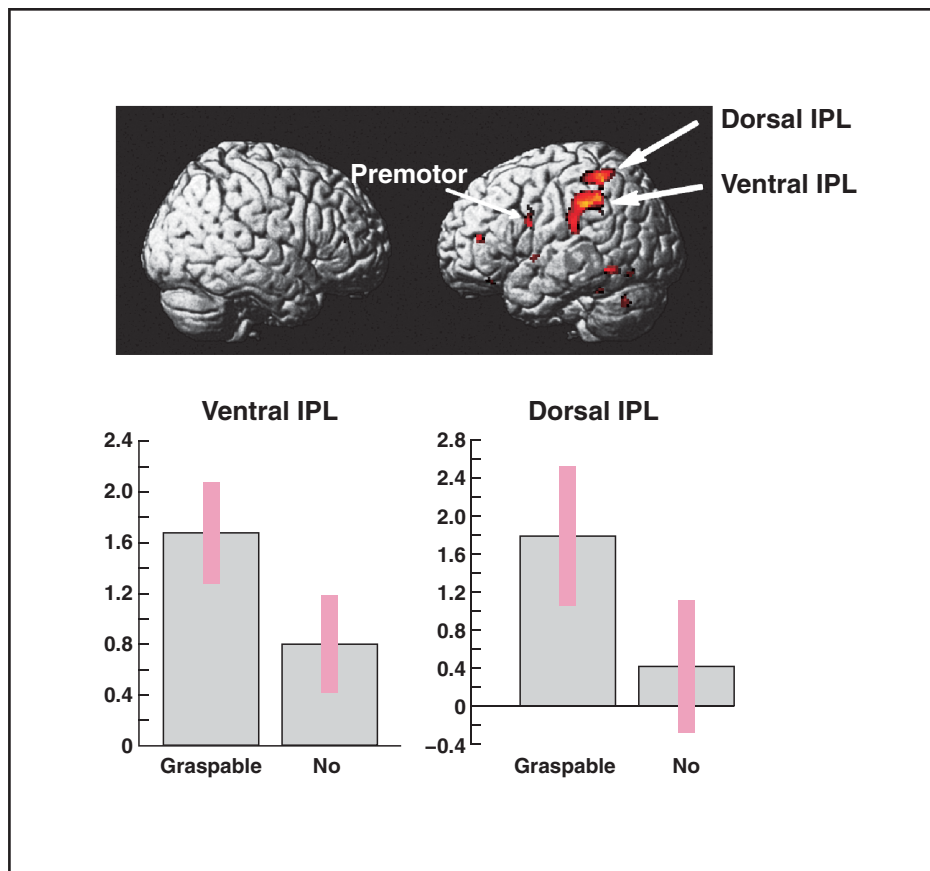


Figure 24-6 Cortical areas activated in the judgment task related to grasping. Graspable objects and non-graspable objects were visually presented to healthy subjects in a random sequence in an event-related functional MRI study. Greater activation was seen in the dorsal and ventral inferior parietal lobule (IPL) of the left parietal lobe and in the left premotor area in the graspable objects (Graspable) compared with the non-graspable objects (No). Cited from Hattori et al., 2009, with permission.

area of the left primary motor cortex. Limb-kinetic apraxia seen in the left hand due to a lesion of the left premotor area is called **sympathetic apraxia**. If the corpus callosum connecting the left and right premotor areas is damaged by an ischemic lesion in the anterior cerebral artery territory, the information from the left to the right premotor area is interrupted, so that the limb-kinetic apraxia occurs only in the left hand. This disconnection syndrome is called **callosal apraxia**. Some patients with a subcortical lesion in the ventral part of the left postcentral gyrus may present with apraxia in both hands because the fiber tract projecting from the left parietal cortex to the left premotor area is damaged just before reaching the left premotor area.

Among neurodegenerative diseases, ideomotor apraxia and limb-kinetic apraxia are commonly seen in patients with corticobasal syndrome. In this condition, other motor abnormalities such as alien hand sign, utilization behavior, and diagnostic apraxia may also be seen. **Alien hand sign** is

BOX 24-3 INABILITY TO OPEN EYES

There are four conditions in which a patient is unable to open the eyes: (1) **ptosis** due to weakness of the levator muscle of the upper eyelids, the most common cause of which is myasthenia gravis; (2) **blepharospasm** as a result of sustained contraction of the orbicularis oculi muscle (focal dystonia); (3) **apraxia of eye opening**; and (4) **functional (psychogenic)**. To distinguish between the last two conditions, it is important to confirm contraction of the frontal muscle. If the patient tries hard to open the eyes, the frontal muscle is expected to contract, and vice versa. Lack of frontal muscle contraction might suggest that the patient is not trying to open the eyes. Apraxia of eye opening is commonly seen in patients with neurodegenerative disorders manifesting with parkinsonism, such as progressive supranuclear palsy and corticobasal syndrome.

a goal-directed but unintended action of one hand that the patient cannot control. If the affected hand interferes with action of the other hand, it is called **diagonistic apraxia**. Patients with **utilization behavior** cannot resist grabbing and using presented objects. Utilization behavior is commonly seen in association with frontal lobe lesions and may be categorized into the primitive reflexes in the wide meaning (see Chapter 10-2E, p. 81) (Schott & Rossor, 2003 for review).

G. TASK-SPECIFIC APRAXIA

Some cortical areas are known to be related to performance of specific tasks. The area including the supramarginal gyrus of the right parietal lobe shows specific apraxia for dressing if damaged (see Figure 24-1). In **dressing apraxia**, the patient is unable to figure out how to put on the given clothes. There seems to be an area in the left parietal cortex that is engaged in calculation. Acalculia, finger agnosia, left-right disorientation, and agraphia form **Gerstmann syndrome** due to a lesion of the left parietal lobe (Box 24-4). In **acalculia**, patients tend to have greater difficulty in written calculation than in mental calculation (see Chapter 23-1C, p. 218). In particular, the patients make mistakes in the position of digits in vertically written calculations.

BOX 24-4 GERSTMANN SYNDROME

The left supramarginal gyrus is involved in various important functions. In addition to writing and calculation, it is engaged in recognizing left and right and identifying fingers. Thus, its lesion causes **left-right disorientation**, **finger agnosia**, **agraphia**, and **acalculia**, a condition called **Gerstmann syndrome** (Vaddiparti et al, 2021). Furthermore, this condition is often accompanied by ideational apraxia and constructional apraxia. There seems to also be a center related to the body image or body schema in the left parietal lobe, and its lesion causes inability to identify parts of one's own body (autotopagnosia).

Whether Gerstmann syndrome is due to a loss of focal cortical functions or disconnection of inter-cortical connectivity has long been controversial, but Rusconi et al. (2010), based on the results of functional neuroimaging studies, proposed that the syndrome might be primarily caused by subcortical disconnection in the parietal lobe.

H. MIRROR NEURON

Rizzolatti and his group found neurons in area 44 (F5 in monkey) of the inferior frontal gyrus that recognize action of other people and coined the term "mirror neuron." These neurons are believed to recognize not just the movement itself but also the goal of action, and even the intention and emotion of others (Rizzolatti & Fogassi, 2014). The unique function of these neurons has drawn special attention in relation to social communication. In fact, functional abnormality of mirror neurons has been proposed in children with autism spectrum disorders such as Asperger syndrome, who have behavioral abnormalities because of difficulty in understanding the actions and emotions of other people. It has also been reported that similar neurons are present in the primary motor cortex, but only 30% of those neurons in the primary motor cortex are oriented to the action goal whereas almost all neurons in area 44 are goal-oriented (Rizzolatti & Craighero, 2004).

I. RECOGNITION

Recognition of each sensory modality is believed to be primarily processed in the secondary sensory area that is located adjacent to each primary sensory area. For the somatosensory modality, areas 5 and 7 of the parietal cortex and the secondary somatosensory areas in the upper bank of the sylvian fissure are involved in somesthetic recognition. As for nociceptive recognition, however, other structures, including insula, amygdala, and anterior cingulate gyrus, are also involved in particular relation to the emotional response (see Figure 20-4, p. 195) (see Chapter 20-1F, p. 194).

BOX 24-5 SYNDROME OF ALICE IN WONDERLAND

A phenomenon in which a part of the body or the whole body looks unreasonably large or small, or deformed, or arranged in abnormal topographic interrelationship to other parts of the body has been long recognized in the field of psychiatry. For this phenomenon, a British psychiatrist, Todd (1955), coined the term "syndrome of Alice in Wonderland" based on the famous novel by Lewis Carroll. This phenomenon has been reported in patients with psychosis or in users of hallucinogens, but it may also occur as an aura in patients with migraine or partial epilepsy. The underlying mechanism is not precisely known, but it is postulated to be caused by epileptogenic abnormality involving the higher somatosensory cortices or the area related to body image (Bayen et al, 2012). It is thought that Carroll himself might have had this syndrome.

BOX 24-6 AMUSIA

Loss of acquired musical abilities by a brain lesion is called **music agnosia** or amusia. For example, the patient becomes unable to distinguish the pitch of sounds, to recognize familiar music, to read scores, to sing familiar songs, to play learned instruments, or to compose. Information processing of music is not simple and requires multiple complex factors, such as melody, time factor, and emotional aspects, and higher cortical functions such as learning, memory, and plasticity (Alossa & Castelli, 2009). Thus, multiple neural circuits are expected to be involved, and no particular structure or hemispheric dominance is known to be responsible for amusia. Detailed analysis of the relationship between the functional deficits and the sites of lesion in patients and functional neuroimaging studies in healthy subjects might contribute to the understanding of musical brain.

J. DISTURBANCE OF AUDITORY RECOGNITION

The secondary auditory cortex is not clearly defined, but it is considered to be around the primary auditory cortex in the **Heschl gyrus** (see Figure 12-2, p. 90). As the results of evoked magnetoencephalographic studies, it is known that sounds of different pitch are received in different areas of the primary auditory cortex (**tonotopic organization**)

BOX 24-7 SAVANT SYNDROME

It is well known that some persons with diffuse cerebral disturbance have extraordinary ability in a special field. The special ability is not just undamaged but is actually outstanding, like a genius. This condition was named “savant,” which means a great scholar, by Down in 1887. This unique condition is mainly seen in men who have congenital brain disorders or autism, but it is also seen in patients with an acquired brain lesion. Examples of the special ability are detailed knowledge about music and sports, absorption in and superlative memory of number plates and maps (splinter skills), exceptional ability and skill in music and arts (talented savant, prodigious savant), and so forth. Regardless of the kind of ability, the savant is especially good at memory. The underlying mechanism of this condition has not been clarified yet, although a plastic change is believed to play a primary role (Trefft, 2009). Application of various noninvasive techniques may help elucidating the mechanism.

BOX 24-8 VENTRILOQUIST EFFECT AND COCKTAIL PARTY EFFECT

It is easier to understand what another person is talking about if one watches his or her mouth. This phenomenon, known as the **ventriloquist effect**, is considered to be a result of integrative recognition of sensory input of different modalities (auditory and visual input in this case). In relation to this, **polysensory areas** are known to exist, for example in the temporoparietal junction where evoked responses are recorded in response to all stimuli of somatosensory, visual, and auditory modalities (Matsushashi et al, 2004).

In a noisy place like a cocktail party, one can still hear what a particular person is talking about if one attends to his or her voice. This **cocktail party effect** is considered to be mediated by a mechanism of **selective attention**. The location in the brain where the selective attention takes place differs depending on the kind of sensation. For the somesthetic modality, selective attention is considered to take place in the secondary somatosensory cortex, while the primary auditory cortex is important for the auditory modality.

(see Chapter 12-1, p. 89). Qualitative recognition of sounds seems to take place in the secondary auditory cortex. Clinically, some patients can hear a sound but cannot tell what the sound is (**auditory agnosia**). Judgment of the spatial source of sounds (where the sound came from) is believed to be recognized based on a subtle difference in the arrival time of sound at each ear and by processing of the time lag in the auditory cortex.

An auditory illusion of sound persisting or repeating after the cessation of an auditory stimulus is called **pallinacousis**. This phenomenon is considered to be coming from the superior temporal gyrus secondary to seizure activity or dysfunction of the area in a postictal state (Fields et al, 2018).

Experience of strong negative emotions such as anger and anxiety in response to everyday sounds is called **misophonia**. A functional neuroimaging study suggested abnormal activation and functional connectivity of the anterior insular cortex (Kumar et al, 2017).

K. HIGHER VISUAL FUNCTIONS AND THEIR DISTURBANCE

Processing of visual information has been extensively investigated. The primary visual area is area 17 (V1) located in

the calcarine sulcus and the adjacent area. There is a precise topographic interrelationship between the retina and V1 (**retinotopic organization**). The secondary visual area (V2) is in areas 18 and 19. Projection from V2 is divided into two pathways: the ventral pathway, which goes to the temporal cortex and is primarily engaged in the processing of the shape of objects, and the dorsal pathway, which goes to the parietal cortex and primarily takes part in the processing of spatial information. Thus, the ventral pathway is called the “what” pathway, and the dorsal pathway is called the “where” pathway (see Chapter 7-1H, p. 44).

If V1 is bilaterally damaged by an acute insult, the patient cannot see at all. This condition, called **cortical blindness**, is distinguished from bilateral optic neuritis by an intact light reflex (see Chapter 8-2B, p. 54). However, visual evoked responses may not be helpful in the distinction because they are expected to be absent in both conditions. In the recovery phase from cortical blindness, patients can see objects but cannot recognize what they are unless they touch the object. This condition is called **visual object agnosia**. Furthermore, the object may look deformed to the patient (**metamorphopsia**).

Face recognition is processed in the fusiform gyrus at the mesial basal temporal cortex bilaterally, and its lesion causes inability to recognize familiar faces (**prosopagnosia**). In this condition, patients can recognize who the person is as soon as they hear the person’s voice. As for the hemispheric dominance, the right hemisphere used to be

claimed to be dominant for this function, but this is still controversial. **Color recognition** is believed to be processed also in a similar region to that for the face recognition. Recognition of movement (**visual motion**) is known to be processed in the posterior part of the superior temporal sulcus (V5, MT in monkey) (Matsumoto et al, 2004a).

In relation to face recognition, the amygdala is believed to play an important role in the **recognition of facial emotion**. Impairment of this function was reported in a unilateral lesion of the amygdala on either side (Cristinzio et al, 2010). In this relation, the ventromedial prefrontal cortex was shown to mediate visual attention during facial emotion recognition (Wolf et al, 2014).

L. VISUOSPATIAL AGNOSIA

In processing visual spatial information, the secondary visual area in the right hemisphere is believed to be dominant, because **visuospatial agnosia** commonly occurs in patients with right parieto-occipital lesions. For example, the patient is unable to count the number of visual targets presented simultaneously (**agnosia of optic counting**), to read a clock (**agnosia of clock reading**), to distinguish between near and far (**agnosia of distance**), or to understand what is going on while looking at a complex visual scene (**simultanagnosia**). It is known that area 31 in the mesial parietal cortex is related to the processing of geographic information, and its damage causes a disturbance of heading in the correct direction (**topographic disorientation**) (Baumann & Mattingley, 2010; Takahashi et al, 1997).

M. POSITIVE SYMPTOMS OF HIGHER VISUAL FUNCTIONS

Some abnormalities of higher visual functions are thought to be positive symptoms. For example, a single object might look as if there is more than one (**polyopia**). Short presentation of an object might be followed by its repeated appearance (**palinopsia**). On the contrary, a patient may be unable to recall an image that has been just shown (**Charcot–Wilbrand syndrome**), although this does not belong to the positive symptoms.

A constant positive visual disturbance described as uncountable tiny dots over the entire visual field is called **visual snow**. It is often associated with other visual symptoms like palinopsia and photophobia (**visual snow syndrome**), and migraine and tinnitus are often seen as a comorbid condition (Puledda et al, 2020). A neuroimaging

BOX 24-9 WILLIAMS SYNDROME

This developmental abnormality of higher brain functions has drawn an increasing amount of attention in relation to the genetics of brain development. It involves infants before the age of 1 year with disturbance of visuospatial recognition, excessive emotional reaction (unusually sociable for a person whom the infant has never met before), stereotypy, auditory hypersensitivity, and phobia about a certain sound. It is a unique condition in which the patient is more than average or even superior in language development, face recognition, interest and talent in music, and sociability. Physically, the child often has cardiovascular abnormality, among others aortic stenosis, and a facial deformity like an elf. Microdeletion of a gene on chromosome 7q11.23 was shown in this condition. A neuroimaging study in this condition reported a smaller volume of the parietal and occipital lobes but a larger volume of the face area of the fusiform gyrus than controls (Golarai et al, 2010).

study suggested structural and functional abnormality of the visual association cortex (Schankin et al, 2020).

A given sensory stimulus might be accompanied by a sensation of another modality in addition to its original sensation. This phenomenon is called **synesthesia**. For example, about 2% of the healthy population sees color when they read characters (grapheme–color synesthetes) (Weiss & Fink, 2009). Three families of sound–color (auditory–visual) synesthesia were studied by whole-exome sequencing, showing genetic heterogeneity (Tilot et al, 2018).

As locations of the primary sensory areas for the olfactory, gustatory, and vestibular sensations are not well defined, the distinction from each secondary sensory area is not precisely understood. As described in each respective chapter (Chapter 6-1, Chapter 11-1C, and Chapter 13-1, respectively), the limbic system including the mesial temporal cortex and insula is considered to be important for these functions.

2. EXAMINATION OF LANGUAGE, PRAXIS, AND RECOGNITION

As for the patient's state of language, praxis, and recognition, general information is usually obtained during the history taking. If any deficit in these functions is confirmed to be present, it provides an important clue to the anatomic diagnosis. Information about the patient's handedness is important before evaluating these higher cortical functions (see p. 230). To assess these cortical functions correctly, it is a prerequisite that the patient does not have any general cognitive impairment or dementia, but in some patients the focal cortical symptoms such as aphasia, apraxia, and agnosia might be superimposed on a general cognitive impairment. Furthermore, in the early clinical stage of Alzheimer disease or frontotemporal degeneration, the patient may present solely with progressive aphasia, apraxia, or agnosia (**primary progressive aphasia**), and the general cognitive decline might appear later in the clinical course (Mesulam et al, 2014). Therefore, this possibility should be kept in mind when the patient presents with any focal cortical symptom with a progressive course.

A. EXAMINATION OF LANGUAGE

Language abnormality, if any, is usually noticed of during the history taking. When spontaneous speech is found to be abnormal, first the examiner should determine whether it is due to dysarthria or aphasia (see Chapter 14-7, p. 107). If

BOX 24-10 LANDAU-KLEFFNER SYNDROME

Children with epilepsy may develop difficulty understanding the spoken language that they had already learned (verbal auditory agnosia) and may become progressively aphasic. This condition is called Landau-Kleffner syndrome. In this condition, the type of epilepsy varies among patients, including simple partial motor seizure, atypical absence, and generalized convulsion. As a result of magnetoencephalographic studies, spikes have been shown to arise near the sylvian fissure on either side. It is postulated that involvement of the language area by epileptogenic activities might cause progressive aphasia (Castillo et al, 2008).

the patient's speech is difficult to understand but grammatically correct, it is more likely to be dysarthria than aphasia. In contrast, if there are grammatical errors, it is more likely to be aphasia. It is also useful to ask patients whether they find it difficult to pronounce the words that they have in mind or to compose a sentence. In general, patients with aphasia can sing songs they have learned before or can make an exclamation like “ouch,” whereas those with dysarthria have difficulty in singing or exclaiming because it involves the final common pathway for speaking.

i. Motor aphasia

Patients with motor aphasia hesitate to talk because they are aware of their own mistakes. This condition is sometimes called nonfluent aphasia. If the patient is asked to repeat a given syllable, word, or sentence, the syllable or word is replaced by different syllable or word, respectively (**paraphasia**) (see p. 232). Motor aphasia is caused by a lesion of Broca's area or adjacent structures in the left inferior frontal gyrus. As patients with motor aphasia also have difficulty writing, it is important to test not only spoken language but also written language.

ii. Sensory aphasia

To test the patient's ability to understand spoken language, it is useful to give a verbal instruction to perform a complex motor task. If the patient cannot do it as instructed, it is useful to see how he or she can imitate a motor task that is visually demonstrated by the examiner. If the patient can imitate, it suggests a problem in verbal comprehension (sensory aphasia), but if the patient cannot imitate,

it might suggest a problem in motor execution (apraxia). The instruction to imitate the given motor task is usually understood even by patients with sensory aphasia, if the instruction is given by the examiner's gesture. If the patient is severely demented, however, this test is not possible.

Patients with typical sensory aphasia are usually talkative, but their speech is totally incomprehensible because they do not understand what they are saying. This condition is called **jargon aphasia**. Repetition of spoken words is also impaired in sensory aphasia. It is also important to test understanding of written language (reading). Alexia with agraphia is caused by a lesion of the area including the angular gyrus of the left parietal lobe (see Box 24-1, p. 233). If the supramarginal gyrus is also damaged, agraphia is seen together with left–right disorientation, finger agnosia, and acalculia (Gerstmann syndrome) (see Box 24-4, p. 236).

iii. Conduction aphasia

In patients who are suspected of suffering from language disturbance and yet show apparently normal performance in the tests of understanding and spontaneous speech, it is necessary to test **repetition**. If the arcuate fasciculus connecting Wernicke's area and Broca's area is disconnected, a difficulty in repeating verbally presented words might predominate the clinical picture (conduction aphasia). In this case, literal paraphasia and verbal paraphasia are commonly seen (see p. 232).

While testing the naming of objects by presenting a series of different pictures, some patients might answer the name of the object just presented on the preceding test. This phenomenon is called **verbal perseveration**. However, perseveration is also seen for motor tasks and may be seen in patients with dementia like Alzheimer disease.

B. EXAMINATION OF PRAXIS

Limb-kinetic apraxia can be noticed while carrying out the physical examination. In contrast, **ideomotor apraxia** might be detected for the first time by specifically asking the patient to make gestures or pantomime of tool use. For example, the patient is asked to salute, or to use a toothbrush, comb, or scissors, or to imitate the use of those tools. The patient with ideomotor apraxia is unable to do these motor tasks or is clumsy in doing them. To test the presence or absence of **ideational apraxia**, it is useful to adopt a motor task that uses two or more kinds of tools in combination. For example, patients are given a piece of paper and a pair of scissors and asked to cut the paper in half, or

to brush their teeth with a toothbrush and toothpaste. To detect **Dressing apraxia** (see p. 236), it is important to ask the patient to actually put on folded pieces of clothing.

Inability to continue a motor task despite the absence of sensorimotor disturbance might be called **motor imper-sistence**. However, the similar condition might also be seen in association with poor motivation or lack of motor attention. Some investigators include inability to perform two different motor tasks simultaneously (**simultanapraxia**) in the concept of motor imper-sistence. Simultanapraxia is believed to occur with right frontal lobe lesions (Sakai et al, 2000).

While examining a series of different motor tasks, the patient might repeat the motor task that he or she did just before. Just like the case of language, this phenomenon is called **perseveration**. Luria's three-step motor sequence test might be applied to test the motor praxis (Weiner et al, 2011).

Gait apraxia is defined as inability to start walking in patients with no bradykinesia or rigidity in the supine position. In most cases of so-called gait apraxia, the disturbance is due to **freezing of gait** or akinesia, because the patient has difficulty in moving the legs also in the supine position. Paradoxical kinesia (see Box 22-1, p. 212) is seen in association with both gait apraxia and freezing of gait.

C. EXAMINATION OF RECOGNITION

Recognition related to the somatosensory system includes stereognosis, graphesthesia, two-point discrimination, baroesthesia, and touch localization (see Chapter 20-1 and Box 20-1, p. 191). For the auditory sense, identification and spatial localization of the source of sound are tested as necessary.

i. Visual recognition

By presenting objects or their pictures, the patient is asked to state the name, size, and color of the objects. However, if the patient has difficulty in naming, by presenting two or more objects or their pictures of different quality (for example, of different color or shape), the patient is asked whether they are the same or different. Likewise, by presenting two visual targets at different distances from the patient's eyes, the patient is asked to judge which is near and which is far. To test visuospatial recognition, by presenting a picture of a clock without digits, the patient is asked to read the time (if impaired, **agnosia of clock reading**). Likewise, by showing the different number of fingers by the examiner's two

BOX 24-11 BALINT SYNDROME

This unique syndrome is characterized by disturbance of visual attention and visually guided praxis. This condition had been described since the end of the 19th century, but in 1909 Balint coined the term “*optische Ataxie*.” A triad of this syndrome is formed by optic ataxia (difficulty in grasping a visually presented target), psychic paralysis of gaze, and disorder of spatial attention (Pisella et al, 2008; Rodriguez-Rivas et al, 2021). Patients with this syndrome keep fixating their gaze upon a spot in space and cannot direct their visual attention to the approaching examiner. When asked to grasp an object, the patient’s hand widely misses the target (Figure 24-7). There is no clear hemianopsia, but the patient tends to neglect one side. This condition is caused by bilateral ischemic lesions of the parieto-occipital junction, which is a watershed area between the middle and posterior cerebral arteries, as a result of hemodynamic change with hypoperfusion (Figure 24-8). In most cases, the predominant lesion is in the right hemisphere, but the original case reported by Balint was on the left.

hands, the patient is asked to state how many fingers were raised in total. Alternatively, by presenting two large circles containing different numbers of small dots, the patient is asked to state the total number of small dots (if impaired, **disturbance of visual counting**). Furthermore, by presenting a scene of cartoon, the patient may be asked to tell what



Figure 24-7 Inability to grasp a visual target seen in a patient with Balint syndrome. With the patient’s permission.

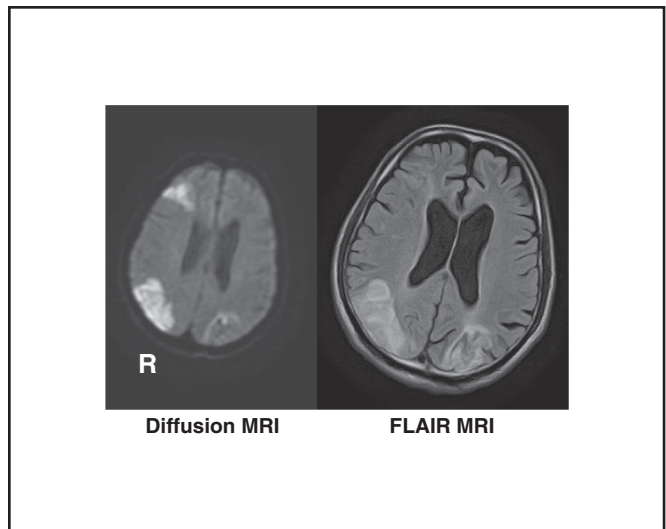


Figure 24-8 MR images of a patient shown in Figure 24-7, showing infarctions in the bilateral parieto-occipital junction areas, predominantly affecting the right.

is going on in the scene. Inability to judge the situation is called **simultanagnosia** (see p. 238).

ii. Hemispatial neglect

When disturbance of visual attention or visual recognition is limited to the left or right side, it is called hemispatial neglect. To test this, the patient may be asked to divide a straight line into two or three equal parts. Alternatively, by presenting a large number of small marks drawn on a sheet of paper, the patient is asked to cross out all the signals by a pencil. In this test, the signals on the neglected side may remain uncrossed. Furthermore, patients with hemispatial neglect may neglect half of their own body, like forgetting to shave one side of the face. As visual attention is primarily controlled by the right hemisphere, hemispatial neglect is commonly seen on the left half. To confirm the presence of hemispatial neglect, it is important to exclude hemianopsia. In practice, however, as the visual field test is often difficult to carry out in this situation because of poor visual attention, it may have to be judged by the confrontation test (see Chapter 7-2B, p. 45).

Patients with a gross vascular lesion in the right hemisphere might deny the presence of left hemiplegia, which is called **anosognosia** (Orfei et al, 2007).

D. USE OF A SIMPLE TEST BATTERY

As the time spent for the whole neurologic examination might be limited in the daily clinical setting, it is useful

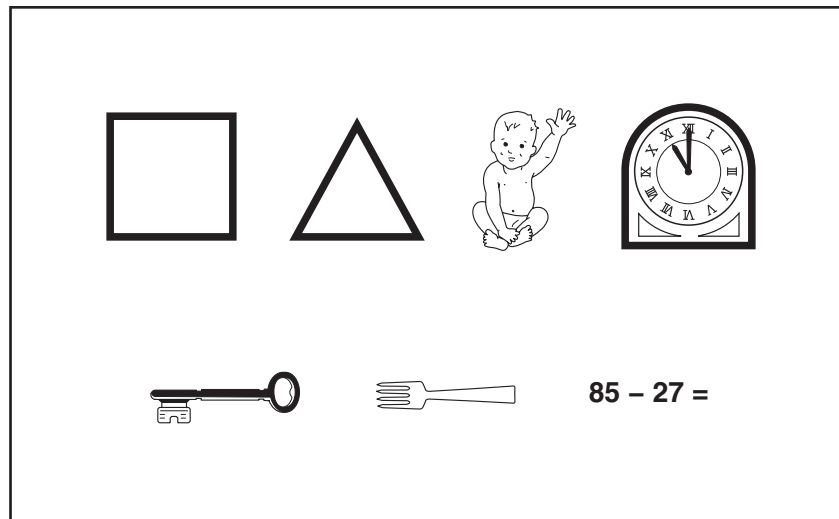


Figure 24-9 A simple battery for testing language, praxis, and recognition in the bedside examination, which was used in the University of Minnesota Hospital when one of the authors (H.S.) was a neurology resident there. The two figures on the top left are for testing recognition of the shape and naming. The picture of a baby is for testing naming and judgment of the left and right. The picture of a clock is used to test naming and visuospatial recognition, and two pictures on the bottom left are for naming and praxis pantomime.

to prepare a simple battery of tests covering most of the important aspects of higher brain functions. A sample battery is shown in Figure 24-9. Of course, this is just for a screening test, and the detailed evaluation must be done as necessary.

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EPILEPSY AND CONVULSION

Epileptic seizure is defined as “transient occurrence of symptoms and/or signs due to abnormal excessive or synchronous neuronal activity in the brain.” It is currently thought that most cases of epileptic seizures are based on organic cerebral lesions. The prevalence of epilepsy is believed to be as high as 0.5% of the population.

1. CLASSIFICATION OF EPILEPTIC SEIZURES

Classification of epileptic seizures was proposed by the International League Against Epilepsy (ILAE) in 1981, and then it was revised twice, first in 1989 and then in 2010 (Berg et al, 2010). They defined “epileptic seizure” as a transient occurrence of signs and/or symptoms due to abnormal excessive or synchronous neuronal activity in the brain, and “epilepsy” as a disorder of the brain characterized by an enduring predisposition to generate epileptic seizures, and by the neurobiologic, cognitive, psychological, and social consequences of this condition (Fisher et al, 2014). Consequently, they proposed revised frameworks for the classification of epilepsies (Scheffer et al, 2017) and for the classification of seizure types (Fisher et al, 2017). In this book, classification of epileptic seizures is explained by using the 2010 classification of the ILAE (Berg et al, 2010) and by taking into account the revised versions of their classification (Fisher et al, 2017; Scheffer et al, 2017) (Box 25-1).

Epileptic seizures are largely classified into three groups: generalized seizures, focal seizures, and unknown. The first two groups are further classified into subgroups mainly based on the clinical manifestations (see Box 25-1). **Generalized epileptic seizures** are defined as seizures occurring in and rapidly engaging bilaterally distributed networks, and **focal epileptic seizures** are seizures occurring within networks limited to one hemisphere and either discretely localized or more widely distributed (Berg et al, 2010). Theoretically, generalized epileptic seizures in the

strict sense are considered to be extremely rare, because even the generalized tonic-clonic seizures may actually start focally and be secondarily generalized. In this case, an epileptogenic focus, if any, may not be detected by the routine laboratory examination. In contrast, typical absence seizure (**petit mal**), which is characterized by a short lapse of consciousness and an electroencephalographic (EEG) burst of bilaterally synchronous high-amplitude 3-Hz rhythmic spike-and-wave discharges (Figure 25-1), may be considered as a typical example of generalized epilepsy.

A. FOCAL EPILEPTIC SEIZURES

Focal seizures are classified into three groups: group A (without impairment of consciousness or awareness), group B (with impairment of consciousness or awareness), and group C (evolving to a bilateral, convulsive seizure). Group A is further classified into group A1 (with observable motor or autonomic components) and group A2 (involving subjective sensory or psychic phenomena only). Group A1 roughly corresponds to **simple partial seizure** in the 1981 ILAE classification (Commission on Classification and Terminology of the International League Against Epilepsy, 1981). Group B roughly corresponds to **complex partial seizure** and group C corresponds to **secondarily generalized seizure** in the 1981 classification. Group A1 (simple partial seizure) may be further divided into neocortical epilepsy and mesial temporal lobe epilepsy, depending on the source of the epileptogenic focus.

A representative form of neocortical focal seizure is **focal motor seizure**, which originates from the primary motor cortex (M1) and manifests itself as tonic and/or clonic convulsion in the corresponding part of the body. A form of focal motor seizure characterized by marching of the convulsion according to the M1 somatotopy is called **Jacksonian seizure**. Focal motor seizure is commonly followed by transient weakness of the involved

**BOX 25-1 CLASSIFICATION OF EPILEPTIC SEIZURES
BY ILAE COMMISSION**

- I. Generalized seizures
 - A. Tonic-clonic (in any combination)
 - B. Absence
 - 1. Typical
 - 2. Atypical
 - 3. Absence with special features
 - Myoclonic absence
 - Eyelid myoclonia
 - C. Myoclonic
 - 1. Myoclonic
 - 2. Myoclonic atonic
 - 3. Myoclonic tonic
 - D. Clonic
 - E. Tonic
 - F. Atonic
- II. Focal seizures
 - A. Without impairment of consciousness or awareness.
 - 1. With observable motor or autonomic components.
This roughly corresponds to the concept of “simple partial seizure.”
 - 2. Involving subjective sensory or psychic phenomena only.
 - B. With impairment of consciousness or awareness. This roughly corresponds to the concept of “complex partial seizure.”
 - C. Evolving to a bilateral, convulsive seizure (involving tonic, clonic, or tonic and clonic components). This expression replaces the term “secondarily generalized seizure.”
- III. Unknown
 - A. Epileptic spasms

(Based on Berg et al, 2010, and modified by Fisher et al, 2017)

extremity (**post-ictal paralysis, Todd paralysis**). Seizures arising from the supplementary motor area (SMA) might show bilateral dystonic posturing and bizarre movements (Figure 25-2).

Simple partial seizure originating from the primary somatosensory cortex (S1) manifests itself with abnormal sensation on the corresponding part of the skin, which might spread according to the S1 somatotopy just like the case of motor seizure. In a focal seizure originating from the SMA or insular cortex, pain might be one of the ictal symptoms. **Painful somatosensory seizures** often originate in

the operculo-insular cortex (Montavont et al, 2015). The focal seizure arising from the occipital cortex may present with a flash of light. In many cases of neocortical partial epilepsy, **cortical dysplasia** is found by detailed neuroimaging studies, which may be a candidate for surgical treatment (Kim et al, 2009).

Epilepsy originating from the mesial temporal cortex (**mesial temporal lobe epilepsy, MTLE**) is the most common form of epilepsy in adults. Clinically MTLE presents with various neurological symptoms: sensory symptoms, autonomic symptoms, abnormal emotion, and abnormal thought or behavior. Sensory symptoms of mesial temporal lobe origin include sensation of abnormal smell (**uncinate fit**), abnormal taste (**gustatory seizure**), and vertigo (**vertiginous seizure, epileptic vertigo**) (Tarnutzer et al, 2015). The most common autonomic symptom is a sensation of nausea ascending from the epigastrium (**epigastric rising sensation**). Other autonomic ictal symptoms include bradycardia, facial pallor, flushing, sweating, mydriasis, ictal apnea, and ictal asystole. Cardiac or autonomic involvement might lead to sudden unexpected death in epilepsy (SUDEP), especially in patients with long-standing and medication-refractory epilepsy (Bermeo-Ovalle et al, 2015; Devinsky et al, 2017; Hampel et al, 2017; Harden et al, 2017; Sveinsson et al, 2017). Apnea might also be an ictal symptom. As a result of the direct stimulation study, the hippocampus and amygdala were suggested to be an epileptogenic zone for **ictal apnea** (Lacuey et al, 2017). By contrast to ictal central apnea, which occurs only in focal epilepsy, post-convulsive central apnea is also seen in generalized epilepsy and may be a clinical biomarker of SUDEP (Vilella et al, 2019).

Emotional symptoms include sudden attacks of fear, anxiety, or pleasure. Sudden attacks of hallucination might be a manifestation of MTLE (**hallucinatory seizure**). In hallucinatory seizure, the patient always sees the same concrete landscape or situation during the attacks. Furthermore, the patient may feel as if a totally new situation has already been seen before (**déjà vu**), or, to the contrary, the patient may feel as if a familiar situation has never been seen before (**jamais vu**).

Attacks characterized by abnormal behavior or actions are called **automatism** or **psychomotor seizure**. Consciousness is impaired in most cases. Typically, the patient first feels an epigastric rising sensation, followed by loss of consciousness. The patient may show chewing movements of the mouth or rubbing movements of the hands but does not respond to verbal call. The patient may even walk or move about during the ictal episode but does

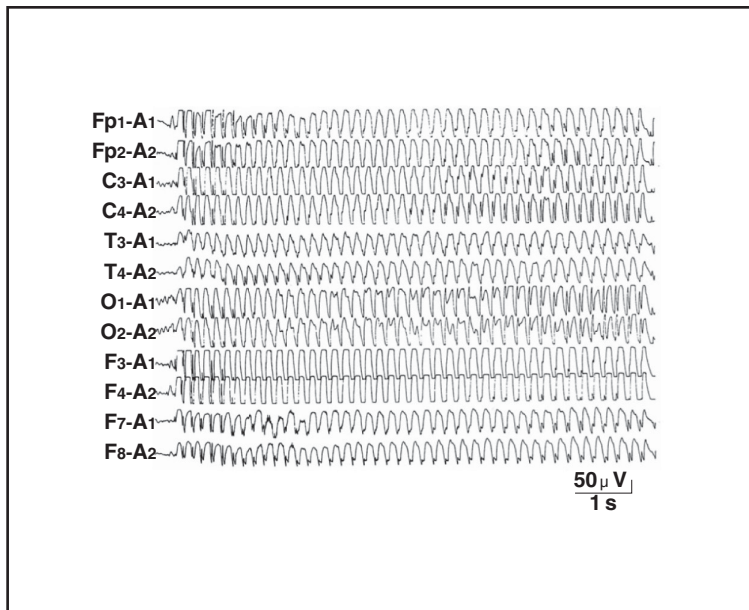


Figure 25-1 EEG of a 13-year-old patient with petit mal during hyperventilation. A burst of bilaterally synchronous rhythmic high-amplitude 3-Hz spike-and-wave discharges is seen.



Figure 25-2 Bilateral dystonic posturing seen in a 21-year-old patient with SMA seizure. By courtesy of Dr. Akio Ikeda of Kyoto University.

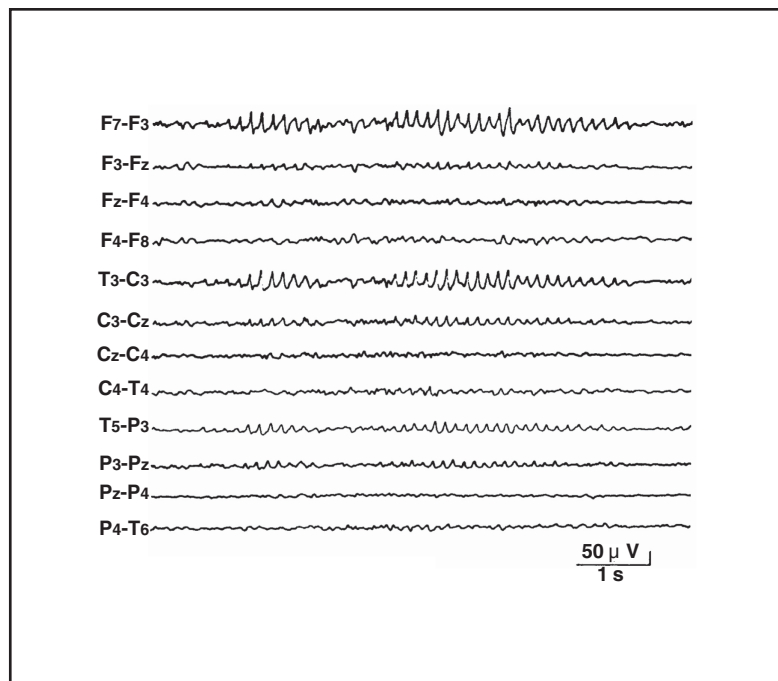


Figure 25-3 Interictal EEG of a patient with psychomotor seizure. Rhythmic spikes are seen at the left temporal leads (F7, T3).

not recall the abnormal behavior at all after the episode. Dystonic posture may be seen in the upper limb contralateral to the epileptic focus and may evolve to a generalized convulsion. After the generalized convulsion, the patient appears obtunded for a short time. EEG often shows spikes in the temporal region even during the intervals (Figure 25-3).

As a result of careful interviews of relatives of patients with MTLE, it was shown that 18.2% of the patients had families who experienced the *déjà vu* phenomenon, suggesting familial MTLE (Perucca et al, 2017).

B. GENERALIZED EPILEPTIC SEIZURES

Most of the simple partial seizures just described may evolve to generalized seizures. Those prodromal symptoms and/or signs occurring before the generalization are called **aura**. Focal seizure symptoms are common in idiopathic generalized epileptic seizures (Seneviratne et al, 2015).

Generalized tonic-clonic seizures are characterized by bilateral tonic extension of the trunk and limbs followed by synchronous muscle jerking associated with loss of consciousness. There may be a loud vocalization during the attacks. Postictally, patients are often lethargic and confused for a various length of time. Ictal and interictal EEG shows a burst of generalized high-amplitude spike-and-waves, but

the burst may not start synchronously and often starts with focal paroxysmal abnormalities.

Typical **absence seizure (petit mal)** is an attack of lapse lasting about 5 seconds. For example, a child suddenly stops moving while eating and cannot respond to the surroundings. Frequent blinks might be seen during the episode. Ictal as well as interictal EEG shows a burst of bilaterally synchronous high-amplitude 3-Hz rhythmic spike-and-wave discharges (see Figure 25-1). The EEG burst is characteristically induced or enhanced by hyperventilation.

Most epileptic seizures occur spontaneously without any precipitating factor, but some seizures are sensitive to a specific stimulus or occur during performance of a certain task. A typical example of the stimulus-sensitive epilepsy is **photosensitive epilepsy** such as the TV anime-induced (animation) epilepsy. In this case, EEG spikes are often induced by photic stimulation, and its sensitivity to the kind of visual stimulus differs among patients (Takahashi, 2008). Task-specific epilepsy occurs during performance of a specific task or game such as playing chess, calculating, and listening to a special kind of music. In the latter case, the attack tends to occur while listening to a particular piece of music by a specific composer.

Autosomal dominant lateral temporal lobe epilepsy is induced by auditory stimuli in children or young adults. Several point mutations of the LGI1 (leucine-rich glioma inactivated-1) gene have been identified, and phenotypes

are also different among families (Kawamata et al, 2010). Following auditory or visual hallucination as an aura, marked psychiatric symptoms appear and in some cases may evolve to generalized convulsion. **Autosomal dominant nocturnal frontal lobe epilepsy** is considered to be an ion channel disorder of the acetylcholine receptor (see Chapter 28-1A, p. 259). Recently, genetic epilepsy has drawn increasing attention. Even focal epilepsy could be genetically inherited, like glucose transporter type 1 (Glut1) defects (Wolking et al, 2014).

2. DIAGNOSIS OF EPILEPSY

As it is rare to encounter an ictal episode in the clinic, its clinical diagnosis primarily depends on the information obtained from the patient or the witnesses. Since an epileptic attack is caused by pathologic hyperexcitation of a certain neuronal group, an important feature of epilepsy is a repeated occurrence of the same symptomatology in all ictal episodes. The kind of ictal symptoms depends on which part of the cerebral cortex is involved by the epileptogenic activity. Although some patients might have more than one epileptogenic focus, the ictal pattern is expected to be stereotyped for each focus. It has to be kept in mind, however, that attacks originating from the same epileptogenic focus might differ in severity, ranging from a mild attack to a severe one. In MTLE, it ranges from aura to psychomotor seizure.

When examining the paroxysmal disorders, it is quite important to obtain information about when and in which circumstances the attacks occur, and about the presence of any precipitating factor. For example, the periodicity of seizure recurrence might be associated with the menstrual cycle (catamenial epilepsy). Furthermore, in patients who are taking anti-seizure drugs, it is important to know whether the patient is taking the medication as prescribed or not. When there is any possibility of **psychogenic non-epileptic seizure (PNES)** (**pseudoseizure, functional seizure, dissociative seizure**), it is useful to carry out long-term video-EEG monitoring in the hospital. In patients with epileptic seizures, EEG and neuroimaging study are always indicated.

Furthermore, since an epileptic seizure might be a manifestation of systemic diseases, the possibility of **symptomatic epilepsy** has to be taken into account. Brain tumors commonly cause convulsive seizures. Seizures and intracerebral hemorrhage are common presentations of brain arteriovenous malformations (AVMs) (Chen et al, 2020).

Systemic lupus erythematosus (SLE) might start with generalized convulsion, and metabolic abnormalities such as electrolyte abnormalities and endocrine disorders might underlie epileptic seizures. Epilepsy is relatively rare in white matter diseases like multiple sclerosis (MS), but an investigation of a large number of Swedish patients with MS disclosed that the cumulative incidence of epilepsy was 3.5%, with a direct link between the severity of MS and epilepsy (Burman & Zelano, 2017).

In children, in addition to a general physical examination, careful observation of the skin for congenital lesions (nevi) is important. Among others, angiofibroma on the face suggests tuberous sclerosis as an underlying cause of seizure (see Figure 3-1, p. 10).

Generalized convulsion may occur in the acute phase of severe head injury, but the first seizure may occur several months after the head trauma (**posttraumatic epilepsy**). Its underlying mechanism might differ among cases, but scar formation following cerebral contusion is considered to be an important factor. In elderly people, epileptic attacks related to strokes are not rare (Box 25-3).

A. EVALUATION OF MEDICALLY INTRACTABLE EPILEPTIC SEIZURES

Medically intractable epileptic seizures are defined as epileptic attacks that occur more than once a month in

BOX 25-2 GELASTIC EPILEPSY (ICTAL LAUGHTER)

Epilepsy characterized by laughter is called gelastic epilepsy or ictal laughter. Laughing in this condition is considered to be unaccompanied by humorous emotion (**mirthless laughter**). Laughter might be either the only ictal symptom or accompanied by other symptoms. It may change to other forms of partial epilepsy or to generalized convulsion during the clinical course. Laughing may be seen as a part of the ictal symptoms due to cortical epilepsy, but its relationship with **hypothalamic hamartoma** has drawn special attention. A study of 100 cases of hypothalamic hamartoma showing ictal laughter in the United States reported the onset of attacks during infancy, mental retardation in 43% of the patients, and precocious puberty in 21% (Parvizi et al, 2011). The hamartoma is commonly located in the posterior hypothalamus at the level of the mammillary bodies (see Figure 31-1, p. 277), and the tumor is especially large in patients associated with mental retardation and precocious puberty.

BOX 25-3 CEREBROVASCULAR DISEASES AND EPILEPSY

The most common cause of the first seizure in elderly people is cerebrovascular disease. A seizure in the acute phase of the first stroke is not rare, although the prevalence varies among reports. A survey of 714 cases of strokes in Italy showed occurrence of seizures during the acute phase in 6.3% of all cases. Among the patients with seizures, 16.2% had intracerebral hemorrhage, 12.5% had cerebral infarction evolving to hemorrhage, and 4.2% had cerebral infarction (Beghi et al, 2011). Regardless of the kind or location of the stroke, epilepsy was particularly common in those patients with strokes involving the cerebral cortex. In contrast, the prevalence of **post-stroke seizures** varies among different reports, but according to the study by Myint et al. (2006), about 10% of the patients with stroke develop seizures within 5 years. Risk factors for cerebral infarction to cause post-stroke seizures include the severity of the acute phase, the severity of residual symptoms, the number and size of infarctions, the degree of cortical damage, and involvement of the hippocampus. A long-term follow-up study of a large number of patients with subarachnoid hemorrhage (SAH) from saccular intracranial aneurysms in Finland showed a cumulative incidence of epilepsy of 12% at 5 years (Huttunen et al, 2015). It was especially common in those patients with SAH associated with intracerebral hemorrhage. In a study of 615 patients with primary intracerebral hemorrhage in Finland, 13.5% developed post-stroke seizures, and early seizures, subcortical hematoma location, large hematoma volume, hematoma evacuation, and lower Glasgow Coma Scale score on admission were the risk factors for post-stroke seizures (Lahti et al, 2017).

patients who are taking the appropriate dose of appropriate kinds of anti-epileptic drugs. In this case, it is useful to admit the patient in a hospital where an epilepsy specialist is available and to confirm the presence of epileptic seizures, to classify the seizure, and to look for the presence of any organic brain lesion (American Clinical Neurophysiology Society, 2008; Kanazawa et al, 2015; Usui et al, 2015). If an organic lesion is discovered or if a single epileptogenic focus is detected, it might be possible to treat the patient surgically after carefully evaluating the functions of the brain areas surrounding the epileptogenic focus (Shibasaki et al, 2007). In MTLE, which is the most common form of epileptic seizures in adults, partial resection of the anterior part of the temporal lobe or selective resection of the amygdala and hippocampus is effective in about two-thirds of

cases (Engel, 2018; Engel et al, 2003). In epileptic seizures of neocortical origin, however, there is no definitive conclusion as to the effect of surgical intervention (Cho et al, 2014; Jcha et al, 2007).

B. PSYCHOGENIC NON-EPILEPTIC SEIZURES

Some seizures may look like epileptic seizures but are not associated with any epileptic discharge of cerebral neurons. This functional disorder, called PNES, is often a diagnostic problem. Generally speaking, during convulsive seizures associated with impairment of consciousness, the patient's eyes are open. Therefore, if the eyes are kept closed during the convulsive seizures, it may be psychogenic. Furthermore, if convulsion seen during the attack looks unusual for the ordinary convulsive seizure, it may make the examiner suspicious of the possibility of PNES. However, autosomal dominant nocturnal frontal lobe epilepsy may manifest itself as extremely bizarre movements (see p. 261). If there is any doubt about the diagnosis, long-term video-EEG monitoring is indicated.

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HEADACHE AND MIGRAINE

1. CLASSIFICATION OF HEADACHE

Headache is one of the most common symptoms presenting in the neurology clinic, and it is caused by a variety of mechanisms. Among the cranial structures, pain receptors are present only in the scalp, its subcutaneous tissue, skull bones, meninges, walls of blood vessels, and cranial nerves possessing the somatosensory fibers. There is no pain receptor in the brain parenchyma. That is why stereotactic surgery can be done without general anesthesia. Headache is classified by the International Headache Society into three groups: (1) **primary headaches**, (2) **secondary headaches**, and (3) painful cranial neuropathies, other facial pains, and other headaches (Box 26-1) (Headache Classification Committee of the International Headache Society [IHS], 2013).

When a patient with headache is seen in the clinic, it is most important not to overlook the secondary or symptomatic headache due to an organic lesion of intra- as well as extra-cranial structures. Headache due to intracranial hemorrhage or tumor used to be called traction headache, and today its diagnosis is relatively easy with the aid of neuroimaging techniques. It is especially important to keep in mind that headache is often caused by ophthalmologic disorders; diseases of the ear, nose, and pharynx; dental and oral diseases; and disorders of the cervical spine. Headache due to cerebrovascular diseases, fever, and hypertension is common and used to be called non-migrainous vascular headache. Headache due to intracranial hypertension is commonly accompanied by nausea and vomiting and is characterized by its aggravation when awakening in the morning (**morning headache, morning vomiting**). This is considered to be due to an increase in the blood carbon monoxide level as a result of nocturnal hypoventilation, which causes an increase in the intracranial arterial perfusion, resulting in intracranial hypertension in the morning.

The **pseudotumor cerebri syndrome (PTCS)** may be primary (**idiopathic intracranial hypertension**) or may arise from an identifiable secondary cause (Friedman et al, 2013). The causes include obesity, endocrine/metabolic abnormalities, drugs, and a high protein concentration in the cerebrospinal fluid, and some investigators include intracranial venous sinus thrombosis in this category.

BOX 26-1 INTERNATIONAL CLASSIFICATION OF HEADACHE

Primary headaches

1. Migraine
2. Tension-type headache
3. Trigeminal autonomic cephalalgias
4. Other primary headache disorders

Secondary headaches

5. Headache attributed to trauma or injury to the head and/or neck
6. Headache attributed to cranial or cervical vascular disorder
7. Headache attributed to non-vascular intracranial disorder
8. Headache attributed to a substance or its withdrawal
9. Headache attributed to infection
10. Headache attributed to disorder of homeostasis
11. Headache or facial pain attributed to disorder of the cranium, neck, eyes, ears, nose, sinuses, teeth, mouth, or other facial or cervical structures
12. Headache attributed to psychiatric disorder

Painful cranial neuropathies, other facial pains, and other headaches

13. Painful cranial neuropathies and other facial pains
14. Other headache disorders

(Headache Classification Committee of the International Headache Society, 2013)

BOX 26-2 POST-LUMBAR PUNCTURE HEADACHE

This is a typical example of headache due to a sudden drop in the intracranial pressure. The headache begins several seconds after standing up from the lying position, reaches its peak within 1 minute, and disappears within 20 seconds after returning to the lying position. The kind and the degree of headache differ greatly among subjects. It is relatively more common in young women. Although the mechanism of post-lumbar puncture headache is not precisely known, it is believed that intracranial pain receptors are pulled downward because of a decreased amount of cerebrospinal fluid (CSF). On the other hand, based on the fact that extradural blood patching is effective, it is postulated that the split injury of the dura caused by its puncture persists, and the CSF leaks out through the split in the upright position.

To prevent this headache, it is important (1) to use as small a puncture needle as possible, (2) to match the slant of the needle tip to the direction of dural fibers, (3) not to rotate the needle after puncture, and (4) to keep the patient supine for at least 2 hours after the puncture. The headache will improve spontaneously with water supplementation and by keeping the patient lying down, but when it is severe, epidural patching of the split with the patient's own blood is effective (Bezov et al, 2010).

Severe headache may occur due to a sudden decrease in the intracranial pressure. A typical example is **post-lumbar puncture headache**, which occurs after a traumatic lesion of the dura mater by lumbar puncture for the purpose of examination or anesthesia (Box 26-2). Headache may be also caused by a spontaneous drop in the intracranial pressure with no preceding lumbar puncture (**spontaneous intracranial hypotension**) (D'Antona et al, 2021). Spontaneous leakage of the cerebrospinal fluid is considered to be its main cause (Schievink et al, 2015). In this condition, cranial magnetic resonance imaging (MRI) shows brain sagging, and the T1-weighted MR image with gadolinium characteristically shows diffuse enhancement of the dura mater, which can be treated by an epidural blood patch (Rukovets, 2018).

2. MIGRAINE AND RELATED DISORDERS

In contrast with the secondary (symptomatic) headaches, the primary headaches may be called functional headaches,

and migraine and tension headache are two representative forms of this category. **Migraine** is characterized by repetition of paroxysmal headaches with no symptoms during the intervals. Migraine is often inherited by autosomal dominant transmission. Many attacks of migraine are preceded by **aura**, most commonly a flashing light in a hemi-field or in the rim of a hemi-field in a crescent shape (**scintillating scotoma**). Aura is believed to be caused by enhanced susceptibility to cortical spreading depression, which increases sensitivity to focal cerebral ischemia (Ferrari et al, 2015). It may present a positive symptom like scintillating scotoma or a negative symptom like hemianopsia. In an MRI study during the visual aura, blood oxygenation level-dependent (BOLD) signal was reduced in the visual cortex during the negative symptom like scotoma without scintillation, and it was increased during a positive symptom like scintillating scotoma (Arnglim et al, 2017). In the following phase, the corresponding artery dilates and activates pain receptors in the arterial wall, which is felt as a throbbing headache. Thus, the kind of aura depends on which artery is involved by the migraine: scintillating scotoma or hemianopsia in the territory of the posterior cerebral artery, blindness of one eye in the ophthalmic artery (**retinal migraine**), and brainstem symptoms in the vertebrobasilar artery (**basilar artery migraine**).

In some attacks of migraine, a neurological deficit seen as an aura might continue during or even after the migraine attack, like **hemiplegic migraine** and **ophthalmoplegic migraine**. These forms of migraine are usually inherited in autosomal dominant fashion. **Familial hemiplegic migraine** is often associated with spinocerebellar ataxia 6 (SCA6) or episodic ataxia, or both, and is known to be a calcium channel disorder (see Chapter 28-1A).

When examining the patient with headache, it is of utmost importance to obtain detailed information about the nature of the headache, including the presence or absence of any precipitating factor such as food and action, time of the day, and relationship to menstruation in case of women. If the headache fluctuates in its severity in synchrony with the arterial pulsation, it is called **throbbing headache** or **pulsating headache**. The pathologic mechanism of migraine has not been fully clarified yet, but pharmacologically a decrease in blood serotonin level is believed to play a primary role. Drugs acting on the serotonin receptor are effective in many cases. For the treatment of migraine, subcutaneous injection of sumatriptan succinate 6 mg, or oral administration of zolmitriptan 2.5–5.0 mg or ergotamine tartrate 1–2 mg is usually effective. If the initial dose of ergotamine tartrate is not effective, it may be

repeated every 30 minutes, but it should not exceed 6 mg in total for one attack.

Recently, there has been a major advance in treatment of migraine headache (Peters, 2019), including drugs that target CGRP, small molecules and monoclonal antibodies against the CGRP receptor and antibodies targeting the CGRP ligand (Caronna & Starling, 2021; Drellia et al, 2021; Ha et al, 2021). They have proven value for prevention and can be used to treat an acute headache as well.

A migraine attack might be complicated by cerebral infarction (**migrainous cerebral infarction**), although it is rare. This condition is relatively common in young women with migraine starting with aura in the posterior cerebral artery territory. If neurological deficits continue after the migraine attack, it is necessary to examine the patient with diffusion MRI (Wolf et al, 2011).

A. CLUSTER HEADACHE

This type of paroxysmal headache begins with severe pain in the deep orbit and is associated with autonomic symptoms such as conjunctival hyperemia, lacrimation, nasal obstruction, nasal discharge, facial sweating, miosis, blepharoptosis, and palpebral edema. This kind of migraine used to be called histamine headache or Horton's headache, but now it is called **cluster headache** or **trigeminal autonomic cephalalgia**. It is common in young men and induced by ethanol and nitroglycerin. An attack lasts 15 minutes to 3 hours, and it tends to occur in repeated clusters over a period of several days to more than a month. Subcutaneous injection of sumatriptan is effective in some cases.

B. REVERSIBLE CEREBRAL VASOCONSTRICTION SYNDROME

This condition is clinically similar to migraine because of repeated attacks of severe headaches over a period of about a week (thunderclap headache), and because it may be associated with focal neurologic symptoms (Chen et al, 2015). It is characterized by an angiographic finding of a bead-like appearance of arteries (Ducros et al, 2007). Some patients with this condition may develop subarachnoid hemorrhage or **reversible posterior leukoencephalopathy** as complications. In this condition, breakdown of the blood-brain barrier was demonstrated by the imaging study (Lee et al, 2017). In a recent population-based study in the United States, the incidence of this condition in adults was approximately 3 per million per year (Magid-Bernstein et al, 2021).

3. TENSION-TYPE HEADACHE

This type of headache is as common as migraine and is considered to be due to abnormal contraction of the muscles surrounding the cranium. It used to be called **muscle contraction headache** and is characterized by a feeling of the head tightened by a headband. It is often influenced by weather and the patient's mood. In clinical practice, it is common to see a combination of tension-type headache and migraine in the same patient.

Recently, a sudden-onset headache that continues without remission within 24 hours had drawn attention as **new daily persistent headache** because it is refractory to treatment (Yamani & Olesen, 2019).

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SLEEP DISORDERS

Sleep disorders are an important topic of clinical neurology because they are much more common than thought, and they may be the result of functional or organic brain disorders. Based on the electroencephalographic (EEG) and behavioral characteristics, sleep is divided into two stages: **rapid eye movement sleep (REM sleep)** and **non-REM sleep (NREM sleep, slow-wave sleep)**. NREM sleep is further divided into three stages: N1, N2, and N3, corresponding to NREM stage 1, stage 2, and stage 3 plus 4 of the Rechtschaffen and Kales classification, respectively (Table 27-1) (Silber et al, 2007). In the REM stage, the EEG is similar to the waking EEG, but it is characterized by the occurrence of rapid eye movements and a decrease or loss of muscle discharge in the jaw muscles. The subject usually dreams during REM sleep. The mechanism of sleep stages has been extensively investigated physiologically as well as pharmacologically.

Sleep disorders are classified into insomnias, sleep-related breathing disorders, hypersomnias, circadian rhythm sleep disorders, parasomnias, and sleep-related movement disorders. As most patients with sleep disorders are not aware of their trouble at night, it is important to obtain information from their family or persons who live together. The patients often visit the clinic with the chief complaint of daytime sleepiness. The most useful test for sleep disorders is **polysomnography (PSG)**, in which EEG, surface electromyography (EMG), electrocardiogram, electro-oculogram, respiration, and blood oxygen saturation are simultaneously monitored throughout the nocturnal sleep.

1. OBSTRUCTIVE SLEEP APNEA-HYPOPNEA SYNDROME

Transient arrest or decrease of breathing during sleep is believed to be relatively common in adults. In this condition, breathing suddenly stops and recovers spontaneously in 10 to 15 seconds, during which time the blood oxygen

saturation is also decreased. PSG during this episode shows loss of nasal airflow despite the presence of thoracic and abdominal excursion. This syndrome is commonly seen in markedly obese persons and is also called **pickwickian syndrome**. **Daytime sleepiness** is common because the patient wakes up frequently at night. Lying in the recumbent position and the use of a mouthpiece during nocturnal sleep might help to some extent, but positive-pressure ventilation is effective in severe cases. As patients with obstructive diseases of the upper airway might present with a similar sleep disorder, a laryngological examination is mandatory.

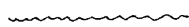
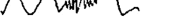
2. CENTRAL SLEEP APNEA-HYPOPNEA SYNDROME

In this type of sleep apnea-hypopnea syndrome, respiratory excursion of the thorax and abdomen is also lost or decreased during the episode as can be documented with the PSG. This phenomenon is commonly seen in patients with various neurodegenerative diseases, and it is especially common in those diseases affecting the brainstem reticular formation. A typical example is **multiple system atrophy**, in which sudden death may occur during sleep (Nishizawa & Shimohata, 2009). This condition is also seen in association with a vascular lesion of the brainstem, and a combined occurrence of central and obstructive types is not infrequent.

3. REM SLEEP BEHAVIOR DISORDERS (RBD)

RBD is characterized by episodes of behavioral abnormalities during REM sleep. Patients with RBD may get up and behave violently during REM sleep. The PSG is also abnormal during the episode. For example, rapid eye movements may be present, but the jaw muscles maintain their discharges (lack of muscle atonia), and periodic

TABLE 27-1 SLEEP STAGES OF ADULTS BASED ON EEG

Stage R & K	Features of EEG	EEG waveforms	Stage AASM
Waking	Waking	C  O 	Stage W
Stage 1	Relatively low voltage, 2~7 Hz activities. Vertex sharp wave (V wave) may appear in the late phase.	C  O 	Stage N1
Stage 2	Sleep spindles and K complex are seen. Slow waves (less than 2 Hz, more than 75 μ V) are seen in less than 20% of the time.	C  O 	Stage N2
Stage 3	Slow waves seen 20~50% of the time. Spindles are seen.	C  O 	Stage N3
Stage 4	Slow waves seen more than 50% of the time. Spindles are seen.	C  O 	
Stage REM	The same as stage 1. Rapid eye movements are seen. Muscle tonus of jaw muscles lost or decreased.	C  O 	Stage R

R & K: Rechtschaffen & Kales, AASM: American Academy of Sleep Medicine.

Slow waves: less than 2 Hz and higher than 75 μ V.

C: central EEG, O: occipital EEG.

(Modified from Silber et al, 2007)

movements may be seen in the lower limbs. Patients with Parkinson disease tend to suffer from insomnia and daytime sleepiness, and it has been shown that RBD is common in Parkinson disease (Arnulf et al, 2008a; Hu et al, 2015). In a report from Germany, 46% of 467 patients with Parkinson disease complaining of sleep disturbance had RBD (Sixel-Döring et al, 2011). In relation to this, it is known that, in Parkinson disease, the Lewy bodies appear also in the locus coeruleus and hypothalamus, which are important for REM sleep (Braak & Del Tredici, 2008) (see Box 16-2, p. 122).

Gaig et al. (2018) reported that patients with anti-IgLON5 antibodies show sleep disorders such as RBD associated with bulbar symptoms, gait abnormalities, oculomotor problems, and cognitive decline.

4. NARCOLEPSY AND RELATED DISORDERS

Narcolepsy is characterized by four cardinal symptoms: irresistible desire to sleep during the day, hypnagogic

hallucination, sleep paralysis, and cataplexy. The symptoms usually start to occur in puberty, and the disease is strongly associated with the major histocompatibility complex HLA-DR2 and HLA-DQB1*0602. The patient with narcolepsy becomes excessively sleepy during the day (**irresistible desire to sleep during the day**). In contrast with normal sleep, which starts from the light non-REM sleep progressing to deep non-REM sleep and then to REM sleep, the sleep of narcoleptic patients begins with the REM sleep. Therefore, the patient dreams from the transition to sleep, which corresponds to the **hypnagogic hallucination**. Furthermore, the patient becomes unable to move as if being firmly bound during the transition between wakefulness and sleep (**sleep paralysis**). **Cataplexy** manifests itself as sudden loss of tone, including falling as soon as the patient starts laughing, which is believed to be due to generalized reflex atonia. In this type of cataplexy, the level of **hypocretin (orexin, ORX)**, which is produced in the hypothalamus, is decreased in the cerebrospinal fluid. For the pathogenesis of this condition, autoimmune

BOX 27-1 COULD SLEEP APNEA SYNDROME CAUSE COGNITIVE IMPAIRMENT?

Might repeated drops in blood oxygen saturation cause dementia in the future? In a multi-institutional study in the United States, 298 women over age 65 who did not have dementia were studied with PSG and were followed up for 3.2 to 6.2 years. Mild cognitive impairment (MCI) was found to be more common in those who had sleep apnea syndrome at the time of the PSG study than those without (44.8% vs. 31.1%) (Yaffe et al, 2011). Another cross-sectional analysis of a large population also reported an association of sleep apnea with worse neurocognitive function (Ramos et al, 2015). An analysis of a large number of participants in the Alzheimer's Disease Neuroimaging Initiative (ADNI) showed that the presence of sleep-disordered breathing was associated with an earlier age at cognitive decline and the use of continuous positive airway pressure (CPAP) was associated with delayed onset of cognitive decline (Osorio et al, 2015). In these kinds of studies, it is expected to be difficult to match various other factors, including past medical history and environmental factors, between the two groups, and further studies might be warranted to answer this important question.

damage of hypocretin-producing cells in the hypothalamus has drawn special attention (Fontana et al, 2010), and various immune-based therapies to slow down or stop the destruction of ORX neurons are in progress (Barateau & Dauvilliers, 2019).

It should be kept in mind that daytime sleepiness can occur as a side effect of drugs. For example, patients with Parkinson disease who are taking dopamine agonists may suddenly become sleepy. However, these patients may also have excessive daytime sleepiness as a part of their disorder (see Chapter 16-2B, p. 121).

5. PERIODIC LIMB MOVEMENT IN SLEEP

Periodic limb movement in sleep (PLMS) is a series of periodic movements of the unilateral or bilateral lower limbs during sleep at a regular interval, commonly once in about 20 seconds. It has drawn special attention because many patients with PLMS also suffer from **restless legs syndrome** (see Chapter 19-10C, p. 186). Patients with restless legs syndrome have an unpleasant sensation in their legs while resting at night and develop a strong urge to move

BOX 27-2 REM SLEEP BEHAVIOR DISORDER (RBD) AS A SYNUCLEINOPATHY

RBD is highly associated with synucleinopathy neurodegeneration, especially Parkinson disease, dementia with Lewy bodies (DLB), multiple system atrophy, and pure autonomic failure (St Louis et al, 2017). Since patients with Parkinson disease who have RBD commonly develop cognitive impairment and hallucination, there has been some controversy as to whether those patients might have DLB (see Chapter 23-2A, p. 223). In fact, in a prospective study in the United States of 234 patients with dementia who were followed up and whose diagnosis was pathologically confirmed, 76% of DLB cases had RBD versus only 4% of other cases (Ferman et al, 2011). Postuma et al. (2009) followed up a number of patients with idiopathic RBD and reported that, after 12 years, about half of them developed neurodegenerative diseases such as Parkinson disease and DLB. Iranzo et al. (2017), by applying dopamine transporter (DAT) imaging to 87 patients with RBD, showed decreased uptake in the putamen in more than half of them, and about further half of them developed clinically defined synucleinopathy after 3 years' follow-up. In fact, this phenoconversion may not be rare (Li et al, 2017; McKenna & Peever, 2017; Meles et al, 2017). In a multicenter, prospective study of 1,280 cases of idiopathic RBD, the overall phenoconversion rate of idiopathic RBD to any neurodegenerative disorder was 73.5% after 12 years (Postuma et al, 2015; Postuma et al, 2019).

the legs. Once they move the legs, the symptom is temporarily released. As dopamine agonists were found to be effective for this condition, its relationship with Parkinson disease has drawn particular attention. However, there is no pathologic evidence suggestive of Parkinson disease in the autopsied cases of restless legs syndrome, and there are no data suggesting progression of this condition to Parkinson disease, although the co-occurrence is greater than chance. The prevalence of PLMS is not precisely known because its diagnosis requires PSG, but it is believed to be high in the elderly population (Natarajan, 2010). These conditions are mostly sporadic, but there are some familial cases, and genetic studies of these two phenomena are under way (Didriksen et al, 2020; Stefansson et al, 2007). A recent multicenter genetic study of restless legs syndrome and Parkinson disease found no evidence for a causal relationship or genetic correlation between the two groups (Estiari et al, 2021).

6. KLEINE-LEVIN SYNDROME

This unique syndrome is characterized by repetitive episodes of severe hypersomnia, cognitive impairment, apathy, derealization, and behavioral disturbances such as hyperphagia and hypersexuality in male teens (Kas et al, 2014). Even during the intervals, the patients tend to be easily tired, need naps, fall asleep rapidly, and have high anxiety/depression scores (Lavault et al, 2015). As there are occasional familial cases and as it is common among Jews, some genetic factors are postulated, but this has not been clarified (Arnulf et al, 2008b).

7. FATAL FAMILIAL INSOMNIA

This disease was first reported by Lugaresi in 1986, and mutation of the prion protein gene was found later (see Box 23-8, p. 225). It occurs in adults and equally affects men and women, but the age of onset varies widely. Clinically it is characterized by intractable insomnia and progressive cognitive decline, but it lacks other symptoms seen in typical Creutzfeldt–Jakob disease, and its clinical course may be relatively long (Krasnianski et al, 2008).

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EPISODIC NEUROLOGICAL DISORDERS RELATED TO ION CHANNELS

Ion channels are transmembrane glycoprotein pores of the membrane of nerve fibers and muscles and are important for controlling the membrane permeability of ions. There are two kinds of ion channels: the **transmitter-gated (ligand-gated) ion channels**, which are controlled by neurotransmitters, and the **voltage-gated ion channels**, which are controlled depending on the amplitude of membrane potentials. Recently many diseases have been categorized as ion channel disorders. Ion channel disorders (**channelopathies**) are either hereditary or autoimmune. Phenotypes of hereditary channelopathies are usually paroxysmal or episodic disorders such as hereditary epilepsies, familial hemiplegic migraine (see Chapter 26-2B, p. 252), startle disease (see Chapter 19-8B, p. 181), episodic ataxia, and periodic paralysis.

Hereditary channelopathies are characterized by **phenotypic heterogeneity**. Namely, mutation of a single gene may manifest different clinical phenomena (**phenotypic overlap**). For example, patients with spinocerebellar ataxia type 6 (SCA6) may also show episodic ataxia or familial hemiplegic migraine, or both. In contrast, mutation of different genes may show a similar clinical phenomenon (**genetic heterogeneity**) (Bernard & Shevell, 2008). Representative forms of hereditary ion channel disorders and autoimmune ion channel disorders of the central nervous system are shown in Tables 28-1 and 28-2, respectively.

1. HEREDITARY CHANNELOPATHIES

A. HEREDITARY CHANNELOPATHIES OF THE CENTRAL NERVOUS SYSTEM

Representative forms of hereditary ion channel disorders of the central nervous system are shown in Table 28-1.

i. Familial hemiplegic migraine

In this form of migraine, hemiplegia or other motor paralysis appears as an aura or during the headache attack. It is usually transmitted with autosomal dominant inheritance, and three forms are known (see Table 28-1). Of these forms, type I is due to mutation of the voltage-gated calcium channel gene (*CACNA1*) and is often overlapped with SCA6 and episodic ataxia type II, which are due to mutations of the same gene (see Chapter 16-2C, p. 129). For treatment, a carbonic anhydrase inhibitor, acetazolamide, is known to be effective.

ii. Episodic ataxia

These autosomal dominant hereditary diseases are characterized by paroxysmal episodes of ataxia and dysarthria, and nine forms are known (Piarroux et al, 2020). Types I and II are most common. In an episode of type I, ataxia and dysarthria last several minutes, and myokymia might be seen during the intervals. The episodes occur spontaneously but may be provoked by startle, emotional change, fatigue, illnesses, or motor action. It is due to mutation of the voltage-gated potassium channel gene (*KCNA1*). Type II occurs in children or young adults, with episodes of ataxia and dysarthria lasting several hours to several days, and there might be interictal nystagmus. It is also provoked by stress, motor action, caffeine, alcohol, intake of carbohydrate-rich food, or fever. It is due to mutation of the voltage-gated calcium channel gene (*CACNA1*), and as already described, it may overlap with familial hemiplegic migraine type I and SCA6. Thus, SCA6 is a unique neurodegenerative disease associated with calcium channelopathy. These two types of episodic ataxia may be treated with anticonvulsants such as phenytoin, carbamazepine, and sodium valproate; acetazolamide; and a

TABLE 28-1 HEREDITARY ION CHANNEL DISORDERS OF THE CENTRAL NERVOUS SYSTEM

Diseases	Inheritance	Ion channel	Chromosome	Gene mutation
Familial hemiplegic migraine	AD			
I		Cav 2.1 α 1	19p13	CACNA1
II		Na, K-ATPase α 2	1q23	ATP1A2
III		Nav 1.1	2q24	SCN1A
Episodic ataxia* ¹	AD			
I		Kv 1.1	12p13	KCNA1
II		Cav 2.1 α 1	19p13	CACNA1
Hereditary startle disease* ²	AD, AR			
I		GlyR α 1	5q33.1	GLRA1
II		GlyR β	4q32.1	GLRB
ADNFLE* ³	AD			
I		AchR α 4	20q13.2-13.3	CHRNA4
III		AchR β 2	1q21.3	CHRNA2
IV		AchR α 2	8p21	CHRNA2
Juvenile myoclonic epilepsy	Variable	GABAR, Ca, Cl		Multiple
BAFME	AD			
I			8q23.3-q24.11	
II			2p11.1-q12.2	
III			5p15.31-p15.1	
IV			3q26.32-q28	
Childhood absence seizure	Variable	GABAR, Cl, Ca		Multiple
Ring chromosome 20	Mostly sporadic		20	Unknown

AD: autosomal dominant, AR: autosomal recessive, R: receptor, v for ion channel: voltage-gated, Gly: glycine,

Ach: acetylcholine, GABA: gamma-amino butyric acid, ADNFLE: autosomal dominant nocturnal frontal lobe epilepsy,

*1: only representative types are shown, *2: two other types are not shown, *3: type II is not shown.

(Modified from Bernard & Shevell, 2008)

potassium channel blocker, 4-aminopyridine. It is also prevented by avoiding the provoking factors mentioned.

iii. Hereditary startle disease, hyperekplexia

As described in the section of myoclonus of brainstem origin (see Chapter 19-8B, p. 181), this is an excessive startle reaction provoked by an unexpected sudden stimulus and is transmitted in either autosomal dominant or recessive

inheritance. Patients with this disorder are noted to be excessively hypertonic in the newborn period. Four types are known, and all types are due to mutation of genes related to the receptor of inhibitory transmitter, glycine. Mutation of the glycine receptor α 1 subunit gene (*GLRA1*) is known for type I, and that of the glycine receptor β subunit gene (*GLRB*) is known for type II (Bode & Lynch, 2014). Recently a French family presenting with exaggerated startle reflex, chorea, and multifocal myoclonus was shown to

TABLE 28-2 AUTOIMMUNE ION CHANNEL DISORDERS

Ion channel	Disease	Cardinal symptoms
Cav (P/Q-type)	Lambert–Eaton syndrome	Muscle weakness
	Paraneoplastic cerebellar degeneration	Cerebellar ataxia
Kv (1.1, 1.2, 1.6)	Acquired neuromyotonia (Isaacs)	Muscle spasm, cramp, hyperhidrosis
	Limbic encephalitis	Amnesia, convulsion, character change
	Morvan syndrome	Neuromyotonia, insomnia
GluR3 (AMPA receptor)	Rasmussen encephalitis	Intractable epilepsy, psychic symptoms, amnesia, hallucination, autonomic disturbance
Nicotinic AChR- $\alpha 1$	Myasthenia gravis	Muscle weakness, easy fatigability
Ganglionic AChR- $\alpha 3$	Autoimmune autonomic ganglionopathy	Orthostatic hypotension, tonic pupil, urinary retention, gastrointestinal symptoms

v: voltage-gated, GluR: glutamate receptor, AChR: acetylcholine receptor.

have mutations of *GRIA3*, which encodes the GluA3 subunit of AMPA receptors (Piard et al, 2020). Clonazepam is effective in most cases.

iv. Epilepsies

Most cases of idiopathic epilepsy due to mutation of a single gene are related to a channel gene of nicotinic acetylcholine receptor or GABA-A receptor, or a gene mutation of the sodium, chloride, calcium, or potassium channel (Heron et al, 2007). Multiple mutations in genes that encode α -subunit sodium channels within the central nervous system have been associated with early-onset epileptic encephalopathies and certain autosomal dominant epilepsy syndromes (Holland et al, 2018).

Autosomal dominant nocturnal frontal lobe epilepsy (ADNFLE) is a unique seizure occurring in childhood at night (see Chapter 25-1B, p. 248). The attack begins with groaning or vocalization and involves abnormal movements such as vigorous or thrusting movements of legs and waist. It rarely progresses to generalized convulsion, but the attacks may occur several times per night. The interictal electroencephalogram (EEG) is often normal and the ictal EEG

may show paroxysmal abnormalities at the midline frontal region, suggesting its origin in the supplementary motor area. Clinically, it is often difficult to distinguish from functional seizures (see Chapter 25-2B, p. 249). Mutation of the gene related to the nicotinic acetylcholine receptor $\alpha 4$ subunit (*CHRNA4*) is known for type I, and likewise $\beta 2$ subunit (*CHRNA2*) for type III, and $\alpha 2$ subunit (*CHRNA2*) for type IV. Carbamazepine is effective for these seizures.

Juvenile myoclonic epilepsy is a form of myoclonus epilepsy affecting persons of puberty, predominantly female, and it may be complicated by generalized tonic-clonic seizures and absence seizures in some cases. It is especially characterized by worsening upon awakening in the morning. Mutations of multiple genes related to the GABA receptor and calcium channel and chloride channel are known in this disorder.

Benign adult familial myoclonus epilepsy (BAFME) is a unique form of autosomal dominant hereditary myoclonus epilepsy reported from Japan and some European countries (see Box 19-12, p. 180). Myoclonus of cortical origin begins in adults, and its progression to generalized convulsion is rather rare. The mutation has been localized on chromosome 8q for type I, 2p for type II, 5p for type III, and 3q for type IV. In Japanese families with BAFME, abnormal expansions of TTTCA and TTTTA repeats in intron 4 of *SAMD12* were found (Ishiura et al, 2018).

Childhood absence seizure is characterized by frequent attacks of absence, each lasting several seconds, affecting infants and children, and it is commonly called **petit mal** (see Chapter 25-1B, p. 247). The EEG typically shows a burst of bilaterally synchronous high-amplitude 3-Hz rhythmic spike-and-wave discharges (see Figure 25-1, p. 246). Mutations of multiple genes related to the GABA receptor, chloride channel, and calcium channel have been reported.

Patients with **ring chromosome 20** show mental retardation, behavioral abnormalities, and epileptic seizures. Seizures of various forms, including nocturnal frontal lobe epilepsy, nonconvulsive status epilepticus, and prolonged confusion, start at around 5 years of age and tend to resist anti-epileptic medication. This condition is also believed to be an ion channel disorder.

Type II of **congenital long QT syndrome** is due to mutation of the cardiac potassium channel gene and is often complicated by epileptic seizures. The seizure in this condition is considered to be a primary ion channel disorder of the brain rather than a seizure secondary to ventricular arrhythmia (Auerbach et al, 2016; Johnson et al, 2009).

B. HEREDITARY CHANNELOPATHIES OF THE PERIPHERAL NERVOUS SYSTEM

Mutations of the genes related to the voltage-gated sodium channel (*SCN9A*) in the dorsal root ganglion and in the sympathetic ganglion are known to cause several kinds of pain disorders. Among those disorders, **erythromalgia** (called **erythromelalgia** in acquired cases) and paroxysmal extreme pain disorder are known to be caused by gain of the channel function, and **congenital insensitivity to pain** is known as an example of loss of channel function (Drenth & Waxman, 2007). Congenital insensitivity to pain is a rare condition transmitted by autosomal recessive inheritance. Patients with this disorder do not feel pain at all since birth, and thus they easily suffer from trauma and burns. Other somatosensory functions are normal, but autonomic disturbances such as anhidrosis are present in many cases (Norcliffe-Kaufmann et al, 2015).

C. HEREDITARY CHANNELOPATHIES OF MUSCLES

Periodic paralysis, a variety of non-dystrophic myotonia, malignant hyperthermia, central core disease, and congenital myasthenic syndrome are known to belong to this category (Table 28-3).

i. Hypokalemic periodic paralysis

This is an autosomal dominant hereditary disease, starting in young age, affecting predominantly males. Generalized weakness usually occurs upon awakening in the morning and lasts for several hours. Any factor lowering the blood potassium level can provoke this episode, including excessive intake of carbohydrate-rich food or salt, stress, drugs, and heavy physical exercise on the preceding day. Mutations of multiple genes related to the voltage-gated calcium channel $\alpha 1$ subunit in the muscle are known for type I, and mutations of multiple genes related to the voltage-gated sodium channel α subunit are known for type II. Avoiding the precipitating factors mentioned is important for preventing the attacks, but correction of the serum potassium level and chronic administration of acetazolamide are effective for the treatment. It is believed that acetazolamide causes metabolic acidosis, which helps move potassium into the muscle fibers. There are no symptoms during the intervals. In Asian male patients, this condition is often accompanied by hyperthyroidism (Vijayakumar et al, 2014).

ii. Hyperkalemic periodic paralysis

This is an autosomal dominant hereditary disease that affects children before puberty with frequent episodes of generalized weakness, each attack lasting less than an hour. During the attack, the serum potassium level is either elevated or normal. During the intervals, myotonia is seen in the face, tongue, and hands, and myopathy might develop in many cases. Mutation of the voltage-gated sodium channel gene on chromosome 17q (*SCN4A*) is known in most cases. In some cases, physical exercise and sugar intake might ameliorate the attacks.

iii. Myotonia congenita

In this disease, stiffness due to disturbance of muscle relaxation (myotonia) (see Chapter 17-6, p. 152) occurs in childhood. Just as in myotonic dystrophy, myotonia will improve by repeated contraction of the involved muscles (warm-up phenomenon). There are two types of this disorder: **Thomsen type**, which is transmitted via autosomal dominant inheritance and is relatively mild, and **Becker type**, which is transmitted via autosomal recessive inheritance and is severe. The first case was reported by Thomsen in 1876, and it is known that Thomsen himself suffered from this disease. For both types, mutations of multiple genes related to the voltage-gated chloride channel (*CLCN1*) in the skeletal muscles are known. It is believed that this gene abnormality impairs the chloride channel gating, which causes decreased chloride conductance and electrical instability of the sarcolemma, resulting in the repetitive continuous electrical discharges. Various drugs, including mexiletine, quinine, lithium carbonate, procainamide, carbamazepine, phenytoin, and acetazolamide, have been found effective for this condition.

iv. Paramyotonia congenita

This autosomal dominant hereditary disease was first described by von Eulenburg in 1886. In this condition, in contrast with ordinary myotonia, in which muscle relaxation is most difficult after the first contraction and becomes easier with the increasing number of repetitions (warm-up phenomenon), relaxation becomes increasingly difficult with repetition of the muscle contraction. It is also characterized by worsening with cold temperatures. For example, cooling of the eyes causes myotonia

TABLE 28-3 HEREDITARY ION CHANNEL DISORDERS OF MUSCLES

Diseases	Inheritance	Ion channel	Chromosome	Gene mutation
Hypokalemic periodic paralysis	AD			
I		Cav α 1	1q31-q32	CACNA1S *2
II		Nav α	17q23-q25	SCN4A *2
Hyperkalemic periodic paralysis	AD	Nav α	17q23-q25	SCN4A
Myotonia congenita		ClvC 1	7q35	CLCN1
Thomsen	AD			
Becker	AR			
Paramyotonia congenita	AD	Nav α	17q23-q25	SCN4A
Andersen–Tawil syndrome	AD	Kir2.1	17q23	KCNJ2
Malignant hyperthermia susceptibility*1	AD	Ryanodine receptor 1	19q13.1	RYR1
Central core disease	AD or AR	Ryanodine receptor 1		Multiple
Congenital myasthenic syndrome	AR	Ach receptors (multiple)		Multiple

ir: inward rectifier, otherwise the same designation as in Table 28-1.

*1: Only the representative type is shown, *2: Typical types are shown.

(Modified from Bernard & Shevell, 2008, and Lehmann-Horn & Jurkat-Rott, 2007)

in the orbicularis oculi muscles, which results in inability to open the eyelids. Extraocular muscles are also occasionally involved. This condition is seen from birth and usually does not progress. In some families with this condition, it may be associated with short-lasting attacks of hyperkalemic periodic paralysis. It is also known that alcohol intake ameliorates the symptoms. Mutations of the genes related to the voltage-gated sodium channel (SCN4A) are known for this condition (Lehmann-Horn & Jurkat-Rott, 2007).

v. Andersen–Tawil syndrome

This syndrome, described by Andersen et al. in 1971 and by Tawil et al. in 1994, is characterized by potassium-responsive periodic paralysis, cardiac arrhythmia, and various anomalies of the head and face (Sansone & Tawil, 2007). The congenital anomalies include hypertelorism, micrognathia, low-set ears, high-arched palate, cleft palate, short stature, clinodactyly, and syndactyly. This is transmitted via autosomal dominant inheritance, and mutation of the voltage-gated potassium channel (KCNJ2) gene on chromosome 17q is known in many cases.

vi. Malignant hyperthermia susceptibility

Malignant hyperthermia is an excessive metabolic reaction induced by simultaneous use of inhalation anesthetics and neuromuscular blocking agents in subjects with this susceptibility. Cardinal symptoms of the reaction are tachypnea, tachycardia, muscle rigidity, acidosis, rhabdomyolysis, and hyperthermia. This susceptibility is transmitted via autosomal dominant inheritance, and six types are known. Mutation of the ryanodine receptor type I gene (RYR1) on chromosome 19q is known in most cases. This receptor is important for muscle contraction by releasing calcium from the sarcoplasmic reticulum of skeletal muscles, and mutation of its gene causes excessive accumulation of calcium in the muscle fibers. Therefore, it is important for anesthesiologists to pay attention to the family history. For treatment, the anesthetics should be discontinued immediately while dantrolene is administered.

vii. Central core disease

This congenital non-progressive myopathy in newborns was first reported by Shy and Magee in 1956. The patients manifest hypotonia, muscle weakness, developmental delay

of motor functions, congenital dislocation of hip joints, anomaly of feet, scoliosis, and joint contracture. Malignant hyperthermia is an important complication of this myopathy. Most cases are transmitted via autosomal dominant inheritance, and like malignant hyperthermia susceptibility, mutation of the ryanodine receptor gene is known.

viii. Congenital myasthenic syndrome

This is a group of diseases affecting newborns and infants with muscle weakness and easy fatigability. Antibody against the acetylcholine receptor is negative in this group. Heterogeneous conditions in terms of pathophysiology and genetics are included in this syndrome, and at least some of these conditions are considered to be caused by ion channel abnormalities.

2. AUTOIMMUNE CHANNELOPATHIES

Recently ion channel disorders mediated by autoimmune pathogenesis have been increasingly recognized (Vernino, 2007) (see Table 28-2).

A. MYASTHENIA GRAVIS

Myasthenia gravis (MG) is a representative disease of autoimmune channelopathies. Clinically it is characterized by muscle weakness and easy fatigability of muscles, often involving the extraocular and bulbar muscles. **Antibodies against the nicotinic acetylcholine receptor** of the postsynaptic membrane of the neuromuscular junction are elevated in 80% to 85% of the patients with the generalized form and in 50% to 60% of the patients with the ocular form. Among the patients with adult-onset MG, **thymoma** is found in 15%, and 30% of the patients with thymoma develop MG. Electrophysiologically, repetitive stimulation of the peripheral nerve at 3 Hz shows decline of the compound muscle action potential (CMAP) by more than 10% (**waning**). Single-fiber electromyography (EMG) is more sensitive than repetitive stimulation but is technically more difficult to do. Intravenous injection of a cholinesterase inhibitor, edrophonium, improves the symptoms and is used as a diagnostic test (**Tensilon test**) (see Chapter 9-5A, p. 65, and Chapter 33-1B, p. 285). The ice pack test can be used also, and as noted earlier, it has a high sensitivity.

In addition to the symptomatic and immunosuppressive treatments, thymectomy plus prednisone is the current

treatment predominantly applied for the patients with generalized non-thymomatous MG (Gronseth et al, 2020; Narayanaswami et al, 2021; Sanders et al, 2016; Wolfe et al, 2016, 2019).

B. LAMBERT-EATON SYNDROME

This is an autoimmune disease against the voltage-gated P/Q type calcium channel (VGCC) in the nerve terminal of the neuromuscular junction and is called **myasthenic syndrome (Lambert-Eaton syndrome)** (Hülsbrink & Hashemolhosseini, 2014). Clinically, weakness of proximal limb muscles and easy fatigability occur predominantly in middle-aged males. Tendon reflexes are reduced or absent when studied at rest without preceding exercise. Autonomic disturbances such as dry mouth, constipation, and impotence may be also seen. In contrast with MG, extraocular muscles are rarely affected in this condition. Furthermore, muscle weakness and reduced tendon reflexes might recover transiently after exercise (contraction) of the corresponding muscle. In relation to this phenomenon, the muscle response to a single electric shock of the corresponding motor nerve is extremely small, but after exercise it is augmented to more than twice the baseline value (**post-tetanic potentiation**). Alternatively, repetitive electrical stimulation of the peripheral nerve innervating the muscle at high frequency (30 to 50 Hz) shows an increment of successive muscle action potentials (**waxing**), although this test is quite painful to the subject. In the myasthenic syndrome, as the release of acetylcholine from the presynaptic membrane is impaired, it is expected that voluntary activation or electrical stimulation of α motor fibers improves the synaptic transmission transiently, thus improving the muscle strength and the tendon reflex.

The serum antibody against the P/Q-type VGCC is detected in 85% of the patients, and about 60% of the adult-onset patients have small cell lung carcinoma. As the P/Q-type VGCC is also present in the cerebellar Purkinje cells, Lambert-Eaton syndrome may be associated with an autoimmune cerebellar lesion called **paraneoplastic cerebellar degeneration (subacute cerebellar degeneration)** (see Chapter 16-2C, p. 131).

A study of 264 cases of biopsy-proven small cell lung carcinoma from Oxford revealed paraneoplastic neurological disorders (PNDs) in 9.4%, most frequently Lambert-Eaton syndrome (3.8%), sensory neuronopathy (1.9%), and limbic encephalitis (1.5%), and the most common antibodies were SOX2 (62.5%), HuD (41.7%), or P/Q VGCC (50%) (Gozzard et al, 2015).

C. ISAACS SYNDROME (NEUROMYOTONIA)

This is due to autoimmune impairment of the voltage-gated potassium channel (VGKC) of the peripheral nerve, which increases its excitability and causes continuous spasm and myokymia, especially in the distal muscles of the extremities (**acquired neuromyotonia**) (Oda et al, 1989). In most patients with Isaacs syndrome, antibodies specific for the VGKC complex are found, but some cases may also have antibodies against GABA_B receptors or Ach receptors (Zhao et al, 2018). The patients may also show generalized muscle stiffness, loss of body weight, and hyperhidrosis. In this condition, the needle EMG recorded from the myotonic muscles shows high-frequency motor unit discharges that suddenly start and gradually disappear after several seconds.

As the VGKC is present also in the central nervous system, the neuromyotonia and autonomic symptoms might be associated with cerebral symptoms such as insomnia and hallucination, which is called **Morvan syndrome**. Among the antibodies against the VGKC, levels of the antibody against CASPR2 are especially elevated in Morvan syndrome (Irani et al, 2012; Nikolaus et al, 2018). Some patients with **limbic encephalitis** (Box 28-1) who have antibodies against the VGKC complex have frequent brief seizures involving posturing of the hemiface and ipsilateral arm associated with paroxysmal EEG abnormality (**faciobrachial dystonic seizures**). In this condition, immunotherapy was found to be effective for seizures and for preventing the subsequent development of cognitive impairment (Irani et al, 2013). **LGII-antibody-associated encephalitis** can cause various forms of seizures (Aurangzeb et al, 2017). It was recently shown that mutant proteins of 14 kDa histidine triad nucleotide-binding protein 1 (HINT1) cause autosomal recessive axonal neuropathy with neuromyotonia (Cortés-Montero et al, 2021).

Rasmussen encephalitis is also considered to be an autoimmune channelopathy of the central nervous system. Clinically it is characterized by predominantly unilateral hemispheric involvement associated with convulsive seizures. Antibodies against the glutamate receptor (GluR3) may be elevated, and a comorbid association with other autoimmune diseases has been reported (Amrom et al, 2014). Recently, **paraneoplastic limbic encephalitis** has drawn particular attention as an autoimmune-related disorder primarily affecting the limbic system (see Box 28-1).

BOX 28-1 LIMBIC ENCEPHALITIS

This is a collection of antibody-mediated encephalitis rather than infectious ones. Following a report of a young woman with what was called “acute non-herpetic encephalitis” from Japan in 2004, this paraneoplastic autoimmune disorder has drawn particular attention in recent years (Iizuka et al, 2008). The disease starts with symptoms of upper respiratory infection and psychic symptoms and is followed by amnesia, epileptic seizures, consciousness disturbance, hypoventilation, autonomic dysfunction, and bizarre involuntary movements such as vigorous dyskinetic movements of the face and mouth. This condition is often associated with **ovarian teratoma**, and levels of antibody against the NR1 or NR2 subunit of the NMDA receptor (NMDAR) are elevated in the serum and cerebrospinal fluid (CSF). It has been shown that those antibodies bind with the NR2B or NR2A subunit of NMDAR in the hippocampus and forebrain (Dalmau, 2016; Dalmau et al, 2008). It has been also shown that the IgG NMDAR antibodies are highly specific for anti-NMDAR encephalitis (Hara et al, 2018). Some patients with limbic encephalitis have antibodies against the GluA1 (GluR1) or GluA2 (GluR2) subunits of the AMPA receptors (Hötberger et al, 2015). Cranial magnetic resonance imaging in this disease shows abnormal signal in the limbic system, including the mesial temporal lobes, which necessitates differentiation from herpes simplex encephalitis. Resection of the ovarian tumor and immunotherapy are effective, but some patients do well without the tumor removal (Iizuka et al, 2008).

A syndrome of limbic encephalitis combined with autonomic disturbance and neuromyotonia is called **Morvan syndrome** (Irani et al, 2012). In limbic encephalitis associated with Hodgkin lymphoma (**Ophelia syndrome**), the metabotropic glutamate receptor 5 (mGluR5) was identified as an autoimmune antigen (Lancaster et al, 2011). Limbic encephalitis is also known to be associated with relapsing polychondritis (Kondo et al, 2014). Autoantibodies to leucine-rich glioma-inactivated 1 (LGII) and contactin-associated protein like-2 (CASPR2) are associated with clinically distinctive syndromes that are highly responsive to immunotherapy, such as limbic encephalitis, faciobrachial dystonic seizure, Morvan syndrome, and neuromyotonia (Michael et al, 2020).

Furthermore, encephalopathy might appear in association with Hashimoto thyroiditis (**Hashimoto encephalopathy**) (Mattozzi et al, 2020). This is considered not to

be caused by hyperthyroidism itself but rather to be autoimmune encephalitis coinciding with autoimmune thyroiditis.

In addition to the hereditary and autoimmune channelopathies already described, the mechanism of ion channel disturbance might be also involved in **diabetic neuropathy**. In patients with diabetic neuropathy, Misawa et al. (2009) found abnormal sodium current at the Ranvier node, which was ascribed to activation of the polyol pathway mediated by aldose reductase and reduced Na/K pump activity.

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FUNCTIONAL (PSYCHOGENIC) NEUROLOGICAL DISEASES

Many neurological symptoms may appear without organic lesions of the nervous system. **Functional (psychogenic) neurological diseases**, also called conversion disorders, are involuntary. There are two voluntary mimics of functional disorders: **factitious disease**, where patients have a psychiatric illness, and **malin-gering**, where persons have no illness and are just lying (Hallett et al, 2012). The term “functional” is beginning to replace “psychogenic” in many circumstances. Negative neurological symptoms like loss of consciousness, blindness, constriction of visual field, motor paralysis, and loss of pain sense are commonly seen in functional diseases, but positive symptoms like convulsion and involuntary movements are more common. Among others, functional movement disorders are not infrequent and have drawn particular attention in the recent years (Czarnecki & Hallett, 2012; Espay et al, 2018; Fahn, 2007; Lang & Voon, 2011; Perez et al, 2021a, 2021b; Shibasaki et al, 2020) (see Chapter 19-10, p. 186). Within the movement disorders, functional tic-like movements might be difficult to distinguish from real tics, and in fact these two may overlap within the patient (Ganos et al, 2019).

1. FUNCTIONAL MOTOR PARALYSIS

Motor paralysis due to conversion disorders is not infrequently encountered. It is characterized by being incongruous with ordinary paralysis and inconsistent in time and involves a tendency to “give way” during manual muscle testing (**give-way weakness**). Furthermore, muscle tone and tendon reflexes are usually normal in functional paralysis.

In cases of asymmetric motor paralysis, it is possible to test how much the patient is cooperating with the muscle testing by taking advantage of the intact or less affected side. For example, in case of unilateral paralysis of the biceps brachii muscle, if the patient is cooperating with

flexion of the paralyzed forearm, the contralateral triceps brachii muscle is expected to contract, thus extending the contralateral forearm. Likewise, in case of unilateral paralysis of the triceps brachii muscle, the patient’s attempt to extend the paralyzed forearm is expected to flex the contralateral forearm. If there is no movement of the contralateral (intact) extremity, it might suggest poor cooperation of the patient. A similar principle can be applied to functional paralysis of the lower extremities as far as it is asymmetric (**Hoover sign**) (Box 29-1).

Conversion disorders may present with complete paraplegia. In this case, muscle tonus and tendon reflexes are expected to be entirely normal, and there are no pathologic reflexes. Functional inability to stand up unsupported and walk normally may be called **astasia-abasia**.

Electrophysiologically, the muscle potentials evoked by transcranial magnetic stimulation of the primary motor cortex (motor evoked potentials, MEP) are expected to

BOX 29-1 HOOVER SIGN

In case of unilateral or asymmetric motor paralysis of the lower extremities, if the patient is requested to raise the affected leg in the supine position and if he or she is cooperating, the contralateral (intact or less affected) leg is expected to be pushed down. Likewise, if the patient is asked to push down the affected leg and if he or she is trying hard, the contralateral leg is expected to be lifted. If there is no movement of the contralateral leg, it might suggest functional origin of the paralysis (**Hoover sign**). This test can also be done in the sitting position. Alternatively, cooperation of the patient with monoplegia of the leg or asymmetric paraplegia can be tested by supporting the patient’s legs with the examiner’s hands from both sides and asking the patient to abduct both legs. If the patient tries hard to abduct both legs, it is expected to find muscle contraction or movement in the intact or less affected leg.

BOX 29-2 FUNCTIONAL GAIT DISORDERS

Functional (psychogenic) gait disorders are not infrequently encountered, and their diagnosis may be difficult because they may look like other disorders due to structural lesions. There are several features in functional gait disorders (Nonnekes et al, 2020): (1) variability in base of support or instability to walk in a straight line, often with excessive arm movements and claims of poor balance, (2) scissoring, (3) weak gait with knees giving way, (4) asymmetrically reduced stance phase, (5) excessive slowness, sometimes with start hesitancy, (6) dragging, and (7) abnormal posturing of the leg or trunk. Thus, those are to be distinguished from (1) ataxia, (2) spasticity, (3) weakness, (4) limping, (5) bradykinesia, (6) hemiparesis, and (7) dystonia, respectively.

be absent in the lower extremities if complete paraplegia is due to an organic lesion. Thus, the presence of normal MEP in patients with complete paraplegia might suggest a functional origin, but it should be kept in mind that, in mild paraplegia due to a mild lesion of the corticospinal tract, MEP could be normal. In contrast, unilateral absence of MEP or delayed conduction of the corticospinal tract in patients with hemiplegia suggests that the hemiplegia is due to an organic lesion.

2. FUNCTIONAL SENSORY LOSS

The most common sensory symptom in functional diseases is complete loss of nociceptive sensation on either half of the body (hemianalgesia). In this case, the nociceptive loss is seen on the whole hemibody from the top of the head to the toes, and there is no reaction to pain or back-off movement away from the nociceptive stimulus. In the thoracic and lumbar segments, the somatosensory innervation crosses the midline and overlaps between the two sides for 1.0 to 1.5 cm from the midline. Therefore, the borderline of the hemianesthesia due to an organic lesion is shifted toward the anesthetic side by 1.0 to 1.5 cm. Therefore, if the borderline of the intact and affected sides is found exactly on the midline of the chest or abdomen, it suggests the possibility of functional origin, although the final judgment should not be based on this finding alone.

Furthermore, as the frontal skull bone and the sternum are each formed of a single bone, a vibratory

stimulus is expected to be felt bilaterally. Therefore, if the vibration sense is unilaterally lost over those bones, it might serve as a clue to the diagnosis of functional disorders.

Electrophysiologically, the somatosensory evoked potentials (SEP) are useful in some cases of functional sensory loss. However, it should be kept in mind that electrical stimulation of the peripheral nerve activates only large fibers and does not reflect nociceptive conduction. For unilateral or asymmetric loss of nociceptive senses, SEP following laser stimulation is useful. If laser evoked potential is normally recorded in patients with complete loss of pain sense, it might suggest its functional origin. To avoid misinterpreting the SEP results due to a technical failure, comparison of the two sides is always important.

3. CONSTRICTION OF VISUAL FIELD

Constriction of the visual field due to conversion disorders is characterized by the fact that the visible angle does not change depending on the distance from the target (**tunnel vision, tubular vision**) (see Chapter 7-2B, p. 46). In healthy subjects, the visible angle is expected to become larger as they move away from the target.

4. COMMON FEATURES OF FUNCTIONAL NEUROLOGICAL DISEASES

Clinical features common to all kinds of functional neurological disorders include (1) **incongruity** with ordinary neurological symptoms, (2) **inconsistency** (variability) of nature, degree and distribution in time, (3) change due to **distraction** to other stimuli or other matters, and (4) the presence of **psychogenic factors** in the social or family background. In case of rhythmic or periodic abnormal movements of a part of the body, if the patient is requested to repeat rhythmic or periodic movements of other parts of the body at a certain rate, the rate of abnormal movements might be entrained into that of the assigned movements (**entrainment**). Regarding the functional positive neurological symptoms, convulsion was described in Chapter 25-2B (p. 249), and involuntary movements were explained in Chapter 18-10B (p. 186). Electrophysiological techniques are often useful for supplementing the clinical impression of functional disorders.

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RELAY CENTER OF ALL SENSORY AND MOTOR FUNCTIONS (THALAMUS)

Although this book is organized according to the functional anatomy of the nervous system, this chapter is especially arranged because the thalamus is the relay center for all kinds of ascending pathways to the cerebral cortex, and also because it is commonly involved by the cerebrovascular diseases.

1. GENERAL STRUCTURE OF THE THALAMUS

The thalamus is a relatively large, egg-shaped structure situated deep in each cerebral hemisphere. Its borders are medially the third ventricle, laterally the internal capsule,

and inferiorly the hypothalamus and midbrain. Each thalamus is divided by the Y-shaped internal medullary lamina into four divisions: the lateral, medial, and anterior nuclear groups and the posterior nucleus (pulvinar) (Figure 30-1). The left and right thalami are connected by the transverse fibers (**massa intermedia**). Lateral to the pulvinar are the lateral geniculate body and the medial geniculate body, which are the relay nuclei for the visual and auditory pathways, respectively (see Chapter 7 and Chapter 12, respectively).

In the **lateral nuclear group**, there are the relay nuclei important for sensory functions and motor control. The relay nuclei for the somatosensory, vestibular, and

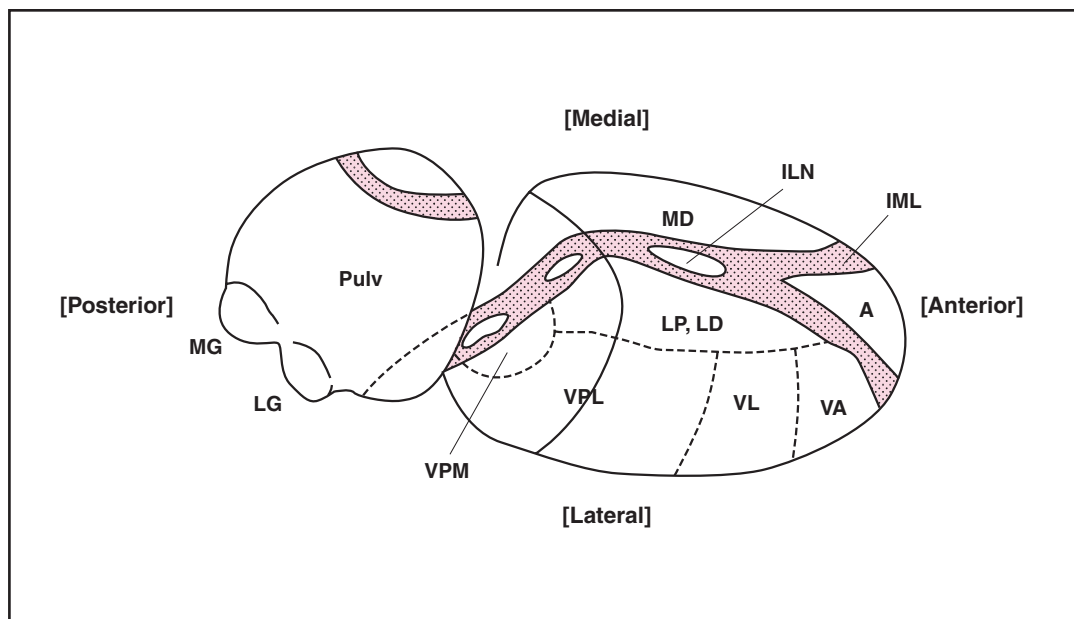


Figure 30-1 Schematic diagram of the thalamus, showing its main nuclei. Lateral upper view of the right thalamus. A: anterior nucleus, ILN: intralaminar nucleus, IML: internal medullary lamina, LD: lateral dorsal nucleus, LG: lateral geniculate body, LP: lateral posterior nucleus, MD: medial dorsal nucleus, MG: medial geniculate body, Pulv: pulvinar, VA: ventral anterior nucleus, VL: ventral lateral (ventrolateral) nucleus, VPL: ventral posterolateral nucleus, VPM: ventral posteromedial nucleus. A complex of VA and VL is also named Voa, Vop, and Vim from anterior to posterior.
Cited from Brodal, 2010, with permission.

gustatory senses are situated in the ventral posterior nuclei, and the motor-related relay nuclei from the basal ganglia and cerebellum are in the lateral nucleus. The main nucleus of the **medial nuclear group** is the medial dorsal nucleus, which is involved in the higher mental functions by projecting to the prefrontal cortex and cingulate gyrus. The **anterior thalamic nucleus** is a relay nucleus of the Papez circuit (see Chapter 23 and Figure 23-2, p. 221), and it is involved in memory function. The **internal medullary lamina** is primarily formed of white matter, but it contains a number of neuronal groups that receive input from the brainstem reticular formation and activate the cerebral cortex via the ascending reticular activating system (see Figure 4-1, p. 25). The posterior nucleus, called the **pulvinar**, is closely connected with the parietal cortex and is primarily engaged in the visual cognitive functions.

In addition to these roles as the relay center of the ascending projections to the cerebral cortex, the thalamus also receives strong input from the cerebral cortex. It is noteworthy that there are bidirectional fiber connections between each area of the thalamus and the corresponding area of the cerebral cortex. The thalamus is also believed to be involved in cognitive and language functions (see Chapter 30-6, p. 274).

The most common cause of thalamic lesions is infarction or hemorrhage. Clinical symptoms caused by **thalamic infarction** are quite variable, because (1) there is a great inter-individual variability in the arterial supply of thalamic nuclei, (2) a single artery supplies multiple thalamic nuclei, and (3) an ischemic lesion might occur in the territory of multiple arteries (Schmahmann, 2003). In **thalamic hemorrhage**, although clinical symptoms vary depending on the size of hemorrhage, motor paralysis and sensory loss develop in the contralateral extremities in the acute stage, and soon thereafter it is accompanied by consciousness disturbance of various degrees. If the lesion extends to affect the hypothalamus and midbrain, pupillary abnormalities and downward medial deviation of the eyes are also seen. If the hemorrhage ruptures into the third ventricle, the patient may fall into coma.

2. VENTRAL POSTERIOR NUCLEUS AS THE SENSORY RELAY CENTER

The tertiary neurons of the somatosensory pathway are located in the ventral posterior nucleus of the thalamus and receive afferent projection from the secondary neurons for both nociceptive and proprioceptive senses. Inputs from the

spinal nerve territories are received in the **ventral posterolateral nucleus (VPL)**, and those from the trigeminal nerve territory reach the **ventral posteromedial nucleus (VPM)**. Thus, the lateral spinothalamic tract and the medial lemniscus project to the VPL (see Figure 20-2, p. 193), and the trigeminothalamic tract and the fibers from the main sensory nucleus of the trigeminal nerve project to the VPM (see Figure 10-1, p. 75). There is somatotopic organization in the VPL. Afferents arising from the caudal part of the body project to the lateral part of the VPL, while those from the rostral part of the body project to the medial part of the VPL, so that its most medial part borders on the VPM. Therefore, as for the somatosensory system, there is somatotopy in all the related nuclei from the spinal cord to the primary sensory cortex.

While the **tactile and deep/proprioceptive afferents** project solely to the VPL and VPM, the **nociceptive afferents** project to wider areas of the thalamus. Namely, in addition to the VPL and VPM, the medial ventral nucleus, the intralaminar nucleus, and the medial dorsal nucleus also receive the nociceptive input.

The tertiary neurons from the VPL and VPM project to each corresponding area of the primary somatosensory cortex (S1) in the postcentral gyrus and partially to the second somatosensory cortex (S2) in the upper bank of the sylvian fissure. Nociceptive inputs from multiple thalamic nuclei (see earlier in the chapter) project not only to the S1 and S2 but also to the amygdala, insula, and cingulate gyrus, which are related to the emotional aspect of nociception (see Figure 20-4, p. 195).

It is known that the ventral posterior nucleus also receives afferents of sense of equilibrium and gustatory sense. The vestibular afferents originating from the vestibular nuclei reach the VPL, and the tertiary neurons project to the area just posterior to the S1 face area in the postcentral gyrus and the parietal insular vestibular cortex located at the parieto-insular junction (see Chapter 13-1, p. 93). The gustatory afferents originating from the solitary nucleus project to the VPM, and the tertiary neurons project to the S1 tongue area and anterior insula like the olfactory sense (see Chapter 11-1C, p. 84).

There is a considerable inter-individual variability in the arterial perfusion of the thalamus. The VPL and VPM receive perfusion from the **thalamogeniculate artery**, a branch of the posterior cerebral artery (see Figure 9-3, p. 59). Therefore, if this artery is occluded by infarction, the somatosensory symptoms predominate. In this case, all the somesthetic modalities are lost or impaired on the contralateral side, but especially deep/proprioception and

stereognosis (ability to recognize the shape of an object by touching it) are completely lost. In contrast, the superficial modalities often become hypersensitive, with unpleasant sensation induced by pin prick that is felt over the wider area of the skin and lasts longer than expected. It is also characterized by intolerable pain that might occur spontaneously or is induced by stimuli. Spontaneous pain is often associated with unpleasant dysesthesia and hyperalgesia to stimuli, a condition called **central pain**. In this case, even a soft breeze presented to the skin might cause unpleasant pain (**hyperpathia**) (see Chapter 20-2B, p. 196, and Box 20-4, p. 198).

Central pain following a stroke (**central poststroke pain**) is commonly seen with an infarction involving the posterior lateral thalamus, but it also occurs in brainstem infarction involving the spinothalamic tract. Central poststroke pain usually involves the whole aspect of the contralateral body, but it may be especially severe on the face and the upper limb. In most cases, the pain starts sometime after the stroke, ranging from 2 weeks to even a few years after the stroke. This kind of pain is usually intractable to medical treatment, but some benefit of electrical stimulation of the motor cortex by subdurally placed electrodes was reported (Hosomi et al, 2015; Sokal et al, 2015).

In an infarction of the posterior cerebral artery territory involving the posterior lateral thalamus, ataxia and involuntary movements might be seen in addition to the previously mentioned sensory symptoms. This condition is called **thalamic syndrome** or **Dejerine–Roussy syndrome** (see Box 20-4). In this case, ataxia is considered to be primarily caused by a partial lesion of the cerebellar relay center (see Chapter 16-2C), but it may also be explained as sensory ataxia due to interruption of the proprioceptive input to the VPL. Any kind of involuntary movement may appear with a thalamic lesion, but chorea, dystonia, and slow rhythmic movements (**myorhythmia**) are especially common with an infarction of the posterior lateral thalamus (Lera et al, 2000).

3. VENTRAL ANTERIOR NUCLEUS AND VENTROLATERAL NUCLEUS AS THE MOTOR THALAMUS

Thalamic nuclei related to motor control (motor thalamus) are located in the ventral anterior part of the lateral nuclear group (see Figure 27-1). There is some confusion about the terminology of these motor nuclei. VA and VL are used in the Jones classification, while the Hassler classification

(Voa, Vop, and Vim) is used for deep brain stimulation, though there is no precise correlation between these two classifications (Krack et al, 2002; Molnar et al, 2005). GABAergic fibers originating from the internal segment of the globus pallidus project to the ventral anterior nucleus (VA) and a part of the ventrolateral nucleus (VL) of the ipsilateral thalamus, which correspond to the anterior oral ventral nucleus (Voa) and the posterior oral ventral nucleus (Vop) in the Hassler classification (see Chapter 16-2B, p. 118). Glutamatergic fibers originating from the cerebellar dentate nucleus project through the superior cerebellar peduncle mainly to the VL or the **ventral intermediate nucleus (Vim)** of the contralateral thalamus (Krack et al, 2002; Molnar et al, 2005) (see Chapter 16-2C, p. 127). However, as a result of recording from the thalamic nuclei during the treatment of tremor with deep brain stimulation, the local field potentials recorded from both Vop and Vim show correlation with the tremor of the contralateral hand at the tremor frequency (Kane et al, 2009; Katayama et al, 2005; Marsden et al, 2000).

The motor thalamus receives blood supply from the **tuberothalamic artery**, which bifurcates from the posterior communicating artery (see Figure 9-3, p. 59). If the projection fibers from the basal ganglia to the motor thalamus are affected by infarction, a variety of symptoms such as hypokinesia, thalamic hand, dystonic posture, chorea, and athetosis might appear in the contralateral upper extremity. In contrast, with a lesion of the cerebellar projection to the motor thalamus, the contralateral hand shows ataxia, which may be combined with action tremor and myoclonus. Negative motor symptoms like ataxia appear from the time of stroke, but the involuntary movements commonly develop sometime after the stroke (Lehéricy et al, 2001; Lera et al, 2000).

Abnormal posture of the hand consisting of flexion of the metacarpophalangeal joints and extension of the interphalangeal joints may appear in a vascular lesion of the thalamus (**thalamic hand**). However, as it is often associated with somatosensory abnormalities, dystonic posture, and other involuntary movements (see earlier in the chapter), the thalamic hand may be modified in various ways.

4. MEDIAL DORSAL NUCLEUS AS A RELAY CENTER TO THE PREFRONTAL CORTEX

The medial dorsal (mediodorsal) nucleus (MD) is situated at the most medial part of the thalamus. It receives input from the hypothalamus, amygdala, ventral pallidum, and

nucleus accumbens and sends projections to the prefrontal cortex. It is believed that there is some functional localization in the MD and its projection to the prefrontal cortex is relatively localized. The prefrontal cortex receives inputs of all sensory modalities, and under the influence of will and emotion, it plays an important role in higher brain functions such as attention, selection of behavior and its planning, execution and inhibition, learning, working memory, and long-term memory. Among others, MD and the mammillothalamic tract compose an important memory circuit (see Chapter 23-1B, p. 217). As MD is closely connected to the hypothalamus, its lesion may cause sleep disturbance, autonomic dysfunction, and endocrine abnormalities (see Chapter 21, p. 204).

MD receives blood supply from the **paramedian thalamic artery**, which bifurcates from the basilar communicating artery. As ischemia in the territory of that artery tends to affect MD bilaterally, consciousness is impaired in the acute stage, and loss of initiative and emotional reaction and marked memory disturbance remain in the chronic stage (see Chapter 23-1B and Figure 23-3, p. 221). Furthermore, it often involves the rostral midbrain and causes extraocular muscle paralysis and abnormalities of ocular movements.

5. ANTERIOR NUCLEUS AND MEMORY

The anterior thalamic nucleus is an important component of the Papez circuit and is related to memory (see Chapter 23-1B, p. 217). The information arising from the hippocampus is transmitted through the fornix to the mammillary body, and then through the mammillothalamic tract to the cingulate gyrus. As for the blood supply, the mammillothalamic tract is supplied by either the tuberothalamic artery or the paramedian thalamic artery, and the anterior thalamic nucleus is mainly supplied by the paramedian thalamic artery. Thus, as the anterior nucleus and MD receive their blood supply from the same artery, infarction of that artery affects both structures and causes consciousness disturbance in the acute stage and marked memory disturbance in the recovery phase.

6. PULVINAR AS A RELAY CENTER TO THE PARIETAL LOBE

The pulvinar, a relatively large structure situated in the posterior thalamus, is connected to the parietal and temporal lobes. In the monkey, the inferior part of the pulvinar receives input from the superior colliculus and projects

to the MT area (see Chapter 7-1F, p. 43). As a result of functional magnetic resonance imaging (MRI) studies in human, it is believed that the pulvinar is involved in visual attention, ocular movement, recognition of facial expression, and language (Benarroch, 2015). The pulvinar receives blood supply from the posterior choroidal artery, thalamogeniculate artery, and paramedian thalamic artery. The pulvinar is known to be affected especially in **variant Creutzfeldt-Jakob disease** (“mad cow” disease), which is revealed as an abnormal signal on MRI (see Box 23-8, p. 225) (Shinde et al, 2009).

Aphasia is known to occur in a lesion of the left thalamus, most likely by affecting the functional connection between the pulvinar and the cortex. Among others, naming, writing, and comprehension of spoken language are impaired in a left thalamic lesion (de Witte et al, 2011; Maeshima & Osawa, 2018).

7. INTRALAMINAR NUCLEUS AS A RELAY CENTER OF THE ASCENDING RETICULAR ACTIVATING SYSTEM

The intralaminar nucleus (ILN) is formed of scattered neuronal groups in the internal medullary lamina. It receives input from the brainstem reticular formation and sends projections to the cerebral cortex. There is some somatotopic organization in the ILN, but each subgroup projects to wider area of the cortex than the specific sensory relay nucleus. For example, the centrolateral nucleus of ILN is known to project to the parietal cortex. The main function of ILN is considered to be increasing the excitability of the sensory cortex and facilitating the cortical reception at each specific receptive area. Electrical stimulation of ILN is known to produce an arousal pattern on electroencephalography (EEG). Multiple neurotransmitters are involved in the reticular activating system (see Chapter 4-1, p. 25).

The ILN also sends a powerful projection to the striatum. Among others, the **centromedian parafascicular nucleus complex (CM-Pf complex)** projects to the putamen and caudate nucleus. It is postulated that the CM-Pf complex is related to pain information processing and motor control, although its functions have not been clarified in detail (Benabid, 2009).

As the ILN is distributed widely in the thalamus, it receives blood supply from multiple arteries, including the posterior choroidal artery, thalamogeniculate artery, and paramedian thalamic artery.

8. RELAY NUCLEUS FOR THE VISUAL AND AUDITORY SYSTEM

The lateral geniculate body (LGB) and the medial geniculate body (MGB) are situated in the posterior lateral part of the thalamus, partially surrounded by the pulvinar. The LGB and the MGB are important relay centers for the visual and auditory system, respectively. The LGB projects to the primary visual cortex in the calcarine fissure by maintaining the retinotopic organization (see Chapter 7-1F, p. 43). The MGB projects to the auditory cortex in the Heschl gyrus by maintaining the tonotopic organization (see Chapter 12-1, p. 89). These two nuclei receive blood supply from the posterior choroidal artery, but the LGB is also supplied by the anterior choroidal artery (see Figure 7-5, p. 41).

As all visual inputs from a hemifield cross at the optic chiasm and reach the contralateral LGB, its lesion causes a contralateral hemianopsia (see Figure 7-6, p. 42). In contrast, as the MGB receives inputs from both ears, its unilateral lesion cannot be detected by an ordinary hearing test (see Figure 12-2, p. 90). However, as the proportion of crossed fibers is known to be larger than that of uncrossed fibers, auditory abnormality due to the MGB lesion might be detected by testing auditory spatial recognition or by recording the auditory evoked response with a magnetoencephalogram (MEG).

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ADJUSTMENT TO THE EXTERNAL ENVIRONMENT AND CONTROL OF THE INTERNAL MILIEU (HYPOTHALAMUS AND NEUROENDOCRINOLOGY)

The hypothalamus is a center for controlling important functions necessary for living, in close relation to the autonomic nervous system, endocrine system, sleep and circadian rhythm, emotion and its related behavior, control of the internal milieu necessary for adjusting to the external environment, and—above all—integration of all those functions.

As for autonomic function, the sympathetic nervous system and the parasympathetic nervous system are clearly distinguishable in the brainstem and spinal cord, but the distinction between the two systems is not clear in the hypothalamus (see Chapter 21-1, p. 204). The autonomic system in the hypothalamus is rather arranged in terms of specific functions.

1. STRUCTURE AND FUNCTION OF THE HYPOTHALAMUS

The hypothalamus is largely divided into the medial part and the lateral part. In the **medial part**, there are the paraventricular nucleus, supraoptic nucleus, and suprachiasmatic nucleus anteriorly; the dorsomedial nucleus, ventromedial nucleus, and infundibular nucleus in the middle; and the posterior nucleus posteriorly (Figure 31-1). Furthermore, the mammillary nucleus is in the mammillary body, which is located at the base of the third ventricle.

In the **lateral part**, there are no distinct nuclear divisions, and it is called the **lateral hypothalamic area** en bloc. The lateral hypothalamic area is penetrated by poorly demarcated fiber tracts called the **medial forebrain bundle**, which projects to the frontal lobe and amygdala.

The paraventricular nucleus, suprachiasmatic nucleus, and mammillary body are clearly distinguishable, but other nuclei are not clearly demarcated and many of those nuclei are connected with each other. The hypothalamus as a whole receives inputs from many brain areas and sends projections to many areas. In addition to multiple neurotransmitters, including acetylcholine, norepinephrine, dopamine, serotonin, histamine, and neuropeptides, the hypothalamus is characterized by the fact that multiple hormones participate in the information transmission.

2. NEUROENDOCRINOLOGY

The hypothalamus is the highest center for controlling the endocrine system. Information from the hypothalamus is transmitted to each endocrine organ via the **pituitary gland (hypophysis)**. The pituitary gland is divided into anterior lobe and posterior lobe.

A. ANTERIOR LOBE OF THE PITUITARY GLAND (ADENOHYPHYSIS)

The anterior lobe of the pituitary gland, also called the adenohypophysis, receives releasing hormones from the hypothalamus via the **hypophyseal portal vascular system**, and the pituitary epithelial cells secrete growth hormone, thyroid-stimulating hormone (TSH), adrenocorticotrophic hormone (ACTH), gonadotropic hormone, and prolactin. These hormones stimulate their respective target organs to produce hormones. There are two kinds of gonadotropic hormones: follicle-stimulating hormone and luteinizing hormone.

There are three kinds of epithelial cells in the anterior lobe of hypothalamus: eosinophilic, basophilic, and chromophobe cells. It is known that the eosinophilic cells produce growth hormone and prolactin, and other hormones are postulated to be produced by basophilic cells.

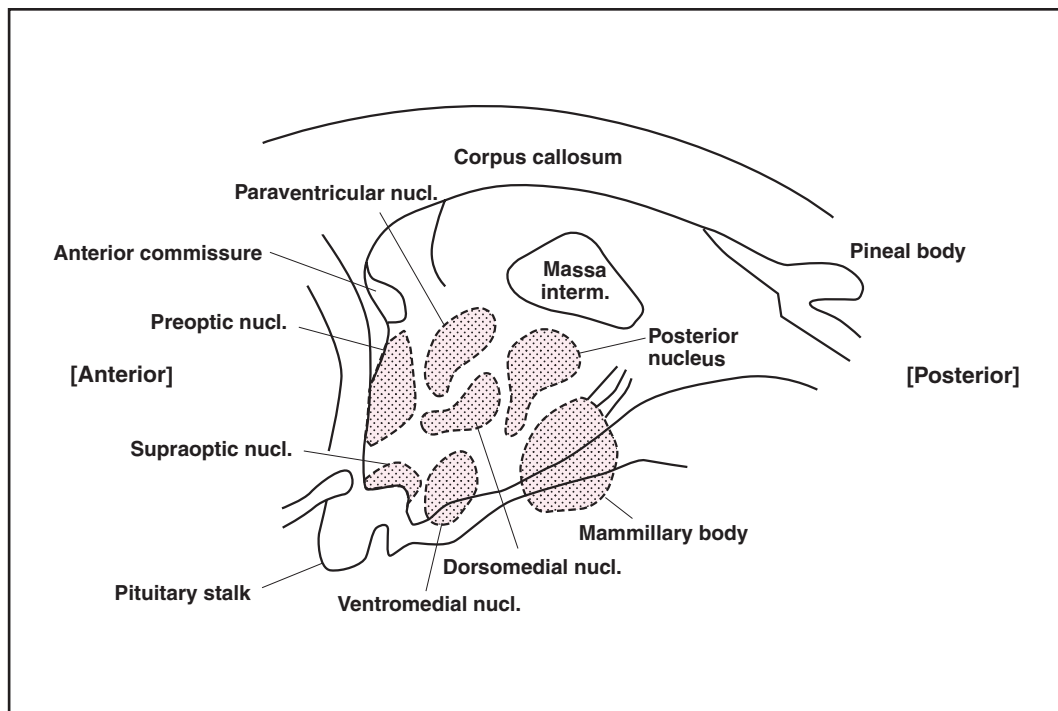


Figure 31-1 Schematic diagram showing main nuclei of the hypothalamus. Sagittal view. nucl.: nucleus, intermed.: intermedia. Cited and modified from Brodal, 2010, with permission.

Eosinophilic pituitary adenoma causes **gigantism** in children and **acromegaly** in adults as a result of excessive production of growth hormone. Furthermore, as a result of excessive production of prolactin, it may cause **galactorrhea** or **galactorrhea–amenorrhea syndrome** in women.

ACTH-producing pituitary adenoma, as a result of excessive production of glucocorticoid, causes central obesity, moon face, buffalo hump, striae and plethora of the skin, hypertension, proximal muscle weakness, and hypokalemia. This condition, called **Cushing disease**, is more common in women than in men. A similar clinical picture may be seen in patients who are receiving corticosteroid treatment, or in association with adrenal tumor and aberrant ACTH-producing tumor. This condition is called **Cushing syndrome**.

The releasing hormones produced in the hypothalamus are also known to be transmitted to brain regions other than the pituitary gland through nerve fiber connections. Those examples are thyrotropin-releasing hormone (TRH), corticotropin-releasing hormone (CRH), and somatostatin.

B. POSTERIOR LOBE OF THE PITUITARY GLAND (NEUROHYPOPHYSIS)

The posterior lobe of the pituitary gland, also called the neurohypophysis, is directly innervated by the hypothalamus

via the **hypothalamohypophyseal tract**. The related hormones are **antidiuretic hormone (ADH)**, which is also called **vasopressin**, and **oxytocin**. These two hormones are produced in the paraventricular nucleus and the supraoptic nucleus, and they are examples of unique hormones that are transmitted to the posterior lobe via the axonal flow of the tract. Thus, cells of these two nuclei are called **neurosecretory cells**. Among these nuclei, ADH is primarily produced in the supraoptic nucleus, and oxytocin is mainly produced in the paraventricular nucleus. ADH acts on the water channel, aquaporin in the kidney, and promotes reabsorption of water, so that it suppresses urinary excretion. Cells of the supraoptic nucleus and other hypothalamic areas are considered to be equipped with receptors sensitive to the osmotic pressure of blood, thus controlling the water content. Furthermore, ADH is known to act on the blood vessels to contract their smooth muscles—which is why it is also called vasopressin.

If ADH secretion is impaired due to a hypothalamic lesion, its antidiuretic action is lost, causing **diabetes insipidus**. In contrast, if excessive ADH is produced by various etiologies, the osmotic pressure and sodium level of blood are decreased because urinary excretion is decreased, and the osmotic pressure of urine is increased. This condition, called **syndrome of inappropriate secretion of ADH (SIADH)**, should be kept in mind when dealing with hyponatremia

of unknown etiology. SIADH is not infrequently seen in patients with neuromyelitis optica who have positive antibody against aquaporin-4 (see Box 17-4, p. 148), and it may appear even as the initial symptom of neuromyelitis optica (Iorio et al, 2011).

Wolfram syndrome is characterized by familial occurrence of diabetes insipidus. This disease is transmitted via autosomal recessive inheritance and affects young people with diabetes insipidus, diabetes mellitus, optic atrophy, and deafness (DIDMOAD). Multiple mutations of the *WFS1* gene are known, and some patients might also show cognitive disturbance and epileptic seizures (Chaussonnet et al, 2011).

Oxytocin is produced mainly in the paraventricular nucleus. As it provokes contraction of the uterine smooth muscles, it plays an important role in obstetric deliveries. Furthermore, oxytocin promotes milk secretion by increasing contraction of the mammary smooth muscles. It is considered that sucking of mammary papilla by the newborn sends the somatosensory input to the paraventricular nucleus, which activates the secretory cells to promote oxytocin and increases the milk secretion.

The pituitary gland may be affected by congenital tumors, especially **craniopharyngioma**, which is more common in children and may present with various endocrine disorders such as small stature, hypothyroidism and diabetes insipidus, and visual disturbance associated with signs of increased intracranial pressure.

3. ADJUSTMENT TO THE EXTERNAL ENVIRONMENT AND CONTROL OF THE INTERNAL MILIEU

The hypothalamus is important for the control of body temperature, feeding and nutrition, circadian rhythm, and sexual functions. Functional localization within the hypothalamus is not clearly known, but it is often named by attaching “center” to each function.

A. CONTROL OF CIRCADIAN RHYTHM

For this function, the **suprachiasmatic nucleus**, located just above the optic chiasm, receives impulses generated by light through a branch of the optic tract (see Chapter 7-1D, p. 42) and activates the pineal gland to promote secretion of

melatonin, which is important for controlling the circadian rhythm.

B. CONTROL OF FEEDING

Based on experimental stimulation of the hypothalamus in animals, it is postulated that the lateral part of the hypothalamus promotes feeding (**feeding center**) and the medial part, especially the ventromedial nucleus, suppresses feeding (**satiety center**). Lesions of the satiety center cause **hyperphagia**. Abnormal behavior characterized by hyperphagia and hypersomnia is known as **Kleine-Levin syndrome** (see Chapter 27-6, p. 258). This syndrome is considered to be caused by a disturbed control mechanism of both circadian rhythm and feeding.

C. CONTROL OF BODY TEMPERATURE

As a result of experimental studies in animals, the anterior part of the hypothalamus is activated when the core temperature is excessively elevated, and it lowers the temperature by dilating the subcutaneous blood vessels and by increasing sweating. In contrast, the posterior part becomes active when the core temperature is excessively low, and it elevates the temperature by promoting shivering and voluntary muscle contraction. Thus, these parts of the hypothalamus are considered to serve as a thermostat. As the hypothalamus is commonly affected by a necrotic lesion associated with small hemorrhages in **Wernicke encephalopathy**, the control mechanism of body temperature is severely impaired, resulting in marked fluctuation of the body temperature (**poikilothermia**) (see Box 23-3, p. 220).

As the **mammillary nucleus** is an important component of memory circuits, its lesion causes marked memory loss and confabulation (**Korsakoff syndrome**) (see Chapter 23-1B, p. 218). Furthermore, although the functional localization is not clear, the hypothalamus is known to control sexual behavior via gonadotropic hormones and the autonomic nervous system.

D. HOMEOSTASIS

As already described, the hypothalamus plays an important role in integrating the control mechanisms of the autonomic nervous system, endocrine system, and behaviors, and it is the control center indispensable for the maintenance and

control of living. The hypothalamus is also considered to be involved in the control of the immunologic system. Control of the interaction between psychological function and somatic activities is another important function of the hypothalamus. Thus, a lesion of the hypothalamus may cause psychosomatic diseases, and that is why the hypothalamus is a target organ for research into psychosomatic medicine.

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INFECTIOUS DISEASES OF THE NERVOUS SYSTEM

Infectious diseases of the nervous system with microorganisms are important because many of them have a favorable prognosis if the appropriate treatment is initiated at an early clinical stage. Some infections present with acute onset of neurological symptoms, while others present more chronically.

1. BACTERIAL INFECTION OF THE NERVOUS SYSTEM

Bacterial infection of the central nervous system (CNS) mainly causes purulent meningitis clinically characterized by fever, headache, impairment of consciousness, and signs of meningeal irritation such as nuchal stiffness. The most frequent causative bacteria of acute meningitis in adults are *Streptococcus pneumoniae*, *Neisseria meningitidis*, and *Hemophilus influenzae* (Box 32-1). Subacute meningitis is caused by other agents such as tuberculosis (Wen et al, 2019) and fungus, including *Cryptococcus* and *Candida albicans*. The features of a subacute duration of illness and a low percentage of neutrophils in the cerebrospinal fluid (CSF) are highly predictive of tuberculous meningitis compared with that caused by other bacteria (Luo et al, 2018). Tuberculous meningitis is often complicated by encephalitis because the bacilli tend to involve the arterial walls supplying the brain parenchyma.

The causes and features of chronic meningeal inflammation or infection differ from those of acute meningitis, and the evaluation includes a search for systemic disorders such as sarcoid, as well as infectious and noninfectious parameningeal lesions, and may require meningeal biopsy (Aksamit, 2021).

2. SPIROCHETE INFECTION OF THE NERVOUS SYSTEM

Syphilis is caused by infection with *Treponema pallidum*. It is usually contracted by sexual contact with another

infected individual. With successful treatment with penicillin, the numbers of typical cases have decreased, but atypical clinical presentations have been reported, especially in patients with HIV infection (Skalnaya et al, 2019).

The CNS is infected by *Treponema*, causing inflammatory pathology in the meninges and brain parenchyma. Clinically it used to be divided into the early form (meningitis and meningovascular disorder) and the late form (general paresis and tabes dorsalis) (Box 32-2). Syphilis is diagnosed by the fluorescent treponemal antibody absorption test (FTA-ABS) (Park et al, 2019). CNS infection is confirmed by CSF VDRL (Venereal Disease Research Laboratory) testing, which is thought to be highly specific for CNS syphilis, although its sensitivity is about 50% to 60% (Tuddenham & Ghanem, 2018).

A spirochete mediated by a tick, *Borrelia burgdorferi*, causes Lyme disease. Lyme disease usually presents with a typical rash, erythema migrans, and subsequently there can be neurological features, including meningitis and cranial and peripheral neuropathy (Lantos et al, 2021; Schoen, 2020). Patients may present with bilateral facial nerve palsy either simultaneously or independently. Bilateral facial nerve palsy is also caused by sarcoidosis and leprosy (see Box 11-3). Bannwarth syndrome consisting of aseptic meningitis, limb weakness, and severe radicular pain may be due to Lyme disease (Dabir & Pawar, 2018).

3. VIRAL INFECTION OF THE NERVOUS SYSTEM

Many kinds of virus cause acute meningoencephalitis or meningitis. The common pathogens are herpes simplex virus and varicella zoster virus. In the United States, West Nile virus, Epstein-Barr virus and HIV are also common (see Box 32-1). Japanese B virus encephalitis is common in Japan.

Zika virus has been endemic in many parts of the world. Zika virus is a single-stranded RNA virus

BOX 32-1 ACUTE MENINGOENCEPHALITIS

Unless it is appropriately treated at early stage, acute infectious meningoencephalitis might leave severe neurological residuals due to internal hydrocephalus and/or cerebral parenchymal lesions. In **aseptic meningitis (viral meningitis)** and **purulent meningitis** due to bacterial infection, the meninges are primarily affected rather than the brain parenchyma. In purulent meningitis, it is important to identify causative bacteria promptly and to give appropriate antibiotics. In adults, *Streptococcus pneumoniae*, *Neisseria meningitidis*, and *Haemophilus influenzae* type B commonly cause acute meningitis (Wall et al, 2021). Causative viruses for aseptic meningitis include coxsackie A and B, echovirus, and mumps. By contrast, subacute meningitis is often caused by tuberculosis and fungi such as *Cryptococcus* and *Candida albicans*.

The most common cause of **acute encephalitis** is virus infection, followed by autoimmune encephalitis. In viral encephalitis, the common pathogens differ depending on the geographic region. The most common pathogen in the United States is herpes simplex virus, followed by varicella-zoster virus, West Nile virus, Epstein–Barr virus, and HIV (Singh et al, 2015). Japanese B virus encephalitis is also common in Japan. Advanced age, immunocompromised state, and coma portend a worse prognosis (Singh et al, 2015). It is important to administer antiviral agents from the early stage, combined with symptomatic treatment for brain edema and convulsive seizures. It should be noted that the prevalence and incidence of autoimmune encephalitis are comparable to those of infectious encephalitis (Dubey et al, 2018).

transmitted by mosquitos and through sexual contact with infected individuals. It causes microcephaly in neonates and polyneuritis (Kassavetis et al, 2016; White et al, 2016).

A new coronavirus, SARS-CoV-2, which causes acute respiratory symptoms, emerged in Wuhan City, China, in December 2019 and rapidly spread worldwide as a pandemic. Most of the patients do not develop severe clinical manifestations in the early stage of the disease, but later in the clinical course some patients rapidly develop acute respiratory distress syndrome (ARDS) and multiple organ failure. In these severe cases, excessive release of cytokine (cytokine storm) was shown to play a role in the pathogenesis (Ye et al, 2020). The primary manifestations are respiratory and cardiac, but neurological symptoms are commonly reported (Romero-Sánchez et al, 2020) (Box 32-3).

Poliomyelitis, or polio for short, has mostly been eradicated from the developed world, but it should not be forgotten. It is caused by an enterovirus affecting anterior horn cells in the spinal cord and motor neurons in the brainstem, leading to flaccid weakness and paralysis (Menant & Gandevia, 2018). Patients may recover partially, but then about 20 years later they have recurrent weakness in the post-polio syndrome.

Since 2012, several clusters of cases presenting with acute flaccid limb weakness with spinal cord gray matter lesions on imaging or evidence of spinal cord motor neuron injury on electrodiagnostic testing have been reported in the United States. This condition was especially common in male children. Prodromal symptoms such as fever and respiratory and gastrointestinal symptoms have been reported. The CSF showed pleocytosis of moderate degree with a lymphocytic predominance. The clinical picture resembled anterior poliomyelitis, but no virus has been detected in the CSF. The condition is now called **acute flaccid myelitis**, and the epidemiologic evidence supports an association with enterovirus D68 and A71 (Fatemi & Chakraborty, 2019; Murphy & Pardo 2020).

4. PARASITIC INFECTION OF THE NERVOUS SYSTEM

Parasitic infections of the CNS are caused by single-celled protozoa and multicellular metazoan (Pitella, 2013). The protozoal infections include malaria, American trypanosomiasis, human African trypanosomiasis, toxoplasmosis, amebiasis, trichinosis, microsporidiasis, and leishmaniasis. The metazoal infections are grouped into flatworms, which include Trematoda and Cestoda, and roundworms or nematodes. The trematodes include schistosomiasis and paragonimiasis. The most important cestode is the tapeworm *Taenia solium*, which causes neurocysticercosis.

The ubiquitous protozoan *Toxoplasma gondii* and the nematode larvae *Toxocara canis* infect significant proportions of the human population (Garcia et al, 2019). Many of them might be asymptomatic, and these parasites might present with clinical manifestations only when the hosts are immunocompromised due to AIDS or receiving immunologic treatments (opportunistic infection) (Brito de Oliveira et al, 2016). In fact, CNS infection with *T. gondii* is the most common cause of brain mass lesions in HIV-infected patients (Marra, 2018) (see Box 23-7, p. 224). It usually presents as one or more abscesses, but it can also cause diffuse encephalitis or ventriculitis. Neuroimaging shows round, isodense or hyperdense lesions with ring

BOX 32-2 GENERAL PARESIS AND TABES DORSALIS

These two conditions become clinically manifest in patients who were infected by *Treponema pallidum* (syphilis) many years before. General paresis or dementia paralytica is due to infection of the brain parenchyma. Clinically it is characterized by dementia, behavior disturbance, and convulsion. Tabes dorsalis is a chronic inflammatory disease of the spinal cord, mainly involving the dorsal root ganglion and the posterior column. Clinically it is characterized by loss of proprioceptive sensation with sensory ataxia, loss of deep pressure pain (Abadie's sign) and tendon reflexes in the lower extremities, lancinating pain, marked hypotonia, and sphincter disturbance. Argyll Robertson pupil (small pupil with loss of light reflex with the preserved convergence reflex, and normal vision) is commonly seen in patients with tabes dorsalis.

enhancement in the hemispheric gray–white matter junction, deep white matter, or basal ganglia.

Other parasitic infections such as malaria and neurocysticercosis are widespread in developing countries (Garcia et al, 2019). **Neurocysticercosis** causes adult-onset seizures, headache, and intracranial hypertension (Garcia, 2018). Neuroimaging gives a clue to diagnosis of this condition. The typical findings are cystic lesions without scolex, enhancing lesions, multilobulated cysts, and calcifications (Del Brutto et al, 2017; Wildman et al, 2020).

5. FUNGAL INFECTION OF THE NERVOUS SYSTEM

Fungal infections of the CNS have become common in immunocompromised hosts. The representative fungi causing the CNS infection are *Cryptococcus neoformans*, *Candida* species, *Aspergillus* species, and *Mucor* species (Bloch & Bailin, 2019; Góralaska et al, 2018). Clinical presentations include meningitis, encephalitis, hydrocephalus, cerebral abscesses, and stroke syndromes. Early diagnosis by demonstration of fungi in the CSF is important because CNS involvement might lead to fatal consequences and also because some of these fungi (*Cryptococcus*, *Candida*, and *Aspergillus*) might respond to treatment with amphotericin B.

6. TRANSMISSIBLE FORM OF PRION DISEASE

As described in Box 23-8 in relation to periodic myoclonus, **prion** diseases are neurodegenerative disorders

BOX 32-3 NEUROLOGICAL MANIFESTATIONS OF COVID-19

COVID-19 involves multiple organs, and the nervous system is not an exception. In fact, the central as well as the peripheral nervous system and muscles are involved. For example, neurological manifestations were reported in more than half of patients hospitalized with this disease in two centers in Spain (Romero-Sánchez et al, 2020). In a prospective study of 4,491 patients with COVID-19 in New York, 13.5% developed a new neurological disorder in a median of 2 days from COVID-19 symptom onset (Frontera et al, 2021). Loss of olfactory sense (anosmia) and loss of taste sense (ageusia) are commonly reported in the early clinical stage (Pellegrino et al, 2020). In relation to this, magnetic resonance imaging (MRI) findings suggestive of edema of the olfactory bulb or cleft have been reported (Eliezer et al, 2020; Laurendon et al, 2020). Cranial nerve palsy with ophthalmoparesis, Miller Fisher syndrome, and lower cranial neuropathy have been reported (Decavel et al, 2020; Dinkin et al, 2020; Gutiérrez-Ortiz et al, 2020). Headache and myalgia are also common complaints in the early stage (Romero-Sánchez et al, 2020). CNS involvement includes meningitis, encephalitis and diffuse leukoencephalopathies, acute disseminated encephalomyelitis or myelitis, and acute cerebrovascular diseases (Ahmad & Rathore, 2020; Asadi-Pooya & Simani, 2020; Ellul et al, 2020; Huang et al, 2020; Kanberg et al, 2020; Kremer et al, 2020; Paterson et al, 2020; Pugin et al, 2020; Virhammar et al, 2020). Neurological manifestations are especially common in patients more severely affected with COVID-19 (Pinna et al, 2020). Patients presenting with involuntary movements such as generalized myoclonus or opsoclonus myoclonus ataxia syndrome have also been reported (Rábano-Suárez et al, 2020; Shah & Desai, 2021). Polyneuropathy like Guillain-Barré syndrome has been reported (Romero-Sánchez et al, 2020).

In a postmortem neuropathologic study of 43 patients with COVID-19 in Germany, the most common finding was pronounced inflammatory changes in the brainstem, but there was no evidence for CNS damage directly caused by the virus (Matschke et al, 2020). An analysis of CSF biomarkers in patients with COVID-19 and neurological symptoms revealed that findings suggestive of typical CNS viral infections were absent (Edén et al, 2021). Excessive release of cytokine (cytokine storm) is believed to aggravate an inflammatory process in multiple organs (Ye et al, 2020). A pathologic study of muscles and nerves in patients with COVID-19 revealed inflammatory/immune-mediated damage likely related to the release of cytokines (Suh et al, 2021). Cerebral infarctions also occur in patients with COVID-19 based on

the disturbance of the coagulation mechanism (coagulopathy) and antiphospholipid antibody (Paterson et al, 2020; Zhang et al, 2020). Whether acute ischemic stroke is more common in patients with COVID-19 than in those with other respiratory infection is still controversial, but the mortality is high in stroke patients with COVID-19 infections (Fridman et al, 2020; Qureshi et al, 2020).

It has become increasingly clear that many patients may go on to develop lasting symptoms (“long COVID”), such as extreme fatigue, muscle and joint pain, breathlessness, heart palpitations, loss or alteration of taste and smell, gastrointestinal distress, and problems with attention, memory and cognition, as stated in a December 7, 2020, editorial in *Nature Medicine*.

characterized by protease-resistant abnormally shaped proteins known as prions, and some of them (kuru, variant Creutzfeldt–Jakob disease [CJD] and iatrogenic CJD) are transmissible (Baldwin & Correll, 2019; Ironside et al, 2017) (see Box 23-8). Pathologically it used to be called progressive spongiform encephalopathy, and clinically it is characterized by progressive dementia, paralysis, and myoclonus. The clinical course is often rapidly progressive, almost resembling subacute encephalitis.

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NEUROLOGICAL EMERGENCY

A **neurological emergency** is defined as a condition that is life-threatening or in which the patient faces a poor functional recovery unless treated promptly. Among others, respiratory paralysis and disturbance of consciousness are life-threatening (Box 33-1). Regarding disturbance of consciousness, diagnosis and treatment of coma will be described in Section 6. Stroke is one of the most common neurological emergencies. This Chapter describes the pathophysiological basis of each emergency condition, the clinical diagnosis and laboratory testing, and emergency treatments.

1. DISORDERS WITH RESPIRATORY PARALYSIS

When faced with breathless patients, it is important to exclude mechanical suffocation such as aspiration before considering paralysis of the respiratory muscles due to neurological disorders. Among neurological disorders, myasthenia gravis, Guillain–Barré syndrome, acute anterior poliomyelitis, brainstem encephalitis, and acute intoxication with puffer poison (tetrodotoxin) and botulinum toxin are important causes of respiratory paralysis (Wijdicks, 2017) (Box 33-2). It is noteworthy that motor neuron diseases, although they are chronic progressive conditions, may cause respiratory paralysis rather acutely.

A. CRISIS OF MYASTHENIA GRAVIS

Symptoms of myasthenia gravis may become suddenly worse (crisis). There are two kinds of crisis: myasthenic crisis and cholinergic crisis.

In **myasthenic crisis**, acetylcholine is insufficient at the nicotinic neuromuscular junction due to worsening of myasthenia in association with upper respiratory infection, insufficient dose of cholinesterase inhibitors, or reduced effectiveness of the drugs. Clinically, excretion

of airway discharge becomes difficult because of weakness of the respiratory as well as bulbar muscles. Some patients who were treated with thymectomy develop postoperative myasthenic crisis. Kanai et al. (2017) reported that postoperative myasthenic crisis can be predicted if three clinical factors—vital capacity less than 80%, disease duration of less than 3 months, and bulbar symptoms immediately before thymectomy—are present.

Cholinergic crisis is caused by excessive doses of cholinesterase inhibitors. Clinically, airway secretion is increased due to increased muscarinic transmission, which worsens the respiration. For either type of crisis, the first procedure to be taken is to aspirate the airway, and if necessary, the airway should be managed by intubation.

The two types of crisis can be differentiated from the history and the physical examination, but its confirmation could be done by the Tensilon test.

B. EDROPHONIUM (TENSILON) TEST

This test is done by intravenous injection of cholinesterase inhibitor to confirm the diagnosis of myasthenia gravis and to distinguish between myasthenic crisis and cholinergic crisis. As the edrophonium increases airway discharge due to increased muscarinic transmission of acetylcholine, it may make the judgment of the test difficult in some cases. In that case, it is useful to inject 0.5 mg of atropine sulfate subcutaneously or intramuscularly before the test to prevent excessive airway discharge. As atropine inhibits only muscarinic transmission and not nicotinic transmission, it does not affect neuromuscular transmission (see Chapter 21-1, p. 204).

To carry out the **Tensilon test**, 10 mg of edrophonium is intravenously injected, but it is effective to inject 2 mg over the first 30 seconds, and if the myasthenic symptoms either improve or worsen, the test can be stopped, and it is not necessary to complete the test. If no change occurs with 2 mg, then the remaining 8 mg may be injected over

BOX 33-1 CAUSES OF NEUROLOGICAL EMERGENCIES

1. Respiratory paralysis
 - a. Crisis of myasthenia gravis
 - b. Guillain–Barré syndrome
 - c. Motor neuron diseases
 - d. Acute anterior poliomyelitis
 - e. Brainstem encephalitis
 - f. Brainstem vascular lesion
 - g. Acute intoxication: puffer (tetrodotoxin), botulinum, organophosphate, etc.
2. Disturbance of consciousness
 - a. Status epilepticus
 - b. Non-epileptic disorders
 - Metabolic and toxic encephalopathies
 - Anoxic encephalopathy
 - Meningoencephalitis
 - Intracranial hemorrhage
 - Cerebral herniation
3. Continuous spasm other than epilepsy
 - a. Tetanus
4. Conditions in which functional recovery is poor unless treated promptly
 - a. Acute compression of spinal cord
 - b. Progressive ischemic cerebrovascular diseases
 - c. Acute meningoencephalitis

another 30 seconds. If muscle weakness improves by this test, it suggests myasthenic crisis, and if it worsens, it suggests cholinergic crisis.

For the immunotherapy of crisis, Mandawat et al. (2010) reported that intravenous immunoglobulin (IVIg) was equally effective with fewer complications and less expense compared with plasmapheresis. As a result of a multicenter analysis of 250 cases requiring mechanical ventilation in Germany, patients who received plasma exchange or immunoadsorption as first-line therapy needed invasive ventilation less often (Neumann et al, 2020).

C. GUILLAIN–BARRÉ SYNDROME

In many cases of Guillain–Barré syndrome, muscle weakness begins in the lower extremities and ascends up to the thoracic level, and may cause dyspnea. In some cases, however, dyspnea might predominate from the early clinical stage. Any pattern of weakness might well occur, including involving cranial muscles, and it is typically associated with a reduction in tendon reflexes. Treatment primarily

BOX 33-2 ACUTE POISONING CAUSING RESPIRATORY PARALYSIS

As intoxication may occur unexpectedly, it is always important to prepare for the emergency. Among others, intoxications with puffer toxin and botulinum toxin are notorious for causing acute respiratory paralysis. As a result of oral intake of the **puffer toxin (tetrodotoxin)**, which is contained in the liver and testis of the puffer fish, the voltage-gated sodium channel is blocked. Clinically, perioral numbness begins several minutes after intake of the toxin, which spreads to the whole body, and flaccid paralysis rapidly develops, causing breathlessness. It can only be treated by artificial support of respiration. Botulism is caused by oral intake of the **botulinum toxin**, which is produced by *Clostridium botulinum*. A typical source of intoxication is homemade canned food prepared at a high altitude, where water boils at a lower temperature. Botulinum toxin severs a protein in the presynaptic nerve terminal that is important for the release of neurotransmitters, and thus inhibits acetylcholine release at the neuromuscular junction. Clinically, first the abducens muscle is paralyzed several hours after intake of the toxin, followed by paralysis of other extraocular muscles and bulbar muscles, and finally respiratory paralysis. For treatment, respiration should be supported first, and then gastric lavage and administration of antitoxin are necessary. It is noteworthy that local injection of this toxin is currently used for treatment of various movement disorders, pain and disorders of the autonomic nervous system, as well as wrinkles.

In intoxication with insecticides containing **organophosphate**, cholinesterase is inhibited so that nicotinic as well as muscarinic cholinergic transmission becomes hyperactive, causing respiratory paralysis and increased airway secretion. For treatment of the acute phase, in addition to respiratory support, atropine sulfate and pralidoxime (PAM) may be administered.

consists of immunosuppressive therapy, but airway management with a respirator may be needed in severe cases. IVIg and plasma exchange have been proven effective (van den Berg et al, 2014). A large case series from the United States showed that, even in the acute stage, the use of plasmapheresis is decreasing while IVIg is increasingly used (Alshekhlee et al, 2008).

D. ACUTE ANTERIOR POLIOMYELITIS

In the past, acute anterior poliomyelitis (polio) due to polio virus infection was the most common cause of acute infectious diseases causing motor paralysis. Clinically it is

characterized by weakness of any of the four extremities (monoplegia), but more extremities may be also involved to a certain extent, and tendon reflexes are diffusely lost. Sensory symptoms may be also seen, although these are rare. Polio has nearly disappeared in most advanced countries as a result of the wide use of vaccination, but it may still be encountered in developing regions (see Chapter 32-3, p. 281).

E. ACUTE BRAINSTEM LESION

As there is a respiratory center in the brainstem, acute respiratory paralysis may be seen in inflammatory or vascular lesions of the brainstem. A representative form of brainstem encephalitis is **Bickerstaff brainstem encephalitis**. Infarction or hemorrhage of brainstem might cause sudden dyspnea. Mechanisms of respiratory abnormalities due to brainstem lesions are described in Section 6-C.

F. AMYOTROPHIC LATERAL SCLEROSIS

As amyotrophic lateral sclerosis is a slowly progressive disease, possible occurrence of dyspnea can be predicted in most cases. In some cases, however, it might show rapid progression at some point during the clinical course, resulting in acute respiratory paralysis. Although it depends on the patient's circumstance, it is advisable to explain the situation to the patient and family beforehand, and to discuss what to do when the respiration becomes very weak.

2. DISORDERS WITH DISTURBANCE OF CONSCIOUSNESS

As discussed in Chapter 4-2 (p. 26), when faced with patients with disturbance of consciousness, it is important to judge whether it is due to a decrease or loss of neuronal excitability of the consciousness center (non-epileptic disorders) or due to its excessive excitation (epileptic disorders). In a disturbance of consciousness associated with convulsive seizures, the patient should be placed in a safe environment with clothes loosened and with the tongue guarded from injury. For adult patients, 10 mg of diazepam can be intravenously injected while watching the respiration. If the seizures still occur repeatedly, more intensive treatment for status epilepticus is warranted (see p. 289).

Patients with non-epileptic disorders presenting with disturbance of consciousness may be examined systematically as shown in Figure 33-1, for example. First, **vital**

signs should be checked, and if necessary, airway support and other primary care should be provided. At the same time, level of consciousness disturbance (**coma scale**) can be evaluated. Then, the eyes, face, and extremities should be observed, looking for any **asymmetry** between the two sides, including conjugate deviation of eyes, asymmetry of nasolabial lines, and asymmetry of posture, muscle tone, strength, tendon reflexes, and plantar reflex of extremities. If any of these items are found to be asymmetric, it highly suggests the presence of a focal organic lesion in the brain parenchyma, and a neuroimaging study with computed tomography (CT) or magnetic resonance imaging (MRI) is indicated. In case of cerebral infarction, the conventional CT or MRI may not show any abnormality within 24 hours after the onset of clinical symptoms. However, special imaging techniques such as diffusion MRI may detect a very early ischemic lesion. If the neuroimaging study does not reveal any abnormality despite clinical evidence of asymmetry, blood and cerebrospinal fluid should be examined for possible inflammatory diseases.

If there is no asymmetry in the eyes, face, or extremities, signs suggestive of **meningeal irritation** should be checked, including nuchal stiffness and Kernig sign (see Chapter 15-1E, p. 107). If nuchal stiffness or Kernig sign is judged to be positive, then a neuroimaging study should be done looking for any evidence of intracranial hemorrhage. In this regard, it should be kept in mind that nuchal stiffness may not be detected until 24 hours after subarachnoid hemorrhage. If there is no evidence of hemorrhage on the neuroimaging study in patients with positive meningeal irritation, the cerebrospinal fluid should be examined for possible meningitis.

If there is no sign suggestive of meningeal irritation, the patient may have a metabolic or toxic disorder, or anoxic encephalopathy. In these cases, an electroencephalogram (EEG) should be obtained, and blood should be tested for electrolytes, glucose level, liver functions including ammonium, renal functions, and various endocrine tests (Box 33-3). Furthermore, it is extremely important to save a blood sample for possible intoxication with drugs or insecticides.

If the EEG shows **triphasic waves**, it most likely suggests metabolic encephalopathy. In particular, when the triphasic waves occur almost continuously, it strongly suggests hepatic encephalopathy (Figure 33-2). In cases of severe anoxic encephalopathy, periodic synchronous discharges (PSD) as seen in Creutzfeldt-Jakob disease may appear on the EEG, but PSD in anoxic encephalopathy does not continue and tends to change rapidly whereas PSD

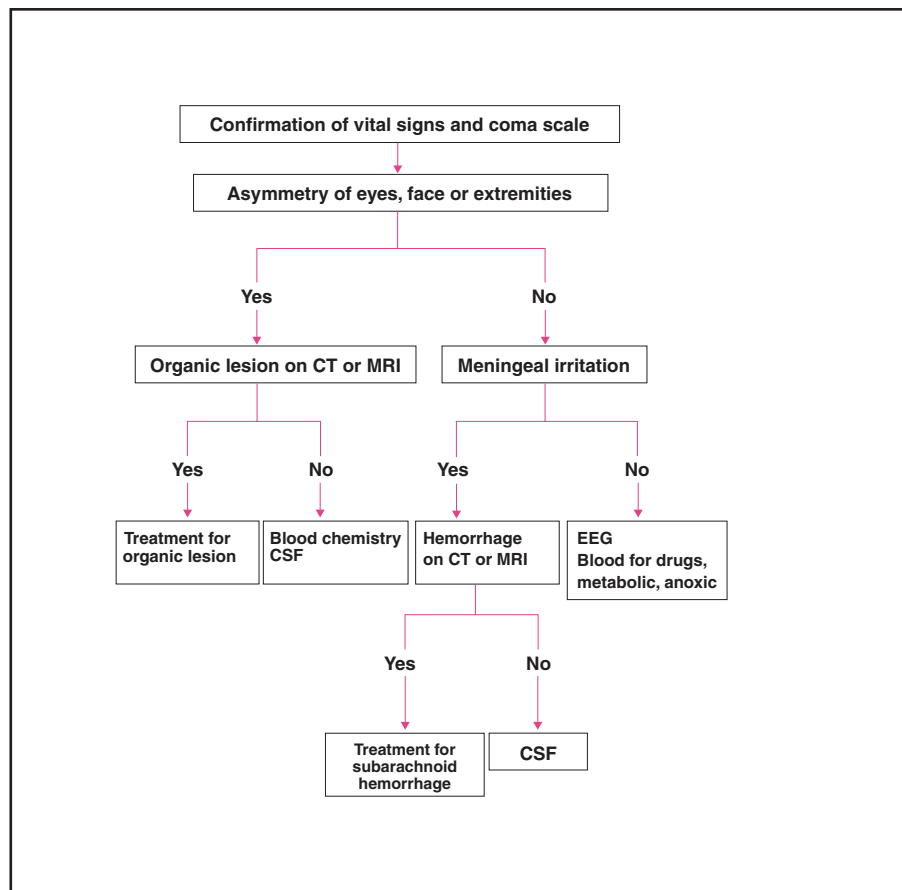


Figure 33-1 Flow chart showing the procedures used to examine patients with disturbance of consciousness. CSF: cerebrospinal fluid.

in Creutzfeldt–Jakob disease persists at least for a certain period of the clinical course.

3. STROKE

As described in Box 1-1, cerebrovascular diseases (CVDs) are largely classified into cerebral infarction, intracerebral hemorrhage, venous sinus thrombosis, and subarachnoid hemorrhage. Among these, cerebral infarction is further classified into atherothrombosis, lacunar infarction, and cardiogenic embolism. There are three forms of atherothrombosis: thrombotic, embolic, and hemodynamic. Among these, CVDs of sudden or acute onset (**stroke**) require emergent attention, because acute ischemic stroke (AIS) improves faster and better if treated with tissue plasminogen activators such as alteplase within 3 hours of the clinical onset, and also because they often present with disturbance of consciousness and disturbance of respiration (Table 33-1). For the thrombolytic therapy of acute ischemic stroke, it is particularly important to exclude intracranial hemorrhage by obtaining emergency neuroimaging

(e.g., CT; Powers et al, 2019). When possible, mechanical thrombectomy should also be undertaken.

As summarized in Table 33-1, each form of stroke has characteristic symptoms and signs and characteristic onset. Atherothrombosis is characterized by sudden or acute onset of neurological deficits such as hemiplegia and speech disturbance. Lacunar infarction occurs in the territory supplied by a penetrating arterial branch in the deep cerebral hemisphere and often presents with pure motor hemiplegia, hemianesthesia without motor paralysis, and dysarthria–clumsy hand syndrome (see Box 2-2). Cardiogenic embolism is characterized by neurological symptoms and signs due to occlusion of more than one artery at once. In an intracerebral hemorrhage, the clinical onset of hemiplegia is sudden and followed by rapid worsening with headache and consciousness disturbance within a few hours. Subarachnoid hemorrhage is characterized by severe headache of sudden onset. It should be kept in mind that nuchal stiffness does not appear until about 24 hours after the initial onset of headache. In an atypical stroke presenting with headache, consciousness disturbance, and convulsion,

BOX 33-3 REYE SYNDROME

Reye syndrome in the narrow sense affects children over 5 years of age with vomiting, disturbance of consciousness, and intracranial hypertension during the recovery stage of infection, and convulsions may occur in severe cases. On laboratory tests, elevation of aminotransferase (AST, ALT) and ammonium levels in blood, hypoglycemia, fatty degeneration of hepatic cells, and diffuse brain edema are found. By contrast, there is no abnormality in the cerebrospinal fluid, which helps to differentiate it from acute meningoencephalitis. Reye syndrome is known to occur commonly in children who received salicylate medication for the preceding infection, but its causative relation has not been completely established. The pathophysiology of Reye syndrome has been poorly elucidated, but it is postulated to be caused by mitochondrial dysfunction of hepatic cells (Mizuguchi et al, 2007).

A similar clinical picture may be seen in infants and adults. Among others, encephalopathy associated with influenza virus infection (**influenza encephalopathy**) is well known (Yamada et al, 2012). Especially in children who have congenital metabolic abnormalities, similar encephalopathy may be provoked by infection or hypoglycemia. Therefore, it may be reasonable to call classical Reye syndrome **Reye disease**, and other similar conditions might be distinguished as **Reye syndrome**.

venous sinus thrombosis should be considered (see Box 1-1) (Idiculla et al, 2020).

4. STATUS EPILEPTICUS

Frequent occurrence of epileptic seizures associated with continuous disturbance of consciousness even during the intervals is called **status epilepticus**. A complex partial seizure like psychomotor seizure may fall into status, but the status of generalized convulsive seizures, in particular, requires emergency treatment. New-onset status epilepticus in the adult population is often cryptogenic, but among the identified causes, autoimmune and paraneoplastic encephalitis is most common (Gaspard et al, 2015).

A. TREATMENT OF CONVULSIVE STATUS EPILEPTICUS

The patient with convulsive status epilepticus should be placed in a safe environment, and vital signs should be

continuously checked. Airway support and blood pressure maintenance should be provided, and intravenous access should be established as necessary. There are a number of pharmacologic approaches to treat status epilepticus (Abend et al, 2015; Betjemann & Lowenstein, 2015). Pharmacologic treatment may start with intravenous injection of 10 mg of diazepam over a period of 30 to 60 seconds. If convulsion recurs, another 10-mg dose of diazepam may be added while paying attention to respiration. Furthermore, 15 to 20 mg of phenytoin per kilogram of body weight may be infused intravenously at a rate of 50 mg per minute. Among the various benzodiazepines, intravenous lorazepam was shown to be most effective (Ameli et al, 2021; Lee, 2020). If convulsions still occur repeatedly, general anesthesia may be needed in an intensive care unit with respiratory support. For anesthesia, midazolam or isoflurane may be used. For laboratory tests, blood should be drawn for complete blood count and chemistry, but an EEG and neuroimaging can be done after the patient's condition is stabilized. During the status, EEG is expected to show marked diffuse abnormality with many movement artifacts, so EEG can be obtained more effectively after the convulsions diminish in their degree and frequency.

B. TETANUS

From the viewpoint of intensive care, tetanus is clinically similar to generalized convulsive status, although they are caused by different mechanisms (Chapter 17-5B, p. 150). Patients with tetanus are extremely sensitive to external stimuli, and even a small sound or touch may cause generalized muscle spasm and opisthotonos. Therefore, it is important to place the patient in a dark, quiet room with intensive care just like in the case of convulsive status epilepticus. Active treatment for the causative organism, *Clostridium tetani*, should be started with chemotherapy with penicillin G and tetanobulin, which is anti-toxin immunoglobulin. As an extremely large dose of tetanobulin has to be injected intramuscularly (3,000 to 6,000 units), intravenous infusion of tetanobulin IH is more practical and recommended.

5. CONDITIONS EXPECTED TO HAVE POOR FUNCTIONAL RECOVERY UNLESS APPROPRIATELY TREATED AT EARLY CLINICAL STAGE

Some conditions, although there may be no respiratory paralysis, no disturbance of consciousness, and no

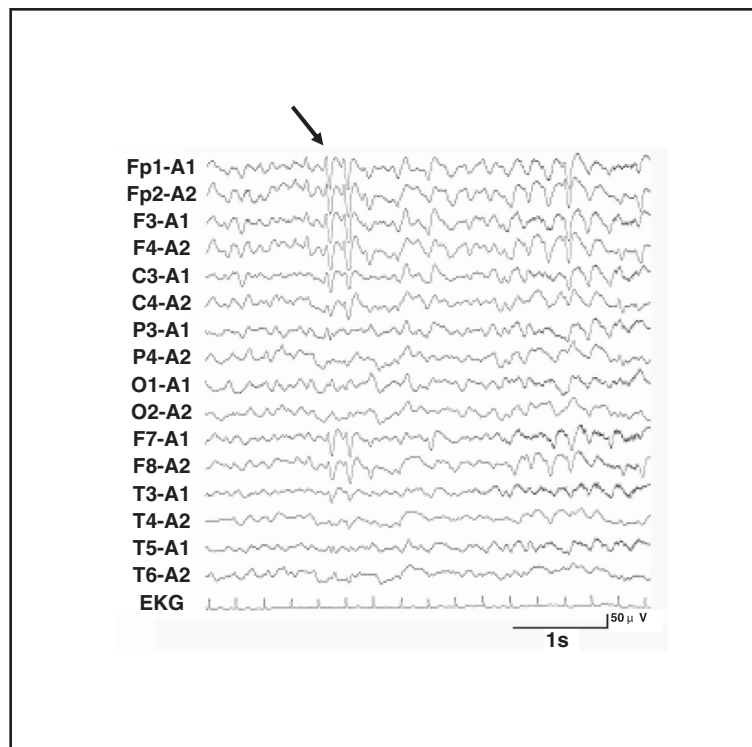


Figure 33-2 EEG of a patient with hepatic encephalopathy, showing lack of the posterior dominant rhythm and diffuse slow waves with frequent burst-like occurrence of high-amplitude slow waves and occasional triphasic waves (arrow).

convulsive status epilepticus, may have to be treated appropriately at an early clinical stage in order to obtain the best functional recovery possible. Those are **acute compression of spinal cord**, **progressive ischemic brain lesion**, and **acute meningoencephalitis** (see Box 33-1).

A. ACUTE COMPRESSION OF SPINAL CORD

If the spinal cord is acutely compressed by a mass or trauma, surgical decompression within 6 hours of compression is considered to be necessary for functional recovery. The degree of compression can be judged clinically, but of course MRI is very useful. To detect abnormalities of the vertebral bone itself, plain X-ray of the spine is also useful. Causes of acute extramedullary compression include vertebral trauma, extramedullary tumor, vertebral fracture due to metastatic tumor, and extradural abscess. **Extramedullary tumors** of the spinal cord differ depending on the patient's age and the vertebral site, but neurofibroma, meningioma, and metastatic tumor are common. Acute myelogenous leukemia might form a mass and compress the spinal cord (Graff-Radford et al, 2012). As a result of the international expansion of human immunodeficiency virus (HIV), tuberculosis is increasing. In some regions, paraplegia might be seen from compression of the

spinal cord by abscess or fracture due to vertebral tuberculosis (**Pott paraplegia**).

B. PROGRESSIVE ISCHEMIC CEREBROVASCULAR DISEASES

In contrast with cerebral embolism, in which symptoms develop abruptly, symptoms in cerebral thrombosis usually take a few hours or even longer to reach their peak after the initial symptom develops. Some cases of cerebral thrombosis, however, show a progressive course even several hours after the onset, in which case active **thrombolytic therapy** is necessary. In particular, progressive ischemia in the vertebrobasilar artery territory requires special attention and a quick approach.

Thrombolytic therapy with **tissue plasminogen activator (tPA)** is widely used for promoting functional recovery. Patients in the acute phase of cerebral infarction are candidates for this therapy if there is no history of hypertension (or mild if any) and no neuroimaging evidence of intracranial hemorrhage. It is important to start this therapy as soon as possible, ideally within 3 hours but practically within 4.5 hours, after the onset of the initial symptom (American Academy of Neurology Practice Advisory, 1996; Minematsu et al, 2013).

TABLE 33-1 CLINICAL FEATURES OF VARIOUS FORMS OF STROKE

	Atherothrombosis	Lacunar infarction	Cardiogenic embolism	Intracerebral hemorrhage	Venous sinus thrombosis	Subarachnoid hemorrhage
Clinical onset	Sudden or acute	Sudden or acute	Sudden	Acute	Acute	Sudden
Symptoms & signs	Hemiplegia, aphasia	Hemiplegia hemisensory	More than one due to multiple lesions	Hemiplegia, coma	Conscious clouding, headache, convulsion	Severe headache, coma, nuchal stiffness
Underlying medical diseases	Hypertension, diabetes mellitus, hyperlipidemia, obesity	Hypertension	Cardiac valvular disease, atrial fibrillation	Hypertension, amyloid angiopathy, arteriovenous malformation	Systemic medical diseases, coagulopathy, endocrine, metabolic	Intracranial aneurysm

6. EXAMINATION OF COMA

All cases of coma require an emergency approach. When a patient is brought to an emergency room in a comatose state, the first thing to do is to check vital signs. If the patient is not breathing or is only poorly breathing, airway support with artificial respiration should be started. After checking the pulse rate and blood pressure, fluid infusion and drugs for maintaining blood pressure may be started as necessary. If no information is available, it is safe to start with an intravenous infusion of 5% glucose solution. Thiamin is always given together with the glucose to prevent the precipitation of Wernicke encephalopathy. If drug use is prevalent in the area, naloxone might be administered as well.

After vital signs are stabilized, then the etiology of the coma may be evaluated (Box 33-4). Usually, at least some information is obtained from the family or the accompanying persons, but if no source of information is available, the physical examination should be started immediately. Comatose patients can be examined according to the general principle shown in Figure 33-1. In practical terms, it is reasonable to pursue the examination focusing on the following four points (A to D).

A. DEGREE OF COMA (COMA SCALING)

To judge whether a patient is in deep coma or not, it is important to apply the most effective painful stimulus. Strong pressure of the supraorbital nerve at the eyebrow by the examiner's thumb, strong pressure of the jaw with the examiner's thumbs from its two sides, or pinching the fingertip with the examiner's thumb and index finger may be adopted. To assess the degree of coma, coma scales shown in Boxes 4-4 and 4-5 may be used.

B. PRESENCE OR ABSENCE OF ASYMMETRY IN NEUROLOGICAL SIGNS

As described in Section 2 (p. 287), asymmetry of neurological signs, if present, provides an important clue to the etiology of coma. First, spontaneous position of the eyes should be observed to determine whether conjugate deviation is present (see Chapter 9-2 for detail, p. 60). Dysconjugate deviation is rare, but if present, it highly suggests a brainstem lesion. The left and right nasolabial lines should be compared for any asymmetry, although this observation is often difficult to make because of intubation. Then the extremities should be checked for any asymmetry of posture, muscle tone, muscle strength, tendon reflex, and plantar reflex between the two sides. For practical purposes, the patient's arms or legs can be lifted by the examiner and then released. The paralyzed or more flaccid side will fall earlier than the intact or less affected side. In fact, in comatose patients, motor paralysis is mostly of the flaccid type. If any of these signs are found to be asymmetric, it suggests the presence of a gross organic lesion in the contralateral cerebral hemisphere causing the coma.

Conjugate deviation of eyes, if present, gives an important clue to the anatomic diagnosis. If the frontal lobe is unilaterally affected, the eyes deviate to the side of the lesion because lateral gaze to the contralateral side is paralyzed. An exception to this rule is thalamic hemorrhage, which may be associated with conjugate deviation of the eyes to the side contralateral to the lesion (at least in the acute phase), which is considered to be a positive neurological symptom. In contrast, a lesion of the unilateral pontine tegmentum may cause conjugate or dysconjugate deviation of the eyes to the side contralateral to the lesion (see Figure 9-5, p. 61).

BOX 33-4 CAUSES OF COMA

1. Associated with meningeal irritation
 - Subarachnoid hemorrhage
 - Ventricular rupture of intracerebral hemorrhage
 - Meningoencephalitis
2. Associated with focal neurological signs
 - Intracerebral hemorrhage
 - Cerebral contusion
 - Cerebral embolism
 - Intracranial subdural hematoma
 - Brain tumor
 - Brain abscess
 - Brainstem hemorrhage
 - Brainstem infarction
3. Associated with neither meningeal irritation nor focal sign
 - Toxic encephalopathy (ethanol, narcotics, carbon monoxide, etc.)
 - Metabolic encephalopathy (diabetes mellitus, hypoglycemia, uremia, hepatic failure)
 - Anoxic encephalopathy
 - Shock (hemorrhage of organs other than brain)
 - Cardiac failure
 - Postictal state

C. EXAMINATION OF BRAINSTEM FUNCTIONS

As the brainstem contains (1) important centers for controlling vital signs, (2) center of consciousness (see Figure 4-1, p. 25), and (3) many reflex arcs that can be clinically tested in comatose patients, testing of brainstem functions plays a key role in the neurological examination of comatose patients.

i. Observation of respiration

The respiratory center is widely distributed in the brainstem, but the medullary respiratory center (located in the ventral lateral part of the medullary reticular formation) and the pontine respiratory center (located in the pontine reticular formation) play main roles (see Figure 5-4, p. 34). The **medullary respiratory center**, also called the ventral respiratory group, receives information about movement of the thorax and lung, and the pH and carbon dioxide content of blood, and controls rhythmicity of the motor neurons innervating the diaphragm and other respiratory muscles. The **pontine respiratory center** is believed to control spontaneous breathing, which is maintained by the medullary respiratory center.

Repetitive occurrence of phases of hyperventilation and apnea is known as **Cheyne–Stokes respiration**. In this case, breathing gradually becomes deeper, and after reaching a peak, it gradually becomes shallower, followed by a short period of apnea, and this cycle is repeated. As the result of these repetitive cycles, it is dominated by hyperventilation and results in respiratory alkalosis. Cheyne–Stokes respiration is seen in association with transtentorial herniation due to a space-occupying hemispheric lesion (see p. 294), with a lesion of the rostral brainstem, or with metabolic encephalopathy.

Continuous hyperventilation, which is associated with a decrease in partial pressure of carbon dioxide (PCO_2) and an increase in partial pressure of oxygen (PO_2) and pH in arterial blood, despite the absence of drug effect or metabolic acidosis, is called **central neurogenic hyperventilation**. This state is commonly seen in association with pontine tumor and is characterized by continuous hyperventilation even during sleep. It is considered to be an excitatory effect of the pontine respiratory center and chemoreceptors (Tarulli et al, 2005).

In contrast with hyperventilation, the condition in which the ventilatory volume per breath is increased (**hyperpnea**) and the respiratory rate per minute is also increased (**tachypnea**) is not necessarily associated with a decrease in PCO_2 . Hyperpnea seen in metabolic acidosis is called **Kussmaul respiration**.

Arrest of respiration in the deep inspiratory phase is called **apneustic breathing**. Under normal conditions, the apneustic center in the pontine respiratory center is suppressed by the pneumotaxic center so that the inspiratory phase can shift to the expiratory phase, and impairment of this mechanism is considered to cause apneustic breathing (El-Khatib et al, 2003).

Irregular breathing in terms of depth and period is called **ataxic breathing**, and it occurs when the medullary respiratory center is impaired. If spontaneous (automatic) breathing is lost due to a medullary lesion, only voluntary breathing is left intact. In this condition, breathing might stop during sleep or decreased vigilance, a condition called **Ondine curse** (Fiedler & Gill, 2020).

As the brainstem respiratory centers are easily inhibited by anesthetics, barbiturates, hunger, and metabolic alkalosis, the possibility of drug effect and those general systemic conditions have to be taken into account when examining respiration.

ii. Brainstem reflexes

As ordinary motor and sensory functions cannot be tested in comatose patients, reflexes are the only objective signs

that can be tested. It is practical to test the brainstem reflexes starting from the midbrain level through the medullary level, namely light reflex, corneal reflex, oculocephalic reflex or doll's eye sign, gag reflex, and cough reflex, in that order.

a. Light reflex

The neural circuit of the light reflex is composed of the optic nerve as its afferent arc and the oculomotor nerve as its efferent arc with the reflex center in the midbrain (see Figure 7-7, p. 42). As comatose patients are usually placed in a well-lit room like an intensive care unit, it is better to test the light reflex after dimming the room. That is because the pupils might be already shrunk because of the room light, which might mislead judgment as to the presence or absence of the light reflex. Even in a bright room, however, it may be still possible to detect the pupillary reaction, if any, just after passively opening the eyelids, because it takes a short but recognizable time for pupils to react to light. Namely, if the pupils shrink after opening the eyelids in a bright room, it might be possible to confirm the presence of the light reflex. At any rate, it is important to use a sufficiently bright light stimulus to test this reflex.

b. Corneal reflex

The neural circuit of the corneal reflex is composed of the first branch of the trigeminal nerve (ophthalmic nerve) as its afferent arc and the facial nerve as its efferent arc with the reflex center in the pons (see Figure 10-3, p. 77). As the eyelids are usually kept closed in comatose patients, the cornea can be stimulated with the tip of a twisted paper string or a wisp of cotton while the eye to be tested is kept open by the examiner's hand. As nociceptive receptors are present in the cornea but not in the bulbar conjunctiva, it is important to stimulate the cornea instead of the sclera. As the stimulated eye is kept open by the examiner, the reaction can be judged by observing contraction of the orbicularis oculi muscle of the non-stimulated side. If the reflex is sufficiently preserved, the reaction of the stimulated eyelids can be felt by the examiner's hand, but when the reaction is weak, it is more easily detected in the contralateral eye. A similar reflex can be elicited by touching the cilia (ciliary reflex). However, as this reflex is less sensitive than the corneal reflex, the corneal reflex must be tested even if the ciliary reflex is absent.

c. Doll's eye sign

The neural circuit of the oculocephalic reflex is composed of the vestibulocochlear nerve (receptors in the semicircular

canal) and cervical nerve (receptors in the cervical muscles) as the afferent arc and the lateral as well as vertical gaze center as the efferent arc. Thus, this reflex involves a neuronal circuit extending from the cervical cord to the midbrain (see Chapter 9-5C, p. 68). By holding the head with the examiner's hands from both sides in the supine position, the head is rotated laterally to test the horizontal reflex and vertically to test the vertical reflex, and the response is judged by observing the position of the eyes relative to the head. The presence of this reflex is judged as a positive doll's eye sign, and it suggests that this reflex circuit is intact throughout the whole length of the brainstem. The doll's eye sign usually appears as a conjugate movement of the two eyes, but in case of brainstem lesions, the reflex may be complicated by extraocular muscle palsy or internuclear ophthalmoplegia. In relation to this, the presence of lateral gaze palsy may be predicted by the presence of conjugate deviation of the eyes before testing the oculocephalic reflex.

d. Gag reflex and cough reflex

The gag reflex and the cough reflex have the reflex center in the medulla oblongata, and both reflexes are important for diagnosis of brain death because they represent activities of the most caudal part of the brainstem (see Chapter 14-2, p. 100, for the reflex circuit). However, as comatose patients are usually intubated for artificial respiration, these two reflexes are often difficult to test. In this case, it is possible to judge the presence or absence of these reflexes by observing reflex vomiting or coughing (bucking) during aspiration of the airway.

e. Caloric test

To diagnose brain death, the caloric test is necessary in addition to the preceding reflexes. For the caloric test, while the head is tilted backward by 60 degrees (downward because of the supine position), 5 to 10 ml of cold water (0 to 10°C) is injected into the external auditory meatus and any movement of eyes thus evoked is observed (see Chapter 13-2, p. 95). The reflex circuit of the caloric response is formed of the semicircular canal and vestibular nerve as its afferent arc and the lateral gaze center of the pons as its efferent arc, thus involving the medulla oblongata and the pons. A normal caloric response consists of deviation of both eyes to the stimulated side associated with horizontal nystagmus with its quick phase directed toward the non-stimulated side. If any one of these responses is observed, it suggests that the reflex circuit is functioning. Before doing this test, it is advisable to confirm that the tympanic membrane is intact by an otoscope.

f. Cilio-spinal reflex

This cutaneous pupillary reflex is elicited by a nociceptive stimulus given to the neck skin, and the response is seen as dilation of pupils. Its afferent arc is the cervical nerve, and the efferent arc is the sympathetic nervous system originating from the cilio-spinal center of Budge at the intermediolateral nucleus of the first thoracic segment (see Figure 8-4, p. 51). How much this reflex circuit involves the brainstem is not well understood. If this reflex circuit is limited to the cervical cord, preservation of this reflex may not rule out brain death. This is because brain death is defined as irreversible death of all neurons in the medulla oblongata and above. However, as far as a possibility of this reflex involving the brainstem structures remains, preservation of this reflex may rule out brain death.

If all of these brainstem reflexes are lost in comatose patients, it suggests that all the brainstem functions are lost. However, a possibility remains that many other neurons not directly related to these reflexes are still preserved. When judging brainstem reflexes, it is of utmost importance to keep in mind that all these reflexes may be completely lost by intake of a large dose of narcotics, during general anesthesia, or in hypothermia. If there is any possibility of these conditions, it is useful to follow up the clinical course by repeating these tests. Needless to say, in a coma due to brainstem lesions, loss of these reflexes may also facilitate localization of the lesion.

D. CEREBRAL HERNIATION

Secondary compression of the brainstem as a result of a gross organic lesion of the cerebral hemisphere(s) causes cerebral herniation, which requires quick attention and treatment. **Central transtentorial herniation** is caused by rapid compression of the thalamus from above bilaterally, and it causes coma quickly. **Uncal herniation** is caused by compression of the midbrain unilaterally from above and causes (1) clouding of consciousness, (2) Cheyne–Stokes respiration (see p. 2), (3) anisocoria and loss or decrease of the light reflex, and (4) decerebrate rigidity/posture. If any or all of these signs appear in patients with a gross hemispheric lesion or brain edema, it suggests that uncal herniation is taking place. Pathologically, uncal herniation is associated with hemorrhagic lesions of various degrees in the midbrain tegmentum, which affects the brainstem reticular formation, causing disturbance of consciousness, respiratory impairment, and postural abnormality.

i. Anisocoria

Anisocoria in uncal herniation is considered to be caused not by direct intramedullary damage of the oculomotor

nerve in the midbrain, but rather by entrapment of the oculomotor nerve between the superior cerebellar artery and the posterior cerebral artery (see Figure 9-3, p. 59). When the compression is relatively mild, anisocoria and a loss or decrease of the light reflex of the dilated pupil may be seen, but paralysis of the extraocular muscles may not be observed. It is most likely because the autonomic fibers are located in the peripheral part of the oculomotor nerve at this extramedullary site, so they are easily affected by external compression. Involvement of the oculomotor nerve in uncal herniation is more commonly seen on the side ipsilateral to the hemispheric lesion, but in some cases it may be seen on the contralateral side.

ii. Decerebrate posture

This abnormal posture consists of tonic extension and internal rotation (pronation) of the upper extremities and extension of the lower extremities, either spontaneously or more often provoked by a nociceptive stimulus. It is associated with increased tone of the extensor muscles (**decerebrate rigidity**). It is considered that the decerebrate posture/rigidity is caused as a result of interruption of the inhibitory pathway projecting to the vestibular nucleus at the level of the midbrain, which increases the tone of the extensor muscles through disinhibition of the vestibulospinal tract (see Chapter 13-1, p. 93). The presence of this phenomenon suggests either a severe lesion in the midbrain or the beginning of uncal herniation.

iii. Tonic neck reflex

This is a postural reflex in response to lateral rotation of the neck, consisting of extension of the upper and lower extremities of the side to which the neck is rotated and flexion of the contralateral upper and lower extremities, resembling the posture of a shot-putter. This phenomenon is seen when there is an asymmetric severe lesion in the brainstem rostral to the vestibular nucleus. Also in this case, passive neck flexion may cause flexion of both upper extremities, and neck extension may cause their extension.

iv. False localizing signs

In some cases of uncal herniation, a Babinski sign and homonymous hemianopsia might be seen on the side of a hemispheric lesion. As these signs are not directly related to the primary hemispheric lesion, they are called **false localizing signs**. Among these two signs, the Babinski sign ipsilateral

to the hemispheric lesion is caused by impairment of the cerebral peduncle contralateral to the hemispheric lesion as a result of midbrain compression. The peduncular pathology in this case is called a **Kernohan notch**. Therefore, the pyramidal signs may be seen bilaterally in this case.

Homonymous hemianopsia on the side ipsilateral to the primary hemispheric lesion is caused by compression of the contralateral posterior cerebral artery as a result of uncus herniation, which causes an ischemic lesion in the occipital lobe contralateral to the primary hemispheric lesion. This sign, however, is not detectable in comatose patients.

In addition to uncus herniation, pyramidal signs might also be seen bilaterally in some other cases of a unilateral cerebral lesion. An example is thalamic hemorrhage, which will compress the contralateral corticospinal tract, causing an ipsilateral Babinski sign, and another example is the presence of an old vascular lesion in the hemisphere contralateral to the new lesion. In this case, distinction from uncus herniation can be easily done from other accompanying signs.

E. MENINGEAL IRRITATION

Nuchal stiffness is a representative sign of meningeal irritation (see Chapter 15-1E, p. 107). This is tested by flexing the neck (lifting the head) in the supine position and is judged by limitation of the neck flexion. If one of the examiner's hands is placed behind the neck while lifting the head using the other hand, the reflex contraction of the nuchal muscles may be palpated in patients with positive nuchal stiffness. This is considered to result from involvement of the dorsal roots of the cervical nerves or the occipital nerves by hemorrhage or inflammation at the site of meningeal penetration. Positive nuchal stiffness suggests subarachnoid hemorrhage or meningitis. However, nuchal stiffness may not be seen within 24 hours after the onset of subarachnoid hemorrhage (see p. 107).

7. JUDGMENT OF BRAIN DEATH

Brain death is defined as a condition in which the functions of all neurons in the cerebral hemispheres and brainstem are irreversibly lost due to organic brain diseases. In patients being evaluated for possible brain death, the cardiovascular center and the respiratory center in the brainstem (see Figure 5-4, p. 34) are not functioning at all, interrupting the central innervation of heart and lung. Even in this condition, however, the heart continues to beat for at least

some time as long as oxygen is supplied. Thus, all candidate patients for brain death are on a respirator. In other words, brain death can be ruled out in patients who are spontaneously breathing, even a little.

The most important prerequisite for diagnosis of brain death is the fact that the patient is not under profound hypothermia or anesthesia and does not have any significant metabolic abnormalities. Clinical diagnosis is based on the confirmation of deep coma and loss of all reflexes that have their reflex centers in the brainstem. Thus, the presence of reflexes that have their centers in the spinal cord does not exclude brain death.

There are substantial differences in perceptions and practices of brain death worldwide (Greer et al, 2020; Lewis et al, 2018; Russell et al, 2019; Varelas et al, 2021 Wahlster et al, 2015; Wijedicks et al, 2010). Regarding the laboratory tests, the implications of an EEG record of electrocerebral inactivity (the so-called flat EEG) differ among diagnostic criteria, but if any cortical activity is detected on EEG, the patient cannot be diagnosed as brain dead. In addition, the caloric test (see p. 95) using ice water and brainstem auditory evoked potentials (BAEP) (see Chapter 12-1, p. 90) are important tests. If there is any response in these tests, it can exclude brain death. However, as the peak I of BAEP is known to be generated in the cochlear nerve, its presence alone does not exclude brain death. If all these tests suggest loss of all brainstem functions, it is recommended to repeat all of them sometime after the first test for reconfirmation, although the time interval varies among different diagnostic criteria.

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DISABILITY, FUNCTIONAL RECOVERY, AND PROGNOSIS

1. ASSESSMENT OF DISABILITY SCALE

Degree of disability and prospect of functional recovery are the most important matters of concern for patients with neurological disorders. After completing the neurologic examination, it is important to consider the severity of disability, although detailed scoring of each activity of daily living can be assessed later. Disability can be evaluated in terms of neurological **impairment** found by the physical examination, **disability** expressed as **activities of daily living (ADLs)**, and the resulting social **handicap**. Scales of disability have been established for some diseases like Parkinson disease (see Box 16-8, p. 127) and multiple sclerosis, but for other diseases, it is useful to evaluate each of the specific ADLs, such as brushing teeth, washing face, combing hair, dressing, use of toilet facilities, taking bath or shower, eating, walking indoors as well as outdoors, and so on.

2. MODE OF TISSUE DAMAGE, FUNCTIONAL DISABILITY, AND RECOVERY

Functional disability due to neurological impairment and the prospect of recovery depend on how the nervous tissue is damaged. A number of factors affect functional disability and recovery (Box 34-1). For general factors, the age of disease onset and the patient's present age, general physical condition, and motivation for recovery are important. In aged subjects, functional recovery is expected to be slower and poorer than in young subjects. In children, functional recovery due to plastic restoration is directly related to the developmental state of the nervous system.

For the neurological factors influencing the disability and functional recovery, the mode of onset of neurological deficits and the mode of tissue damage should be taken into account (see Figure 1-2, p. 3). Even for a relatively small lesion, the more acutely the tissue is damaged, the

more conspicuous are the resulting neurological deficits. In contrast, even for a gross lesion, the symptoms and signs are relatively mild if their onset is insidious and followed by a slowly progressive course. For example, a patient with an intracranial meningioma occupying a large space in the cranial cavity may be almost unaware of symptoms. In contrast, neurodegenerative diseases may progress very slowly, but the disability might be quite severe in the advanced stage. Even a small lesion, if it occurs in a region where multiple important structures are closely located, like in the brainstem, might cause various neurological deficits.

The distribution of lesions shows a characteristic histologic pattern specific for each disease group (see Figure 1-1, p. 2). However, even lesions in the same brain region might show different clinical pictures depending on the mode of tissue damage. The size of the lesion is another important factor, but it depends on where in the brain the lesion is located and how quickly it has evolved.

To predict the prognosis, it is important to take into account the laboratory data; however, these results, including neuroimaging, are not necessarily correlated with the degree of functional disability. Thus, it is necessary to take into account the whole clinical picture to judge the patient's prognosis.

A. GRAY MATTER LESION VERSUS WHITE MATTER LESION

When considering the functional disability and recovery in patients with an organic lesion of the cerebral hemisphere, whether the lesion primarily affects the gray matter or the white matter is one of the key factors (Table 34-1). Generally speaking, a cortical lesion tends to manifest distinct focal neurological deficits, whereas focal neurological deficit is rare and relatively indistinct, if any, in a white matter lesion. This is because a large number of neurons are located close together in the narrow cortical area and because neurons recover poorly once they

BOX 34-1 FACTORS INFLUENCING THE DEGREE OF FUNCTIONAL DISABILITY AND ITS RECOVERY IN NEUROLOGICAL DISORDERS

1. General factors
 - A. Age of onset and present age
 - B. General physical condition
 - C. Motivation for recovery
2. Neurological factors
 - A. Mode of onset
 - B. Distribution of lesion at tissue level
 - C. Size of lesion
 - D. Number of lesions
 - E. Gray matter lesion versus white matter lesion
 - F. Axonal damage versus myelin sheath damage

are damaged. In contrast, the white matter is formed of myelinated nerve fibers whose myelin sheath may regenerate to a certain extent. However, a disconnection syndrome due to a lesion of the cerebral white matter may cause focal neurological deficit due to interruption of functional connectivity between different cortical areas (see Chapter 24-1, p. 230). Furthermore, a progressive diffuse lesion of the cerebral white matter, such as those due to various kinds of leukodystrophy and ischemic white matter lesions (see Chapter 23-2C, p. 225), may cause severe dementia and spastic quadriplegia. Convulsive seizures and involuntary movements, myoclonus in particular, are commonly seen in cortical lesion but are relatively rare in white matter lesions.

In relation to this, it is important to note that, in the cerebral hemispheres, the gray matter is in the cortex as well as in the deep structures (thalamus and basal ganglia) while the medullary part is formed of the white matter; however, in the spinal cord, the gray matter is present only in the central portion and all the peripheral parts are occupied by the fiber tracts, namely the white matter.

TABLE 34-1 FEATURES OF CLINICAL SYMPTOMS AND SIGNS; COMPARISON BETWEEN GRAY MATTER AND WHITE MATTER LESIONS

	Gray matter	White matter
Neurological deficits	Distinct	Indistinct*
Focal sign	Common	Rare
Convulsive seizures	Common	Rare
Myoclonus	Common	Rare

*An exception is a diffuse white matter lesion of cerebral hemispheres, which may cause severe dementia and spastic quadriplegia.

B. AXONAL DAMAGE VERSUS MYELIN DAMAGE

Closely related to the issue of gray matter versus white matter damage, the issue of axonal damage versus myelin sheath damage is an important factor at the cellular level (Table 34-2). Generally speaking, in disorders in which axons are primarily damaged, neurological deficits tend to be severe and functional recovery is incomplete, but in conditions in which the myelin sheath is primarily damaged, neurological deficits are relatively mild and functional recovery is relatively good. However, in multiple sclerosis, which is a representative form of demyelinating diseases, the clinical course is initially characterized by remissions and relapses, but in the long run a progressive course associated with the secondary axonal damage is common (Filippi et al, 2013). Moreover, in the neuromyelitis optica spectrum disorders, which are relatively common in Asian populations, the spinal cord is severely affected involving axons with severe functional disability (see Box 17-4, p. 148).

In cerebral contusion due to head trauma and spinal cord injury, both axons and the myelin sheath are damaged, causing severe disability and poor functional recovery.

3. MECHANISMS OF FUNCTIONAL RECOVERY

Mechanisms by which the impaired neurological functions improve have not been fully clarified, and its molecular mechanisms are an important topic of modern neuroscience.

If the nervous system is only functionally impaired without structural damage even at the tissue level, the impaired function is expected to recover without leaving any residual. Even if there is some tissue damage, if it is only mildly affected with preserved capacity to recover, then its function is also expected to recover along with the recovery of the tissue damage. If the nervous system is severely damaged, there are two main hypotheses for possible recovery, as follows.

TABLE 34-2 CHARACTERISTICS OF NEUROLOGICAL IMPAIRMENT; COMPARISON OF AXONAL DAMAGE AND MYELIN DAMAGE

Lesion	Axon	Myelin sheath	Axon + myelin sheath
Degree of disability	Severe	Relatively mild	Severe
Recovery	Incomplete	Relatively good	Poor
Example	Strokes	Multiple sclerosis	Contusion

A. PLASTIC REORGANIZATION

The nervous system is considered to have some capacity to reorganize its structure, function, and network in response to internal or external stimuli. For example, **collateral sprouting** may grow from the remaining part of the damaged axon or from other intact axons so that it could form synaptic contacts with the target neuron and produce some functional recovery (Figure 34-1). In this case, it is expected to take some time for the synaptic reorganization to occur.

Plastic changes are considered to contribute to the recovery of neurological deficits, but on the other hand they may also cause positive neurological symptoms that may even be hazardous to the patient. For example, patients with thalamic infarction involving the distal portion of the cerebellothalamic pathway (superior cerebellar peduncle) may develop tremor in the contralateral hand a few months after the stroke. For another example, patients with a thalamic vascular lesion involving the somatosensory afferent pathway may develop unpleasant pain (**thalamic pain**) and **hyperpathia** in the contralateral upper extremity at some time after the stroke (Chapter 20-2B, p. 196; Box 20-4, p. 198). In relation to the plastic reorganization of fiber tracts, diffusion tensor tractography might be used to delineate disruption of fiber tracts in the white matter.

In recent years, clinical attempts have been made to apply the scientific principle of plastic reorganization to promote functional recovery in patients with trauma and vascular lesions of the central nervous system, psychiatric disorders, substance use and habituation, developmental abnormalities of children, and neurodegenerative disorders. Examples are brain stimulation therapy, pharmacological approach, learning, and rehabilitation (Cramer et al, 2011; Grefkes & Fink, 2011).

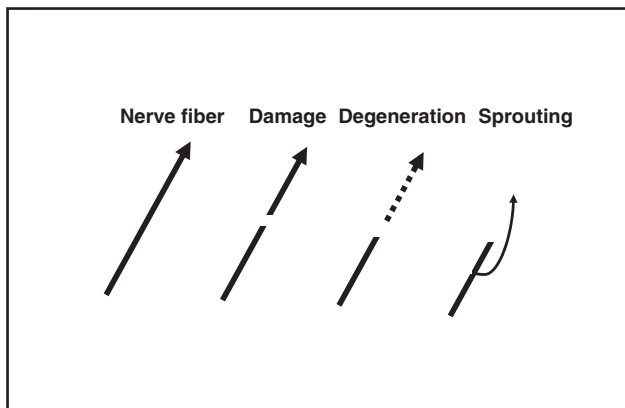


Figure 34-1 Schematic diagram of a nerve fiber showing possible mechanism of damage and regeneration.

B. MOBILIZATION OF RESERVE FUNCTIONS

Another hypothesis about recovery of impaired function is mobilization of a nervous system that is not primarily related to that particular function (**reserve functions**). For example, about 85% of fibers of the direct corticospinal tract originating from the motor cortex cross at the pyramidal decussation at the medulla oblongata and innervate the extremities contralateral to the motor cortex as the lateral corticospinal tract, while the remaining 15% of fibers descend through the ipsilateral anterior column as the anterior corticospinal tract (see Chapter 16-1A, p. 109, and Figure 16-2, p. 110). The mobilization theory assumes that the uncrossed fiber tract is not actively participating in actual movement in the healthy condition, but once the crossed fiber tract is impaired, the uncrossed fiber tract becomes actively involved in the movement of the paralyzed extremities.

Alternatively, a nervous system that is closely related to the lost function but is involved in other functions in the healthy condition may be activated to compensate for the lost function when needed. For example, if the corticospinal tract originating from the primary motor cortex is damaged and causes paralysis of the corresponding part of the body, fiber tracts originating from the secondary motor cortices (supplementary motor area and premotor area) and from the primary somatosensory cortex are activated to supplement the function of the damaged tract (see Figure 11-3, p. 85). These mechanisms are being demonstrated by non-invasive techniques such as transcranial magnetic stimulation and functional magnetic resonance imaging (MRI).

With the recent remarkable advances in the field of cell transplantation techniques, histologic replacement of the damaged tissue with embryonic stem cells (ES cells) or induced pluripotent stem cells (iPS cells) may be the way of the future.

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HOW TO PLAN LABORATORY TESTS

It is reasonable and important to plan the necessary laboratory tests after carefully considering the site of the lesion, its etiology, and the most likely diagnosis and differential diagnoses based on the history taking and physical examination. For example, when faced with a patient complaining of symptoms suggestive of a cerebral lesion, it is a poor practice to immediately obtain cranial magnetic resonance imaging (MRI). It is necessary to consider the expected site of lesion and pathogenesis before ordering the MRI in order to reasonably correlate the clinical picture and the neuroimaging data. The quality and size of MRI abnormality may not necessarily correlate with the severity of neurological impairment. Moreover, from a viewpoint of limiting medical expenses and the limited size of hospital staff, the number of laboratory tests should be minimized.

In most neurological diseases, the site of the lesion and the etiology can be diagnosed by the history taking and physical examination, but some special conditions require laboratory tests. For examples, in **primary central nervous system vasculitis** (Salvarani et al, 2007), **primary lymphoma** of the central nervous system (Porter et al, 2008), and **granulomatous inflammation**, there is no specific clinical manifestation and no specific cerebrospinal fluid (CSF) abnormality. Therefore, cerebral angiography may be needed in primary vasculitis, and brain biopsy may be needed for primary lymphoma and granulomatous inflammation. Needless to say, laboratory tests are necessary to confirm the clinical diagnosis of disorders that might have specific abnormalities, such as specific antibodies and causative bacteria or viruses in the blood or CSF.

1. FUNCTIONAL NEUROIMAGING STUDIES AND ELECTROPHYSIOLOGIC STUDIES

Functional neuroimaging studies of regional cerebral blood flow with the use of single-photon emission

computed tomography (SPECT) or positron emission tomography (PET) are commonly used for clinical purposes. A question often asked is whether a decrease in the regional cerebral blood flow is correlated with a decrease in the electrical activity of neurons in that area. It has been shown in animal experiments that the hemodynamic response is significantly correlated with neuronal activity, especially local field potential (synaptic activity), within a certain range, but the hemodynamic response tends to be more widespread in space and lasts longer than the neuronal activity (Shibasaki, 2008). For example, in patients with a thalamic lesion, blood flow might be found decreased in the ipsilateral frontal cortex on a SPECT study. In this case, however, judgment of frontal hypo-function should be made with caution. This is an example of the neurophysiologic changes that occur distant to a focal brain lesion (diaschisis). The concept of **diaschisis**, proposed by von Monakow in 1914, nearly disappeared from the mainstream of clinical neuroscience, but with the recent advances in neuroimaging studies of functional connectivity, functional diaschisis has drawn increasing attention of many investigators (Carrera & Tononi, 2014).

By contrast with the central nervous system, electrophysiologic studies of the peripheral nerves and muscles provide useful information on their functions. The state of impulse conduction can be directly assessed by various techniques, and this data will greatly contribute to the clinical diagnosis and evaluation of the treatment effect (Kimura, 2013). Recently nerve ultrasound has been increasingly applied to the clinical diagnosis of peripheral neuropathy (Telleman et al, 2019).

2. LUMBAR PUNCTURE

Lumbar puncture is commonly used to diagnose neurological disorders, but indications should be carefully considered and its overuse should be avoided. With the development of neuroimaging techniques, indications for

lumbar puncture are limited to infectious diseases such as meningitis and meningoencephalitis, autoimmune inflammation of the central nervous system and polyradiculitis, and some metabolic abnormalities.

CSF is produced mainly from the choroid plexus. The mean CSF volume is 150 ml (25 ml in the ventricles and 125 ml in the subarachnoid space), and CSF is renewed about four times every 24 hours (Sakka et al, 2011). The opening pressure at the time of lumbar puncture in the relaxed recumbent position is in the range of 60 to 140 mm of CSF. Normal CSF contains fewer than five lymphocytes or mononuclear cells per mm³ and total protein of 15 to 45 mg/dl.

It is especially important not to abuse the **Queckenstedt test**. This test is used to observe elevation of the CSF pressure measured during the lumbar puncture in response to transient compression of the unilateral or bilateral jugular vein, which is expected to suddenly elevate the intracranial venous pressure in the normal condition. Failure of the CSF pressure to increase with this maneuver suggests obstruction of the spinal canal somewhere between the foramen magnum and the level of lumbar puncture (usually the

intervertebral space L3-L4). Therefore, the only purpose of this test is to prove that there is a blockage of the spinal canal, and there is no rationale at all for this test in patients with intracranial pathology. Furthermore, even in the spinal diseases, mechanical compression of the spinal cord can be easily detected by neuroimaging studies. It is especially important to avoid this maneuver in patients with intracranial hypertension, because it might suddenly cause cerebral herniation (see Chapter 33-6D, p. 294).

3. GENETIC TESTS

The genetic background of many neurodegenerative diseases has been elucidated, and clinicians caring for these patients often face ethical and practical decisions as to how much genetic testing should be performed (Roberts & Uhlmann, 2013). To avoid unnecessary genetic tests and expenses, it is important to evaluate the clinical diagnosis and family history carefully before requesting the genetic screening tests. The types of genetic tests available are rapidly changing, so

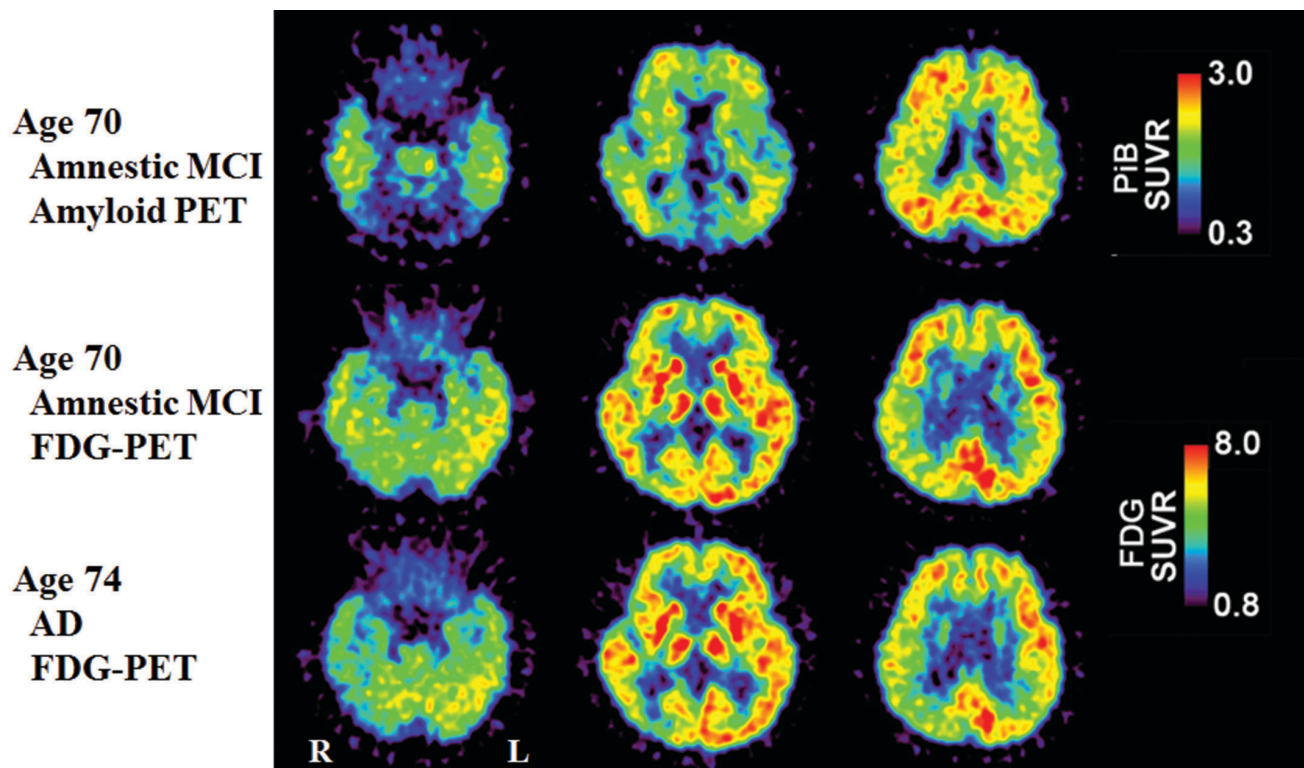


Figure 35-1 Amyloid imaging with Pittsburgh Compound B (top) and glucose metabolism imaging with fluorodeoxyglucose (FDG) (middle) with PET in a 70-year-old patient with amnesic mild cognitive impairment (MCI) who developed Alzheimer disease (AD) at age 74 (bottom). Glucose metabolism was decreased in the right temporoparietal region at age 70, and more so at age 74. Amyloid deposition was seen in bilateral temporoparietal and frontal regions at age 70 (top).

By courtesy of Dr. Kenji Ishii, Tokyo Metropolitan Institute of Gerontology.

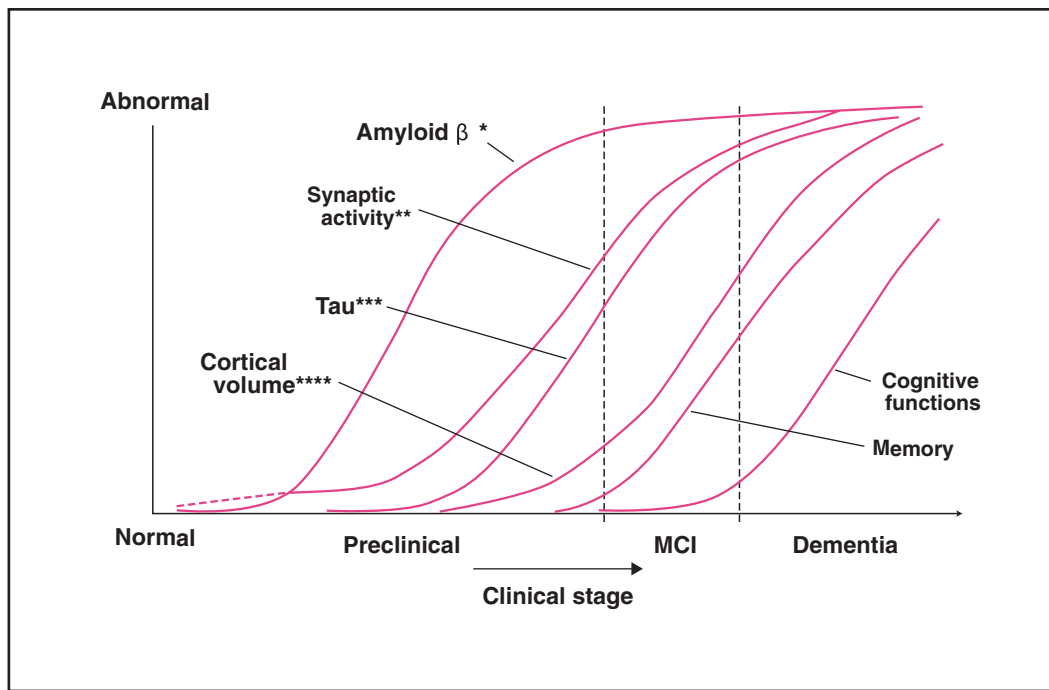


Figure 35-2 Time relationship between appearance of various biomarkers and clinical stage of Alzheimer disease. *, deposit of amyloid β in brain shown by measurement of $A\beta_{42}$ in CSF or amyloid imaging with PET; **, synaptic activity measured by ^{18}F -deoxyglucose (FDG)-PET or fMRI; ***, tau or phosphorylated tau in CSF; ****, measured by MRI. MCI: mild cognitive impairment. Cited from Sperling et al., 2011, with permission.

identification of the most appropriate test requires careful consideration.

4. USE OF LABORATORY TESTS TO PREVENT DISEASE

If it becomes possible to predict the occurrence of a neurodegenerative disease by laboratory tests before the initial symptom appears, preventing it might be possible.

For example, it was postulated that selegiline, the selective inhibitor of monoamine oxidase B, might have a **neuroprotective** effect in Parkinson disease, although this has not been confirmed (Olanow, 2009).

For the purpose of neuroprotection, it is necessary to find a disease-specific abnormality by noninvasive techniques a sufficiently long time in advance of the clinical onset. One way would be to use functional or chemical imaging of the brain. For example, attempts have been made to predict the progression of mild cognitive impairment (MCI) to

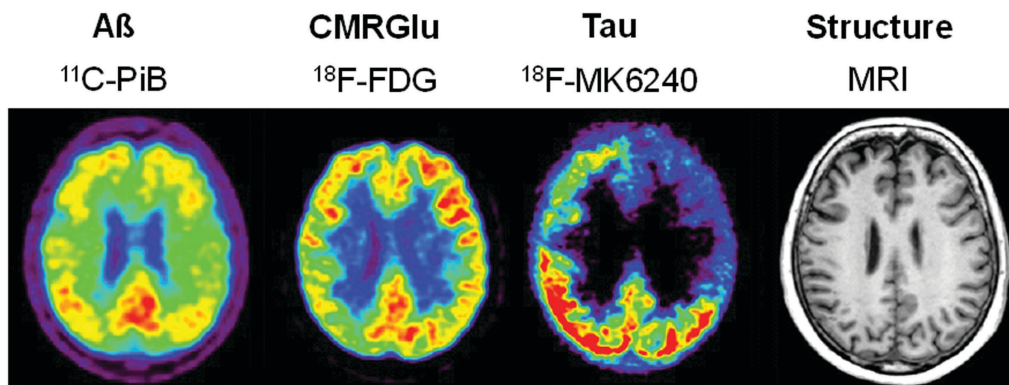


Figure 35-3 Accumulation of tau in the cerebral cortex shown by PET with ^{18}F -MK6240, in comparison with PET image of amyloid ($A\beta$) and cerebral glucose metabolic rate (CMR Glu) in a patient with mild Alzheimer disease. By courtesy of Dr. Kenji Ishii, Tokyo Metropolitan Institute of Gerontology.

Alzheimer disease (AD) by using amyloid imaging with PET (Kulichikhin et al, 2021; Sintini & Whitwell, 2021; Wolk et al, 2009) (Figure 35-1). It has now also become possible to image the neurofibrillary pathology in AD by tau imaging with PET (Okamura et al, 2014). Chemically, for example, there are studies of amyloid β 42, tau, and phosphorylated tau in the CSF for differentiating various kinds of dementia and for predicting the prognosis of MCI (Schoonenboom et al, 2012). It has been shown that amyloid PET and CSF biomarkers equally identified early AD (Palmqvist et al, 2015). In AD, it is known that deposition of amyloid in the brain, abnormality of synaptic functions, and tau-mediated degeneration are seen in advance of the clinical onset (Figures 35-2 and 35-3). Plasma neurofilament light or plasma phospho-tau 217 may be also a useful biomarker for neurodegenerative diseases like AD (Lewczuk et al, 2018; Palmqvist et al, 2020). If these markers are found to be elevated in subjects who carry the apolipoprotein E (APOE) ϵ 4 gene, it might suggest a high possibility that those subjects will develop dementia in the future (Bateman et al, 2012). It is hoped that these lines of research will advance further and lead to the prevention of intractable diseases.

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AFTERWORD: FOR THOSE WHO WISH TO STUDY NEUROLOGY

In a video symposium held during the annual congress of American Academy of Neurology in 2007, distinguished neurologists reviewed why they chose neurology for their specialty. The main reasons common to most of the speakers were (1) the diversity and mysteriousness of neurological symptoms and signs, (2) a thrilling process of thinking about sites of lesion and etiology based on neurological symptoms and signs, and (3) a close connection with rapidly advancing neuroscience. Basics of clinical neurology are the bedside observation of patients, and in fact many important research endeavors of neurology and neuroscience have originated from careful clinical observation.

Therefore, clinical neurology starts from comprehension of the symptoms and their time course, followed by confirmation of sign(s) found by neurologic examination, and then consideration of the most probable diagnosis based on the symptoms and signs. No matter how laboratory tests are advanced, neglect of symptoms and signs is not effective for clinical practice and research. In this sense, this book was arranged to serve as a bridge from basic sciences such as anatomy, physiology, pharmacology, and molecular biology to the understanding of clinical symptomatology.

As some of the statements in this book are based on the authors' own thoughts and experiences, different views, if any, are welcomed.

As clinical aspects of neurology were emphasized throughout this book, we might have given the impression that the history taking and the neurologic examination are almighty. However, most information on the neurological symptoms and signs is based on the observation and vast experience of many neurologists over a period of 150 years, and some of the symptoms and signs might have just been handed down from predecessors to successors. Therefore, some of the currently known neurological signs may not necessarily be meaningful (Johnston & Hauser, 2011). Since the brain is so complex, there is always more to learn about the generating mechanisms and diagnostic significance of neurological signs.

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